EGA IV, SECTIONS 16–18: SOURCE-ALIGNED ENGLISH READER

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Contents

§16. DIFFERENTIAL INVARIANTS. DIFFERENTIALLY SMOOTH MORPHISMS

In this section we present, in global form, some notions of differential calculus that are particularly useful in algebraic geometry. We pass over many developments classical in differential geometry (connections, infinitesimal transformations associated with a vector field, jets, etc.), although these notions admit particularly natural formulations in the setting of schemes. We likewise pass over here the phenomena special to characteristic $p > 0$ (some of which are studied, in the affine setting, in (**0**, 21)). For further material on the differential formalism for preschemes, the reader may consult Exposés II and VII of [**?**], as well as later chapters of this treatise. **IV-4** | 5

16.1. Normal invariants of an immersion

(16.1.1). Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be two ringed spaces, and let $f = (\psi, \theta) : Y \to X$ be a morphism of ringed spaces (**0**, 4.1.1) such that the homomorphism

$$\theta^{\#} : \psi^*(\mathscr{O}_X) \longrightarrow \mathscr{O}_Y$$

is surjective, so that $\mathscr{O}_Y$ is identified with a sheaf of quotient rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$. We can then endow $\psi^*(\mathscr{O}_X)$ with the $\mathscr{I}_f$-preadic filtration.

Definition (16.1.2). — The $\mathscr{O}_Y$-augmented sheaf of rings $\psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ is called the n-th *normal invariant* of f; the ringed space $(Y, \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1})$ is called the n-th *infinitesimal neighbourhood* of Y for the morphism f and is denoted by $Y_f^{(n)}$ or simply $Y^{(n)}$. The sheaf of graded rings associated to the sheaf of filtered rings $\psi^*(\mathscr{O}_X)$

$$\mathscr{G}r_\bullet(f) = \bigoplus_{n \geqslant 0} (\mathscr{I}_f^n / \mathscr{I}_f^{n+1}) \tag{16.1.2.1}$$

is called the sheaf of graded rings *associated with* f. The sheaf $\mathscr{G}r_1(f) = \mathscr{I}_f/\mathscr{I}_f^2$ is called the *conormal sheaf* of f (and is also denoted by $\mathscr{N}_{Y/X}$ when no confusion can result).

It is clear that the $\mathscr{O}_{Y^{(n)}} = \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$ (also denoted $\mathscr{O}_{Y_f^{(n)}}$) form a projective system of sheaves of rings on Y; for $n \leqslant m$, the transition homomorphism $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ identifies $\mathscr{O}_{Y^{(n)}}$ with the quotient of $\mathscr{O}_{Y^{(m)}}$ by the power $(\mathscr{I}_f/\mathscr{I}_f^{n+1})^m$ of the *augmentation ideal* of $\mathscr{O}_{Y^{(n)}}$, the kernel of $\phi_{0n} : \mathscr{O}_{Y^{(n)}} \to \mathscr{O}_Y$. The $Y^{(n)}$ therefore form an inductive system of ringed spaces, all having underlying space Y, and we have canonical morphisms of ringed spaces $h_n : Y^{(n)} \to X$ equal to (ψ, θ_n), where $\theta_n^{\#}$ is the canonical morphism $\psi^*(\mathscr{O}_X) \to \psi^*(\mathscr{O}_X)/\mathscr{I}_f^{n+1}$. It is clear that the sheaf $\mathscr{G}r_\bullet(f)$ is a sheaf of graded algebras over the sheaf of rings $\mathscr{O}_Y = \mathscr{G}r_0(f)$, and that the $\mathscr{G}r_k(f)$ are $\mathscr{O}_Y$-modules. **IV-4** | 6

As with every sheaf of filtered rings, we have a *canonical surjective homomorphism* of graded $\mathscr{O}_Y$-algebras

$$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_\bullet(f) \tag{16.1.2.2}$$

which coincides in degrees 0 and 1 with the identities.

Examples (16.1.3). —

(i) Suppose that X is a locally ringed space, Y is reduced to a single point y (endowed with a ring $\mathscr{O}_y$) and that, if $x = \psi(y)$, $\theta^{\#} : \mathscr{O}_x \to \mathscr{O}_y$ is a *surjective* homomorphism of rings having as kernel the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_x$. So the $\mathscr{O}_{Y^{(n)}}$ are identified with the rings $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$ and $\mathscr{G}r_\bullet(f)$ with the graded ring associated with the local ring $\mathscr{O}_x$ endowed with the $\mathfrak{m}_x$-preadic filtration.

(ii) Suppose that Y is a closed subset of an open subspace U of X and that $\mathscr{O}_Y$ is induced on Y by a quotient sheaf $\mathscr{O}_U/\mathscr{I}$, where $\mathscr{I}$ is an ideal of $\mathscr{O}_U$ such that $\mathscr{I}_x = \mathscr{O}_x$ for every $x \notin Y$; if X is a locally ringed space, suppose further that $\mathscr{I}_x \neq \mathscr{O}_x$ for $x \in Y$, so that $(Y, \mathscr{O}_Y)$ is again a locally ringed space.

Let $\psi_0 : Y \to U$ be the canonical injection and denote by $\theta_0 : \mathscr{O}_U \to (\psi_0)_*(\mathscr{O}_Y)$ the homomorphism such that $\theta_0^\#$ is the canonical homomorphism $\psi_0^*(\mathscr{O}_U) = \mathscr{O}_U|Y \to (\mathscr{O}_U/\mathscr{I})|Y$, so that $j_0 = (\psi_0, \theta_0) : Y \to U$ is a morphism of ringed spaces (and of locally ringed spaces if X is a locally ringed space); if $i : U \to X$ is the canonical injection (morphism of ringed spaces), $j = i \circ j_0$ is the morphism (ψ, θ) of Y to X where $\psi : Y \to X$ is the canonical injection and $\theta : \mathscr{O}_X \to \psi_*(\mathscr{O}_Y)$ is the homomorphism such that $\theta^\# = \theta_0^\#$. Since $\theta^\#$ is surjective we can apply the previous definitions; $\mathscr{O}_{Y^{(n)}}$ is equal to $\psi_0^*(\mathscr{O}_U/\mathscr{I}^{n+1})$, and we have $(\psi_0)_*(\mathscr{O}_{Y^{(n)}}) = \mathscr{O}_U/\mathscr{I}^{n+1}$, and $\mathscr{G}r_n(j) = \mathscr{G}r_n(j_0) = \psi_0^*(\mathscr{I}^n/\mathscr{I}^{n+1}) = j_0^*(\mathscr{I}^n/\mathscr{I}^{n+1})$.

(16.1.4). The example (16.1.3, (ii)) shows that in general the $\mathscr{O}_{Y^{(n)}}$ are *not canonically endowed with a structure of an $\mathscr{O}_Y$-module*, or *a fortiori* with a structure of an $\mathscr{O}_Y$-algebra. The data of such a structure is equivalent to the data of a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation morphism ϕ_{0n}; it is also equivalent to the data of a morphism of ringed spaces $(1_Y, \lambda_n) : Y^{(n)} \to Y$ left inverse to the canonical morphism $(1_Y, \phi_{0n}) : Y \to Y^{(n)}$.

Proposition (16.1.5). — *Let $f = (\psi, \theta) : Y \to X$ be an immersion of preschemes. Then:*

(i) *$\mathscr{G}r_\bullet(f)$ is a quasi-coherent graded $\mathscr{O}_Y$-algebra.* **IV-4** | 7

(ii) *The $Y^{(n)}$ are preschemes, canonically isomorphic to subpreschemes of X.*

(iii) *Every homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$, right inverse to the augmentation homomorphism ϕ_{0n}, makes $\mathscr{O}_{Y^{(n)}}$ and the $\mathscr{O}_{Y^{(k)}}$ for $k \leqslant n$ quasi-coherent $\mathscr{O}_Y$-algebras; the $\mathscr{O}_Y$-module structures induced from the above structures on the $\mathscr{G}r_k(f)$ for $k \leqslant n$ coincide with the ones defined in* (16.1.2).

Proof. (i) Since the question is local on X and Y, we can reduce to the case where Y is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$; since $\mathscr{O}_Y$ is the restriction to Y of $\mathscr{O}_X/\mathscr{I}$, assertion (i) is evident, and $Y^{(n)}$ is the closed subprescheme of X defined by the quasi-coherent ideal $\mathscr{I}^{n+1}$ of $\mathscr{O}_X$. Finally, to prove (iii), note that the data of λ_n makes the ideal $\mathscr{I}/\mathscr{I}^{n+1}$ of the augmentation ϕ_{0n} and its quotients $\mathscr{I}/\mathscr{I}^{k+1}$ ($1 \leqslant k \leqslant n$) $\mathscr{O}_Y$-modules; it suffices to prove by induction on k that the $\mathscr{I}/\mathscr{I}^{k+1}$ are quasi-coherent $\mathscr{O}_Y$-modules and that the quotient $\mathscr{O}_Y$-module structure thereby induced on $\mathscr{I}^k/\mathscr{I}^{k+1}$ is the same as that defined in (16.1.2). The second assertion is immediate, $\mathscr{I}^k/\mathscr{I}^{k+1}$ being killed by $\mathscr{I}/\mathscr{I}^{n+1}$; the first follows by induction on k, since it is trivial for $k = 1$ and $\mathscr{I}/\mathscr{I}^{k+1}$ is an extension of $\mathscr{I}/\mathscr{I}^k$ by $\mathscr{I}^k/\mathscr{I}^{k+1}$ (1.4.17). ☐

Corollary (16.1.6). — *Under the general hypotheses of* (16.1.5), *if the immersion f is locally of finite presentation then the $\mathscr{G}r_n(f)$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

Proof. Indeed, with the notation from the proof of (16.1.5), $\mathscr{I}$ is an ideal of finite type of $\mathscr{O}_X$ (1.4.7), therefore the $\mathscr{I}^n/\mathscr{I}^{n+1}$ are $\mathscr{O}_Y$-modules of finite type, hence the conclusion. ☐

Corollary (16.1.7). — *Under the general hypotheses of* (16.1.5), *let $g : X \to Y$ be a morphism of preschemes, left inverse to f. Then, for every n, the composite morphism $(1, \lambda_n) : Y^{(n)} \xrightarrow{h_n} X \xrightarrow{g} Y$ defines a homomorphism of sheaves of rings $\lambda_n : \mathscr{O}_Y \to \mathscr{O}_{Y^{(n)}}$ right inverse to the augmentation ϕ_{0n}, making $\mathscr{O}_{Y^{(n)}}$ a quasi-coherent $\mathscr{O}_Y$-algebra; via these homomorphisms, the transition homomorphisms $\phi_{nm} : \mathscr{O}_{Y^{(m)}} \to \mathscr{O}_{Y^{(n)}}$ ($n \leqslant m$) are homomorphisms of $\mathscr{O}_Y$-algebras. Also, if g is locally of finite type, then the $\mathscr{O}_{Y^{(n)}}$ are quasi-coherent $\mathscr{O}_Y$-modules of finite type.*

Proof. The first assertion follows immediately from the definitions and (16.1.5). On the other hand, if g is locally of finite type, then f is locally of finite presentation (1.4.3, (v)); the $\mathscr{G}r_n(f)$ being then quasi-coherent $\mathscr{O}_Y$-modules of finite type by (16.1.6), the same goes for the $\mathscr{O}_Y$-modules $\mathscr{I}/\mathscr{I}^{n+1}$, being extensions of a finite number of the $\mathscr{G}r_k(f)$ (**III**, 1.4.17). ☐

Proposition (16.1.8). — *Let X be a locally Noetherian prescheme and let $j : Y \to X$ be an immersion. Then the $Y^{(n)}$ are locally Noetherian preschemes, the $\mathscr{G}r_n(j)$ are coherent $\mathscr{O}_Y$-modules, and $\mathscr{G}r_\bullet(j)$ is a coherent sheaf of rings on the space Y.*

Proof. Everything is local on X and Y, so we reduce to the case where X is affine and j is a closed immersion. All the assertions are then evident except the last; this follows from the fact that, if A is a Noetherian ring and $\mathfrak{I}$ is an ideal of A, then $\mathrm{gr}^{\bullet}_{\mathfrak{I}}(A)$ is a Noetherian ring, taking into account the exactness of the functor ψ^* and (**0**, 5.3.7). □

Proposition (16.1.9). — *Let X be a prescheme, let $j : Y \to X$ be an immersion locally of finite presentation, and let y be a point of Y. The following conditions are equivalent:* **IV-4** | 8

(a) *There exists an open neighbourhood U of y in Y such that $j|U$ is a homeomorphism of U onto an open subset of X.*

(b) *There is an integer $n > 0$ such that the canonical homomorphism*

$$(\phi_{n-1,n})_y : \mathscr{O}_{Y^{(n)},y} \longrightarrow \mathscr{O}_{Y^{(n-1)},y}$$

is bijective.

(c) *There is an integer $n > 0$ such that $(\mathscr{G}r_n(j))_y = 0$.*

In addition, if the integer n satisfies (b) *or* (c)*, then there is a neighbourhood V of y in Y such that $\mathscr{G}r_m(j)|V = 0$ for $m \geqslant n$ and that $\phi_{nm}|V : \mathscr{O}_{Y^{(m)}}|V \to \mathscr{O}_{Y^{(n)}}|V$ is bijective for $m \geqslant n$.*

Proof. Since the question is local on Y, we can restrict ourselves to the case where j is a closed immersion, with Y defined by a quasi-coherent ideal *of finite type* $\mathscr{I}$ of $\mathscr{O}_X$. The equivalence of (b) and (c), for a given n, is immediate; also, since $\mathscr{I}^n/\mathscr{I}^{n+1}$ is an $\mathscr{O}_X$-module of finite type, there is an open neighbourhood U of y in X such that $\mathscr{I}^n|U = \mathscr{I}^{n+1}|U$ (**0**, 5.2.2), and hence $\mathscr{I}^n|U = \mathscr{I}^m|U$ for $m \geqslant n$, proving the last assertions. To prove that (a) implies (b), we can restrict ourselves to the case where the underlying space of Y is equal to the underlying space of X and where $\mathscr{I}$ is generated by finitely many sections over X: since $\mathscr{I}$ is contained in the nilradical $\mathscr{N}$ of $\mathscr{O}_X$ (**I**, 5.1.2), it is nilpotent, which proves (b). Finally, to prove that (b) implies (a), we can likewise restrict ourselves to the case where $\mathscr{I}^n = \mathscr{I}^{n+1}$; therefore, for every $y \in Y$, since $\mathscr{I}_y \subset \mathfrak{m}_y$, maximal ideal of $\mathscr{O}_{X,y}$, we must have $\mathscr{I}^n_y = 0$ because of Nakayama's lemma, since $\mathscr{I}_y$ is an ideal of finite type. The set of $x \in X$ such that $\mathscr{I}^n_x = 0$ is therefore an open subset U of X containing Y (**0**, 5.2.2); since on the other hand $\mathscr{I}_x \neq 0$ for $x \notin Y$, we must have $U = Y$. □

Corollary (16.1.10). — *For the restriction of the immersion j to a neighbourhood of y in Y to be an open immersion (in other words, for j to be a* local isomorphism *at the point y), it is necessary and sufficient that $(\mathscr{G}r_1(j))_y = (\mathscr{N}_{Y/X})_y = 0$.*

Proof. The condition is clearly necessary, and the previous reasoning applied to $n = 1$ proves that it is sufficient. □

Remark (16.1.11). —

(i) Under the conditions of the definition (16.1.1), the projective limit of the projective system $(\mathscr{O}_{Y^{(n)}}, \phi_{nm})$ of sheaves of rings on Y is called the *normal invariant of infinite order* of f, and is sometimes denoted by $\mathscr{O}_{Y^{(\infty)}}$. When X is a locally Noetherian prescheme and $j : Y \to X$ is a closed immersion, so that Y is a closed subprescheme of X defined by a coherent ideal $\mathscr{I}$, the sheaf $\mathscr{O}_{Y^{(\infty)}}$ is precisely the *formal completion* of $\mathscr{O}_X$ along Y (**I**, 10.8.4), and $Y^{(\infty)} = (Y, \mathscr{O}_{Y^{(\infty)}})$ is the formal prescheme obtained by *completing* X along Y (**I**, 10.8.5). In all cases, one may say that $Y^{(\infty)}$ is the *formal neighbourhood* of Y in X (for the morphism f). In the particular case we have just considered, it is the formal prescheme that is the inductive limit of the infinitesimal neighbourhoods of order n.

(ii) Note that for a morphism of preschemes $f = (\psi, \theta) : Y \to X$, it can happen that the homomorphism $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_Y$ is surjective without f being a local immersion and without f being injective. We obtain an example by taking Y to be a sum of preschemes Y_λ, all isomorphic to $\mathrm{Spec}(\mathscr{O}_x)$, where $x \in X$, and taking f to be the morphism equal to the canonical morphism on each Y_λ. **IV-4** | 9

16.2. Functorial properties of the normal invariants of an immersion

(16.2.1). Let $f = (\psi, \theta) : Y \to X$ and $f' = (\psi', \theta') : Y' \to X'$ be two morphisms of ringed spaces such that $\theta^{\#}$ and $\theta'^{\#}$ are surjective; consider a commutative diagram of morphisms of ringed spaces

(16.2.1.1)
$$\begin{tikzcd} Y \arrow[r, "f"] & X \\ Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \end{tikzcd}$$

Let $u = (\rho, \lambda), v = (\sigma, \mu)$. We have $\rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X))$, and hence a commutative diagram of homomorphisms of sheaves of rings on Y':

$$\begin{tikzcd} \rho^*(\psi^*(\mathscr{O}_X)) = \psi'^*(\sigma^*(\mathscr{O}_X)) \arrow[r, "\psi'^*(\mu^{\#})"] \arrow[d, "\rho^*(\theta^{\#})"'] & \psi'^*(\mathscr{O}_{X'}) \arrow[d, "\theta'^{\#}"] \\ \rho^*(\mathscr{O}_Y) \arrow[r, "\lambda^{\#}"'] & \mathscr{O}_{Y'} \end{tikzcd}$$

from which, if $\mathscr{I}$ and $\mathscr{I}'$ are the kernels of $\theta^{\#}$ and $\theta'^{\#}$, we conclude that $\psi'^*(\mu^{\#})(\rho^*(\mathscr{I})) \subset \mathscr{I}'$, by the exactness of the functor ρ^*. It follows immediately that, for every integer n, $\psi'^*(\mu^{\#})(\rho^*(\mathscr{I}^n)) \subset \mathscr{I}'^n$, which shows that $\psi'^*(\mu^{\#})$ defines, on passing to the quotients, a homomorphism of sheaves of rings

(16.2.1.2)
$$\nu_n : \rho^*(\psi^*(\mathscr{O}_X)/\mathscr{I}^{n+1}) \longrightarrow \psi'^*(\mathscr{O}_{X'})/\mathscr{I}'^{n+1}$$

and therefore a morphism of ringed spaces $w_n = (\rho, \nu_n) : Y'^{(n)} \to Y^{(n)}$ (which, for $n = 0$, is none other than u). It follows immediately from this definition that the diagrams

$$\begin{tikzcd} Y^{(n)} \arrow[r, "h_{mn}"] & Y^{(m)} \arrow[r, "h_m"] & X \\ Y'^{(n)} \arrow[u, "w_n"] \arrow[r, "h'_{mn}"'] & Y'^{(m)} \arrow[u, "w_m"] \arrow[r, "h'_m"'] & X' \arrow[u, "v"'] \end{tikzcd} \qquad (n \leqslant m)$$

(where the horizontal arrows are the canonical morphisms (16.1.2)) are commutative.

By passage to the quotients via the morphisms (16.2.1.2), and taking into account the exactness of the functor ρ^*, we obtain a di-homomorphism of graded algebras (relative to the homomorphism $\lambda^{\#} : \rho^*(\mathscr{O}_Y) \to \mathscr{O}_{Y'}$) **IV-4** | 10

(16.2.1.3)
$$\mathrm{gr}(u) : \rho^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f')$$

(or, if one prefers, a ρ-morphism (**0**, 3.5.1) $\mathscr{G}r_\bullet(f) \to \mathscr{G}r_\bullet(f')$), and, in particular, a di-homomorphism of conormal sheaves

$$\mathrm{gr}_1(u) : \rho^*(\mathscr{G}r_1(f)) \longrightarrow \mathscr{G}r_1(f').$$

It is also immediate that these homomorphisms give rise to a commutative diagram

(16.2.1.4)
$$\begin{tikzcd} \rho^*(\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{G}r_1(f))) \arrow[r] \arrow[d, "\mathbf{S}(\mathrm{gr}_1(u))"'] & \rho^*(\mathscr{G}r_\bullet(f)) \arrow[d, "\mathrm{gr}(u)"] \\ \mathbf{S}^\bullet_{\mathscr{O}_{Y'}}(\mathscr{G}r_1(f')) \arrow[r] & \mathscr{G}r_\bullet(f') \end{tikzcd}$$

where the horizontal arrows are the canonical homomorphisms (16.1.2.2).

Finally, if we have a commutative diagram of morphisms of ringed spaces

$$\begin{tikzcd}
Y \arrow[r, "f"] & X \\
Y' \arrow[u, "u"] \arrow[r, "f'"'] & X' \arrow[u, "v"'] \\
Y'' \arrow[u, "u'"] \arrow[r, "f''"'] & X'' \arrow[u, "v'"']
\end{tikzcd}$$

where $f'' = (\psi'', \theta'')$ is such that $\theta''^{\#}$ is surjective, and if w'_n and w''_n are defined, respectively, from u', v' and from $u'' = u \circ u', v'' = v \circ v'$, then $w''_n = w_n \circ w'_n$, as follows immediately from the definitions and from (**0**, 3.5.5); we have also $\mathrm{gr}(u'') = \mathrm{gr}(u') \circ \rho'^*(\mathrm{gr}(u))$ if $u' = (\rho', \lambda')$. Therefore we can say that $Y^{(n)}$ and $\mathscr{G}r_\bullet(f)$ *depend functorially* on f.

Proposition (16.2.2). — *With the notation and hypotheses of (16.2.1), suppose also that f, f', u, and v are morphisms of preschemes. We have:*

(i) *The morphisms $w_n : Y'^{(n)} \to Y^{(n)}$ are morphisms of preschemes.*

(ii) *If $Y' = Y \times_X X'$, with u and f' the canonical projections, and if f is an immersion or v is flat, then $Y'^{(n)} = Y^{(n)} \times_X X'$.*

(iii) *If $Y' = Y \times_X X'$ and if v is flat (resp. if f is an immersion), the homomorphism*

$$\mathrm{Gr}(u) = \mathrm{gr}(u) \otimes 1 : \mathscr{G}r_\bullet(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_\bullet(f')$$

is bijective (resp. surjective).

Proof.

(i) The hypotheses immediately imply that, for every $y' \in Y'$, $\rho^*_{y'}(\theta^{\#}_{\psi'(y')})$ is a *local* homomorphism (**I**, 1.6.2), so w_n is a morphism of preschemes (**I**, 2.2.1). **IV-4** | 11

(ii) and (iii) If f is an immersion, we may restrict to the case where f is a closed immersion, with Y defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_X$ and $Y^{(n)}$ by the ideal $\mathscr{I}^{n+1}$; the assertions then follow from (**I**, 4.4.5).

Second, suppose that v is flat; we can restrict ourselves to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(A')$ are affines, A' being a flat A-module; so $Y' = \mathrm{Spec}(B')$ where $B' = B \otimes_A A'$; moreover, if $\mathfrak{I}$ is the kernel of the homomorphism $A \to B$, the kernel $\mathfrak{I}'$ of $A' \to B'$ is identified with $\mathfrak{I} \otimes_A A'$ by flatness, and

$$\mathfrak{I}'^n / \mathfrak{I}'^{n+1} = (\mathfrak{I}^n / \mathfrak{I}^{n+1}) \otimes_A A'.$$

It follows at once from (**0**, 4.3.3) that the $\mathscr{O}_{Y'}$-module $\mathscr{I}'^n / \mathscr{I}'^{n+1}$ is equal to

$$\begin{aligned}
&\psi'^*(\sigma^*((\mathfrak{I}^n/\mathfrak{I}^{n+1})^\sim) \otimes_{\sigma^*(\mathscr{O}_X)} \mathscr{O}_{X'}) = \\
&\quad \psi'^*(\sigma^*((\mathfrak{I}^n/\mathfrak{I}^{n+1})^\sim)) \otimes_{\psi'^*(\sigma^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'}) = \rho^*(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^*(\psi^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'})
\end{aligned}$$

and in particular for $n = 0$, we have

$$\mathscr{O}_{Y'} = \rho^*(\mathscr{O}_Y) \otimes_{\rho^*(\psi^*(\mathscr{O}_X))} \psi'^*(\mathscr{O}_{X'})$$

whence we obtain a canonical isomorphism from $\mathscr{I}'^n / \mathscr{I}'^{n+1}$ onto

$$\rho^*(\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\rho^*(\mathscr{O}_Y)} \mathscr{O}_{Y'} = (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$$

which proves (iii). Set now $C_n = \Gamma(Y, \mathscr{O}_{Y^{(n)}})$ and $C'_n = \Gamma(Y', \mathscr{O}_{Y'^{(n)}})$. Since $Y^{(n)}$ and $Y'^{(n)}$ are affine schemes (16.1.5), the kernel $\mathfrak{K}_n$ (resp. $\mathfrak{K}'_n$) of the homomorphism $C_n \to C_{n-1}$ (resp. $C'_n \to C'_{n-1}$) is $\Gamma(Y, \mathscr{I}^n/\mathscr{I}^{n+1})$ (resp. $\Gamma(Y', \mathscr{I}'^n/\mathscr{I}'^{n+1})$); therefore we can deduce from the above results that $\mathfrak{K}'_n = \mathfrak{K}_n \otimes_A A'$. Now, we have a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathfrak{K}_n \otimes_A A' \arrow[r] \arrow[d, "r"] & C_n \otimes_A A' \arrow[r] \arrow[d, "s_n"] & C_{n-1} \otimes_A A' \arrow[r] \arrow[d, "s_{n-1}"] & 0 \\
0 \arrow[r] & \mathfrak{K}'_n \arrow[r] & C'_n \arrow[r] & C'_{n-1} \arrow[r] & 0
\end{tikzcd}$$

where the left vertical arrow is bijective and both rows are exact (since A' is a flat A-module). It follows by induction that s_n is bijective for every n: this is true by hypothesis for $n = 0$, and the general case follows from the five lemma. This proves the second assertion of (ii).

□

Corollary (16.2.3). — *Let $g : X \to Y$ and $u : Y' \to Y$ be two morphisms of preschemes, let $X' = X \times_Y Y'$, and let $g' : X' \to Y'$ and $v : X' \to X$ be the canonical projections. Let $f : Y \to X$ be a Y-section of X (and hence an immersion), and let $f' = f_{(Y')} : Y' \to X'$ be the Y'-section of X' deduced from f by the base change u. We have:*

(i) *The morphism $w_n : {Y'_{f'}}^{(n)} \to Y_f^{(n)}$ corresponding to f, f', u, v (16.2.1) and the canonical morphism $h'_n : {Y'_{f'}}^{(n)} \to X'$ jointly identify ${Y'_{f'}}^{(n)}$ with the product $Y_f^{(n)} \times_X X'$.*

(ii) *If we endow $\mathscr{O}_{Y_f^{(n)}}$ (resp. $\mathscr{O}_{{Y'_{f'}}^{(n)}}$) with the structure of an $\mathscr{O}_Y$-algebra defined by g (resp. with the structure of an $\mathscr{O}_{Y'}$-algebra defined by g') (16.1.5, (iii)), then the homomorphism of $\mathscr{O}_{Y'}$-algebras*

(16.2.3.1) $$\rho^*(\mathscr{O}_{Y_f^{(n)}}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{O}_{{Y'_{f'}}^{(n)}}$$

induced by the homomorphism ν_n (16.2.1.2) is bijective. Also, the homomorphism of $\mathscr{O}_{Y'}$-modules **IV-4** | 12

(16.2.3.2) $$\mathrm{Gr}_1(u) : \mathscr{G}r_1(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_1(f')$$

is bijective.

Proof.

(i) First note that $f' : Y' \to X'$ and $u : Y' \to Y$ jointly identify Y' with the product $Y \times_X X'$ (for the structure morphisms $f : Y \to X$ and $v : X' \to X$) (14.5.12.1). The conclusion of (i) now follows from (16.2.2, (ii)), the morphism g being an immersion.

(ii) The commutative diagram

$$\begin{tikzcd}
Y_f^{(n)} \arrow[d, "h_n"] & {Y'_{f'}}^{(n)} \arrow[l, "w_n"'] \arrow[d, "h'_n"] \\
X \arrow[d, "g"] & X' \arrow[l, "v"'] \arrow[d, "g'"] \\
Y & Y' \arrow[l, "u"']
\end{tikzcd}$$

identifies ${Y'_{f'}}^{(n)}$ with the product $Y_f^{(n)} \times_X X'$ and identifies X' with the product $X \times_Y Y'$; hence, by (**I**, 3.3.9), the morphisms $g' \circ h'_n$ and w_n identify ${Y'_{f'}}^{(n)}$ with the product $Y_f^{(n)} \times_Y Y'$. Since $Y_f^{(n)}$ (resp. ${Y'_{f'}}^{(n)}$) is the affine prescheme over Y (resp. over Y') associated with the $\mathscr{O}_Y$-algebra $\mathscr{O}_{Y_f^{(n)}}$ (resp. the $\mathscr{O}_{Y'}$-algebra $\mathscr{O}_{{Y'_{f'}}^{(n)}}$), the bijectivity of the canonical homomorphism (16.2.3.1) follows from (**II**, 1.5.2). Finally, the canonical homomorphism (16.2.3.1) is compatible with the augmentations $\mathscr{O}_{Y_f^{(n)}} \to \mathscr{O}_Y$ and $\mathscr{O}_{{Y'_{f'}}^{(n)}} \to \mathscr{O}_{Y'}$; since $\mathscr{O}_{Y_f^{(n)}}$ is the direct sum (as an $\mathscr{O}_Y$-module) of $\mathscr{O}_Y$ and the augmentation ideal $\mathscr{I}/\mathscr{I}^{n+1}$, it follows that the canonical homomorphism (16.2.3.1), restricted to $(\mathscr{I}/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'}$, maps this module bijectively onto $\mathscr{I}'/\mathscr{I}'^{n+1}$. For $n = 1$, this shows that $\mathrm{Gr}_1(u)$ is bijective.

□

We note that, under the hypotheses of (16.2.3), the homomorphisms $\mathrm{Gr}_n(u)$ are *surjective* in view of the above, but are not bijective in general for $n \geqslant 2$. However:

Corollary (16.2.4). — *Under the hypotheses of (16.2.3), suppose that $u : Y' \to Y$ is a flat morphism (resp. that the $\mathscr{G}r_n(f)$ are flat $\mathscr{O}_Y$-modules for $n \leqslant m$). Then the homomorphism*

$$\mathrm{Gr}_n(u) : \mathscr{G}r_n(f) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \longrightarrow \mathscr{G}r_n(f')$$

is bijective for all n (resp. for $n \leqslant m$).

Proof. If u is flat, then $v : X' \to X$ is flat by base change, and in this case we already know that $\mathrm{Gr}(u)$ is bijective (16.2.2, (iii)). If the $\mathscr{G}r_n(f)$ are flat for $n \leqslant m$, then we first see by induction on n that the same holds for $\mathscr{I}/\mathscr{I}^{n+1}$ for $n \leqslant m$, because of the exact sequences

$$0 \longrightarrow \mathscr{I}^n/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^{n+1} \longrightarrow \mathscr{I}/\mathscr{I}^n \longrightarrow 0$$

(**0**, 6.1.2); in addition, we have the commutative diagram **IV-4** | 13

$$\begin{tikzcd}
0 \arrow[r] & (\mathscr{I}^n/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^{n+1}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & (\mathscr{I}/\mathscr{I}^{n}) \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y'} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathscr{I}'^n/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^{n+1} \arrow[r] & \mathscr{I}'/\mathscr{I}'^{n} \arrow[r] & 0
\end{tikzcd}$$

in which the rows are exact (the first by flatness (**0**, 6.1.2)) and the last two vertical arrows are bijective by (16.2.2, (ii)); hence the conclusion. □

Remarks (16.2.5). —

(i) The reasoning of (16.2.2, (i)) still applies when the maps in (16.2.1.1) are morphisms of *locally ringed spaces* ($\mathbf{Err}_{\mathbf{III}}$, (**I**, 1.8.2)).

(ii) In (16.2.2, (ii)), the conclusion is no longer necessarily valid if we only suppose that v and f are morphisms of preschemes (f satisfying the condition of (16.1.1)). For example (with the notation of the proof of (16.2.2, (ii))), it can happen that $\mathfrak{J} = 0$ but the kernel $\mathfrak{J}'$ of $A' \to B' = B \otimes_A A'$ is not zero and that $B' \neq 0$, in which case we have $Y^{(n)} = Y$ for all n, but $Y'^{(n)} \neq Y'$. We have an example of this by taking $A = \mathbf{Z}$, $B = \mathbf{Q}$, $A' = \prod_{h=1}^{\infty}(\mathbf{Z}/m^h\mathbf{Z})$ where $m > 1$.

(16.2.6). Consider the special case of diagram (16.2.1.1) in which $X' = X$, v is the identity, X is a prescheme, Y is a subprescheme of X, Y' is a subprescheme of Y, and $f, u, f' = f \circ u$ are the canonical injections; by tensoring with $\mathscr{O}_{Y'}$ over $\rho^*(\mathscr{O}_Y)$, the di-homomorphism (16.2.1.3) gives a homomorphism of graded $\mathscr{O}_{Y'}$-algebras

(16.2.6.1) $$u^*(\mathscr{G}r_\bullet(f)) \longrightarrow \mathscr{G}r_\bullet(f').$$

On the other hand, we identify $\mathscr{O}_Y$ with $\psi^*(\mathscr{O}_X)/\mathscr{I}_f$ and $\mathscr{O}_{Y'}$ with $\rho^*(\mathscr{O}_Y)/\mathscr{I}_u$; since ρ^* is exact, we have $\rho^*(\mathscr{O}_Y) = \rho^*(\psi^*(\mathscr{O}_X))/\rho^*(\mathscr{I}_f) = \psi'^*(\mathscr{O}_X)/\rho^*(\mathscr{I}_f)$, and since $\mathscr{O}_{Y'}$ is also identified with $\psi'^*(\mathscr{O}_X)/\mathscr{I}_{f'}$, we see that $\mathscr{I}_u = \mathscr{I}_{f'}/\rho^*(\mathscr{I}_f)$. Thus, for every integer n, there is a canonical homomorphism $\mathscr{I}_{f'}^n/\mathscr{I}_{f'}^{n+1} \to \mathscr{I}_u^n/\mathscr{I}_u^{n+1}$, and hence a canonical homomorphism of graded $\mathscr{O}_{Y'}$-algebras

(16.2.6.2) $$\mathscr{G}r_\bullet(f') \longrightarrow \mathscr{G}r_\bullet(u).$$

Proposition (16.2.7). — *Let X be a prescheme, let Y be a subprescheme of X, let Y' be a subprescheme of Y, and let $j : Y' \to Y$ be the canonical injection. We then have an exact sequence of conormal sheaves ($\mathscr{O}_{Y'}$-modules)*

(16.2.7.1) $$j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0$$

where the arrows are the degree 1 components of the canonical homomorphisms (16.2.6.1) and (16.2.6.2).

Proof. Since the question is local, we may restrict to the case where $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(A/\mathfrak{J})$, and $Y' = \mathrm{Spec}(A/\mathfrak{K})$, where $\mathfrak{J}$ and $\mathfrak{K}$ are ideals of A such that $\mathfrak{J} \subset \mathfrak{K}$; everything reduces to seeing that the sequence of canonical morphisms $\mathfrak{J}/\mathfrak{K}\mathfrak{J} \to \mathfrak{K}/\mathfrak{K}^2 \to (\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2 \to 0$ is exact, **IV-4** | 14
which is immediate given that the image of $\mathfrak{J}/\mathfrak{K}\mathfrak{J}$ in $\mathfrak{K}/\mathfrak{K}^2$ is $(\mathfrak{J} + \mathfrak{K}^2)/\mathfrak{K}^2$ and that $(\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2$ is identified with $\mathfrak{K}/(\mathfrak{J} + \mathfrak{K}^2)$. □

It is easy to give examples where the sequence (16.2.7.1) extended on the left by 0 is not exact; with the above notation, it suffices to take $A = k[T]$, $\mathfrak{J} = AT^2$, $\mathfrak{K} = AT$, because then $(\mathfrak{J} + \mathfrak{K}^2)/\mathfrak{K}^2 = 0$ and $\mathfrak{J}/\mathfrak{K}\mathfrak{J} \neq 0$. See however (16.9.13) and (19.1.5) for some cases where the extended sequence is indeed exact.

16.3. Fundamental differential invariants of morphisms of preschemes

Definition (16.3.1). — Let $f : X \to S$ be a morphism of preschemes, and let $\Delta_f : X \to X \times_S X$ be the corresponding diagonal morphism, which is an immersion ((**I**, 5.3.9) and $\mathbf{Err_{III}}$, 10). We denote by $\mathscr{P}^n_f$ or $\mathscr{P}^n_{X/S}$, and call the *sheaf of principal parts of order n of the S-prescheme X*, the $\mathscr{O}_X$-augmented sheaf of rings that is the n-th normal invariant of Δ_f (16.1.2). We will also write $\mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S} = \varprojlim_n \mathscr{P}^n_{X/S}$, $\mathscr{G}r_n(\mathscr{P}_f) = \mathscr{G}r_n(\mathscr{P}_{X/S}) = \mathscr{G}r_n(\Delta_f)$ (16.1.2); the $\mathscr{O}_X$-module $\mathscr{G}r_1(\Delta_f)$, the augmentation ideal of $\mathscr{P}^1_{X/S}$, is also denoted by Ω^1_f or $\Omega^1_{X/S}$, and is called the $\mathscr{O}_X$-module of 1-*differentials* of f, or of X with respect to S, or of the S-prescheme X.

It follows from this definition that $\mathscr{P}^0_{X/S}$ is canonically identified with $\mathscr{O}_X$ (16.1.2).
We have (16.1.2.2) a canonical surjective homomorphism of graded $\mathscr{O}_X$-algebras

(16.3.1.1) $$\mathbf{S}^\bullet_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}).$$

It also follows from Definition (16.3.1) that for every open U of X we have $\mathscr{P}^n_{f|U} = \mathscr{P}^n_f|U$, $\mathscr{P}^\infty_{f|U} = \mathscr{P}^\infty_f|U$, $\mathscr{G}r_n(\mathscr{P}_{f|U}) = \mathscr{G}r_n(\mathscr{P}_f)|U$, and $\Omega^1_{f|U} = \Omega^1_f|U$ (in other words, the notions introduced are *local* on X).

(16.3.2). Let p_1, p_2 be the two canonical projections of the product $X \times_S X$; since Δ_f is an X-section of $X \times_S X$ for both p_1 and p_2, *each* of these morphisms defines, for every n, a homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ that is right inverse to the augmentation $\mathscr{P}^n_{X/S} \to \mathscr{O}_X$ (16.1.7); equivalently, this defines on $\mathscr{P}^n_{X/S}$ *two structures of quasi-coherent augmented $\mathscr{O}_X$-algebra*; the corresponding $\mathscr{O}_X$-module structures on $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are the same. We also have, by passing to the limit, two $\mathscr{O}_X$-algebra structures on $\mathscr{P}^\infty_{X/S}$.

(16.3.3). The morphism $s = (p_2, p_1)_S : X \times_S X \to X \times_S X$ is an *involutive automorphism* of $X \times_S X$, called the *canonical symmetry*, such that

(16.3.3.1) $$p_1 \circ s = p_2, \qquad p_2 \circ s = p_1, \qquad s \circ \Delta_f = \Delta_f.$$

If we set $s = (\rho, \lambda)$, $p_i = (\pi_i, \mu_i)$ $(i = 1, 2)$, and $\Delta_f = (\delta, \nu)$, then $\lambda^\#$ is an isomorphism from $\rho^*(\pi_1^*(\mathscr{O}_X))$ onto $\pi_2^*(\mathscr{O}_X)$, and $\delta^*(\lambda^\#)$ leaves invariant $\delta^*(\mathscr{O}_{X\times_S X})$ and the kernel $\mathscr{I}$ of the homomorphism $\nu^\# : \delta^*(\mathscr{O}_{X\times_S X}) \to \mathscr{O}_X$. Therefore:

Proposition (16.3.4). — *The homomorphism $\sigma = \delta^*(\lambda^\#)$ induced from s (and also called the* canonical symmetry*) is an involutive automorphism of the projective system $(\mathscr{P}^n_{X/S})$ of $\mathscr{O}_X$-augmented sheaves of* **IV-4** | 15
rings, and as a result also of the projective limit $\mathscr{P}^\infty_{X/S}$. This automorphism interchanges the two $\mathscr{O}_X$-algebra structures on $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$.

(16.3.5). In what follows, the two $\mathscr{O}_X$-algebra structures defined on the $\mathscr{P}^n_{X/S}$ and on $\mathscr{P}^\infty_{X/S}$ will play very different roles: *we will now agree, unless said otherwise, that when $\mathscr{P}^n_{X/S}$ or $\mathscr{P}^\infty_{X/S}$ is considered as an $\mathscr{O}_X$-algebra, it is the algebra structure induced by p_1.*

For every open U of X and every section $t \in \Gamma(U, \mathscr{O}_X)$, we will simply denote by $t.1$ or even t the image of t under the structure morphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^n_{X/S})$ (resp. $\Gamma(U, \mathscr{O}_X) \to \Gamma(U, \mathscr{P}^\infty_{X/S})$) (that is to say, the homomorphism corresponding to p_1).

Definition (16.3.6). — We denote by d^n_f, or $d^n_{X/S}$ (resp. d^∞_f, or $d^\infty_{X/S}$), or simply d^n (resp. d^∞), the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_f = \mathscr{P}^n_{X/S}$ (resp. $\mathscr{O}_X \to \mathscr{P}^\infty_f = \mathscr{P}^\infty_{X/S}$) induced by p_2. For every open U of X, and every $t \in \Gamma(U, \mathscr{O}_X)$, $d^n t$ (resp. $d^\infty t$) is called the *principal part of order n* (resp. *principal part of infinite order*) of t. We set $dt = d^1 t - t$, and we say that dt is the differential of t (an element of $\Gamma(U, \Omega^1_{X/S})$, also denoted $d_{X/S}(t)$).

It follows immediately [1] from this definition that we have

$$d(t_1t_2) = t_1dt_2 + t_2dt_1 \tag{16.3.6.1}$$

for every t_1, t_2 in $\Gamma(U, \mathscr{O}_X)$; in other words, d is a *derivation* of the ring $\Gamma(U, \mathscr{O}_X)$ with values in the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

In all the notation introduced in (16.3.1) and (16.3.6), we will sometimes replace S by A when $S = \mathrm{Spec}(A)$.

(16.3.7). Suppose in particular that $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine schemes, B then being an A-algebra. Then Δ_f corresponds to the canonical surjective homomorphism $\pi : B \otimes_A B \to B$ such that $\pi(b \otimes b') = bb'$, with kernel $\mathfrak{J} = \mathfrak{J}_{B/A}$ (**0**, 20.4.1); $\mathscr{P}^n_f$ is the structure sheaf of the prescheme $\mathrm{Spec}(P^n_{B/A})$, where

$$P^n_{B/A} = (B \otimes_A B)/\mathfrak{J}^{n+1};$$

$\mathscr{G}r_\bullet(\mathscr{P}_f)$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the graded B-module

$$\mathrm{gr}^\bullet_{\mathfrak{J}}(B \otimes_A B) = \bigoplus_{n \geqslant 0} (\mathfrak{J}^n/\mathfrak{J}^{n+1});$$

in particular $\Omega^1_f = \Omega^1_{X/S}$ is the quasi-coherent $\mathscr{O}_X$-module corresponding to the B-module of 1-differentials of B over A, $\Omega^1_{B/A}$ (**0**, 20.4.3). The projection morphisms $p_1 : X \times_S X \to X$ and $p_2 : X \times_S X \to X$ correspond to the two homomorphisms of rings $j_1 : B \to B \otimes_A B$ and $j_2 : B \to B \otimes_A B$ such that $j_1(b) = b \otimes 1$ and $j_2(b) = 1 \otimes b$; thus (by the convention of (16.3.5)), $P^n_{B/A}$ is always considered as a B-algebra via the composite homomorphism $B \xrightarrow{j_1} B \otimes_A B \to P^n_{B/A}$; the ring homomorphism $B \xrightarrow{j_2} B \otimes_A B \to P^n_{B/A}$ is denoted by $d^n_{B/A}$ and corresponds to $d^n_{X/S}$ acting on $\Gamma(X, \mathscr{O}_X)$; for every $t \in B$, dt is equal to $d_{B/A}t$, defined in (**0**, 20.4.6) (cf. $\mathbf{Err_{IV}}$, 11).

If $\pi_n : B \otimes_A B \to P^n_{B/A}$ is the canonical homomorphism, then the preceding definitions give

$$\pi_n(b \otimes b') = b \cdot \pi_n(1 \otimes b') = b \cdot d^n_{B/A}(b') \quad \text{for } b \in B, b' \in B. \tag{16.3.7.1}$$

Proposition (16.3.8). — *The image of the canonical homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$.* **IV-4 | 16**

Proof. We immediately reduce to the case where $X = \mathrm{Spec}(B)$ and $S = \mathrm{Spec}(A)$ are affine; the proposition then follows from (16.3.7.1), since π_n is surjective. We note that in general $d^n_{X/S}$ *is not surjective* (even for $n = 1$). □

Proposition (16.3.9). — *Suppose that $f : X \to S$ is a morphism locally of finite type. Then the $\mathscr{P}^n_f$ and the $\mathscr{G}r_n(\mathscr{P}_f)$ are quasi-coherent $\mathscr{O}_X$-modules of finite type.*

Proof. This follows from (16.1.6) and from the fact that Δ_f is locally of finite presentation (**I**, 4.3.1). □

[1] [Translator's note.] Locally, this is (**0**, 20.1.1).

16.4. Functorial properties of differential invariants

(16.4.1). Consider a commutative diagram of morphisms of preschemes

(16.4.1.1)
$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "w"] \end{tikzcd}$$

We deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & X' \arrow[l, "u"'] \arrow[d, "\Delta_{f'}"] \\ X \times_S X & X' \times_{S'} X' \arrow[l, "v"] \end{tikzcd}$$

where v is the composite morphism (**I**, 5.3.5) and (**I**, 5.3.15):

(16.4.1.2)
$$X' \times_{S'} X' \xrightarrow{(p'_1,p'_2)_S} X' \times_S X' \xrightarrow{u \times_S u} X \times_S X.$$

Thus u and v give, as explained in (16.2.1), homomorphisms of augmented sheaves of rings

(16.4.1.3)
$$\nu_n : \rho^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{X'/S'}$$

(where we put $u = (\rho, \lambda)$); these homomorphisms form a projective system and therefore give, in the limit, a homomorphism of augmented sheaves of rings

(16.4.1.4)
$$\nu_\infty : \rho^*(\mathscr{P}^\infty_{X/S}) \longrightarrow \mathscr{P}^\infty_{X'/S'};$$

on the other hand, by passing to the quotients, the homomorphisms ν_n give rise to a di-homomorphism of graded algebras (relative to $\lambda^\#$):

(16.4.1.5)
$$\mathrm{gr}(u) : \rho^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'}).$$

(16.4.2). If we have a commutative diagram **IV-4** | 17

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "u"'] \arrow[d, "f'"] & X'' \arrow[l, "u'"'] \arrow[d, "f''"] \\ S & S' \arrow[l, "w"] & S'' \arrow[l, "w'"] \end{tikzcd}$$

we deduce a commutative diagram

$$\begin{tikzcd} X \arrow[d, "\Delta_f"'] & X' \arrow[l, "u"'] \arrow[d, "\Delta_{f'}"] & X'' \arrow[l, "u'"'] \arrow[d, "\Delta_{f''}"] \\ X \times_S X & X' \times_{S'} X' \arrow[l, "v"] & X'' \times_{S''} X'' \arrow[l, "v'"] \end{tikzcd}$$

where v' is defined from u', w', f', f'' as v is from u, w, f, f'. We verify immediately that if $u'' = u \circ u'$ and $w'' = w \circ w'$, then the composite morphism $v \circ v'$ is equal to the morphism v'' deduced from u'', w'', f, f'' as v is deduced from u, w, f, f'. If we set $u' = (\rho', \lambda')$ and $u'' = (\rho'', \lambda'')$, it follows from (16.2.1) that the homomorphism $\nu''_n : \rho''^*(\mathscr{P}^n_{X/S}) \to \mathscr{P}^n_{X''/S''}$ is equal to the composite

$$\rho'^*(\rho^*(\mathscr{P}^n_{X/S})) \xrightarrow{\rho'^*(\nu^\#_n)} \rho'^*(\mathscr{P}^n_{X'/S'}) \xrightarrow{\nu'_n} \mathscr{P}^n_{X''/S''},$$

and there are analogous transitivity properties for the homomorphisms (16.4.1.4) and (16.4.1.5), allowing us to say that $\mathscr{P}^n_{X/S}$, $\mathscr{P}^\infty_{X/S}$, and $\mathscr{G}r_\bullet(\mathscr{P}_{X/S})$ *depend functorially on* f.

(16.4.3). We verify immediately (for example, by reducing to the affine case using (16.3.7)) that, with the notation of (16.4.1), the diagram

(16.4.3.1)
$$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d] & \mathscr{O}_{X'} \arrow[d] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

is commutative, where the vertical arrows define the algebra structures chosen in (16.3.5) (that is, those coming from the first projections); the same goes for the diagram

(16.4.3.2)
$$\begin{tikzcd} \rho^*(\mathscr{O}_X) \arrow[r, "\lambda^\#"] \arrow[d, "\rho^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ \rho^*(\mathscr{P}^n_{X/S}) \arrow[r, "\nu_n"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

the vertical arrows here define the algebra structures coming from the second projections; besides, **IV-4** | 18
if σ and σ' are the canonical symmetries corresponding to f and f' (16.3.4), we have

$$\nu_n \circ \rho^*(\sigma) = \sigma' \circ \nu_n$$

which carries one of the preceding diagrams to the other. We therefore deduce from (16.4.3.1) a canonical homomorphism of *augmented $\mathscr{O}_{X'}$-algebras*

(16.4.3.3)
$$P^n(u) : u^*(\mathscr{P}^n_{X/S}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \mathscr{P}^n_{X'/S'}$$

and it follows from (16.4.3.2) that the diagram

(16.4.3.4)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "u^*(d^n_{X/S})"'] & \mathscr{O}_{X'} \arrow[d, "d^n_{X'/S'}"] \\ u^*(\mathscr{P}^n_{X/S}) \arrow[r, "P^n(u)"'] & \mathscr{P}^n_{X'/S'} \end{tikzcd}$$

is commutative. We thus obtain a homomorphism of *graded $\mathscr{O}_{X'}$-algebras*

(16.4.3.5)
$$\mathrm{Gr}_\bullet(u) : u^*(\mathscr{G}r_\bullet(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X'/S'})$$

and in particular a homomorphism of $\mathscr{O}_{X'}$-modules

(16.4.3.6)
$$\mathrm{Gr}_1(u) : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/S'}$$

giving rise to a commutative diagram

(16.4.3.7)
$$\begin{tikzcd} \mathscr{O}_{X'} \arrow[r, "\mathrm{id}"] \arrow[d, "d_{X/S}\otimes 1"'] & \mathscr{O}_{X'} \arrow[d, "d_{X'/S'}"] \\ \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \arrow[r] & \Omega^1_{X'/S'} \end{tikzcd}$$

(16.4.4). When $S = \mathrm{Spec}(A)$, $S' = \mathrm{Spec}(A')$, $X = \mathrm{Spec}(B)$, $X' = \mathrm{Spec}(B')$ are affine, so that we have a commutative diagram of ring homomorphisms

$$\begin{tikzcd} B \arrow[r] & B' \\ A \arrow[r] \arrow[u] & A' \arrow[u] \end{tikzcd}$$

the image of $\mathfrak{J}_{B/A}$ in $B' \otimes_{A'} B'$ is contained in $\mathfrak{J}_{B'/A'}$, and the homomorphism ν_n corresponds to the homomorphism of rings $P^n_{B/A} \to P^n_{B'/A'}$ induced from the homomorphism $B \otimes_A B \to B' \otimes_{A'} B'$ by passing to quotients. The homomorphism (16.4.3.6) corresponds to the homomorphism defined in (**0**, 20.5.4.1), and the commutative diagram (16.4.3.7) to the diagram (**0**, 20.5.4.2).

Proposition (16.4.5). — *Suppose that $X' = X \times_S S'$, f' and u the canonical projections. Then the canonical homomorphisms $P^n(u)$ (16.4.3.3) and $\mathrm{Gr}_1(u)$ (16.4.3.6) are bijective.* **IV-4** | 19

Proof. We have $X' \times_{S'} X' = (X \times_S X) \times_S S'$, and it suffices to apply (16.2.3, (ii)), replacing g by the first projection $p_1 : X \times_S X \to X$ and f by the diagonal Δ_f. □

We note that under the hypotheses of (16.4.5) the homomorphism $\mathrm{Gr}_\bullet(u)$ (16.4.3.5) is *surjective*, but not bijective in general. However (16.2.4):

Corollary (16.4.6). — *Under the hypotheses of (16.4.5), suppose in addition that $w : S \to S'$ is flat (resp. that $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are flat $\mathscr{O}_X$-modules for $n \leqslant m$); then the homomorphism*

$$\mathrm{Gr}_n(u) : u^*(\mathscr{G}r_n(\mathscr{P}_{X/S})) \longrightarrow \mathscr{G}r_n(\mathscr{P}_{X'/S'})$$

is bijective for each n (resp. for $n \leqslant m$).

Proof. Indeed, if w is flat, then so is $v : X' \times_{S'} X' \to X \times_S X$, so the conclusion follows from (16.2.4). □

(16.4.7). Let S be a prescheme, let $\mathscr{E}$ be a quasi-coherent $\mathscr{O}_S$-module, and set $X = \mathbf{V}(\mathscr{E})$ (**II**, 1.7.8), the vector bundle associated with $\mathscr{E}$, equal to $\mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$. Let $f : X \to S$ be the structure morphism. For every open U of S and every section $t \in \Gamma(U, \mathscr{E})$, t is identified with a section of $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ over U; let t' be its image in $\Gamma(f^{-1}(U), \mathscr{O}_X) = \Gamma(U, f_*(\mathscr{O}_X)) = \Gamma(U, \mathbf{S}_{\mathscr{O}_S}(\mathscr{E}))$, and set

$$\delta(t) = d^n_{X/S}(t') - t' \in \Gamma(f^{-1}(U), \mathscr{P}^n_{X/S}); \tag{16.4.7.1}$$

it is clear that δ is a di-homomorphism of modules (corresponding to the homomorphism of rings $\Gamma(U, \mathscr{O}_S) \to \Gamma(f^{-1}(U), \mathscr{O}_X)$) from $\Gamma(U, \mathscr{E})$ to $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$, whose image moreover belongs to the augmentation ideal of $\Gamma(f^{-1}(U), \mathscr{P}^n_{X/S})$. We deduce (by varying U) a canonical homomorphism of $\mathscr{O}_X$-algebras

$$f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{P}^n_{X/S} \tag{16.4.7.2}$$

and in view of the preceding remark, if $\mathscr{K}$ is the kernel ideal of the augmentation $\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \to \mathscr{O}_S$, the image of $\mathscr{K}^{n+1}$ under (16.4.7.2) is zero; hence, after factoring by $\mathscr{K}^{n+1}$, we finally obtain a canonical homomorphism

$$\delta_n : f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E})/\mathscr{K}^{n+1}) \longrightarrow \mathscr{P}^n_{X/S}. \tag{16.4.7.3}$$

Proposition (16.4.8). — *Under the conditions of (16.4.7), the homomorphisms δ_n are bijective and form a projective system of isomorphisms; we deduce an isomorphism of graded $\mathscr{O}_X$-algebras*

$$f^*(\mathbf{S}^\bullet_{\mathscr{O}_S}(\mathscr{E})) \longrightarrow \mathscr{G}r_\bullet(\mathscr{P}_{X/S}). \tag{16.4.8.1}$$

Proof. The fact that homomorphisms (16.4.7.3) form a projective system follows immediately from their definition. To prove they are isomorphisms, it suffices to prove that (16.4.8.1) is an isomorphism, since the filtrations on both sides of (16.4.7.3) are finite (Bourbaki, *Alg. comm.*, chap. III, §2, no 8, Corollary 3 to Theorem 1). To do this, consider the split exact sequence of $\mathscr{O}_S$-modules **IV-4** | 20

$$0 \longrightarrow \mathscr{E} \xrightarrow{\ u\ } \mathscr{E} \oplus \mathscr{E} \xrightarrow{\ v\ } \mathscr{E} \longrightarrow 0 \tag{16.4.8.2}$$

where, for every pair of sections s, t of $\mathscr{E}$ over an open U of S, we take $u(s) = (-s, s)$ and $v(s,t) = s + t$. We have

$$X \times_S X = \mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E}) \otimes_{\mathscr{O}_S} \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})) = \mathrm{Spec}(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}))$$

((**II**, 1.4.6) and (**II**, 1.7.11)), and the diagonal morphism $X \to X \times_S X$ corresponds (**II**, 1.2.7) to the homomorphism of $\mathscr{O}_S$-algebras $\mathbf{S}(v) : \mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E}) \to \mathbf{S}_{\mathscr{O}_S}(\mathscr{E})$ (**II**, 1.7.4); thus, if $\mathscr{I}$ is the kernel of this homomorphism, then

$$\mathscr{P}^n_{X/S} = f^*(\mathbf{S}_{\mathscr{O}_S}(\mathscr{E} \oplus \mathscr{E})/\mathscr{I}^{n+1}).$$

The proposition now will be a consequence of the following lemma: □

Lemma (16.4.8.3). — *Let Y be a ringed space, $0 \to \mathscr{F}' \xrightarrow{u} \mathscr{F} \xrightarrow{v} \mathscr{F}'' \to 0$ an exact sequence of $\mathscr{O}_Y$-modules such that each point $y \in Y$ has an open neighbourhood V such that the sequence $0 \to \mathscr{F}'|V \to \mathscr{F}|V \to \mathscr{F}''|V \to 0$ is split. Let $\mathscr{I}$ be the kernel ideal of $\mathbf{S}(v)$:*

$$\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}) \longrightarrow \mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}''),$$

and let $\mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$ be the graded $\mathscr{O}_Y$-algebra associated to the $\mathscr{O}_Y$-algebra $\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})$ endowed with the $\mathscr{I}$-preadic filtration. *Then the homomorphism of graded $\mathscr{O}_Y$-algebras*

$$\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'') \longrightarrow \mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})) \tag{16.4.8.4}$$

(where the left-hand side is the graded tensor product of symmetric $\mathscr{O}_Y$-algebras endowed with their canonical gradings (**II**, 1.7.4) *and* (**II**, 2.1.2)*), induced by the canonical injection*

$$\mathscr{F}' \longrightarrow \mathscr{I} = \mathrm{gr}^1_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F})),$$

is bijective.

Proof. The injection $\mathscr{F}' \to \mathscr{I}$ indeed canonically gives a homomorphism of graded $\mathscr{O}_Y$-algebras $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \to \mathrm{gr}^\bullet_{\mathscr{I}}(\mathbf{S}_{\mathscr{O}_Y}(\mathscr{F}))$, and since the right-hand side is by definition a graded $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$-algebra, we obtain the canonical homomorphism (16.4.8.4) by tensoring the preceding one with $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$. To prove the lemma we can, being a local problem, restrict to the case where $\mathscr{F} = \mathscr{F}' \oplus \mathscr{F}''$, u and v the canonical homomorphisms. Then the graded algebra $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F})$ is canonically identified with the graded tensor product $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$ (**II**, 1.7.4), and it is immediate that $\mathscr{I}$ is therefore the ideal $\mathscr{I}' \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where $\mathscr{I}'$ is the augmentation ideal of $\mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}')$, that is to say the (direct) sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant 1$. We conclude that $\mathscr{I}^n = \mathscr{I}'^n \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, where this time $\mathscr{I}'^n$ is the direct sum of the $\mathbf{S}^m_{\mathscr{O}_Y}(\mathscr{F}')$ for $m \geqslant n$; we therefore have $\mathscr{I}^n/\mathscr{I}^{n+1} = \mathbf{S}^n_{\mathscr{O}_Y}(\mathscr{F}') \otimes_{\mathscr{O}_Y} \mathbf{S}^\bullet_{\mathscr{O}_Y}(\mathscr{F}'')$, which proves that (16.4.8.4) is bijective. □

Having proved the lemma, it remains to see that the homomorphism (16.4.8.1) is indeed the image under f^* of the homomorphism (16.4.8.4) corresponding to the exact sequence (16.4.8.2); this follows readily from the definitions of u (16.4.8.2) and δ (16.4.7.1), together with the definitions of the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ and of $d^n_{X/S}$ (16.3.5) and (16.4.3.6). **IV-4** | 21

In particular:

Corollary (16.4.9). — *Under the conditions of (16.4.7), we have a canonical isomorphism*

$$\mathrm{gr}_1(\delta) : f^*(\mathscr{E}) \simeq \Omega^1_{X/S}. \tag{16.4.9.1}$$

Corollary (16.4.10). — *If $S = \mathrm{Spec}(A)$, $\mathscr{E} = \mathscr{O}_S^m$, so that*

$$X = \mathrm{Spec}(A[T_1, \ldots, T_m]),$$

then $\mathscr{P}^n_{X/S}$ is canonically identified with the $\mathscr{O}_X$-algebra corresponding to the quotient $A[T_1, \ldots, T_m]$-algebra
$A[T_1, \ldots, T_m, U_1, \ldots, U_m]/\mathfrak{K}^{n+1}$, where the U_i ($1 \leqslant i \leqslant m$) are m new indeterminates and $\mathfrak{K}$ is the ideal generated by $U_1, \ldots, U_m$.

We thus recover in particular the structure of $\Omega^1_{X/S}$ in this case (**0**, 20.5.13).

Moreover, $d^n_{X/S}$ then sends a polynomial $F(T_1, \ldots, T_m)$ to the class modulo $\mathfrak{K}^{n+1}$ of $F(T_1 + U_1, \ldots, T_m + U_m)$, as follows from the definition (16.4.7.1).

Proposition (16.4.11). — *Let $f : X \to S$ be a morphism, let $g : S \to X$ be an S-section of X, and let $S^{(n)}$ be the n-th infinitesimal neighbourhood of S with respect to the immersion g (16.1.2). Then there exists a unique isomorphism of $\mathscr{O}_S$-algebras*

$$\varpi_n : g^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{O}_{S_g^{(n)}} \tag{16.4.11.1}$$

(via the $\mathscr{O}_S$-algebra structure on $\mathscr{O}_{S_g^{(n)}}$ defined by f (16.1.7)), making the diagram

(16.4.11.2)
$$\begin{tikzcd} \mathscr{O}_S = g^*(\mathscr{O}_X) \arrow[rr, "\lambda_n"] \arrow[dr, "g^*(d^n_{X/S})"'] & & \mathscr{O}_{S_g^{(n)}} \\ & g^*(\mathscr{P}^n_{X/S}) \arrow[ur, "\varpi_n"'] & \end{tikzcd}$$

commutative (where λ_n is the structure morphism).

Proof. In light of (**I**, 5.3.7), where we replace X, Y, S by S, X, S respectively and f by g, the diagrams

(16.4.11.3)
$$\begin{tikzcd} S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\ X \arrow[r, "(g\circ f, 1_X)_S"'] & X \times_S X \end{tikzcd} \qquad \begin{tikzcd} S \arrow[r, "g"] \arrow[d, "g"'] & X \arrow[d, "\Delta_f"] \\ X \arrow[r, "(1_X, g\circ f)_S"'] & X \times_S X \end{tikzcd}$$

identify S with the product of the $(X \times_S X)$-preschemes X and X via the morphisms Δ_f and $(g \circ f, 1_X)_S$ (resp. $(1_X, g \circ f)_S$). On the other hand, the diagrams **IV-4** | 22

(16.4.11.4)
$$\begin{tikzcd} X \arrow[r, "(g\circ f, 1_X)_S"] \arrow[d, "f"'] & X \arrow[d, "p_1"] \\ S \arrow[r, "g"'] & X \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "(1_X, g\circ f)_S"] \arrow[d, "f"'] & X \arrow[d, "p_2"] \\ S \arrow[r, "g"'] & X \end{tikzcd}$$

identify X with the product of the X-preschemes S and $X \times_S X$ via the morphisms g and p_1 (resp. p_2) (a special case of the associativity formula (**I**, 3.3.9.1)). We can say that Δ_f, considered as an X-section of $X \times_S X$ (relative to p_1 or p_2) plays the role of a *universal section* for the S-sections of X: each such section g is indeed obtained from it by the *base change* $(g \circ f, 1_X)_S : X \to X \times_S X$. The definition of the homomorphism ϖ_n and the fact that it is bijective therefore follow from these remarks and from (16.2.3, (ii)) applied to the first diagram (16.4.11.4). The commutativity of diagram (16.4.11.2) likewise follows from (16.2.3, (ii)), this time applied to the second diagram (16.4.11.4). To explain ϖ_n, we can restrict ourselves to the case where g is a closed immersion: Indeed, for every $s \in S$, there is an open neighbourhood W of s in S such that $g(W)$ is closed in an open set U of X, and it is clear that $g|W$ is a W-section of the morphism $U \cap f^{-1}(W) \to W$, the restriction of f. We can then suppose that S is a closed subprescheme of X defined by a quasi-coherent ideal $\mathscr{K}$. Then the preceding definitions show that if W is an open of S, t is a section of $\mathscr{O}_X$ over $f^{-1}(W)$, $\varpi_n(d^n t|W)$ is equal to the canonical image of t in $\Gamma(W, (\mathscr{O}_X/\mathscr{K}^{n+1})|W)$. The uniqueness of ϖ_n then follows since the image of $\mathscr{O}_X$ under $d^n_{X/S}$ generates the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$ (16.3.8). □

Corollary (16.4.12). — *Let k be a field, X a k-prescheme, x a point of X* rational over k. *Then $(\mathscr{P}^n_{X/k})_x \otimes_{\mathscr{O}_x} k(x)$ is canonically isomorphic (as an augmented $k(x)$-algebra) to $\mathscr{O}_x/\mathfrak{m}_x^{n+1}$.*

Proof. It suffices to consider the unique k-section g of X such that $g(\mathrm{Spec}(k)) = \{x\}$. □

Corollary (16.4.13). — *Let $f : X \to S$ be a morphism, s a point of S, $X_s = X \times_S \mathrm{Spec}(k(s))$ the fibre of f in s. If $x \in X_s$ is* rational over $k(s)$, *$(\mathscr{P}^n_{X/S})_x \otimes_{\mathscr{O}_s} k(s)$ is canonically isomorphic to $\mathscr{O}_{X_s,x}/\mathfrak{m}'^{n+1}_x$, where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x}$; more precisely, this isomorphism sends $(d^n t)_x \otimes 1$* (where t is a section of $\mathscr{O}_X$ over an open neighbourhood of x in X) *to the class of $t_x \otimes 1$* modulo $\mathfrak{m}'^{n+1}_x$.

Proof. This follows from (16.4.5) and (16.4.12). □

The preceding corollaries justify the terminology "sheaf of principal parts of order n".

Proposition (16.4.14). — *Let $\rho : A \to B$ be a homomorphism of rings, and let S be a multiplicative subset of B. Then the canonical homomorphisms*

$$S^{-1}P^n_{B/A} \longrightarrow P^n_{S^{-1}B/A} \tag{16.4.14.1}$$

deduced from the canonical homomorphisms $P^n_{B/A} \to P^n_{S^{-1}B/A}$ (16.4.4), form a projective system and are bijective. **IV-4** | 23

Proof. It suffices to remark that $S^{-1}((B \otimes_A B)/\mathfrak{I}^{n+1}) = S^{-1}(B \otimes_A B)/(S^{-1}\mathfrak{I})^{n+1}$ by flatness, and that $S^{-1}(B \otimes_A B) = (S^{-1}B) \otimes_A (S^{-1}B)$ (**I**, 1.3.4). ☐

Corollary (16.4.15). — *The notation being that of (16.4.14), let R be a multiplicative subset of A such that $\rho(R) \subset S$. Then we have canonical isomorphisms*

$$S^{-1}P^n_{B/A} \simeq P^n_{S^{-1}B/R^{-1}A} \tag{16.4.15.1}$$

forming a projective system.

Proof. It evidently suffices to define canonical isomorphisms

$$P^n_{S^{-1}B/A} \simeq P^n_{S^{-1}B/R^{-1}A} \tag{16.4.15.2}$$

that is to say that we reduce to the case where $\rho(R)$ consists of *invertible* elements of B. But then the isomorphism (16.4.15.2) is simply induced by the canonical isomorphism $B \otimes_A B \to B \otimes_{R^{-1}A} B$ by passing to quotients (**I**, 1.5.3). ☐

Corollary (16.4.16). — *Let $f : X \to S$ be a morphism of preschemes, x a point of X, $s = f(x)$. Then we have canonical isomorphisms*

$$(\mathscr{P}^n_{X/S})_x \simeq P^n_{\mathscr{O}_x/\mathscr{O}_s} \tag{16.4.16.1}$$

forming a projective system.

We deduce corresponding isomorphisms for the associated graded rings, and in particular a canonical isomorphism

$$(\Omega^1_{X/S})_x \simeq \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}. \tag{16.4.16.2}$$

Corollary (16.4.17). — *Let k be a field, K the field of rational functions $k(T_1, \dots, T_r)$. Then, for every integer n, the homomorphism from $K[U_1, \dots, U_r]$ (U_i indeterminates) to $P^n_{K/k}$ which sends U_i to $d^nT_i - T_i.1$ is surjective and induces an isomorphism from the quotient $K[U_1, \dots, U_r]/\mathfrak{m}^{n+1}$ (where $\mathfrak{m}$ is the ideal generated by the U_i) to $P^n_{K/k}$.*

Proof. This follows from (16.4.8), (16.4.10) and (16.4.14), where we take $A = k$, $B = k[T_1, \dots, T_r]$ and $S = B - \{0\}$. ☐

We thus recover the fact that the dT_i form a basis of the K-vector space $\Omega^1_{K/k}$ (**0**, 20.5.10).

(16.4.18). Let $f : X \to Y$, $g : Y \to Z$ be two morphisms of preschemes, and consider the canonical homomorphisms of augmented $\mathscr{O}_X$-algebras (16.4.3.3)

$$g_{X/Y/Z} : \mathscr{P}^n_{X/Z} \longrightarrow \mathscr{P}^n_{X/Y} \tag{16.4.18.1}$$

$$f_{X/Y/Z} : f^*(\mathscr{P}^n_{Y/Z}) \longrightarrow \mathscr{P}^n_{X/Z}. \tag{16.4.18.2}$$

Then $g_{X/Y/Z}$ is surjective, and its kernel is the sheaf of ideals generated by the image under $f_{X/Y/Z}$ of the augmentation ideal of $f^*(\mathscr{P}^n_{Y/Z})$.

Proof. First note that $g_{X/Y/Z}$ corresponds to the case in (16.4.3.3) where $X' = X$, $S' = Y$ and $S = Z$, $u = 1_X$, and $f_{X/Y/Z}$ to the case where we replace X', X, S, S' by X, Y, Z, Z respectively and u, f by f, g respectively. **IV-4** | 24

We have a commutative diagram (**I**, 3.5)

(16.4.18.3)
$$\begin{tikzcd} X \arrow[r, "\Delta_f"] \arrow[dr, "f"'] & X \times_Y X \arrow[r, "j"] \arrow[d, "p"] & X \times_Z X \arrow[d, "f \times_Z f"] \\ & Y \arrow[r, "\Delta_g"'] & Y \times_Z Y \end{tikzcd}$$

where $j = (1_X, 1_X)_Z$ is an immersion, $j \circ \Delta_f = \Delta_{g \circ f}$, and p is the structure morphism. Since we can restrict ourselves to the case where X, Y and Z are affine, we can suppose that the immersions Δ_f, Δ_g and j are closed, so that $\mathscr{O}_X$ and $\mathscr{O}_{X \times_Y X}$ are identified respectively with $\mathscr{O}_{X \times_Z X}/\mathscr{I}$ and $\mathscr{O}_{X \times_Z X}/\mathscr{L}$, where $\mathscr{L} \supset \mathscr{I}$ are two quasi-coherent ideals corresponding respectively to the immersions $\Delta_{g \circ f}$ and j. The $\mathscr{O}_X$-algebra $\mathscr{P}^n_{X/Z}$ is identified with $\mathscr{O}_{X \times_Z X}/\mathscr{I}^{n+1}$, and $\mathscr{P}^n_{X/Y}$ is identified with $\mathscr{O}_{X \times_Y X}/(\mathscr{I}/\mathscr{L})^{n+1}$, which is to say with $\mathscr{O}_{X \times_Z X}/(\mathscr{I}^{n+1} + \mathscr{L})$, and therefore with the quotient of $\mathscr{P}^n_{X/Z}$ by $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$. But we know (*loc. cit*) that if p and j make $X \times_Y X$ the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$, so if $\mathscr{O}_Y$ is identified to $\mathscr{O}_{Y \times_Z Y}/\mathscr{K}$, where $\mathscr{K}$ is the ideal corresponding to Δ_g, $\mathscr{L}$ is equal to $(f \times_Z f)^*(\mathscr{K}).\mathscr{O}_{X \times_Z X}$ (**I**, 4.4.5). Since $(\mathscr{I}^{n+1} + \mathscr{L})/\mathscr{I}^{n+1}$ is the ideal of $\mathscr{P}^n_{X/Z}$ generated by the image of $\mathscr{L}$, we deduce the proposition. □

Corollary (16.4.19). — *With the notation of (16.4.18), we have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.19.1)
$$f^*(\Omega^1_{Y/Z}) \xrightarrow{f_{X/Y/Z}} \Omega^1_{X/Z} \xrightarrow{g_{X/Y/Z}} \Omega^1_{X/Y} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 5.7.1).

Proposition (16.4.20). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ a closed immersion, $\mathscr{K}$ the quasi-coherent sheaf of ideals of $\mathscr{O}_Y$ corresponding to j. It follows that $\mathscr{P}^n_{X/Y} = \mathscr{O}_X = \mathscr{O}_Y/\mathscr{K}$, the canonical homomorphism $j_{X/Y/Z} : j^*(\mathscr{P}^n_{Y/Z}) \to \mathscr{P}^n_{X/Z}$ is surjective, and its kernel is the ideal of $j^*(\mathscr{P}^n_{Y/Z})$ generated by $j^*(\mathscr{O}_Y \cdot d^n_{Y/Z}(\mathscr{K}))$ (it should be noted that $d^n_{Y/Z}(\mathscr{K})$ is a subsheaf of abelian groups of $\mathscr{P}^n_{Y/Z}$, but not an $\mathscr{O}_Y$-module in general).*

Proof. We know (**I**, 5.3.8) that the diagonal $\Delta_j : X \to X \times_Y X$ is an isomorphism, from which the first assertion follows. If ϖ_1 and ϖ_2 are the two canonical homomorphisms of algebras $\mathscr{O}_Y \to \mathscr{P}^n_{Y/Z}$ corresponding respectively to the two canonical projections p_1, p_2 of $Y \times_Z Y \to Y$, recall that by definition ((16.3.5) and (16.3.6)) ϖ_1 is the structure homomorphism of the $\mathscr{O}_Y$-algebra $\mathscr{P}^n_{Y/Z}$ and $\varpi_2 = d^n_{Y/Z}$. The $\mathscr{O}_X$-algebra $j^*(\mathscr{P}^n_{Y/Z})$ is therefore identified with $\mathscr{P}^n_{Y/Z}/\varpi_1(\mathscr{K})\mathscr{P}^n_{Y/Z}$ and its quotient by the ideal generated by $j^*(d^n_{Y/Z}(\mathscr{K}))$ to $\mathscr{P}^n_{Y/Z}/(\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K}))\mathscr{P}^n_{Y/Z}$. Now note that we have a commutative diagram **IV-4** | 25

$$\begin{tikzcd} Y \arrow[d, "\Delta_f"'] & X \arrow[l, "j"'] \arrow[d, "\Delta_{f \circ j}"] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j \times_Z j"] \end{tikzcd}$$

identifying X with the product of the $(Y \times_Z Y)$-preschemes Y and $X \times_Z X$ (**I**, 5.3.7). Since $j \times_Z j$ is an immersion, we therefore deduce from this remark and from (16.2.2) that if $\Delta^n_{Y/Z}$ and $\Delta^n_{X/Z}$ denote the infinitesimal neighbourhoods of order n of Y and X by the canonical immersions Δ_f and $\Delta_{f \circ j}$ respectively, then we have a diagram

$$\begin{tikzcd} \Delta^n_{Y/Z} \arrow[d] & \Delta^n_{X/Z} \arrow[l] \arrow[d] \\ Y \times_Z Y & X \times_Z X \arrow[l, "j \times_Z j"] \end{tikzcd}$$

making $\Delta^n_{X/Z}$ the product of the $(Y \times_Z Y)$-preschemes $\Delta^n_{Y/Z}$ and $X \times_Z X$. We can also say that $\mathscr{P}^n_{X/Z}$ is identified with the sheaf of rings $\mathscr{P}^n_{Y/Z} \otimes_{\mathscr{O}_{Y\times_Z Y}} \mathscr{O}_{X\times_Z X}$. But we see immediately (for example, by restricting to the affine case) that $\mathscr{O}_{X\times_Z X} = \mathscr{O}_{Y\times_Z Y}/(p_1^*(\mathscr{K}) + p_2^*(\mathscr{K}))\mathscr{O}_{Y\times_Z Y}$. Therefore $\mathscr{P}^n_{X/Z}$ is identified with the quotient of $\mathscr{P}^n_{Y/Z}$ by the ideal generated by the image in $\mathscr{P}^n_{Y/Z}$ of $p_1^*(\mathscr{K}) + p_2^*(\mathscr{K})$. But by definition this ideal is generated by $\varpi_1(\mathscr{K}) + \varpi_2(\mathscr{K})$. □

Corollary (16.4.21). — *Let $f : Y \to Z$ be a morphism, $j : X \to Y$ an immersion. We have an exact sequence of quasi-coherent $\mathscr{O}_X$-modules*

(16.4.21.1)
$$\mathscr{N}_{X/Y} \longrightarrow j^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow 0.$$

When X, Y, Z are affine, we recover the exact sequence (**0**, 20.5.12.1).

Corollary (16.4.22). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{P}^n_{X/S}$ and $\Omega^1_{X/S}$ are quasi-coherent $\mathscr{O}_X$-modules of finite presentation.*

Proof. We immediately reduce to the case where $S = \operatorname{Spec}(A)$ is affine, $X = \operatorname{Spec}(B)$, where $B = A[T_1, \dots, T_r]/\mathfrak{K}$, $\mathfrak{K}$ being an ideal of finite type of $C = A[T_1, \dots, T_r]$. We then apply (16.4.20) with $Z = S$, $Y = \operatorname{Spec}(C)$ and $\mathscr{K} = \widetilde{\mathfrak{K}}$. Then $j^*(\mathscr{P}^n_{Y/Z})$ is a free $\mathscr{O}_X$-module of finite rank (16.4.10) and the hypothesis on $\mathfrak{K}$ implies that $j^*(\mathscr{O}_Y.d^n_{Y/Z}(\mathscr{K}))$ generates a quasi-coherent $\mathscr{O}_X$-module of finite type; hence the conclusion. □

Proposition (16.4.23). — *Let X, Y be two S-preschemes, $Z = X \times_S Y$ their product, $p : X \times_S Y \to X$ and $q : X \times_S Y \to Y$ the canonical projections. Then the canonical homomorphism*

(16.4.23.1)
$$p_{Z/X/S} \oplus q_{Z/Y/S} : p^*(\Omega^1_{X/S}) \oplus q^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{(X\times_S Y)/S}$$

is bijective.

Proof. The commutative diagram **IV-4 | 26**

$$\begin{tikzcd}
Y \arrow[d, "g"'] & X \times_S Y \arrow[l, "q"'] \arrow[d, "h"'] & X \times_S Y \arrow[l, "\mathrm{id}"'] \arrow[d, "p"] \\
S & S \arrow[l, "\mathrm{id}"] & X \arrow[l, "f"]
\end{tikzcd}$$

gives us a factorization of the canonical *isomorphism* $P^n(p)$ (16.4.5)

$$p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y}$$

and similarly, switching X with Y, we have a factorization of the *isomorphism* $P^n(q)$

$$q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/X}.$$

This proves that the canonical homomorphism (16.4.18.1)

$$p_{Z/X/S} : p^*(\mathscr{P}^n_{X/S}) \longrightarrow \mathscr{P}^n_{Z/S} \quad (\text{resp. } q_{Z/Y/S} : q^*(\mathscr{P}^n_{Y/S}) \longrightarrow \mathscr{P}^n_{Z/S})$$

is *injective*, and that the kernel of the canonical surjective homomorphism (16.4.18.2)

$$\mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/Y} \quad (\text{resp. } \mathscr{P}^n_{Z/S} \longrightarrow \mathscr{P}^n_{Z/X})$$

is complementary to the image of $p_{Z/X/S}$ (resp. $q_{Z/Y/S}$). On the other hand, by (16.4.18), this kernel is generated by the image under $q_{Z/Y/S}$ (resp. $p_{Z/X/S}$) of the augmentation ideal of $q^*(\mathscr{P}^n_{Y/S})$ (resp. $p^*(\mathscr{P}^n_{X/S})$). We conclude the proposition by considering the case $n = 1$. □

We immediately generalize (16.4.23) to the case of a product of any finite number of S-preschemes.

Remarks (16.4.24). —

(i) We will see (17.2.3) that when the morphism $f : X \to Y$ in (16.4.18) is *smooth*, the homomorphism $f_{X/Y/Z}$ in (16.4.19.1) is locally *left invertible* and in particular injective. Similarly, when the morphism $f \circ j : X \to Z$ of (16.4.20) is *smooth*, the homomorphism on the left in (16.4.21.1) is locally *left invertible* and *a fortiori* injective (17.2.5). In Chapter V, we will also give a variant, in the case of modules over a prescheme, of the "imperfection modules" studied in (**0**, 20.6), and the exact sequences where they occur.

(ii) Let X be a topological space, $\mathscr{A}$ a sheaf of rings over X and $\mathscr{B}$ an $\mathscr{A}$-algebra over X. Then it is clear that

$$U \longmapsto P^n_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{A})} \quad (U \text{ open in } X)$$

is a presheaf of augmented $\Gamma(U,\mathscr{B})$-algebras, and therefore the associated sheaf $\mathscr{P}^n_{\mathscr{B}/\mathscr{A}}$ is an augmented $\mathscr{B}$-algebra. In the particular case where X is a prescheme, $f = (\psi,\theta) : X \to S$ a morphism of preschemes, it follows easily from (16.4.16) and from the exactness of the functor $\varinjlim$ that $\mathscr{P}^n_{X/S}$ is canonically isomorphic to $\mathscr{P}^n_{\mathscr{O}_X/\psi^*(\mathscr{O}_S)}$. It follows that the formalism developed in the present paragraph could be considered as a particular case **IV-4** | 27
of a differential formalism for ringed spaces endowed with a sheaf of algebras over the structure sheaf. However, we did not start with this point of view, which is less intuitive and less convenient for applications. It also seems that, for various kinds of "varieties", the "global" constructions of the $\mathscr{P}^n$ analogous to those we have used here are also better suited for applications.

16.5. Relative tangent sheaves and bundles; derivations.

(16.5.1). Let $f = (\psi,\theta) : X \to S$ be a morphism of ringed spaces. For every $\mathscr{O}_X$-module $\mathscr{F}$, an *S-derivation* (or (X/S)*-derivation*, or f*-derivation*) *from* $\mathscr{O}_X$ *to* $\mathscr{F}$ is any homomorphism of *sheaves of additive groups* $D : \mathscr{O}_X \to \mathscr{F}$ satisfying the following conditions:

(a) for every open V of X, and every pair of sections (t_1, t_2) of $\mathscr{O}_X$ over V, we have

$$D(t_1t_2) = t_1D(t_2) + D(t_1)t_2; \tag{16.5.1.1}$$

(b) for every open V of X, every section t of $\mathscr{O}_X$ over V, and every section s of $\mathscr{O}_S$ over an open U of S such that $V \subset f^{-1}(U)$, we have

$$D((s|V)t) = (s|V)D(t). \tag{16.5.1.2}$$

It is clear that this amounts to saying that, for all $x \in X$, the homomorphism of additive groups $D_x : \mathscr{O}_x \to \mathscr{F}_x$ is an $\mathscr{O}_{f(x)}$*-derivation*.

Another interpretation consists of considering the $\mathscr{O}_X$-algebra $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$ as equal to $\mathscr{O}_X \oplus \mathscr{F}$, the algebra structure being defined by the condition that for every open V of X, the product of two sections of $\mathscr{O}_X$ (resp. of a section of $\mathscr{O}_X$ and a section of $\mathscr{F}$) over V is defined by the ring structure of $\Gamma(V,\mathscr{O}_X)$ (resp. the $\Gamma(V,\mathscr{O}_X)$-module structure on $\Gamma(V,\mathscr{F})$), and the product of two sections of $\mathscr{F}$ over V is chosen to be zero; then $\mathscr{F}$ is an ideal of $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, the kernel of the canonical augmentation $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F}) \to \mathscr{O}_X$, and to say that D is an S-derivation from $\mathscr{O}_X$ to $\mathscr{F}$ means that $\iota_{\mathscr{O}_X} + D$ is an $\mathscr{O}_S$-*homomorphism of algebras* from $\mathscr{O}_X$ to $\mathscr{D}_{\mathscr{O}_X}(\mathscr{F})$, which, composed with the augmentation, gives $\iota_{\mathscr{O}_X}$.

The S-derivations from $\mathscr{O}_X$ to $\mathscr{F}$ clearly form a $\Gamma(X,\mathscr{O}_X)$*-module* $\mathrm{Der}_S(\mathscr{O}_X,\mathscr{F})$.

When $\mathscr{F} = \mathscr{O}_X$, an S-derivation of $\mathscr{O}_X$ to itself is simply called an *S-derivation of* $\mathscr{O}_X$.

Proposition (16.5.2). — *Let A be a ring, B an A-algebra, L a B-module; let $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, $\mathscr{F} = \widetilde{L}$. Then the map $D \mapsto \Gamma(D)$ which sends every S-derivation D from $\mathscr{O}_X$ to $\mathscr{F}$ to the map $\Gamma(D) : t \mapsto D(t)$ from B to L, is an isomorphism of B-modules from $\mathrm{Der}_S(\mathscr{O}_X,\mathscr{F})$ to $\mathrm{Der}_A(B,L)$ (cf.* (**0**, 20.1.2)*).*

Proof. This follows immediately from the given interpretation of S-derivations in terms of homo- **IV-4** | 28
morphisms of algebras, analogous to the interpretation given in (**0**, 20.1.6), and from the canonical correspondence between homomorphisms of $\mathscr{O}_X$-algebras and homomorphisms of B-algebras ((**I**, 1.3.13) and (**I**, 1.3.8)). ☐

Proposition (16.5.3). — *Let $f = (\psi,\theta) : X \to S$ be a morphism of preschemes.*

(i) *The differential $d_{X/S} : \mathscr{O}_X \to \Omega^1_{X/S}$ (16.3.6) is an S-derivation.*

(ii) *For every $\mathscr{O}_X$-module $\mathscr{F}$, the map $u \mapsto u \circ d_{X/S}$ is an isomorphism of $\Gamma(X, \mathscr{O}_X)$-modules*

$$\mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \simeq \mathrm{Der}_S(\mathscr{O}_X, \mathscr{F}). \tag{16.5.3.1}$$

Proof. Assertion (i) has already been noted in (16.3.6). On the other hand, it follows immediately (in light of (**0**, 20.4.8)) that $u \mapsto u \circ d_{X/S}$ is injective, by considering the restrictions of both sides to a stalk $\mathscr{O}_x$ and using (16.4.16.2). To see that the homomorphism (16.5.3.1) is surjective, consider an S-derivation $D : \mathscr{O}_X \to \mathscr{F}$; for every affine open $V = \mathrm{Spec}(B)$ of X, such that $f(V)$ is contained in an affine open $U = \mathrm{Spec}(A)$ of S, $D_V : B \to \Gamma(V, \mathscr{F})$ is an A-derivation, and therefore there exists a *unique* B-homomorphism $u_V : \Omega^1_{B/A} \to \Gamma(V, \mathscr{F})$ such that $D_V = u_V \circ d_{B/A}$ (**0**, 20.4.8); in addition, the uniqueness of u_V shows immediately that for an affine open $W \subset V$ we have $u_W = u_V|W$, and therefore the u_V define a homomorphism of $\mathscr{O}_X$-modules $u : \Omega^1_{X/S} \to \mathscr{F}$ answering the question. □

(16.5.4). With the notation of (16.5.1), for every open U of X, $\mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a $\Gamma(U, \mathscr{O}_X)$-module and it is clear that the map $U \mapsto \mathrm{Der}_S(\mathscr{O}_U, \mathscr{F}|U)$ is a presheaf; in fact, it is even a *sheaf* (and therefore an $\mathscr{O}_X$-module), in light of the pointwise characterization of S-derivations, seen in (16.5.1). This $\mathscr{O}_X$-module is denoted by $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ and is called the *sheaf of S-derivations from $\mathscr{O}_X$ to $\mathscr{F}$*, and what we have just seen is also expressed by the following corollary:

Corollary (16.5.5). — *For every $\mathscr{O}_X$-module $\mathscr{F}$, the homomorphism of $\mathscr{O}_X$-modules induced by $u \mapsto u \circ d_{X/S}$*

$$\mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{F}) \longrightarrow \mathscr{D}er_S(\mathscr{O}_X, \mathscr{F}) \tag{16.5.5.1}$$

is bijective.

Corollary (16.5.6). — (i) *If the morphism $f : X \to S$ is locally of finite presentation and if $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module.*

(ii) *If in addition S is locally Noetherian and if $\mathscr{F}$ is coherent, then $\mathscr{D}er_S(\mathscr{O}_X, \mathscr{F})$ is a coherent $\mathscr{O}_X$-module.*

Proof. The assertion (i) follows from the isomorphism (16.5.5.1), from (16.4.22), and (**I**, 1.3.12); the assertion (ii) follows from (**0**, 5.3.5). □

(16.5.7). We set

$$\mathfrak{G}_{X/S} = \mathscr{H}om_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \mathscr{D}er_S(\mathscr{O}_X, \mathscr{O}_X), \tag{16.5.7.1}$$

and say that it is the *sheaf of S-derivations of $\mathscr{O}_X$*, or even the *tangent sheaf of X relative to S*: it is therefore the *dual* of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. If f is locally of finite presentation, $\mathfrak{G}_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module; if in addition S is locally Noetherian, then $\mathfrak{G}_{X/S}$ is coherent (16.5.6). **IV-4** | 29

(16.5.8). Suppose in particular that $\Omega^1_{X/S}$ is a *locally free* $\mathscr{O}_X$-module (of finite rank) (which will be the case when f is *smooth* (17.2.3)); then $\mathfrak{G}_{X/S}$ is a locally free $\mathscr{O}_X$-module of the same rank as $\Omega^1_{X/S}$ at each point. More specifically, suppose that $\Omega^1_{X/S}$ is of rank n at a point x; then there are n sections s_i ($1 \leqslant i \leqslant n$) of $\mathscr{O}_X$ over an affine neighbourhood U of x such that the canonical images of the ds_i in $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$ form a basis of this $k(x)$-vector space; by virtue of Nakayama's lemma, the germs $(ds_i)_x = d(s_i)_x$ of the ds_i at the point x form a basis of the $\mathscr{O}_x$-module $(\Omega^1_{X/S})_x$, and therefore, by restricting U, we can suppose that the ds_i form a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$. So the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \mathfrak{G}_{X/S})$ is dual to the above; we denote by $(D_i)_{1 \leqslant i \leqslant n}$ or $\left(\frac{\partial}{\partial s_i}\right)_{1 \leqslant i \leqslant n}$ the dual basis of $(ds_i)_{1 \leqslant i \leqslant n}$, so that, by (16.5.3), we have

$$D_i s_j = \langle D_i, ds_j \rangle = \left\langle \frac{\partial}{\partial s_i}, ds_j \right\rangle = \delta_{ij} \quad \text{(Kronecker's symbol).} \tag{16.5.8.1}$$

Every $\Gamma(S, \mathscr{O}_S)$-derivation of the $\Gamma(S, \mathscr{O}_S)$-algebra $\Gamma(U, \mathscr{O}_X)$ is therefore written uniquely as

$$D = \sum_{i=1}^{n} a_i D_i = \sum_{i=1}^{n} a_i \frac{\partial}{\partial s_i},$$

where the a_i ($1 \leqslant i \leqslant n$) are sections of $\mathscr{O}_X$ over U. For every section $g \in \Gamma(U, \mathscr{O}_X)$, if we put $dg = \sum_{i=1}^n c_i ds_i$, then we have $c_i = \langle D_i, dg\rangle = D_i g$ by virtue of (16.5.8.1), in other words,

$$dg = \sum_{i=1}^{n} (D_i g) ds_i = \sum_{i=1}^{n} \frac{\partial g}{\partial s_i} ds_i. \tag{16.5.8.2}$$

(16.5.9). Let D_1, D_2 be two S-derivations of $\mathscr{O}_X$. For every open U of X, if D_1^U, D_2^U are the corresponding derivations of the ring $\Gamma(U, \mathscr{O}_X)$, the *bracket*

$$[D_1^U, D_2^U] = D_1^U \circ D_2^U - D_2^U \circ D_1^U$$

is also a derivation of this ring, and therefore the $\psi^*(\mathscr{O}_S)$-endomorphism of $\mathscr{O}_X$

$$[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1 \tag{16.5.9.1}$$

is also an S-derivation; as we immediately check that this bracket satisfies the Jacobi identity, we have thus defined on $\mathrm{Der}_S(\mathscr{O}_X, \mathscr{O}_X)$ a $\Gamma(S, \mathscr{O}_S)$-*Lie algebra* structure. Since the definition of this structure commutes with the restriction to an open of X, we thus see that $\mathfrak{G}_{X/S}$ is canonically equipped with a $\psi^*(\mathscr{O}_S)$-*Lie algebra* structure. Note that the mapping $(D_1, D_2) \mapsto [D_1, D_2]$ is *not* $\Gamma(X, \mathscr{O}_X)$-*bilinear*.

(16.5.10). For every base change $g : S' \to S$, if we set $X' = X \times_S S'$, then we have seen in (16.4.5) that there is a canonical isomorphism

$$\Omega^1_{X/S} \otimes_S S' \simeq \Omega^1_{X'/S'}, \tag{16.5.10.1}$$

from which we deduce, by (16.5.10.1), a canonical homomorphism (Bourbaki, *Alg.*, chap. II, 3rd ed., **IV-4** | 30
§5, n°3)

$$\mathfrak{G}_{X/S} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} \longrightarrow \mathfrak{G}_{X'/S'}, \tag{16.5.10.2}$$

which is neither injective nor surjective in general. However:

Proposition (16.5.11). — (i) *If $g : S' \to S$ is a flat morphism and if f is locally of finite type (resp. locally of finite presentation), then the homomorphism* (16.5.10.2) *is injective (resp. bijective).*
(ii) *If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of finite type, then the homomorphism* (16.5.10.2) *is bijective.*

Proof. The assertion (ii) follows from Bourbaki, *Alg.*, chap. II, 3rd ed., §5, n°3, prop. 7. The assertion (i) follows similarly from Bourbaki, *Alg. Comm.*, chap. I, §2, n°10, prop. 11 and from the fact that if f is locally of finite type (resp. locally of finite presentation), then $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of finite type (resp. of finite presentation) ((16.3.9) and (16.4.22)). ☐

(16.5.12). Since $\Omega^1_{X/S}$ is a quasi-coherent $\mathscr{O}_X$-module, we can consider the vector bundle over X defined by $\Omega^1_{X/S}$ (**II**, 1.7.8)

$$T_{X/S} = \mathbf{V}(\Omega^1_{X/S}) \tag{16.5.12.1}$$

which is called the *tangent bundle of X relative to S*. We have therefore a canonical bijection (**II**, 1.7.9)

$$\Gamma(T_{X/S}/S) \simeq \mathrm{Hom}_{\mathscr{O}_X}(\Omega^1_{X/S}, \mathscr{O}_X) = \Gamma(X, \mathfrak{G}_{X/S})$$

by definition of $\mathfrak{G}_{X/S}$, and we can replace X by an open set U of X in this isomorphism; so we can say that the *tangent sheaf* of X relative to S is isomorphic to the *sheaf of germs of S-sections* of the tangent bundle of X relative to S. If $f : X \to Y$ is an S-morphism, we saw (16.4.19) that we have a canonical homomorphism $f_{X/Y/S} : f^*(\Omega^1_{Y/S}) \to \Omega^1_{X/S}$, which, having in mind that

$$\mathbf{V}(f^*(\Omega^1_{Y/S})) = \mathbf{V}(\Omega^1_{Y/S}) \times_Y X \quad (\mathbf{II}, 1.7.11)\,,$$

gives us an X-morphism $T_{X/S}(f) : T_{X/S} \to T_{Y/S} \times_Y X$. If $g : Y \to Z$ is a second S-morphism, we have $T_{X/S}(g \circ f) = (T_{Y/S}(g) \times 1_X) \circ T_{X/S}(f)$ (**0**, 20.5.4.1).

It follows from (16.5.10.1) and from (**II**, 1.7.11) that for every base change $g : S' \to S$ we have a canonical isomorphism

$$T_{X'/S'} \simeq T_{X/S} \times_S S' = T_{X/S} \times_X X'. \tag{16.5.12.2}$$

(16.5.13). For every point $x \in X$, we define the *tangent space of X at the point x* (relative to S) to be the set of points in the fibre $T_{X/S} \times_X \operatorname{Spec}(k(x))$ that are *rational over* $k(x)$, that is, the set

(16.5.13.1) $$T_{X/S}(x) = \operatorname{Hom}_{k(x)}(\Omega^1_{X/S} \otimes_{\mathscr{O}_x} k(x), k(x)),$$

which is the *dual* of the $k(x)$-vector space $\Omega^1_{\mathscr{O}_x/\mathscr{O}_s}/\mathfrak{m}_x \cdot \Omega^1_{\mathscr{O}_x/\mathscr{O}_s}$. When $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module of *finite type*, then $T_{X/S}(x)$ is a vector space of finite rank over $k(x)$, and for every base change $g : S' \to S$, and every point $x' \in X' = X \times_S S'$ over x, we have a canonical isomorphism **IV-4** | 31

(16.5.13.2) $$T_{X'/S'}(x') \simeq T_{X/S} \otimes_{k(x)} k(x').$$

If x is *rational over* $k(s)$, where $s = f(x)$ (so that $k(s) \to k(x)$ is an isomorphism), it follows from (16.4.13) that we have a canonical isomorphism

(16.5.13.3) $$T_{X/S}(x) = T_{X_s/k(s)}(x) = \operatorname{Hom}_{k(s)}(\mathfrak{m}'_x/\mathfrak{m}'^2_x, k(x)),$$

where $\mathfrak{m}'_x$ is the maximal ideal of $\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$. In the case where S is the spectrum of a field k, we recover the definition of the Zariski tangent space of a point $x \in X$ *rational over* k, as the dual of $\mathfrak{m}_x/\mathfrak{m}_x^2$.

Let Y be a second S-prescheme and let $g : Y \to X$ be an S-morphism; then we have a canonical homomorphism of $\mathscr{O}_Y$-modules (16.4.19)

(16.5.13.4) $$g_{Y/X/S} : g^*(\Omega^1_{X/S}) \longrightarrow \Omega^1_{Y/S}.$$

Now note that if $y \in Y$ and $x = g(y)$, then we have

$$g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y) = (\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)) \otimes_{k(x)} k(y)$$

and consequently, if $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module *of finite type*, then we can identify

$$\operatorname{Hom}_{k(y)}(g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y), k(y))$$

with $T_{X/S}(x) \otimes_{k(x)} k(y)$. We therefore deduce from the homomorphism (16.5.13.4) a homomorphism of $k(y)$*-vector spaces*

(16.5.13.5) $$T_y(g) : T_{Y/S}(y) \longrightarrow T_{X/S}(x) \otimes_{k(x)} k(y)$$

called the *linear map tangent to g at the point y*. When y is *rational over* $k(s)$, we can identify $k(s)$, $k(y)$, and $k(x)$, and $T_y(g)$ is then a homomorphism of $k(s)$-vector spaces $T_{Y/S}(y) \to T_{X/S}(x)$; also note that in this case, $g^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_Y} k(y)$ is identified with $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x)$, and the above homomorphism is therefore defined without any finiteness conditions on $\Omega^1_{X/S}$ and it is none other than the homomorphism $T_{Y/S}(g)$ (16.5.12) restricted to the fibre of $T_{Y/S}$ at the point y.

(16.5.14). The interpretation of derivations of an A-algebra B to a B-module L, given in (**0**, 20.1.1), translates to the language of preschemes in the following way.

Consider two morphisms of preschemes $f : X \to S$, $g : Y \to S$, and a closed subprescheme Y_0 of Y defined by a *square-zero* ideal $\mathscr{J}$ of $\mathscr{O}_Y$ (so that Y and Y_0 have the *same underlying topological space*). Suppose we are given an S-morphism $u_0 : Y_0 \to X$, so that we have a commutative diagram

(16.5.14.1) $$\begin{tikzcd} X \arrow[d, "f"'] & Y_0 \arrow[l, "u_0"'] \arrow[d, "j"] \\ S & Y \arrow[l, "g"] \arrow[ul, dashed, "u"'] \end{tikzcd}$$

and we seek *S-morphisms* $u : Y \to X$ such that $u_0 = u \circ j$ (in other words, we ask whether the preceding diagram can be completed by the dotted arrow u while remaining *commutative*). **IV-4** | 32

For that, consider an affine open $U = \operatorname{Spec}(C)$ of Y; its inverse image $j^{-1}(U)$ is the affine open $U_0 = \operatorname{Spec}(C/\mathfrak{L})$, where $\mathfrak{L} = \Gamma(U, \mathscr{J})$, a *square-zero* ideal in C; suppose that U is small enough so that $u_0(U_0)$ is contained in an affine open $V = \operatorname{Spec}(B)$ of X and that $g(U) = f(u_0(U_0))$ is contained in an affine open $W = \operatorname{Spec}(A)$ of S, so that B and C are A-algebras and $u_0|U_0$ corresponds to an A-homomorphism ψ from B to $C/\mathfrak{L}$; Let $P(U_0)$ be the set of restrictions $u|U$ of the sought morphisms; it corresponds canonically to the A-algebra homomorphisms $\phi : B \to C$ such that the *composite* $B \xrightarrow{\phi} C \to C/\mathfrak{L}$ *is equal to* ψ. So we know (**0**, 20.1.1) that the set of such homomorphisms is either empty or of the form $\phi_1 + \mathrm{Der}_A(B, \mathfrak{L})$; when $P(U_0)$ is not empty, the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ acts by addition on $P(U_0)$, which is therefore an *affine space* for the additive group $\mathrm{Der}_A(B, \mathfrak{L})$ (or even a *principal homogeneous space* (or *torsor*) *under* $\mathrm{Der}_A(B, \mathfrak{L})$).

Now notice that, since $\mathfrak{L}$ is equipped with a B-module structure via ψ, we have an *isomorphism* $v \mapsto v \circ d_{B/A}$ of $\mathrm{Hom}_B(\Omega^1_{B/A}, \mathfrak{L})$ onto $\mathrm{Der}_A(B, \mathfrak{L})$ (**0**, 20.4.8). Besides, as $\mathfrak{L}$ is square-zero, therefore a $(C/\mathfrak{L})$-module, every B-homomorphism $v : \Omega^1_{B/A} \to \mathfrak{L}$ can be considered as a $(C/\mathfrak{L})$-homomorphism $\Omega^1_{B/A} \otimes_B (C/\mathfrak{L}) \to \mathfrak{L}$. As $\mathscr{J}$ is square-zero, it can be considered as a quasi-coherent $\mathscr{O}_{Y_0}$-module; let's introduce the $\mathscr{O}_{Y_0}$-module

(16.5.14.2)
$$\mathscr{G} = \mathscr{H}\!om(u_0^*(\Omega^1_{X/S}), \mathscr{J});$$

it follows from the fact that $\Omega^1_{B/A} = \Gamma(V, \Omega^1_{X/S})$ (16.3.7) that we can write $\mathrm{Der}_A(B, \mathfrak{L}) = \Gamma(U_0, \mathscr{G})$.

As $P(U_0)$ is defined as a set of S-morphisms $U \to X$, it is clear that $U_0 \mapsto P(U_0)$ is a *sheaf of sets* $\mathscr{P}$ on Y_0. We can use this fact to prove that the map $h : \Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ defining the torsor structure on $P(U_0)$ is independent of choice of V and W and also that, if $U' \subset U$ is a second affine open of Y, U'_0 its inverse image in Y_0, then the diagram

(16.5.14.3)
$$\begin{tikzcd} \Gamma(U_0, \mathscr{G}) \times P(U_0) \arrow[r, "h"] \arrow[d] & P(U_0) \arrow[d] \\ \Gamma(U'_0, \mathscr{G}) \times P(U'_0) \arrow[r, "h'"'] & P(U'_0) \end{tikzcd}$$

is commutative (the vertical arrows being the restrictions). In light of the above remark, we reduce to proving the commutativity of the above diagram when h is defined as such from affine opens V, W and h' from affine opens $V' \subset V$ and $W' \subset W$. But because of the preceding description of h, **IV-4** | 33
this follows from the commutativity of the diagram (**0**, 20.5.4.2).

The mapping $\Gamma(U_0, \mathscr{G}) \times P(U_0) \to P(U_0)$ therefore defines a homomorphism of *sheaves of sets* $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for all open sets U_0 for which $\Gamma(U_0, \mathscr{P}) \neq \varnothing$, $m_{U_0} : \Gamma(U_0, \mathscr{G}) \times \Gamma(U_0, \mathscr{P}) \to \Gamma(U_0, \mathscr{P})$ is an external law defining in $\Gamma(U_0, \mathscr{P})$ a torsor structure for the group $\Gamma(U_0, \mathscr{G})$.

(16.5.15). In general, when we are given a sheaf of sets $\mathscr{P}$ over a topological space Z, a sheaf of groups $\mathscr{G}$ (not necessarily commutative), and a homomorphism of sheaves of sets $m : \mathscr{G} \times \mathscr{P} \to \mathscr{P}$ such that, for every open $U \subset Z$ such that $\Gamma(U, \mathscr{P}) \neq \varnothing$, $m_U : \Gamma(U, \mathscr{G}) \times \Gamma(U, \mathscr{P}) \to \Gamma(U, \mathscr{P})$ makes $\Gamma(U, \mathscr{P})$ a *torsor* under the group $\Gamma(U, \mathscr{G})$, then we say that $\mathscr{P}$ is a *pseudo-torsor* (or *formally principal homogeneous sheaf*) under the sheaf of groups $\mathscr{G}$. We say that $\mathscr{P}$ is a *torsor* (or *principal homogeneous sheaf*) under $\mathscr{G}$ [2] if in addition $\Gamma(U, \mathscr{P}) \neq \varnothing$ for every open $U \neq \varnothing$ in a suitable basis for the topology of Z.

For the general theory of torsors, we refer to **[?]**; we will limit ourselves to recalling the canonical correspondence between isomorphism classes of torsors (for a *given* $\mathscr{G}$) and elements of the cohomology set $\mathrm{H}^1(Z, \mathscr{G})$. Consider a torsor $\mathscr{P}$ under $\mathscr{G}$ and an open cover (U_λ) of Z such that $\Gamma(U_\lambda, \mathscr{P}) \neq \varnothing$ for every λ; denote by p_λ an element of $\Gamma(U_\lambda, \mathscr{P})$. For every pair of indices λ, μ such that $U_\lambda \cap U_\mu \neq \varnothing$, there then exists a unique element $\gamma_{\lambda\mu}$ of $\Gamma(U_\lambda \cap U_\mu, \mathscr{G})$ such that $\gamma_{\lambda\mu} \cdot (p_\mu | U_\lambda \cap U_\mu) = p_\lambda | U_\lambda \cap U_\mu$; in addition, if λ, μ, ν are three indices such that $U_\lambda \cap U_\mu \cap U_\nu \neq \varnothing$, then the restrictions $\gamma'_{\lambda\mu}$, $\gamma'_{\mu\nu}$, $\gamma'_{\lambda\nu}$ of $\gamma_{\lambda\mu}$, $\gamma_{\mu\nu}$, $\gamma_{\lambda\nu}$ to $U_\lambda \cap U_\mu \cap U_\nu$ satisfy the condition $\gamma'_{\lambda\nu} = \gamma'_{\lambda\mu}\gamma'_{\mu\nu}$; in other words, $(\lambda, \mu) \mapsto \gamma_{\lambda\mu}$ is a 1-*cocycle* of the cover (U_λ) with values in $\mathscr{G}$. If, for every λ, p'_λ is a second element of $\Gamma(U_\lambda, \mathscr{P})$, then there exists a unique element $\beta_\lambda \in \Gamma(U_\lambda, \mathscr{G})$ such that $p'_\lambda = \beta_\lambda \cdot p_\lambda$, and the 1-cocycle $(\gamma'_{\lambda\mu})$ corresponding to the family (p'_λ) is given by $\gamma'_{\lambda\mu} = \beta_\lambda \gamma_{\lambda\mu} \beta_\mu^{-1}$, that is, it is *cohomologous* to $\gamma_{\lambda\mu}$. Conversely, the data of a 1-cocycle $(\gamma_{\lambda\mu})$ defines, for every pair (λ, μ), an automorphism $\theta_{\lambda\mu}$ of the sheaf *of sets* $\mathscr{G} | U_\lambda \cap U_\mu$, namely the right translation by $\gamma_{\lambda\mu}$, and the fact that it is a cocycle shows that we can *glue* the sheaves of sets $\mathscr{G} | U_\lambda$ via the automorphisms $\theta_{\lambda\mu}$ (**0**, 3.3.1); we thus obtain a torsor under $\mathscr{G}$, denoted $\mathscr{P}$, and if we

[2] [Trans.] This is nowadays more commonly called a $\mathscr{G}$-torsor rather than a torsor *under* $\mathscr{G}$.

take for p_λ the unit section over U_λ, then the corresponding 1-cocycle is none other than the given 1-cocycle $(\gamma_{\lambda\mu})$; in addition, if we replace $(\gamma_{\lambda\mu})$ by a 1-cocycle $\gamma'_{\lambda\mu} = \beta_\lambda\gamma_{\lambda\mu}\beta_\mu^{-1}$ cohomologous to it, then we check immediately that the torsor obtained is isomorphic to $\mathscr{P}$.

In particular, if $(\gamma_{\lambda\mu})$ is a 1-*coboundary*, in other words of the form $\gamma_{\lambda\mu} = \beta_\lambda\beta_\mu^{-1}$, then the torsor $\mathscr{P}$ obtained is *isomorphic to* $\mathscr{G}$ (considered as a torsor under itself by left translations); we say in this case that $\mathscr{P}$ is *trivial*, and the converse is evident.

In particular, it follows from (**III**, 1.3.1) that we have:

Proposition (16.5.16). — *Let Z be an affine scheme, $\mathscr{G}$ a quasi-coherent $\mathscr{O}_Z$-module; then every torsor under $\mathscr{G}$ is trivial.*

Returning to the problem considered in (16.5.14), we thus obtain: **IV-4** | 34

Proposition (16.5.17). — *Let X, Y be two S-preschemes, Y_0 a closed subprescheme of Y defined by a quasi-coherent ideal $\mathscr{I}$ of $\mathscr{O}_Y$ such that $\mathscr{I}^2 = 0$, $j : Y_0 \to Y$ the canonical injection. Let $u_0 : Y_0 \to X$ be an S-morphism, and $\mathscr{P}$ the sheaf of sets on Y such that, for every open U of Y, $\Gamma(U, \mathscr{P})$ is the set of S-morphisms $u : U \to X$ such that $u_0|U_0 = u \circ (j|U_0)$, where $U_0 = j^{-1}(U)$. Then $\mathscr{P}$ has the structure of a pseudo-torsor under the $\mathscr{O}_{Y_0}$-module $\mathscr{G} = \mathscr{H}om_{\mathscr{O}_{Y_0}}(u_0^*(\Omega^1_{X/S}), \mathscr{I})$.*

In particular:

Corollary (16.5.18). — *With the notation of (16.5.17), suppose that Y is affine and $\Omega^1_{X/S}$ is of finite presentation; if there is an open cover (U_α) of Y, and, for every index α, an S-morphism $v_\alpha : U_\alpha \to X$ such that, if $U^0_\alpha = j^{-1}(U_\alpha)$, we have $v_\alpha \circ (j|U^0_\alpha) = u_0|U^0_\alpha$, then there is an S-morphism $u : Y \to X$ such that $u \circ j = u_0$.*

Proof. Indeed, $\mathscr{G}$ is a quasi-coherent $\mathscr{O}_{Y_0}$-module (**I**, 1.3.12); by (16.5.16) and the fact that Y_0 is then affine, the sheaf $\mathscr{P}$, which is by hypothesis a torsor under $\mathscr{G}$, and not only a pseudo-torsor, is *trivial*; if w is an isomorphism from $\mathscr{G}$ to $\mathscr{P}$ (as torsors under $\mathscr{G}$), the image under w of the zero section of $\mathscr{G}$ is the S-morphism sought. □

16.6. Sheaf of p-differentials and exterior differentials.

(16.6.1). Let $f : X \to S$ be a morphism of preschemes. We define the *sheaf of p-differentials of X relative to S* (p an integer) to be the *pth exterior power* (**0**, 4.1.5) of the $\mathscr{O}_X$-module $\Omega^1_{X/S}$, denoted by

(16.6.1.1) $$\Omega^p_{X/S} = \bigwedge^p(\Omega^1_{X/S}).$$

So we have $\Omega^0_{X/S} = \mathscr{O}_X$, and $\Omega^p_{X/S} = 0$ for $p < 0$; the $\Omega^p_{X/S}$ are the homogeneous components of the exterior algebra of $\Omega^1_{X/S}$

(16.6.1.2) $$\Omega^\bullet_{X/S} = \bigwedge(\Omega^1_{X/S}) = \bigoplus_{p\in\mathbf{Z}} \bigwedge^p(\Omega^1_{X/S}),$$

which is therefore a quasi-coherent graded anti-commutative $\mathscr{O}_X$-algebra whose elements of degree 1 are square-zero. For every affine U of X, we have $\Gamma(U, \Omega^\bullet_{X/S}) = \bigwedge(\Gamma(U, \Omega^1_{X/S}))$, where $\Gamma(U, \Omega^1_{X/S})$ is considered as a $\Gamma(U, \mathscr{O}_X)$-module.

When $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, so that B is an A-algebra, we have (**0**, 4.1.5) $\Omega^p_{X/S} = (\Omega^p_{B/A})^\sim$, where $\Omega^p_{B/A} = \bigwedge^p \Omega^1_{B/A}$.

Theorem (16.6.2). — *There is one and only one endomorphism d of the sheaf of additive groups $\Omega^\bullet_{X/S}$ with the following properties:*

(i) $d \circ d = 0$.
(ii) *For every open set U of X and every section $f \in \Gamma(U, \mathscr{O}_X)$ we have $df = d_{X/S}f$.* **IV-4** | 35
(iii) *For every open set U of X, every pair of integers p, q and every pair of sections $\omega'_p \in \Gamma(U, \Omega^p_{X/S})$, $\omega''_q \in \Gamma(U, \Omega^q_{X/S})$, we have*

(16.6.2.1) $$d(\omega'_p \wedge \omega''_q) = (d\omega'_p) \wedge \omega''_q + (-1)^p \omega'_p \wedge d\omega''_q.$$

Also, d is an endomorphism of graded $\psi^*(\mathscr{O}_X)$-modules of degree $+1$.

Proof. Suppose that we have proved the existence of an endomorphism d. For every affine open U of X, every section of $\Omega^p_{X/S}$ over U is (by (ii)) a linear combination of finitely many elements of the form $g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)$, where g and the f_i are sections of $\mathscr{O}_X$ over U (**0**, 20.4.7). The conditions (i) and (iii) then show, by induction on p, that we necessarily have

$$d(g(df_1 \wedge df_2 \wedge \cdots \wedge df_p)) = dg \wedge df_1 \wedge df_2 \wedge \cdots \wedge df_p. \tag{16.6.2.2}$$

This therefore proves the *uniqueness* of d and the last claim of the theorem. By virtue of this uniqueness property, to show the existence of d, we can restrict ourselves to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affines. Now (Bourbaki, *Alg.*, chap. III, 3rd ed., §10), to define an A-antiderivation D of degree $+1$ on an exterior algebra $\bigwedge(M)$ (where M is a B-module and B an A-algebra), with values in a *graded anticommutative* A-algebra $C = \bigoplus_{n=0}^{\infty} C_n$ whose elements of degree 1 are square-zero, it suffices to choose arbitrarily an A-derivation D_0 from B to C_1 and an A-homomorphism D_1 from M to C_2; there then exists a unique A-antiderivation D from $\bigwedge(M)$ to C agreeing with D_0 on B and with D_1 on M.

In the present case, D_0 is necessarily equal to $d_{B/A}$ by (ii); we reduce to seeing, having (16.6.2.2) in mind, that there is an A-homomorphism u of $\Omega^1_{B/A}$ in $\Omega^2_{B/A}$ such that

$$u(g.df) = dg \wedge df \tag{16.6.2.3}$$

for all f, g in B; it suffices to show that there is an A-homomorphism $v : B \otimes_A \Omega^1_{B/A} \to \Omega^2_{B/A}$ such that

$$v(g.\omega) = dg \wedge \omega \tag{16.6.2.4}$$

for $g \in B$ and $\omega \in \Omega^1_{B/A}$. Finally, since $\Omega^1_{B/A} = \mathfrak{I}/\mathfrak{I}^2$ (where $\mathfrak{I} = \mathfrak{I}_{B/A}$ is the kernel of the canonical homomorphism $B \otimes_A B \to B$) and $\Omega^1_{B/A}$ is generated by elements of the form $g.df$, it is enough to define an A-homomorphism $w : B \otimes_A (B \otimes_A B) \to \Omega^2_{B/A}$ such that

$$w(g' \otimes g \otimes f) = dg' \wedge (g.df) \tag{16.6.2.5}$$

and such that w is zero on the image of $B \otimes_A \mathfrak{I}^2$. Since the right-hand side of (16.6.2.5) is A-trilinear in g', g, and f, the existence of w satisfying (16.6.2.5) is immediate. Since, on the other hand, $\mathfrak{I}$ is generated by elements of the form $1 \otimes x - x \otimes 1$ ($x \in B$), we reduce to checking that when $z = (1 \otimes x - x \otimes 1)(1 \otimes y - y \otimes 1)$ we have $w(g' \otimes z) = 0$. Now, since $z = 1 \otimes (xy) + (xy) \otimes 1 - x \otimes y - y \otimes x$, formula (16.6.2.4) shows that it is enough to see that we have $d(xy) - x.dy - y.dx = 0$, which is to say that d is a derivation. **IV-4** | 36

It remains to be shown that d satisfies the condition (i). Now, the square of an antiderivation is a derivation (Bourbaki, *loc. cit.*), and since $\Omega^\bullet_{B/A}$ is generated by $\Omega^1_{B/A}$ as a B-algebra, it is enough to verify that $d(dz) = 0$ when $z \in B$ and when $z \in \Omega^1_{B/A}$; in the first case, this follows from the formula (16.6.2.3) when $g = 1$; for the second, we can restrict ourselves to the case where $z = g.df$ with f, g in B, and then we have, because of (16.6.2.1) and (16.6.2.3),

$$d(d(g.df)) = d(dg \wedge df) = (d(dg)) \wedge (df) - (dg) \wedge (d(df)) = 0.$$

□

Definition (16.6.3). — The antiderivation d defined in (16.6.2) (also denoted by $d_{X/S}$) is called the *exterior differential* on X (relative to S).

Proposition (16.6.4). — *For every base change $g : S' \to S$, if we put $X' = X \times_S S'$, the canonical morphism*

$$\Omega^\bullet_{X/S} \otimes_S S' \longrightarrow \Omega^\bullet_{X'/S'} \tag{16.6.4.1}$$

deduced from the isomorphism (16.5.10.1) is bijective. Also, if s is a section of $\Omega^\bullet_{X/S}$ over an open set U of X, $s \otimes 1$ its inverse image, section of $\Omega^\bullet_{X'/S'}$ over the inverse image U' of U in X', we have $d_{X'/S'}(s \otimes 1) = d_{X/S}(s) \otimes 1$.

Proof. The first claim is immediate, the formation of the exterior algebra of a module commutes with extending the scalar ring. To prove the second, we can, because of (16.6.2.2), restrict ourselves to the case where $s \in \Gamma(U, \mathscr{O}_X)$, and in this case the claim has already been proven (16.4.3.7). □

(16.6.5). Suppose that $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module of rank n at a point x, so that we have n sections $s_i \in \Gamma(U, \mathscr{O}_X)$ such that the ds_i form a basis for the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$ (16.5.8). Then, for every integer $p \geqslant 1$, the p-differentials $ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}$ (for $i_1 < i_2 < \cdots < i_p$ in $[1, n]$) form a basis of $\binom{n}{p}$ elements of $\Gamma(U, \Omega^p_{X/S})$ over $\Gamma(U, \mathscr{O}_X)$. Also the formula (16.6.2.2) shows that for every section $g \in \Gamma(U, \mathscr{O}_X)$, we have

$$d(g.ds_{i_1} \wedge ds_{i_2} \wedge \cdots \wedge ds_{i_p}) = \sum_k (-1)^r \frac{\partial g}{\partial s_k} ds_{i_1} \wedge \cdots \wedge ds_{i_r} \wedge ds_k \wedge ds_{i_{r+1}} \wedge \cdots \wedge ds_{i_p} \tag{16.6.5.1}$$

where, on the right-hand side, k ranges over the $n - p$ indices distinct from the i_h, and i_r is the largest index $< k$.

We note that the relation $d(dg) = 0$ for every section $g \in \Gamma(U, \mathscr{O}_X)$ expresses itself in the form

$$D_i(D_j g) = D_j(D_i g) \quad \text{for } i \neq j;$$

in other words, the derivations D_i defined in (16.5.7) commute with each other.

16.7. The $\mathscr{P}^n_{X/S}(\mathscr{F})$.

(16.7.1). Let $f : X \to S$ be a morphism of preschemes, $\mathscr{F}$ an $\mathscr{O}_X$-module. We denote by $X^{(n)}_{\Delta_f}$ the *nth infinitesimal neighbourhood* of X for the diagonal morphism $\Delta_f : X \to X \times_S X$, by $h_n : X^{(n)}_{\Delta_f} \to X \times_S X$ **IV-4 | 37** the canonical morphism (16.1.2), and consider the two composite morphisms

$$p_1^{(n)} : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_1} X, \quad p_2^{(n)} : X^{(n)}_{\Delta_f} \xrightarrow{h_n} X \times_S X \xrightarrow{p_2} X$$

so that, by definition, $p_1^{(n)}$ corresponds to the homomorphism of sheaves of rings $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ which we have chosen to define the $\mathscr{O}_X$-algebra structure on $\mathscr{P}^n_{X/S}$ (16.3.5), and $p_2^{(n)}$ to the homomorphism of sheaves of rings $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (16.3.6). Since $X^{(n)}_{\Delta_f}$ and X have the same underlying space, we can write

$$\mathscr{P}^n_{X/S} = (p_1^{(n)})_*((p_2^{(n)})^*(\mathscr{O}_X)). \tag{16.7.1.1}$$

More generally, we define

$$\mathscr{P}^n_{X/S}(\mathscr{F}) = (p_1^{(n)})_*((p_2^{(n)})^*(\mathscr{F})). \tag{16.7.1.2}$$

so that $\mathscr{P}^n_{X/S} = \mathscr{P}^n_{X/S}(\mathscr{O}_X)$; by definition, $\mathscr{P}^n_{X/S}(\mathscr{F})$ is an $\mathscr{O}_X$-module.

(16.7.2). Returning to the definition of the inverse image of modules on ringed spaces (**0**, 4.3.1), and bearing in mind that $X^{(n)}_{\Delta_f}$ and X have the same underlying space, we see that definition (16.7.1.2) can be written in the form

$$\mathscr{P}^n_{X/S}(\mathscr{F}) = \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F}, \tag{16.7.2.1}$$

where it must be kept in mind that, in interpreting the symbol $\otimes$, $\mathscr{P}^n_{X/S}$ is endowed with the $\mathscr{O}_X$-module structure defined by *the homomorphism of sheaves of rings* $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$. It follows immediately from such formula (or directly from (16.7.1.2)) that $\mathscr{P}^n_{X/S}(\mathscr{F})$ is canonically equipped with a $\mathscr{P}^n_{X/S}$-module structure.

Proposition (16.7.3). — (i) *The functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$ from the category of $\mathscr{O}_X$-modules to the category of $\mathscr{P}^n_{X/S}$-modules is right exact, and commutes with arbitrary inductive limits; it is exact when $\mathscr{P}^n_{X/S}$ is flat.*

(ii) *If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module (resp. of finite type, resp. of finite presentation), then $\mathscr{P}^n_{X/S}(\mathscr{F})$ is quasi-coherent (resp. of finite type, resp. of finite presentation).*

Proof. The claims from (i) follow from (16.7.2.1) and the consideration of the symmetry of $\mathscr{P}^n_{X/S}$ (16.3.4). The claims from (ii) follow from the right exactness of the functor $\mathscr{F} \mapsto \mathscr{P}^n_{X/S}(\mathscr{F})$. □

(16.7.4). The two $\mathscr{O}_X$-module structures on $\mathscr{P}^n_{X/S}$ define two commuting $\mathscr{O}_X$-module structures on $\mathscr{P}^n_{X/S}(\mathscr{F})$, and hence an $\mathscr{O}_X$-bimodule structure. It is convenient to denote the structure coming from the structure homomorphism $\mathscr{O}_X \to \mathscr{P}^n_{X/S}$ (chosen in (16.3.5)) on the *left* and the one coming from the homomorphism $d^n_{X/S} : \mathscr{O}_X \to \mathscr{P}^n_{X/S}$ on the *right*. In other words, for every open U of X, and every triple $a \in \Gamma(U, \mathscr{O}_X)$, $b \in \Gamma(U, \mathscr{P}^n_{X/S})$, $t \in \Gamma(U, \mathscr{F})$, we have by definition

(16.7.4.1) $$a(b \otimes t) = (ab) \otimes t, \quad (b \otimes t)a = (b.d^n a) \otimes t = b \otimes (at) = (d^n a).(b \otimes t).$$

The $\mathscr{O}_X$-module structure coming from the definition (16.7.1.2) is therefore, under these conventions, the *left* $\mathscr{O}_X$-module structure. If $\mathscr{F}$ is a quasi-coherent $\mathscr{O}_X$-module, then the same is true for **IV-4** | 38
$\mathscr{P}^n_{X/S}(\mathscr{F})$ for any one of its $\mathscr{O}_X$-module structures. If also $\mathscr{F}$ is of finite type (resp. of finite presentation) and $f : X \to S$ is locally of finite type (resp. locally of finite presentation), $\mathscr{P}^n_{X/S}(\mathscr{F})$ is (for any one of its $\mathscr{O}_X$-module structures) of finite type (resp. of finite presentation), which is a consequence of (16.3.9) and (16.4.22).

(16.7.5). The definition (16.7.2.1) entails the existence of a homomorphism of sheaves of commutative groups

(16.7.5.1) $$d^n_{X/S,\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (\text{also denoted } d^n_{X/S})$$

such that, in the notations of (16.7.4), we have

(16.7.5.2) $$d^n_{X/S,\mathscr{F}}(t) = 1 \otimes t$$

and consequently, because of (16.7.4.1)

(16.7.5.3) $$d^n_{X/S,\mathscr{F}}(at) = (1 \otimes t)a = (d^n_{X/S,\mathscr{F}}(t)).a$$

(16.7.5.4) $$d^n_{X/S,\mathscr{F}}(at) = (d^n_{X/S,\mathscr{F}}(a)).(1 \otimes t) = (d^n_{X/S,\mathscr{F}}(a))(d^n_{X/S,\mathscr{F}}(t)).$$

Therefore it is $\mathscr{O}_X$-linear for the *right* $\mathscr{O}_X$-module structure on $\mathscr{P}^n_{X/S}(\mathscr{F})$, and *semilinear* (relative to the automorphism σ (16.3.4)) for the *left* $\mathscr{O}_X$-module structure.

Proposition (16.7.6). — *The left $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}(\mathscr{F})$ is generated by the image of $\mathscr{F}$ under the homomorphism $d^n_{X/S,\mathscr{F}}$.*

Proof. This is an immediate consequence of (16.7.5.3) and of the particular case $\mathscr{F} = \mathscr{O}_X$ (16.3.8). □

(16.7.7). The canonical homomorphisms of sheaves of rings

$$\phi_{nm} : \mathscr{P}^m_{X/S} \longrightarrow \mathscr{P}^n_{X/S}$$

for $n \leqslant m$ (16.1.2) define, because of (16.7.2.1), canonical homomorphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F}) \quad (n \leqslant m)$$

which are homomorphisms of $\mathscr{O}_X$-bimodules in light of (16.1.6) and (16.7.4.1); also we have commutative diagrams

$$\begin{tikzcd}
\mathscr{P}^m_{X/S}(\mathscr{F}) \arrow[rr] & & \mathscr{P}^n_{X/S}(\mathscr{F}) \\
& \mathscr{F} \arrow[ul, "d^m_{X/S,\mathscr{F}}"] \arrow[ur, "d^n_{X/S,\mathscr{F}}"'] &
\end{tikzcd}$$

We have therefore a projective system of $\mathscr{O}_X$-bimodules $(\mathscr{P}^n_{X/S}(\mathscr{F}))$, and we define

(16.7.7.1) $$\mathscr{P}^\infty_{X/S}(\mathscr{F}) = \varprojlim \mathscr{P}^n_{X/S}(\mathscr{F}).$$

Also, this shows that the homomorphisms (16.7.5.1) form a projective system of homomorphisms, and therefore define a canonical homomorphism

(16.7.7.2) $$d^\infty_{X/S,\mathscr{F}} : \mathscr{F} \longrightarrow \mathscr{P}^\infty_{X/S}(\mathscr{F}).$$

IV-4 | 39

(16.7.8). Let $\mathscr{F}, \mathscr{G}$ be two $\mathscr{O}_X$-modules; it follows immediately from the definition (16.7.2.1) that we have a canonical isomorphism of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}) \simeq \mathscr{P}^n_{X/S}(\mathscr{F}) \otimes_{\mathscr{P}^n_{X/S}} \mathscr{P}^n_{X/S}(\mathscr{G}) \tag{16.7.8.1}$$

(Bourbaki, *Alg.*, chap. II, 3^{rd} ed., § 5, no. 1, prop. 3).

We conclude in particular (or see directly from definition (16.7.2.1)) that if $\mathscr{F}$ has an $\mathscr{O}_X$-algebra structure (not necessarily associative), $\mathscr{P}^n_{X/S}(\mathscr{F})$ has a canonical $\mathscr{P}^n_{X/S}$-algebra structure; the latter is associative (resp. commutative, resp. unital, resp. a Lie algebra) if $\mathscr{F}$ is so. Moreover, the canonical homomorphisms $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ for $n \leqslant m$ (16.7.7) are then algebra dihomomorphisms; similarly, (16.7.5.1) is then an $\mathscr{O}_X$-algebra homomorphism when $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with the $\mathscr{O}_X$-algebra structure arising from its *right* $\mathscr{O}_X$-module structure.

With the same notation, we also have a canonical homomorphism of $\mathscr{P}^n_{X/S}$-modules

$$\mathscr{P}^n_{X/S}(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})) \longrightarrow \mathscr{H}om_{\mathscr{P}^n_{X/S}}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{P}^n_{X/S}(\mathscr{G})) \tag{16.7.8.2}$$

(Bourbaki, *Alg.*, chap. II, 3^{rd} ed., § 5, no. 3), which is bijective when $\mathscr{P}^n_{X/S}$ is an $\mathscr{O}_X$-module *locally free of finite type* (*loc. cit.*, prop. 7).

(16.7.9). Suppose we are in the situation described in (16.4.1); then from the canonical homomorphism $P^n(u)$ (16.4.3.3) we deduce immediately a canonical homomorphism of $\mathscr{O}_{X'}$-bimodules

$$u^*(\mathscr{P}^n_{X/S}(\mathscr{F})) \longrightarrow \mathscr{P}^n_{X'/S'}(u^*(\mathscr{F})). \tag{16.7.9.1}$$

We leave it to the reader to extend to this homomorphism the properties established in (16.4) for the case $\mathscr{F} = \mathscr{O}_X$.

Remark (16.7.10). — The definition of $\mathscr{P}^n_{X/S}(\mathscr{F})$ in the form (16.7.1.2) still makes sense when $\mathscr{F}$ is a sheaf of sets (the inverse image of a sheaf of sets by $p_2^{(n)}$ being defined in (**0**, 3.7.1)); a variant of this definition allows one to define the "jet scheme" (relative to S) of an arbitrary X-prescheme.

16.8. Differential operators. [3]

Definition (16.8.1). — Let $f = (\psi, \theta) : X \to S$ be a morphism of preschemes, $\mathscr{F}, \mathscr{G}$ two $\mathscr{O}_X$-modules, n an integer $\geqslant 0$. We say that a morphism $D : \mathscr{F} \to \mathscr{G}$ of sheaves of additive groups is a differential operator of order $\leqslant n$ (relative to S) if there is a homomorphism of $\mathscr{O}_X$-modules $u : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G}$ (where $\mathscr{P}^n_{X/S}(\mathscr{F})$ is equipped with its left $\mathscr{O}_X$-module structure (16.7.4)) such that $D = u \circ d^n_{X/S,\mathscr{F}}$.

It is clear, because of the existence of canonical morphisms

$$\mathscr{P}^m_{X/S}(\mathscr{F}) \longrightarrow \mathscr{P}^n_{X/S}(\mathscr{F})$$

for $n \leqslant m$ (16.7.7), that a differential operator of order $\leqslant n$ is a differential operator of order $\leqslant m$ **IV-4** | 40
for $n \leqslant m$. If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, then, for every open set U of X, $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$.

We say that a homomorphism $D : \mathscr{F} \to \mathscr{G}$ is a *differential operator* (relative to S) if, for every $x \in X$, there is an open neighbourhood U of x and an integer $n \geqslant 0$ such that $D|U : \mathscr{F}|U \to \mathscr{G}|U$ is a differential operator of order $\leqslant n$. The *order* of a differential operator D is the infimum of the integers n such that D is a differential operator of order $\leqslant n$ (and therefore $+\infty$ if there is no such integer); this order is always finite if X is *quasi-compact*. The differential operators of order 0 are exactly the homomorphisms of $\mathscr{O}_X$-modules $\mathscr{F} \to \mathscr{G}$; the operators of order < 0 are *zero* by convention. For $n \geqslant 0$, a differential operator of order $\leqslant n$ is not in general a homomorphism of $\mathscr{O}_X$-modules, but is always a homomorphism of $\psi^*(\mathscr{O}_S)$-modules.

When $\mathscr{F} = \mathscr{O}_X$, a differential operator of order $\leqslant 1$ from $\mathscr{O}_X$ to $\mathscr{G}$ can be written uniquely in the form $v + D$, where $v : \mathscr{O}_X \to \mathscr{G}$ is an $\mathscr{O}_X$-homomorphism and D is an S-*derivation* (16.5.1) from $\mathscr{O}_X$ to $\mathscr{G}$; this follows from the structure of $P^1_{B/A}$ (**0**, 20.4.8).

[3]For a more general formalism, see Exposé VII of [**?**] (due to P. Gabriel).

(16.8.2). To describe in a more precise manner a differential operator of order $\leqslant n$, $D : \mathscr{F} \to \mathscr{G}$, it suffices, for every open set U of X whose image in S is contained in an affine open set V, to characterize the homomorphism $D = D_U : \Gamma(U, \mathscr{F}) \to \Gamma(U, \mathscr{G})$. If we put $\Gamma(V, \mathscr{O}_S) = A$, $\Gamma(U, \mathscr{O}_X) = B$, so that B is an A-algebra, we have $\Gamma(U, \mathscr{P}^n_{X/S}) = (B \otimes_A B)/\mathfrak{I}^{n+1}$, where, for brevity, $\mathfrak{I} = \mathfrak{I}_{B/A}$. Also put $M = \Gamma(U, \mathscr{F})$, $N = \Gamma(U, \mathscr{G})$; then the definition of D means that for every pair (U, V) satisfying the above, the A-homomorphism $D : M \to N$ factors through

$$M \longrightarrow ((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M \xrightarrow{v} N$$

where the first arrow is the canonical homomorphism $t \mapsto 1 \otimes t$, and v is a B-homomorphism, the B-module structure on its source coming from the first factor B (whereas, in forming the tensor product over B, the B-module structure on $(B \otimes_A B)/\mathfrak{I}^{n+1}$ comes from the second factor B). Note also that the B-module $((B \otimes_A B)/\mathfrak{I}^{n+1}) \otimes_B M$ is isomorphic to $(B \otimes_A M)/\mathfrak{I}^{n+1}(B \otimes_A M)$, where $B \otimes_A M$ is regarded as a $(B \otimes_A B)$-module and its B-module structure comes from the homomorphism $b \mapsto b \otimes 1$ from B to $B \otimes_A B$. Let D' then be the B-homomorphism from $B \otimes_A M$ to N such that $D'(b \otimes t) = bD(t)$; the factorization condition on D is equivalently that D' vanish on the B-submodule $\mathfrak{I}^{n+1}(B \otimes_A M)$.

(16.8.3). It is clear that the set of differential operators of order $\leqslant n$ from $\mathscr{F}$ to $\mathscr{G}$ forms an additive group, denoted by $\mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G})$; when $\mathscr{F} = \mathscr{G} = \mathscr{O}_X$, we also write $\mathrm{Diff}^n_{X/S}$ instead of $\mathrm{Diff}^n_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$.

We saw in (16.8.1) that for two open sets $U \supset V$ of X, there is a canonical restriction homomorphism

$$\mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U) \longrightarrow \mathrm{Diff}^n_{V/S}(\mathscr{F}|V, \mathscr{G}|V)$$

from which we deduce that $U \mapsto \mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U)$ is a presheaf of additive groups; in fact, it is **IV-4** | 41
actually a *sheaf*, since for an open set U varying in X, the homomorphisms $u \mapsto u \circ d^n_{U/S, \mathscr{F}|U}$ are *isomorphisms* of sheaves of additive groups

(16.8.3.1) $$\mathrm{Hom}_{\mathscr{O}_U}(\mathscr{P}^n_{U/S}(\mathscr{F}|U), \mathscr{G}|U) \simeq \mathrm{Diff}^n_{U/S}(\mathscr{F}|U, \mathscr{G}|U),$$

because of the fact that the image of $\mathscr{F}$ by $d^n_{X/S,\mathscr{F}}$ generates $\mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.6). We denote this sheaf by $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$, and we have:

Proposition (16.8.4). — *The isomorphisms* (16.8.3.1) *define an isomorphism of sheaves of additive groups*

(16.8.4.1) $$\mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}) \simeq \mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}).$$

When $\mathscr{F} = \mathscr{G} = \mathscr{O}_X$, we also write $\mathscr{D}\!\mathit{iff}^n_{X/S}$ instead of $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{O}_X, \mathscr{O}_X)$; it follows from (16.8.4) that $\mathscr{D}\!\mathit{iff}^n_{X/S}$ is canonically identified with the *dual* of the $\mathscr{O}_X$-module $\mathscr{P}^n_{X/S}$; so we also write $\langle t, D \rangle$ instead of $u(t)$ if t is a section of $\mathscr{P}^n_{X/S}$ over an open set and if u is a homomorphism from $\mathscr{P}^n_{X/S}$ to $\mathscr{O}_X$ corresponding to D.

(16.8.5). Since $\mathscr{P}^n_{X/S}(\mathscr{F})$ has an $\mathscr{O}_X$-bimodule structure (16.7.4), we canonically obtain an $\mathscr{O}_X$-bimodule structure on $\mathscr{H}\!\mathit{om}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G})$, and hence on $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ by (16.8.4.1). More precisely, to the left $\mathscr{O}_X$-module structure on $\mathscr{P}^n_{X/S}(\mathscr{F})$ corresponds, because of (16.8.1), the left $\mathscr{O}_X$-module structure on $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ explained as follows: for every open set U of X, every section $a \in \Gamma(U, \mathscr{O}_X)$ and every differential operator $D : \mathscr{F}|U \to \mathscr{G}|U$, aD is the differential operator which, for every section $t \in \Gamma(U, \mathscr{F})$, makes correspond the section

(16.8.5.1) $$(aD)(t) = a(D(t))$$

of $\Gamma(U, \mathscr{G})$. Similarly, the right $\mathscr{O}_X$-module structure on $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ is made explicit as follows: under the same notations as above, Da is the operator which, to every $t \in \Gamma(U, \mathscr{F})$, makes correspond the section

(16.8.5.2) $$(Da)(t) = D(at).$$

Proposition (16.8.6). — *If $f : X \to S$ is a morphism locally of finite presentation, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module of finite presentation and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_X$-module, then $\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G})$ is a quasi-coherent $\mathscr{O}_X$-module for either of the structures defined in* (16.8.5).

Proof. The proposition follows from the fact that, under these hypotheses, $\mathscr{P}^n_{X/S}(\mathscr{F})$ is a quasi-coherent $\mathscr{O}_X$-module of finite presentation (16.7.4), together with (**I**, 1.3.12). ☐

(16.8.7). The set of differential operators (of unspecified order (16.8.1)) is denoted by $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$; we also see as in (16.8.3) that $U \mapsto \mathrm{Diff}_{U/S}(\mathscr{F}|U,\mathscr{G}|U)$ is a sheaf of additive groups, which we will denote by $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$. It is immediate that $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$ is the union of the increasing filtered family of its subsheaves $\mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G})$; if X is quasi-compact, $\mathrm{Diff}_{X/S}(\mathscr{F},\mathscr{G})$ is similarly the union of its subgroups $\mathrm{Diff}^n_{X/S}(\mathscr{F},\mathscr{G})$ (16.8.1). The $\mathscr{O}_X$-bimodule structures on the $\mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G})$ therefore induce an $\mathscr{O}_X$-bimodule structure on $\mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$, again described by (16.8.5.1) and (16.8.5.2). **IV-4** | 42

Note that, for $n \leqslant m$, we have a commutative diagram

(16.8.7.1)
$$\begin{array}{ccc} \mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G}) & \xrightarrow{\sim} & \mathscr{D}iff^n_{X/S}(\mathscr{F},\mathscr{G}) \\ \downarrow & & \downarrow \\ \mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^m_{X/S}(\mathscr{F}),\mathscr{G}) & \xrightarrow{\sim} & \mathscr{D}iff^m_{X/S}(\mathscr{F},\mathscr{G}) \end{array}$$

where the horizontal arrows are the isomorphisms (16.8.4.1) and the left vertical arrow comes from the canonical homomorphism $\mathscr{P}^m_{X/S}(\mathscr{F}) \to \mathscr{P}^n_{X/S}(\mathscr{F})$ (16.7.7). For every open set U of X, we then equip $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F})) = \varprojlim \Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F}))$ with the projective-limit topology of the discrete topologies on the groups $\Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F}))$. This makes $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F}))$ a topological $\Gamma(U,\mathscr{O}_X)$-bimodule, so that $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ becomes a sheaf with values in the category of topological commutative groups (**0**, 3.2.6). Then, by [**?**, II, 1.11], the inductive limit of the system of sheaves of commutative groups $\big(\mathscr{H}om_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}),\mathscr{G})\big)$ is precisely the sheaf of germs of *continuous* homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$ (the latter equipped with the discrete topology): indeed, the continuous homomorphisms from $\Gamma(U,\mathscr{P}^\infty_{X/S}(\mathscr{F}))$ to the discrete group $\Gamma(U,\mathscr{G})$ correspond bijectively to the inductive systems of group homomorphisms $\Gamma(U,\mathscr{P}^n_{X/S}(\mathscr{F})) \to \Gamma(U,\mathscr{G})$. We can furthermore express (16.8.4) by saying there is a canonical isomorphism

$$\mathscr{H}om.\mathrm{cont}_{\mathscr{O}_X}(\mathscr{P}^\infty_{X/S}(\mathscr{F}),\mathscr{G}) \simeq \mathscr{D}iff_{X/S}(\mathscr{F},\mathscr{G})$$

where the left-hand side denotes the sheaf of germs of continuous homomorphisms from $\mathscr{P}^\infty_{X/S}(\mathscr{F})$ to $\mathscr{G}$.

Proposition (16.8.8). — *Let $\mathscr{F},\mathscr{G}$ be two $\mathscr{O}_X$-modules, $D : \mathscr{F} \to \mathscr{G}$ a homomorphism of $\psi^*(\mathscr{O}_S)$-modules, n an integer $\geqslant 0$. The following conditions are equivalent:*

- *(a) D is a differential operator of order $\leqslant n$.*
- *(b) For all sections a of $\mathscr{O}_X$ over an open set U, the homomorphism $D_a : \mathscr{F}|U \to \mathscr{G}|U$ such that, for every section t of $\mathscr{F}$ over an open set $V \subset U$, we have*

(16.8.8.1)
$$D_a(t) = D(at) - aD(t)$$

is a differential operator of order $\leqslant n-1$.

- *(c) For every open set U of X, every family $(a_i)_{1\leqslant i\leqslant n+1}$ of $n+1$ sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, we have the identity*

(16.8.8.2)
$$\sum_{H\subset I_{n+1}} (-1)^{\mathrm{Card}(H)}\Big(\prod_{i\in H} a_i\Big)D\Big(\Big(\prod_{i\notin H} a_i\Big)t\Big) = 0$$

(where I_{n+1} is the interval $1 \leqslant i \leqslant n+1$ of $\mathbb{N}$).

IV-4 | 43

Proof. Let us first prove the equivalence of (a) and (c). By definition, to prove that D is a differential operator of order $\leqslant n$, it suffices to prove this for the restriction $D|U : \mathscr{F}|U \to \mathscr{G}|U$ to every affine open subset U of X; on the other hand, property (c) holds for every open subset U of X if it holds for every affine open subset. We may therefore restrict ourselves to the case where $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine. By (16.8.2) (whose notation we retain), condition (a) then means that the A-homomorphism $D' : B \otimes_A M \to N$ defined by $D'(b \otimes t) = bD(t)$ vanishes on $\mathfrak{I}^{n+1}(B \otimes_A M)$; by (**0**, 20.4.4), this is equivalent to saying that D' vanishes on every element of the form

$$(\prod_{i=1}^{n+1}(a_i \otimes 1 - 1 \otimes a_i)).(1 \otimes t)$$

where $a_i \in B$ and $t \in M$. Now this element can be written as $\sum_{H \subset I_{n+1}}(\prod_{i \in H} a_i) \otimes ((\prod_{i \notin H} a_i)t)$, and its image under D' is the left-hand side of (16.8.8.2); this proves the equivalence of (a) and (c).

Let us now prove the equivalence of (b) and (c). We argue by induction on n, the assertion being trivial for $n = 0$. Writing a_{n+1} in place of a in condition (b), we see from the induction hypothesis that condition (b) means that, for every family $(a_i)_{1 \leqslant i \leqslant n}$ of n sections of $\mathscr{O}_X$ over U and every section t of $\mathscr{F}$ over U, one has

$$\sum_{H' \subset I_n}(-1)^{\mathrm{Card}(H')}(\prod_{i \in H'} a_i)D_{a_{n+1}}((\prod_{i \notin H'} a_i)t) = 0.$$

But if in this relation we replace $D_{a_{n+1}}$ by its definition (16.8.8.1), we see at once that, up to sign, we obtain the left-hand side of (16.8.8.2); hence the conclusion. $\square$

Proposition (16.8.9). — *If $D : \mathscr{F} \to \mathscr{G}$ is a differential operator of order $\leqslant n$, and $D' : \mathscr{G} \to \mathscr{H}$ a differential operator of order $\leqslant n'$, then $D' \circ D : \mathscr{F} \to \mathscr{H}$ is a differential operator of order $\leqslant n + n'$.*

Proof. By hypothesis, we can write $D = u \circ d^n_{X/S,\mathscr{F}}$ and $D' = v \circ d^{n'}_{X/S,\mathscr{G}}$, where $u : \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \to \mathscr{G}$ and $v : \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G} \to \mathscr{H}$ are $\mathscr{O}_X$-homomorphisms. It is therefore enough to show that the composite homomorphism of sheaves of additive groups

$$\mathscr{F} \xrightarrow{d^n_{X/S,\mathscr{F}}} \mathscr{P}^n_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{u} \mathscr{G} \xrightarrow{d^{n'}_{X/S,\mathscr{G}}} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

factors as

$$\mathscr{F} \xrightarrow{d^{n+n'}_{X/S,\mathscr{F}}} \mathscr{P}^{n+n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{F} \xrightarrow{w} \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{G}$$

where w is an $\mathscr{O}_X$-homomorphism. It suffices to prove the following lemma.

Lemma (16.8.9.1). — *There is one and only one $\mathscr{O}_X$-homomorphism*

(16.8.9.2) $$\delta : \mathscr{P}^{n+n'}_{X/S} \longrightarrow \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}) = \mathscr{P}^{n'}_{X/S} \otimes_{\mathscr{O}_X} \mathscr{P}^n_{X/S}$$

making the following diagram commute **IV-4** | 44

(16.8.9.3)
$$\begin{tikzcd} \mathscr{O}_X \arrow[r, "d^{n+n'}_{X/S}"] \arrow[d, "d^n_{X/S}"'] & \mathscr{P}^{n+n'}_{X/S} \arrow[d, "\delta"] \\ \mathscr{P}^n_{X/S} \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}) \end{tikzcd}$$

We will then have, indeed, a commutative diagram deduced from (16.8.9.3) by tensorization with $\mathscr{F}$

$$\begin{tikzcd} \mathscr{F} \arrow[r, "d^{n+n'}_{X/S,\mathscr{F}}"] \arrow[d, "d^n_{X/S,\mathscr{F}}"'] & \mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \arrow[d, "\delta\otimes 1"] \\ \mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \end{tikzcd}$$

and on the other hand, we verify immediately from the definition (16.7.5) that the diagram

$$\begin{tikzcd}
\mathscr{P}^n_{X/S}(\mathscr{F}) \arrow[r, "u"] \arrow[d, "d^{n'}_{X/S,\mathscr{P}^n_{X/S}(\mathscr{F})}"'] & \mathscr{G} \arrow[d, "d^{n'}_{X/S,\mathscr{G}}"] \\
\mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \arrow[r, "1\otimes u"'] & \mathscr{P}^{n'}_{X/S}(\mathscr{G})
\end{tikzcd}$$

is commutative. We finish the proof by taking w to be the composite $\mathscr{O}_X$-homomorphism

$$\mathscr{P}^{n+n'}_{X/S}(\mathscr{F}) \xrightarrow{\delta\otimes 1} \mathscr{P}^{n'}_{X/S}(\mathscr{P}^n_{X/S}(\mathscr{F})) \xrightarrow{1\otimes u} \mathscr{P}^{n'}_{X/S}(\mathscr{G}).$$

□

Proof of Lemma 16.8.9.1. It remains to prove Lemma (16.8.9.1). In view of (16.7.6), which proves the uniqueness of δ, we are reduced to the case where $S = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine. Putting $\mathfrak{I} = \mathfrak{I}_{B/A}$, it suffices to define a canonical homomorphism of B-modules

$$\phi : (B\otimes_A B)/\mathfrak{I}^{n+n'+1} \longrightarrow ((B\otimes_A B)/\mathfrak{I}^{n'+1}) \otimes_B ((B\otimes_A B)/\mathfrak{I}^{n+1})$$

the B-module structures on both sides being induced by the first factor B. Recall that in the tensor product on the right, $(B\otimes_A B)/\mathfrak{I}^{n'+1}$ must be regarded as a right B-module by its second B factor, **IV-4** | 45
and $(B\otimes_A B)/\mathfrak{I}^{n+1}$ as a left B-module by its first B factor (16.7.2). This is in turn equivalent to defining a homomorphism of B-modules

$$\phi_0 : B\otimes_A B \longrightarrow ((B\otimes_A B)/\mathfrak{I}^{n'+1}) \otimes_B ((B\otimes_A B)/\mathfrak{I}^{n+1})$$

and proving that it vanishes on $\mathfrak{I}^{n+n'+1}$. Such a homomorphism is defined at once by the condition

$$\phi_0(b\otimes b') = \pi_{n'}(b\otimes 1)\otimes \pi_n(1\otimes b') \quad \text{for } b, b' \text{ in } B$$

with the notation of (16.3.7). Moreover, it is immediate that ϕ_0 is a homomorphism of *rings*. Now we may write

$$\phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes\pi_n(1\otimes 1) + \pi_{n'}(1\otimes b)\otimes\pi_n(1\otimes 1) - \pi_{n'}(1\otimes 1)\otimes\pi_n(1\otimes b)$$

and we have

$$\pi_{n'}(1\otimes b)\otimes\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)b\otimes\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes b\pi_n(1\otimes 1) = \pi_{n'}(1\otimes 1)\otimes\pi_n(b\otimes 1)$$

whence, finally,

$$\phi_0(b\otimes 1 - 1\otimes b) = \pi_{n'}(b\otimes 1 - 1\otimes b)\otimes\pi_n(1\otimes 1) + \pi_{n'}(1\otimes 1)\otimes\pi_n(b\otimes 1 - 1\otimes b). \tag{16.8.9.4}$$

A product of $n+n'+1$ factors of the form (16.8.9.4) is therefore necessarily zero, since this is so for a product of $n+1$ factors of the form $\pi_n(b\otimes 1 - 1\otimes b)$ and for a product of $n'+1$ factors of the form $\pi_{n'}(b\otimes 1 - 1\otimes b)$. The conclusion therefore follows from (**0**, 20.4.4). □

Corollary (16.8.10). — *The sheaf $\mathscr{D}\!\mathit{iff}_{X/S}(\mathscr{O}_X,\mathscr{O}_X)$ (also denoted $\mathscr{D}\!\mathit{iff}_{X/S}$) is canonically endowed with a structure of sheaf of rings, and the $\mathscr{D}\!\mathit{iff}^n_{X/S}$ form an increasing filtration compatible with this structure.*

In particular, $\mathscr{D}\!\mathit{iff}^0_{X/S}$ is a sheaf of subrings of $\mathscr{D}\!\mathit{iff}_{X/S}$, which is canonically identified with $\mathscr{O}_X$ (16.8.1). The formulas (16.8.5.1) and (16.8.5.2) show that the structure of $\mathscr{O}_X$-bimodule of $\mathscr{D}\!\mathit{iff}_{X/S}$ comes from the multiplication on the left and on the right by sections of $\mathscr{O}_X$ considered as a sheaf of subrings of $\mathscr{D}\!\mathit{iff}_{X/S}$.

Remarks (16.8.11). — (i) Suppose that $\mathscr{F} = \oplus_{\lambda\in L}\mathscr{F}_\lambda$; then it is clear from (16.7.2.1) that $\mathscr{P}^n_{X/S}(\mathscr{F}) = \oplus_{\lambda\in L}\mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$. Since the functor $\mathscr{F}\mapsto\Gamma(U,\mathscr{F})$ commutes with the formation of arbitrary direct sums, $d^n_{X/S,\mathscr{F}}$ is the homomorphism whose restriction to each $\mathscr{F}_\lambda$ is $d^n_{X/S,\mathscr{F}_\lambda} : \mathscr{F}_\lambda \to \mathscr{P}^n_{X/S}(\mathscr{F}_\lambda)$; it follows at once that

$$\operatorname{Diff}^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\lambda\in L}\operatorname{Diff}^n_{X/S}(\mathscr{F}_\lambda,\mathscr{G}),$$

and therefore also (**0**, 3.2.6)

$$\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F},\mathscr{G}) = \prod_{\lambda\in L}\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}_\lambda,\mathscr{G}).$$

Moreover, if $\mathscr{G} = \prod_{\mu \in M} \mathscr{G}_\mu$ (**0**, 3.2.6), we have

$$\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}) = \prod_{\mu \in M} \mathrm{Hom}_{\mathscr{O}_X}(\mathscr{P}^n_{X/S}(\mathscr{F}), \mathscr{G}_\mu),$$

Every homomorphism u from $\mathscr{P}^n_{X/S}(\mathscr{F})$ to $\mathscr{G}$ corresponds bijectively to the family of its composites $u_\mu : \mathscr{P}^n_{X/S}(\mathscr{F}) \to \mathscr{G} \to \mathscr{G}_\mu$. Thus **IV-4** | 46

$$\mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\mu \in M} \mathrm{Diff}^n_{X/S}(\mathscr{F}, \mathscr{G}_\mu),$$

and consequently also

$$\mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}) = \prod_{\mu \in M} \mathscr{D}\!\mathit{iff}^n_{X/S}(\mathscr{F}, \mathscr{G}_\mu).$$

(ii) So far, we have encountered differential operators $\mathscr{F} \to \mathscr{G}$ almost exclusively when $\mathscr{F}$ and $\mathscr{G}$ are locally free of finite rank; in that case, by (i), their structure is reduced locally to that of the sheaf $\mathscr{D}\!\mathit{iff}_{X/S}$. The latter will be studied below, in a particular case, in (16.11).

16.9. Regular and quasi-regular immersions

Definition (16.9.1). — Let X be a ringed space. We say that an ideal $\mathscr{I}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular) if, for every point $x \in \mathrm{Supp}(\mathscr{O}_X/\mathscr{I})$, there is an open neighbourhood U of x in X and a regular sequence (**0**, 15.2.2) (resp. quasi-regular sequence (**0**, 15.2.2)) of elements of $\Gamma(U, \mathscr{O}_X)$ which generates $\mathscr{I}|U$.

A regular (resp. quasi-regular) sequence of sections of $\mathscr{O}_X$ over U which generates $\mathscr{I}|U$ is called a *regular system* (resp. *quasi-regular system*) of generators of $\mathscr{I}|U$.

Definition (16.9.2). — Let $j : Y \to X$ be an immersion of preschemes, and let U be an open subset of X such that $j(Y) \subset U$ and j is a closed immersion of Y into U. We say that j is regular (resp. quasi-regular) if the closed subprescheme $j(Y)$ of U associated to j is defined by a regular (resp. quasi-regular) ideal of $\mathscr{O}_U$ (condition independent of the choice of U).

We say that a subprescheme Y of a prescheme X is *regularly immersed* (resp. *quasi-regularly immersed*) if the canonical injection $j : Y \to X$ is a regular immersion (resp. quasi-regular immersion). If Y is a closed subprescheme and $\mathscr{I}$ is the ideal of $\mathscr{O}_X$ that defines Y, it is the same as asking $\mathscr{I}$ to be *regular* (resp. *quasi-regular*).

For example, if A is an *integral* ring and f is a nonzero element of A, the closed subprescheme $V(f)$ of $\mathrm{Spec}(A)$ (isomorphic to $\mathrm{Spec}(A/(f))$) is *regularly immersed* in $\mathrm{Spec}(A)$.

Every regular ideal is quasi-regular (**0**, 15.2.2); every regular immersion is quasi-regular (cf. (16.9.11) for a converse).

Proposition (16.9.3). — *Let X be a ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$, $(f_i)_{1 \leqslant i \leqslant m}$ a finite sequence of sections of $\mathscr{O}_X$ over X generating $\mathscr{I}$. For (f_i) to be a quasi-regular sequence* (**0**, 15.2.2), *it is necessary and sufficient that the following conditions be satisfied:*

(i) *The canonical images of f_i in $\mathscr{I}/\mathscr{I}^2$ form a basis of this $\mathscr{O}_X/\mathscr{I}$-module.*
(ii) *The canonical surjective homomorphism* (16.1.2.2)

$$\mathbf{S}^\bullet_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}\!r^\bullet_{\mathscr{I}}(\mathscr{O}_X)$$

is bijective.

Moreover, if this is so, every sequence $(f_i')_{1 \leqslant i \leqslant n}$ of n sections of $\mathscr{I}$ over X which generates $\mathscr{I}$ is quasi-regular.

IV-4 | 47

Proof. The two conditions in the statement merely express the definition given in (**0**, 15.2.2), taking into account the definition of the canonical homomorphisms (16.2.1.1). The last claim follows from the fact that, if a module M over a commutative ring A admits a basis of n elements, every system of n generators of M is a basis (Bourbaki, *Alg. comm.*, chap. II, §3, cor. 5 of th. 1). ☐

Corollary (16.9.4). — *Let X be a locally ringed space, $\mathscr{I}$ an ideal of $\mathscr{O}_X$. For $\mathscr{I}$ to be quasi-regular, it is necessary and sufficient that the following conditions hold:*

(i) *$\mathscr{I}$ is of finite type.*
(ii) *$\mathscr{I}/\mathscr{I}^2$ is a locally free $\mathscr{O}_X/\mathscr{I}$-module.*
(iii) *The canonical homomorphism*

$$\mathbf{S}^{\bullet}_{\mathscr{O}_X/\mathscr{I}}(\mathscr{I}/\mathscr{I}^2) \longrightarrow \mathscr{G}r^{\bullet}_{\mathscr{I}}(\mathscr{O}_X) \tag{16.9.4.1}$$

is bijective.

Proof. The necessity of the conditions follows immediately from (16.9.3). To see that they are sufficient, it is enough, by (16.9.3), to prove the following: if, at a point $x \in \operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$, there are an open neighbourhood U of x in X and n sections f_i $(1 \leqslant i \leqslant n)$ of $\mathscr{I}$ over U whose canonical images in $\mathscr{I}/\mathscr{I}^2$ form a basis of $(\mathscr{I}/\mathscr{I}^2)|U$ over $(\mathscr{O}_X/\mathscr{I})|U$, then there is an open neighbourhood $V \subset U$ of x such that the $f_i|V$ generate $\mathscr{I}|V$. Now, by hypothesis, $\mathscr{I}_x \neq \mathscr{O}_x$, so $\mathscr{I}_x$ is contained in the maximal ideal of $\mathscr{O}_x$. Since $\mathscr{I}_x$ is an $\mathscr{O}_x$-module of finite type and the classes of $(f_i)_x$ in $\mathscr{I}_x/\mathscr{I}_x^2$ generate this $(\mathscr{O}_x/\mathscr{I}_x)$-module, Nakayama's lemma shows that the $(f_i)_x$ generate $\mathscr{I}_x$. Since $\mathscr{I}$ is of finite type, we conclude by (**0**, 5.2.2). □

Corollary (16.9.5). — *Let X be a locally ringed space, $\mathscr{I}$ a quasi-regular ideal of $\mathscr{O}_X$, $(f_i)_{1\leqslant i\leqslant n}$ a sequence of sections of $\mathscr{I}$ over X, x a point of* $\operatorname{Supp}(\mathscr{O}_X/\mathscr{I})$. *The following conditions are equivalent:*

(a) *There is an open neighbourhood U of x in X such that the $f_i|U$ form a quasi-regular sequence of elements of $\Gamma(U, \mathscr{O}_X)$ generating $\mathscr{I}|U$.*
(b) *The $(f_i)_x$ form a system of generators of $\mathscr{I}_x$ whose size is as small as possible.*
(b′) *The $(f_i)_x$ form a minimal set of generators of $\mathscr{I}_x$.*
(c) *If $\bar{f}_i$ is the canonical image of f_i in $\Gamma(X, \mathscr{I}/\mathscr{I}^2)$, the $(\bar{f}_i)_x$ form a basis for the $(\mathscr{O}_x/\mathscr{I}_x)$-module $\mathscr{I}_x/\mathscr{I}_x^2$.*

Proof. By hypothesis, $\mathscr{O}_x$ is a local ring, $\mathscr{I}_x$ an ideal of finite type of $\mathscr{O}_x$ contained in the maximal ideal of $\mathscr{O}_x$; the equivalence of (b), (b′) and (c) follows from Nakayama's lemma (Bourbaki, *Alg. comm.*, chap. II, §3, n°2, prop. 5). It is clear that (a) implies (c) because of (16.9.3); on the other hand, from (**0**, 5.2.2) it follows that if condition (c) is satisfied (and therefore so is (b)), there is a neighbourhood U of x in X such that $(\mathscr{I}/\mathscr{I}^2)|U$ has constant rank equal to n, and that the $f_i|U$ generate $\mathscr{I}|U$; it suffices now to apply the last assertion of (16.9.3) to U. □

Remarks (16.9.6). — (i) Under the general hypothesis of (16.9.5), for the sequence (f_i) to generate $\mathscr{I}$, it is not enough that the $(\bar{f}_i)_y$ form a basis of the $(\mathscr{O}_y/\mathscr{I}_y)$-module $(\mathscr{I}_y/\mathscr{I}_y^2)$ for all
$y \in X$. An example is obtained by taking $X = \operatorname{Spec}(A)$, where A is a Dedekind ring, and **IV-4** | 48
$\mathscr{I} = \widetilde{\mathfrak{I}}$, where $\mathfrak{I}$ is a *non-principal* prime ideal of A. Then $\mathscr{I}_y/\mathscr{I}_y^2 = 0$ at every point y other than the point $x \in X$ corresponding to $\mathfrak{I}$, while $\mathscr{I}_x/\mathscr{I}_x^2$ has rank 1 over the field $\mathscr{O}_x/\mathscr{I}_x$; moreover, $\mathscr{I}$ is plainly a regular ideal.

(ii) In (16.9.5), one cannot replace "quasi-regular" by "regular", even when X is a prescheme (cf. (16.9.12)). Indeed, let B denote the ring of germs at 0 in $\mathbb{R}$ of infinitely differentiable functions. It has a maximal ideal $\mathfrak{m}$ generated by the germ t of the identity map of $\mathbb{R}$ at 0, and the intersection $\mathfrak{n}$ of the $\mathfrak{m}^k$ for $k > 0$ is nonzero. Now let A be the quotient ring $B[T]/\mathfrak{n}TB[T]$, and let f_1, f_2 be the canonical images in A of the elements t and T of $B[T]$. The sequence (f_1, f_2) is *regular* in A: indeed, f_1 is not a zero divisor in A, because the relation $tP[T] \in \mathfrak{n}TB[T]$, for a polynomial $P \in B[T]$, implies that the products of t with the coefficients of P belong to $\mathfrak{n}$; it follows at once that the coefficients themselves belong to $\mathfrak{n}$, and hence $P[T] \in \mathfrak{n}TB[T]$. Since B/tB is isomorphic to $\mathbb{R}$, A/f_1A is isomorphic to the polynomial ring $\mathbb{R}[T]$ and is therefore integral; the image of f_2 in A/f_1A, being equal to T, is consequently not a zero divisor, proving the assertion. However, f_2 is a zero divisor in A, since for every nonzero element $x \in \mathfrak{n}$, the image of x in A is nonzero whereas the image of xT is zero. We conclude that the sequence (f_2, f_1) is *not regular* in A; on the other hand, the ideal $\mathfrak{I} = f_1A + f_2A$ is distinct from A, so the conditions (b), (b′) and (c) of (16.9.5) do not imply the condition (a) when we replace "quasi-regular" by "regular".

(16.9.7). If $X = \operatorname{Spec}(A)$ is an affine scheme, we say that an ideal $\mathfrak{I}$ of A is *regular* (resp. *quasi-regular*) if the ideal $\mathscr{J} = \widetilde{\mathfrak{I}}$ of $\mathscr{O}_X$ is regular (resp. quasi-regular). We note that this notion is *local* and does not imply the existence of a *system of generators* of $\mathfrak{I}$ forming a regular (resp. quasi-regular) sequence in A, as Example (16.9.6, (i)) shows; it does, however, imply this when A is *local* (16.9.5).

The proposition (16.9.4) can be translated in terms of quasi-regular immersions in the following manner:

Proposition (16.9.8). — *Let $j : Y \to X$ be a morphism of preschemes; for j to be a quasi-regular immersion, it is necessary and sufficient that j satisfies the following conditions:*

(i) j is an immersion locally of finite presentation.
(ii) The conormal sheaf $\mathscr{G}r^1(j) = \mathscr{N}_{Y/X}$ (16.1.2) is a locally free $\mathscr{O}_Y$-module.
(iii) The canonical homomorphism

$$\mathbf{S}^{\bullet}_{\mathscr{O}_Y}(\mathscr{G}r^1(j)) \longrightarrow \mathscr{G}r^{\bullet}(j)$$

(16.1.2.2) is bijective.

Proof. Since the question is local on Y, we may restrict ourselves to the case where j is the canonical injection of a closed subprescheme Y of X; the translation of (16.9.4) into (16.9.8) then follows from the description of $\mathscr{G}r^1(j)$ and $\mathscr{G}r^{\bullet}(j)$ in terms of the ideal $\mathscr{I}$ of $\mathscr{O}_X$ defining Y (16.1.3, (ii)). □

Corollary (16.9.9). — *Let Y be a prescheme, X a Y-prescheme, and $j : Y \to X$ a Y-section of X, so that the n-th normal invariant $\mathscr{A}^{(n)}$ of j (16.1.2) is an augmented $\mathscr{O}_Y$-algebra (16.1.7); take $\mathscr{A}^{(\infty)} = \varprojlim \mathscr{A}^{(n)}$. For j to be a quasi-regular immersion, it is necessary and sufficient that j be locally of finite presentation and that every $y \in Y$ admit an affine open neighbourhood U with coordinate ring C such that $\mathscr{A}^{(\infty)}|U$ is isomorphic, as an augmented topological $\mathscr{O}_U$-algebra, to $\mathscr{O}_U[[T_1, \ldots, T_n]]$.*

Proof. We may reduce to the case where j is a closed immersion by restricting to a sufficiently small neighbourhood of y (see the argument of (16.4.11)); then $\mathscr{O}_Y$ is identified with the quotient algebra $\mathscr{O}_X/\mathscr{I}$, and the canonical surjective homomorphism $\mathscr{O}_X \to \mathscr{O}_Y$ admits a right inverse (16.1.7). We **IV-4** | 49
may therefore suppose that $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$ are affine, with B an augmented A-algebra whose augmentation ideal $\mathfrak{I}$ is of finite type. Since $\mathscr{A}^{(n)}$ is then identified with $(B/\mathfrak{I}^{n+1})^{\sim}$, the corollary follows from the equivalence of (b) and (c) in (**0**, 19.5.4), since $B/\mathfrak{I} = A$. □

We note that, in the affine case under consideration, the fact that j is a quasi-regular immersion is also equivalent, by (**0**, 19.5.4), to saying that B is a *formally smooth* A-algebra for the $\mathfrak{I}$-preadic topology.

We also note that the condition that j is locally of finite presentation is always satisfied when $X \to Y$ is locally of finite type (1.4.3, (v)).

Proposition (16.9.10). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, $j : Y \to X$ the canonical injection, y a point of Y.*

(i) *For there to exist an open neighbourhood U of y in X such that the restriction $Y \cap U \to U$ of j be a regular immersion, it is necessary and sufficient that the kernel $\mathscr{I}_y$ of the surjective homomorphism $\mathscr{O}_{X,y} \to \mathscr{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathscr{O}_{X,y}$.*
(ii) *For the immersion j to be regular, it is necessary and sufficient that it is quasi-regular.*

Proof. (i) We can reduce to the case where Y is a subprescheme defined by a *coherent* ideal $\mathscr{I}$ of $\mathscr{O}_X$. The condition is clearly necessary. Conversely, if $\mathscr{I}_y$ is generated by a regular sequence $(s_i)_y$, where the s_i are sections of $\mathscr{I}$ over an open neighbourhood U of y in X, we may suppose that the s_i generate $\mathscr{I}|U$ (**0**, 5.5.2) and form a regular sequence (**0**, 15.2.4), whence the assertion.

(ii) The fact that a quasi-regular immersion is regular follows from (i) and the identification of quasi-regular and regular sequences in $\mathscr{O}_{X,y}$ formed by elements of the maximal ideal (**0**, 15.1.11).

□

When, without any Noetherian hypothesis on X, the kernel $\mathscr{I}_y$ of $\mathscr{O}_{X,y} \to \mathscr{O}_{Y,y}$ is generated by a regular sequence of elements of $\mathscr{O}_{X,y}$, we say that the immersion j is *regular at the point y*.

Corollary (16.9.11). — *Let X be a locally Noetherian prescheme; then every quasi-regular ideal of $\mathscr{O}_X$ is regular.*

Remarks (16.9.12). — (i) We note that a regular immersion is not in general a flat morphism, and hence *a fortiori* is not a regular morphism in the sense of (6.8.1).

(ii) Let A be a local Noetherian ring; it follows immediately from (16.9.4) and from (**0**, 17.1.1) that for A to be *regular*, it is necessary and sufficient that its maximal ideal $\mathfrak{m}$ is *quasi-regular* (or regular, which amounts to the same thing given that A is Noetherian). For an affine Noetherian scheme X to be *regular*, it is necessary and sufficient that for every closed point $x \in X$, the canonical injection $\operatorname{Spec}(k(x)) \to X$ be a *regular* immersion.

Proposition (16.9.13). — *Let X be a locally Noetherian prescheme, Y a subprescheme of X, Y' a subprescheme of Y, such that the canonical injection $j : Y' \to Y$ is regular. Then the sequence of $\mathscr{O}_{Y'}$-modules*

(16.9.13.1) $$0 \longrightarrow j^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X} \longrightarrow \mathscr{N}_{Y'/Y} \longrightarrow 0$$

is exact; furthermore, for every $x \in X$, there is an open neighbourhood U of x such that the restrictions to U of the homomorphisms of (16.9.13.1) form a split exact sequence. **IV-4** | 50

Let us first prove the following lemma:

Lemma (16.9.13.2). — *Let A be a ring, $\mathfrak{I}$ an ideal of A, $A' = A/\mathfrak{I}$, $(f_i)_{1\leqslant i\leqslant r}$ a sequence of elements of A which is A'-regular, $\mathfrak{K} = \sum_i f_i A$, $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, $\mathfrak{K}' = \sum_i f_i A'$, so that $C = A/\mathfrak{L}$ is isomorphic to $A'/\mathfrak{K}'$. For every integer $n > 0$, and every integer $N \geqslant n$, we have the relation*

(16.9.13.3) $$\mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I}\mathfrak{K}^n + (\mathfrak{I} \cap \mathfrak{K}^N).$$

Proof. It is plainly enough to prove that every element of the left-hand side belongs to the right-hand side, and an induction reduces us to the case $N = n + 1$. An element of the left-hand side of (16.9.13.3), being in $\mathfrak{K}^n$, can be written as $P(f_1, \ldots, f_r)$, where $P \in A[T_1, \ldots, T_r]$ is homogeneous of degree n. If f'_i is the canonical image of f_i in A', the hypothesis $P(f_1, \ldots, f_r) \in \mathfrak{I}$ means that $P(f'_1, \ldots, f'_r) = 0$. But $P(f'_1, \ldots, f'_r) \in \mathfrak{K}'^n$, so the canonical image of $P(f'_1, \ldots, f'_r)$ in $\mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is zero. Now the hypothesis that (f_i) is A'-regular implies that the canonical homomorphism $\mathbf{S}^n_C(\mathfrak{K}'/\mathfrak{K}'^2) \to \mathfrak{K}'^n/\mathfrak{K}'^{n+1}$ is bijective (**0**, 15.1.9); we conclude that the coefficients of P are in $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$. It follows at once that $P(f_1, \ldots, f_r) \in \mathfrak{I}\mathfrak{K}^n + \mathfrak{K}^{n+1}$, and since $P(f_1, \ldots, f_r) \in \mathfrak{I}$, we finally have $P(f_1, \ldots, f_r) \in \mathfrak{I}\mathfrak{K}^n + (\mathfrak{I} \cap \mathfrak{K}^{n+1})$, which proves the lemma. □

Taking the quotients of both sides of (16.9.13.3) by $\mathfrak{I}\mathfrak{K}^n$, we see that the relations (16.9.13.3), for $N \geqslant n$, imply

(16.9.13.4) $$(\mathfrak{I} \cap \mathfrak{K}^n)/\mathfrak{I}\mathfrak{K}^n \subset \bigcap_{N\geqslant n} \mathfrak{K}^N \cdot (A/(\mathfrak{I}\mathfrak{K}^n)).$$

We deduce the following corollary.

Corollary (16.9.13.5). — *Suppose that the hypotheses of (16.9.13.2) are satisfied and also that the ring A is Noetherian and $\mathfrak{K}$ is contained in the radical of A. Then for every integer $n > 0$,*

(16.9.13.6) $$\mathfrak{I} \cap \mathfrak{K}^n = \mathfrak{I}\mathfrak{K}^n.$$

Proof. Indeed, the second member of (16.9.13.4) is then zero, given that $A/\mathfrak{I}\mathfrak{K}^n$ is an A-module of finite type (Bourbaki, *Alg. comm.*, chap. III, §3, n°3, prop. 6). □

Taking in particular $n = 2$ in (16.9.13.6), and observing that $\mathfrak{L}^2 = \mathfrak{I}^2 + \mathfrak{I}\mathfrak{K} + \mathfrak{K}^2 = \mathfrak{I}\mathfrak{L} + \mathfrak{K}^2$, we have $\mathfrak{I}\mathfrak{L} \subset \mathfrak{I} \cap \mathfrak{L}^2$ and therefore

$$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I}\mathfrak{L} + (\mathfrak{I} \cap \mathfrak{K}^2) = \mathfrak{I}\mathfrak{L} + \mathfrak{I}\mathfrak{K}^2 = \mathfrak{I}\mathfrak{L},$$

in other words,

(16.9.13.7) $$\mathfrak{I} \cap \mathfrak{L}^2 = \mathfrak{I}\mathfrak{L},$$

which may also be expressed by saying that the canonical homomorphism

$$\mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow (\mathfrak{I} + \mathfrak{L}^2)/\mathfrak{L}^2$$

is bijective.

Proof (16.9.13). Having demonstrated the lemmas, let us prove the first claim of (16.9.13): It is clearly enough to prove that, at a point $x \in Y'$, the sequence of stalks of the sheaves appearing in (16.9.13.1) is exact. Put $A = \mathscr{O}_{X,x}$. We may write $\mathscr{O}_{Y,x} = A' = A/\mathfrak{I}$, where $\mathfrak{I}$ is an ideal contained in the maximal ideal of A, and then $\mathscr{O}_{Y',x} = A'/\mathfrak{K}'$, where $\mathfrak{K}'$ is generated by an A'-*regular* sequence of elements of A' which are the images of elements of an A-regular sequence in A belonging to the maximal ideal of A. If $\mathfrak{K}$ is the ideal generated by the latter elements and $\mathfrak{L} = \mathfrak{I} + \mathfrak{K}$, then $\mathscr{O}_{Y',x} = A/\mathfrak{L}$; since we are in the situation of (16.9.13.5), the canonical homomorphism $\mathfrak{I}/\mathfrak{I}\mathfrak{L} \to (\mathfrak{I} + \mathfrak{L}^2)/\mathfrak{L}^2$ is bijective. But this shows that the sequence **IV-4 | 51**

$$0 \longrightarrow \mathfrak{I}/\mathfrak{I}\mathfrak{L} \longrightarrow \mathfrak{L}/\mathfrak{L}^2 \longrightarrow (\mathfrak{L}/\mathfrak{I})/(\mathfrak{L}/\mathfrak{I})^2 \longrightarrow 0$$

is exact (see the proof of (16.2.7)), and the modules appearing in this sequence are precisely the stalks at x of the sheaves in (16.9.13.1). The second claim follows from the fact that $\mathscr{N}_{Y'/Y}$ is a locally free $\mathscr{O}_{Y'}$-module (16.9.8) and Bourbaki, *Alg.*, chap. II, 3rd ed., §1, n°11, prop. 21. □

16.10. Differentially smooth morphisms.

Definition (16.10.1). — We say that a morphism of preschemes $f : X \to S$ is differentially smooth (or that X is differentially smooth over S) if it satisfies the following conditions:

(i) $\Omega^1_{X/S}$ is a locally projective $\mathscr{O}_X$-module, which is to say that every point $x \in X$ admits an affine open neighbourhood U such that $\Gamma(U, \Omega^1_{X/S})$ is a projective $\Gamma(U, \mathscr{O}_X)$-module (not necessarily of finite type).

(ii) The canonical morphism (16.3.1.1)

$$\mathbf{S}^{\bullet}_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_{\bullet}(\mathscr{P}_{X/S})$$

is bijective.

In particular, if $\Omega^1_{X/S}$ is locally free of finite rank, the $\mathscr{P}^n_{X/S}$ are locally free of finite rank $\mathscr{O}_X$-modules (being extensions of such modules).

We say that f is *differentially smooth at a point* $x \in X$ if there is an open neighbourhood U of x in X such that $f|U$ is differentially smooth.

We will see later (17.12.4) that a smooth morphism is differentially smooth, which justifies the terminology; but the converse is not true; indeed, a *monomorphism* $f : X \to S$ is differentially smooth, since $\Omega^1_{X/S} = 0$ because of (**I**, 5.3.8), and consequently the surjective homomorphism (16.3.1.1) is clearly bijective; however a monomorphism is not even necessarily flat, neither *a fortiori* smooth. Let us limit ourselves to proving the following proposition:

Proposition (16.10.2). — *Let A be a ring, B a formally smooth A-algebra for the discrete topology* (**0**, 19.3.1). *Then* $\mathrm{Spec}(B)$ *is differentially smooth over* $\mathrm{Spec}(A)$.

Proof. Indeed, $B \otimes_A B$ is then (with the discrete topology) a formally smooth B-algebra (by one or the other canonical homomorphisms $b \mapsto b \otimes 1$, $b \mapsto 1 \otimes b$ of B to $B \otimes_A B$) (**0**, 19.3.5, (iii)); therefore $B \otimes_A B$ is a formally smooth A-algebra with the discrete topology (**0**, 19.3.5, (ii)). Taking $\mathfrak{I} = \mathfrak{I}_{B/A}$, it follows that $B \otimes_A B$ is a formally smooth A-algebra for the $\mathfrak{I}$-preadic topology (**0**, 19.3.8); since by hypothesis $B = (B \otimes_A B)/\mathfrak{I}$ is a formally smooth A-algebra for the discrete topology, the proposition follows from the equivalence of (a) and (b) in (**0**, 19.5.4). □ **IV-4 | 52**

Proposition (16.10.3). — *For a morphism $f : X \to S$ to be differentially smooth, it is necessary and sufficient that for every $x \in X$, there is an affine open neighbourhood of x, of ring A, such that $\Gamma(U, \mathscr{P}^{\infty}_{X/S})$ is an augmented topological A-algebra isomorphic to the completion $\widehat{B}$, where $B = \mathbf{S}_A(V)$, V being a projective A-module and B being endowed with the B^+-preadic topology (where B^+ is the augmentation ideal). If $\Omega^1_{X/S}$ is locally free of finite rank, we can replace $\widehat{B}$ with the ring of formal series $A[[T_1, \ldots, T_n]]$.*

Proof. The notion of a differentially smooth morphism is clearly local on X, so we reduce to the case where $S = \mathrm{Spec}(B)$, $X = \mathrm{Spec}(C)$. Consider $C \otimes_B C$ as a C-algebra (by the first factor); put $\mathfrak{I} = \mathfrak{I}_{C/B}$ and endow $C \otimes_B C$ with the $\mathfrak{I}$-preadic topology; we can apply to the topological C-algebra $C \otimes_B C$ and to the ideal $\mathfrak{I}$ of $C \otimes_B C$ the equivalence of (b) and (c) of (**0**, 19.5.4), given that $(C \otimes_B C)/\mathfrak{I} = C$ is clearly a formally smooth C-algebra for the discrete topologies. The topology of $\Gamma(U, \mathscr{P}^{\infty}_{X/S})$ is clearly the projective limit topology of this ring (16.1.11). □

We note that the integer n of the proposition (16.10.3) is the *rank* of $\Omega^1_{X/S}$ in the point x. We shall see (17.13.5) that when f is differentially smooth and locally of finite type, n is equal to the dimension of the fibre $f^{-1}(f(x))$.

Proposition (16.10.4). — *Let $f : X \to S$, $g : S' \to S$ be two morphisms, and take $X' = X \times_S S'$, $f' = f_{(S')} : X' \to S'$.*

(i) *If f is differentially smooth, the same is true for f'.*

(ii) *Conversely, if g is faithfully flat and quasi-compact, and if f' is differentially smooth and $\Omega^1_{X'/S'}$ is a finite type $\mathscr{O}_{X'}$-module, f is differentially smooth and $\Omega^1_{X/S}$ is a finite type $\mathscr{O}_X$-module.*

Proof. Indeed, if f is differentially smooth, the $\mathscr{G}r_n(\mathscr{P}_{X/S})$ are *flat* $\mathscr{O}_X$-modules; therefore, by (16.4.6), the homomorphisms $\mathscr{G}r_n(\mathscr{P}_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}_{X'/S'})$ are bijective for every n. In view of the commutativity of diagram (16.2.1.3), it follows from Definition (16.10.1) that f' is differentially smooth. On the other hand, if g is faithfully flat and quasi-compact, it also follows from (16.4.6) that $\mathscr{G}r_n(\mathscr{P}_{X/S}) \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \to \mathscr{G}r_n(\mathscr{P}_{X'/S'})$ is bijective for every n. Suppose also that f' is differentially smooth and $\Omega^1_{X'/S'}$ is of finite rank. Since the canonical projection $X' \to X$ is a faithfully flat and quasi-compact morphism, it follows first from (2.5.2) that $\Omega^1_{X/S}$ is an $\mathscr{O}_X$-module locally free of finite rank, then from (2.2.7) that the canonical homomorphism (16.3.1.1) is bijective, and therefore f is differentially smooth. □

Proposition (16.10.5). — *For a morphism locally of finite type $f : X \to S$ to be differentially smooth, it is necessary and sufficient that the diagonal immersion $\Delta_f : X \to X \times_S X$ be quasi-regular.*

IV-4 | 53

Proof. Being a local problem, we can reduce to the case where S and X are affines, and therefore the diagonal subprescheme of $X \times_S X$ is closed. The hypothesis that f is locally of finite type implies that Δ_f is locally of finite presentation (**I**, 4.3.1), therefore the diagonal prescheme of $X \times_S X$ is defined by an ideal $\mathscr{I}$ of finite type, and $\Omega^1_{X/S} = \mathscr{I}/\mathscr{I}^2$ is an $\mathscr{O}_X$-module of finite type. The proposition is now immediate from the comparison of the conditions of (16.10.1) and (16.9.4). □

Remark (16.10.6). — Let $f : X \to S$ be a morphism such that the $\mathscr{O}_X$-module $\Omega^1_{X/S}$ is locally free of finite rank. It follows from (**0**, 20.4.7) that every $x \in X$ has an open neighbourhood such that there is a finite family $(z_\lambda)_{\lambda \in L}$ of sections of $\mathscr{O}_X$ over U for which $(dz_\lambda)_{\lambda \in L}$ forms a *basis* of the $\Gamma(U, \mathscr{O}_X)$-module $\Gamma(U, \Omega^1_{X/S})$.

16.11. Differential operators on a differentially smooth S-prescheme

(16.11.1). Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda \in L}$ a family of sections of $\mathscr{O}_X$ over U such that the dz_λ form a system of generators of $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. Let m be an integer or the symbol ∞, and take, for every λ

(16.11.1.1) $$\zeta_\lambda = \delta z_\lambda = d^m z_\lambda - z_\lambda \in \Gamma(U, \mathscr{P}^m_{X/S}).$$

We will also use the usual notations from analysis; for every $\mathbf{p} = (p_\lambda) \in \mathbb{N}^{(L)}$ (so that $p_\lambda = 0$ except for a finite number of indices), we put

(16.11.1.2) $$|\mathbf{p}| = \sum_\lambda p_\lambda, \quad \mathbf{p}! = \prod_\lambda (p_\lambda!)$$

(16.11.1.3) $$\binom{\mathbf{p}}{\mathbf{q}} = \mathbf{p}!/(\mathbf{q}!(\mathbf{p}-\mathbf{q})!), \quad \text{for } \mathbf{p}, \mathbf{q} \text{ in } \mathbb{N}^{(L)}, \mathbf{q} \leqslant \mathbf{p}$$

and we adopt the convention that $\binom{\mathbf{p}}{\mathbf{q}} = 0$ when $\mathbf{q} \not\leqslant \mathbf{p}$,

(16.11.1.4) $$\mathbf{z}^{\mathbf{p}} = \prod_\lambda (z_\lambda)^{p_\lambda}, \quad \boldsymbol{\zeta}^{\mathbf{p}} = \prod_\lambda (\zeta_\lambda)^{p_\lambda}.$$

We therefore have, with this notation

(16.11.1.5) $$d^m(\mathbf{z}^{\mathbf{p}}) = (d^m(\mathbf{z}))^{\mathbf{p}} = (\boldsymbol{\zeta} + \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q} \leqslant \mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}} \mathbf{z}^{\mathbf{p}-\mathbf{q}} \boldsymbol{\zeta}^{\mathbf{q}}$$

$$\boldsymbol{\zeta}^{\mathbf{p}} = (d^m\mathbf{z} - \mathbf{z})^{\mathbf{p}} = \sum_{\mathbf{q}\leqslant\mathbf{p}} (-1)^{|\mathbf{p}-\mathbf{q}|}\binom{\mathbf{p}}{\mathbf{q}}\mathbf{z}^{\mathbf{p}-\mathbf{q}}d^m(\mathbf{z}^{\mathbf{q}}). \tag{16.11.1.6}$$

Since the dz_λ generate $\Omega^1_{X/S}$, are the images of the δz_λ, and the canonical morphism (16.3.1.1) is surjective, we conclude that, for finite m, the δz_λ generate the $\mathscr{O}_U$-algebra $\mathscr{P}^m_{U/S}$ (Bourbaki, *Alg. comm.*, chap. III, §2, n°8, cor. 2 to th. 1). Therefore the $\boldsymbol{\zeta}^{\mathbf{p}}$ (for $|\mathbf{p}| \leqslant m$) generate the $\mathscr{O}_U$-module $\mathscr{P}^m_{U/S}$. A differential operator $D \in \mathrm{Diff}^m_{U/S}$ is consequently entirely determined by the values $\langle\boldsymbol{\zeta}^{\mathbf{p}}, D\rangle$ for $|\mathbf{p}| \leqslant m$, or, equivalently by (16.11.1.5) and (16.11.1.6), by the values of $\langle d^m(\mathbf{z}^{\mathbf{p}}), D\rangle =$ **IV-4** | 54
$D(\mathbf{z}^{\mathbf{p}})$ for $|\mathbf{p}| \leqslant m$; more precisely, it follows from (16.11.1.5) that

$$D(\mathbf{z}^{\mathbf{p}}) = \langle d^m(\mathbf{z}^{\mathbf{p}}), D\rangle = \sum_{\mathbf{q}\leqslant\mathbf{p}} \binom{\mathbf{p}}{\mathbf{q}}\langle\boldsymbol{\zeta}^{\mathbf{q}}, D\rangle\mathbf{z}^{\mathbf{p}-\mathbf{q}}. \tag{16.11.1.7}$$

Theorem (16.11.2). — *Let $f : X \to S$ be a morphism, U an open set of X, $(z_\lambda)_{\lambda\in L}$ a family of sections of $\mathscr{O}_X$ over U such that the family $(dz_\lambda)_{\lambda\in L}$ generates $\Omega^1_{X/S}|U = \Omega^1_{U/S}$. The following conditions are equivalent:*

(a) $f|U$ is differentially smooth and (dz_λ) is a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$.
(b) There is a family $(D_{\mathbf{p}})_{\mathbf{p}\in\mathbb{N}^{(L)}}$ of differential operators of $\mathscr{O}_U$ to itself verifying the conditions

$$D_{\mathbf{p}}(\mathbf{z}^{\mathbf{q}}) = \binom{\mathbf{q}}{\mathbf{p}}\mathbf{z}^{\mathbf{q}-\mathbf{p}}, \quad (\mathbf{p}, \mathbf{q} \textit{ in } \mathbb{N}^{(L)}). \tag{16.11.2.1}$$

Also, when these conditions are satisfied, the family $(D_{\mathbf{p}})$ is uniquely determined by the conditions (16.11.2.1) and satisfies the relations

$$D_{\mathbf{q}} \circ D_{\mathbf{p}} = D_{\mathbf{p}} \circ D_{\mathbf{q}} = \frac{(\mathbf{p}+\mathbf{q})!}{\mathbf{p}!\mathbf{q}!}D_{\mathbf{p}+\mathbf{q}} \quad (\mathbf{p}, \mathbf{q} \textit{ in } \mathbb{N}^{(L)}). \tag{16.11.2.2}$$

Finally, if L is finite, for every integer m, the $D_{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ form a basis of the $\mathscr{O}_U$-module $\mathscr{D}\mathit{iff}^m_{U/S}$, in other words, every differential operator of order $\leqslant m$ on U can be written uniquely as

$$D = \sum_{|\mathbf{p}|\leqslant m} a_{\mathbf{p}}D_{\mathbf{p}}$$

where the $a_{\mathbf{p}}$ are sections of $\mathscr{O}_X$ over U.

Proof. Note first that because of (16.11.1.6) and (16.11.1.5), we verify immediately that the conditions (16.11.2.1) are equivalent to

$$\langle\boldsymbol{\zeta}^{\mathbf{p}}, D_{\mathbf{q}}\rangle = \delta_{\mathbf{pq}} \quad \text{(Kronecker's symbol)}. \tag{16.11.2.3}$$

The existence of the family $(D_{\mathbf{p}})$ verifying these conditions implies first (by taking $|\mathbf{p}| = 1$) that the dz_λ are linearly independent, and therefore form a basis of the $\mathscr{O}_U$-module $\Omega^1_{U/S}$. Then, for every integer $m \geqslant 1$, we deduce similarly from (16.11.2.3) that the $\boldsymbol{\zeta}^{\mathbf{p}}$ such that $|\mathbf{p}| \leqslant m$ are linearly independent; it follows that the canonical homomorphism (16.3.1.1) is injective, and therefore bijective, which proves that (b) implies (a). The converse follows immediately from Definition (16.10.1): the fact that the $\boldsymbol{\zeta}^{\mathbf{p}}$ form a basis of $\mathscr{P}^m_{U/S}$ for $|\mathbf{p}| \leqslant m$ implies the existence and uniqueness of a family of homomorphisms $u_{\mathbf{q},m} : \mathscr{P}^m_{U/S} \to \mathscr{O}_U$ ($|\mathbf{q}| \leqslant m$) such that $\langle\boldsymbol{\zeta}^{\mathbf{p}}, u_{\mathbf{q},m}\rangle = \delta_{\mathbf{pq}}$ for $|\mathbf{p}| \leqslant m$, $|\mathbf{q}| \leqslant m$. For a given value of $\mathbf{q}$, the differential operators corresponding to $u_{\mathbf{q},m}$ for $m \geqslant |\mathbf{q}|$ are identified with the same operator $D_{\mathbf{q}}$. This proves that (a) implies (b), and also that the family $(D_{\mathbf{q}})$ is uniquely determined and that, if L is finite, the $D_{\mathbf{p}}$ with $|\mathbf{p}| \leqslant m$ form a basis of the $\mathscr{O}_U$-module $\mathscr{D}\mathit{iff}^m_{U/S}$, the dual of $\mathscr{P}^m_{U/S}$. Finally, the relations (16.11.2.2) follow immediately by evaluating the three operators in question on the $\mathbf{z}^{\mathbf{r}}$ and using the fact that the $\boldsymbol{\zeta}^{\mathbf{r}}$ for $|\mathbf{r}| \leqslant m$ generate $\mathscr{P}^m_{U/S}$. □

IV-4 | 55

Remarks (16.11.3). — (i) The fact that, by (16.11.2.2), the $D_{\mathbf{p}}$ commute pairwise naturally does not imply that the $\mathscr{O}_U$-algebra $\mathscr{D}\mathit{iff}_{U/S}$ is commutative: the $D_{\mathbf{p}}$ do not commute with multiplication by sections of $\mathscr{O}_U$ unless $\mathbf{p} = 0$.

(ii) The indices $\mathbf{p}$ such that $|\mathbf{p}| = 1$ are the $\boldsymbol{\epsilon}_\lambda = (\epsilon_{\lambda\mu})_{\mu \in L}$ where $\epsilon_{\lambda\mu} = 0$ if $\mu \neq \lambda$ and $\epsilon_{\lambda\lambda} = 1$; when L is finite, the operators $D_{\boldsymbol{\epsilon}_\lambda}$ are exactly the S-derivations D_λ introduced in (16.5.7). We note that in general (contrary to what happens in classical analysis), it is not true that a differential operator of any order can be written as a linear combination of powers of D_i (cf. (16.12)).

(iii) For every integer $r \geqslant 1$, we can define the notion of *differentially smooth up to order* r by replacing in (16.10.1) the condition (ii) by the condition that the homomorphisms

$$\mathbf{S}^m_{\mathscr{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathscr{G}r_m(\mathscr{P}_{X/S})$$

are bijective *for every* $m \leqslant r$. The argument of (16.11.2) proves also that if, in the condition (a), we replace "differentially smooth" by "differentially smooth up to order r", this condition is equivalent to (b) by restricting ourselves to $\mathbf{p} \in \mathbb{N}^{(L)}$, $\mathbf{q} \in \mathbb{N}^{(L)}$ such that $|\mathbf{p}| \leqslant r$, $|\mathbf{q}| \leqslant r$.

16.12. Case of characteristic zero: Jacobian criterion for differentially smooth morphisms.

(16.12.1). We say that a prescheme X is *of characteristic* p (p equal to zero or a prime number) if, for every affine open set U of X, the ring $\Gamma(U, \mathscr{O}_X)$ is of characteristic p (**0**, 21.1.1). It follows from (**0**, 21.1.3) that for X to be of characteristic 0, it is necessary and sufficient that for every *closed* point x of X, the residue field $k(x)$ is of characteristic 0, or even that X can be given a structure of $\mathbb{Q}$-prescheme (necessarily unique).

Theorem (16.12.2). — *Let X be a prescheme of characteristic 0, $f : X \to S$ a morphism. If $\Omega^1_{X/S}$ is a locally free $\mathscr{O}_X$-module (not necessarily of finite type), f is differentially smooth.*

Proof. The problem being local on X, we can suppose there is a family (z_λ) of sections of $\mathscr{O}_X$ over X such that the (dz_λ) is a basis for the $\mathscr{O}_X$-module $\Omega^1_{X/S}$. Applying the criterion (16.11.2), it is enough for the operators

$$D_{\mathbf{p}} = (\mathbf{p}!)^{-1} \prod_\lambda D_\lambda^{p_\lambda}$$

(where the D_λ are the coordinate forms corresponding to the basis (dz_λ)) to verify the relations (16.11.2.1), which is a consequence of the fact that the D_λ are derivations. □

(16.12.3). The theorem above is not true if we discard the hypothesis that X is of characteristic 0. For example, if $S = \operatorname{Spec}(k)$, where k is a field of characteristic $p > 0$, $X = \operatorname{Spec}(K)$ where $K = k(\alpha)$ where $\alpha \notin k$, $\alpha^p \in k$, we verify immediately that $\Omega^1_{X/S}$ has rank 1, and that the morphism $X \to S$ **IV-4** | 56
is differentially smooth up to order $p - 1$ (16.11.3, (iii)), but not up to order p. However, the proof of (16.12.2) proves that if $\Omega^1_{X/S}$ is locally free, and if $n!1_{\mathscr{O}_X}$ is invertible in $\Gamma(X, \mathscr{O}_X)$, then X is differentially smooth over S up to order n.

§17. Smooth morphisms, unramified (or net) morphisms, and étale morphisms.

In this section, we return to the notions studied in (**0**, 19), expressed in the geometric language of schemes and from the global point of view, for preschemes locally of finite presentation over a given base prescheme.

Most of the results (except those of (17.7), (17.8), (17.9), (17.13), and (17.16)) reduce, moreover, to variants of properties already encountered in (**0**, 19).

For more specific results on étale morphisms, the reader should consult (18).

17.1. Formally smooth morphisms, formally unramified morphisms, formally étale morphisms.

Definition (17.1.1). — Let $f : X \to Y$ be a morphism of preschemes. We say that f is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) if, for all affine schemes Y', all closed subschemes Y'_0 of Y' defined by a nilpotent ideal $\mathscr{J}$ of $\mathscr{O}_{Y'}$, and every morphism $Y' \to Y$, the map

(17.1.1.1) $$\mathrm{Hom}_Y(Y', X) \longrightarrow \mathrm{Hom}_Y(Y'_0, X)$$

induced by the canonical map $Y'_0 \to Y'$, is *surjective* (resp. *injective*, resp. *bijective*).

One also says that X is *formally smooth* (resp. *formally unramified*, resp. *formally étale*) over Y.

It is clear that for f to be formally étale, it is necessary and sufficient for f to be formally smooth and formally unramified.

Remark (17.1.2). —

(i) Suppose that $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, so that f comes from a homomorphism of rings $\phi : A \to B$. According to (**0**, 19.3.1) and (**0**, 19.10.1), saying that f is formally smooth (resp. formally unramified, resp. formally étale) means that, via ϕ, B is a *formally smooth* (resp. *formally unramified*, resp. *formally étale*) A-algebra, for the *discrete* topologies on A and B.

(ii) To verify that f is formally smooth (resp. formally unramified, resp. formally étale), we can, in Definition (17.1.1), restrict to the case where $\mathscr{J}^2 = 0$. To see this, if f satisfies the corresponding condition of Definition (17.1.1) in the particular case $\mathscr{J}^2 = 0$, and if we have $\mathscr{J}^n = 0$, then we consider the closed subscheme Y'_j of Y' defined by the sheaf of ideals $\mathscr{J}^{j+1}$ for $0 \leqslant j \leqslant n-1$, so that Y'_j is a closed subscheme of Y'_{j+1} defined by a square-zero sheaf of ideals; the hypotheses imply that each of the maps

$$\mathrm{Hom}_Y(Y'_{j+1}, X) \longrightarrow \mathrm{Hom}_Y(Y'_j, X) \quad (0 \leqslant j \leqslant n-1)$$

is surjective (resp. injective, resp. bijective); by composition, we conclude that the same holds for (17.1.1.1). **IV** | 57

(iii) Note that the properties of the morphism f defined in (17.1.1) are properties of the *representable functor* ($\mathbf{0_{III}}$, 8.1.8)

$$Y' \longmapsto \mathrm{Hom}_Y(Y', X)$$

from the category of Y-preschemes to the category of sets; they keep a meaning for *any* contravariant functor with the same domain and codomain, representable or not.

(iv) Assume that the morphism f is formally unramified (resp. formally étale); consider an *arbitrary* Y-prescheme Z and a closed subprescheme Z_0 of Z defined by a *locally nilpotent* sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_Z$. Then the map

(17.1.2.1) $$\mathrm{Hom}_Y(Z, X) \longrightarrow \mathrm{Hom}_Y(Z_0, X)$$

induced by the canonical injection $Z_0 \to Z$, is still injective (resp. bijective). To see this, let (U_α) be an affine open covering of Z such that the sheaves of ideals $\mathscr{J}|U_\alpha$ are nilpotent, and for each α, let U_α^0 be the inverse image of U_α in Z_0, which is the closed subprescheme of U_α defined by $\mathscr{J}|U_\alpha$. Let $f_0 : Z_0 \to X$ be a Y-morphism; by hypothesis, for each α, there is at most one (resp. one and only one) Y-morphism $f_\alpha : U_\alpha \to X$ whose restriction to Z_0 coincides with $f_0|U_\alpha$. We immediately conclude that if f_α and f_β are defined, then, for each affine open $V \subset U_\alpha \cap U_\beta$, we have $f_\alpha|V = f_\beta|V$, as the restrictions of these morphisms to the inverse image V_0 of V in Z_0 coincide. There is therefore at most one (resp. one and only one) Y-morphism $f : Z \to X$ whose restriction to Z_0 coincides with f_0.

Proposition (17.1.3). —

(i) *A monomorphism of preschemes is formally unramified; an open immersion is formally étale.*

(ii) *The composition of two formally smooth (resp. formally unramified, resp. formally étale) morphisms is formally smooth (resp. formally unramified, resp. formally étale).*

(iii) *If $f : X \to Y$ is a formally smooth (resp. formally unramified, resp. formally étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are two formally smooth (resp. formally unramified, resp. formally étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.*
(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if $g \circ f$ is formally unramified, then so is f.*
(vi) *If $f : X \to Y$ is a formally unramified morphism, then so is $f_{\mathrm{red}} : X_{\mathrm{red}} \to Y_{\mathrm{red}}$.*

Proof. According to (**I**, 5.5.12), it suffices to prove (i), (ii), and (iii). The assertions in (i) are both trivial. To prove (ii), consider two morphisms $f : X \to Y$, $g : Y \to Z$, an affine scheme Z', a closed subscheme Z'_0 of Z' defined by a nilpotent ideal, and a morphism $Z' \to Z$. Suppose that f and g are formally smooth, and consider a Z-morphism $u_0 : Z'_0 \to X$; the hypothesis on g implies that **IV** | 58
there exists a Z-morphism $v : Z' \to Y$ such that $f \circ u_0 = v \circ j$ (where $j : Z'_0 \to Z'$ is the canonical injection); the hypothesis on f then implies that there exists a morphism $u : Z' \to X$ such that $f \circ u = v$ and $u \circ j = u_0$, therefore $(g \circ f) \circ u$ is equal to the given morphism $Z' \to Z$ and $u \circ j = u_0$, which proves that $g \circ f$ is formally smooth; we argue the same way when we suppose that f and g are formally unramified.

Finally, to prove (iii), let $X' = X_{S'}$, $Y' = Y_{S'}$, $f' = f_{S'}$; consider an affine scheme Y'', a closed subscheme Y''_0 defined by a nilpotent sheaf of ideals, and a morphism $g : Y'' \to Y'$ making Y'' a Y'-prescheme; we then know by (**I**, 3.3.8) that $\mathrm{Hom}_{Y'}(Y'', X')$ is canonically identified with $\mathrm{Hom}_Y(Y'', X)$, and $\mathrm{Hom}_{Y'}(Y''_0, X')$ with $\mathrm{Hom}_Y(Y''_0, X)$, and the conclusion follows immediately from Definition (17.1.1). □

We note that a *closed immersion* is not necessarily formally smooth.

Proposition (17.1.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is formally unramified. Then, if $g \circ f$ is formally smooth (resp. formally étale), so is f.*

Proof. Let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent sheaf of ideals, $h : Y' \to Y$ a morphism, $j : Y'_0 \to Y'$ the canonical injection, and $u_0 : Y'_0 \to X$ a Y-morphism, so that $f \circ u_0 = h \circ j$. Suppose that $g \circ f$ is formally smooth; then there exists a morphism $u : Y' \to X$ such that $u \circ j = u_0$ and $(g \circ f) \circ u = g \circ h$. But these two relations imply that $f \circ u$ and h are Z-morphisms from Y' to Y such that $(f \circ u) \circ j = h \circ j$; by virtue of the hypothesis that g is formally unramified, we get that $f \circ u = h$, in other words that u is a Y-morphism; thus f is formally smooth. Taking into account (17.1.3, (v)), this proves the proposition. □

Corollary (17.1.5). — *Suppose that g is formally étale; then, for $g \circ f$ to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that f is.*

Proof. This follows from (17.1.4) and (17.1.3, (ii) and (v)). □

Proposition (17.1.6). — *Let $f : X \to Y$ be a morphism of preschemes.*

(i) *Let (U_α) be an open covering of X and, for each α, let $i_\alpha : U_\alpha \to X$ be the canonical injection. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each $f \circ i_\alpha$ is.*
(ii) *Let (V_λ) be an open covering of Y. For f to be formally smooth (resp. formally unramified, resp. formally étale), it is necessary and sufficient that each of the restrictions $f^{-1}(V_\lambda) \to V_\lambda$ of f is.*

Proof. First note that (ii) is a consequence of (i): if $j_\lambda : V_\lambda \to Y$ and $i_\lambda : f^{-1}(V_\lambda) \to X$ are the canonical injections, then the restriction $f_\lambda : f^{-1}(V_\lambda) \to V_\lambda$ of f is such that $j_\lambda \circ f_\lambda = f \circ i_\lambda$; if f is formally smooth (resp. formally unramified), then so is $f \circ i_\lambda$ since i_λ is formally étale (17.1.3); but since j_λ is formally étale, this means that f_λ is formally smooth (resp. formally unramified), by virtue of (17.1.5). Conversely, if all the f_λ are formally smooth (resp. formally unramified), the same applies to $j_\lambda \circ f_\lambda$ (17.1.3), so also to f in virtue of (i).

If we take into account that the i_α are formally étale, everything comes down to proving that if **IV** | 59
the $f \circ i_\alpha$ are formally smooth (resp. formally unramified), then the same applies to f.

Therefore let Y' be an affine scheme, Y'_0 a closed subscheme of Y' defined by a nilpotent ideal $\mathscr{J}$, which we may assume to satisfy $\mathscr{J}^2 = 0$ (17.1.2, (ii)), and finally let $g : Y' \to Y$ be a morphism. Suppose we are given a Y-morphism $u_0 : Y'_0 \to X$; denote by W_α (resp. W^0_α) the prescheme induced by Y' (resp. Y'_0) on the open subset $u_0^{-1}(U_\alpha)$ (we recall that Y' and Y'_0 share the *same underlying topological space*). Let us first suppose that the $f \circ i_\alpha$ are *formally unramified*, and show that, if u' and

u'' are two Y-morphisms from Y' to X whose restrictions to Y'_0 coincide, then we have $u' = u''$. Indeed, taking into account (17.1.2, (iv)), the hypothesis that the $f \circ i_\alpha$ are formally unramified implies that for all α, we have $u'|W_\alpha = u''|W_\alpha$, since the restrictions of both Y-morphisms to W^0_α coincide. Hence the conclusion follows.

Now suppose that the $f \circ i_\alpha$ are *formally smooth* and prove the existence of a Y-morphism $u : Y' \to X$ whose restriction to Y'_0 is u_0. Now, since Y' is an *affine scheme*, we can apply (16.5.17), the hypotheses of which are satisfied, and the conclusion of which precisely proves the existence of u. ☐

We can therefore say that the notions introduced in (17.1.1) are *local* on X and Y, which always allows, in virtue of (17.1.2, (i)), to be reduced to the study of formally smooth (resp. formally unramified, resp. formally étale) *algebras*.

17.2. General properties of differentials

Proposition (17.2.1). — *For a morphism $f : X \to Y$ to be formally unramified, it is necessary and sufficient that $\Omega^1_f = 0$ (what we still write $\Omega^1_{X/Y} = 0$ (16.3.1)).*

Proof. Taking into account (17.1.6), we reduce to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, and the conclusion then follows from (**0**, 20.7.4) and the interpretation of $\Omega^1_{X/Y}$ in this case (16.3.7). ☐

Corollary (17.2.2). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms. For f to be formally unramified, it is necessary and sufficient that the canonical morphism (16.4.19)*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is surjective.

Proof. This is an immediate consequence of (17.2.1) and the exact sequence (16.4.19.1). ☐

Proposition (17.2.3). — *Let $f : X \to Y$ be a formally smooth morphism.*

(i) *The $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is locally projective (16.10.1). If f is locally of finite type, then $\Omega^1_{X/Y}$ is locally free and of finite type.*

(ii) *For all morphisms $g : Y \to Z$, the sequence (16.4.19) of $\mathscr{O}_X$-modules*

(17.2.3.1) $$0 \longrightarrow f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.3.1) form a split *exact sequence.*

IV | 60

Proof.

(i) We know (16.3.9) that if f is locally of finite type, then Ω^1_f is an $\mathscr{O}_X$-module of finite type. To prove that, in all cases, it is locally projective, we can reduce, by virtue of (17.1.6), to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, and the result follows from the hypothesis on f and from (**0**, 20.4.9) and (**0**, 19.2.1).

(ii) Again, we can restrict to the case where X, Y, and Z are affine (17.1.6), and the conclusion in this case follows from the interpretation of the sheaves of modules in the sequence (17.2.3.1) and from (**0**, 20.5.7).

☐

Corollary (17.2.4). — *If $f : X \to Y$ is formally étale, then, for all morphisms $g : Y \to Z$, the canonical homomorphism of $\mathscr{O}_X$-modules*

$$f^*(\Omega^1_{Y/Z}) \longrightarrow \Omega^1_{X/Z}$$

is bijective.

Proof. This follows from the exactness of the sequence (17.2.3.1) and from the fact that we then have $\Omega^1_{X/Y} = 0$ (17.2.1). ☐

Proposition (17.2.5). — *Let $f : X \to Y$ be a morphism, X' a subprescheme of X such that the composite morphism $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) is formally smooth. Then the sequence of $\mathscr{O}_{X'}$-modules* (16.4.21)

$$0 \longrightarrow \mathscr{N}_{X'/X} \longrightarrow \Omega^1_{X/Y} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} \longrightarrow \Omega^1_{X'/Y} \longrightarrow 0 \tag{17.2.5.1}$$

is exact; moreover, for each $x \in X$, there exists an open neighbourhood U of x such that the restrictions to U of the homomorphisms in (17.2.5.1) *form a* split *exact sequence.*

Proof. By virtue of (17.1.6), we reduce to the case where $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(B)$ are affine, and $X' = \operatorname{Spec}(B/\mathfrak{J})$, where $\mathfrak{J}$ is an ideal of B. The conormal sheaf $\mathscr{N}_{X'/X}$ then corresponds to the B-module $\mathfrak{J}/\mathfrak{J}^2$ (16.1.3), and the conclusion follows from (**0**, 20.5.14). □

Proposition (17.2.6). — *Let X and Y be two preschemes, $f : X \to Y$ a morphism locally of finite type. The following conditions are equivalent:*

(a) *f is a monomorphism.*
(b) *f is radicial and formally unramified.*
(c) *For each $y \in Y$, the fibre $f^{-1}(y)$ is empty or $k(y)$-isomorphic to* $\operatorname{Spec}(k(y))$ *(in other words, it is reduced to a single point z such that $k(y) \to \mathscr{O}_z/\mathfrak{m}_y\mathscr{O}_z$ is an isomorphism).*

Proof. The fact that (a) implies (c) follows from (8.11.5.1). It is clear that (c) implies that f is radicial; let us prove that it also follows from (c) that $\Omega^1_{X/Y} = 0$, which will prove that (c) implies (b) (17.2.1). Note that the $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is quasi-coherent of finite type (16.3.9). It follows from (**I**, 9.1.13.1) that, for $(\Omega^1_{X/Y})_x = 0$, it is necessary and sufficient that if we set $Y_1 = \operatorname{Spec}(k(y))$, $X_1 = f^{-1}(y) = X \times_Y Y_1$, then we have $(\Omega^1_{X_1/Y_1})_x = 0$; but as the morphism $f_1 : X_1 \to Y_1$ induced by f is formally unramified by virtue of the hypothesis (c) (17.1.3), the conclusion follows from (17.2.1). Finally, let us prove that (b) implies (a); for this, consider the diagonal morphism $g = \Delta_f : X \to X \times_Y X$; since f is radicial, g is surjective (1.8.7.1); on the other hand, $\Omega^1_{X/Y}$ is by definition the conormal sheaf $\mathscr{G}r_1(g)$ of the immersion g (16.3.1), and to say that f is formally unramified therefore means that $\mathscr{G}r_1(g) = 0$ (17.2.1). In addition, g is locally of finite presentation (1.4.3.1); therefore the hypothesis **IV | 61** $\mathscr{G}r_1(g) = 0$ implies that g is an open immersion (16.1.10); being surjective, this immersion is an isomorphism, hence f is a monomorphism (**I**, 5.3.8). □

17.3. Smooth morphisms, unramified morphisms, étale morphisms

Definition (17.3.1). — We say that a morphism $f : X \to Y$ is *smooth* (resp. *unramified*, or *net* [4] resp. *étale*) if it is locally of finite presentation and formally smooth (resp. formally unramified, resp. formally étale).

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over* Y.

We will see later (17.5.2) that this definition of a smooth morphism coincides with the definition already given in (6.8.1); until then, we will exclusively use definition (17.3.1).

It is clear that saying that f is étale means that it is *both* smooth and unramified.

Remarks (17.3.2). —

(i) Note that definition (17.3.1) can be phrased using only the functor

$$Y' \longmapsto \operatorname{Hom}_Y(Y', X)$$

considered in (17.1.2, (iii)) because to say that f is locally of finite presentation is equivalent to saying that the preceding functor *commutes with projective limits of affine schemes* (8.14.2).

(ii) Let A be a ring and B an A-algebra. We say that B is a *smooth* (resp. *unramified*, resp. *étale*) A-algebra if the corresponding morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is smooth (resp. unramified, resp. étale). It is equivalent to saying that B is an A-algebra *of finite presentation* (1.4.6) that is furthermore formally smooth (resp. formally unramified, resp. formally étale) for the discrete topologies.

[4] The words "net" and "formally net" seem preferable to the established terminology "unramified" (resp. "formally unramified") and will be used almost exclusively beginning with Chapter V. In this chapter, we have retained the old terminology so as not to conflict with (**0**, 19.10).

(iii) It follows from (17.1.6) and the definition of a morphism locally of finite presentation (1.4.2) that the notion of a smooth (resp. unramified, resp. étale) morphism is *local on X and on Y*.

Proposition (17.3.3). —

(i) *An open immersion is étale. For an immersion to be unramified, it is necessary and sufficient for it to be locally of finite presentation.*

(ii) *The composition of two smooth (resp. unramified, resp. étale) morphisms is smooth (resp. unramified, resp. étale).*

(iii) *If $f : X \to Y$ is a smooth (resp. unramified, resp. étale) S-morphism, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for any base extension $S' \to S$.*

(iv) *If $f : X \to X'$ and $g : Y \to Y'$ are smooth (resp. unramified, resp. étale) S-morphisms, then so is $f \times_S g : X \times_S Y \to X' \times_S Y'$.* **IV** | 62

(v) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms; if g is locally of finite type and if $g \circ f$ is unramified, then f is unramified.*

Proof. This follows from (1.4.3) and (17.1.3). □

Proposition (17.3.4). — *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and suppose that g is unramified. Then, if $g \circ f$ is smooth (resp. unramified, resp. étale), so is f.*

Proof. As g and $g \circ f$ are locally of finite presentation, so is f (1.4.3, (v)); the conclusion thus follows from (17.1.4) and (17.1.3, (v)). □

Corollary (17.3.5). — *Suppose that g is étale; then, for f to be smooth (resp. unramified, resp. étale) it is necessary and sufficient that $g \circ f$ is.*

Proof. This follows from (17.3.4) and (17.3.3, (ii)). □

Proposition (17.3.6). — *Let $g : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation. For an S-morphism $f : X \to Y$ to be unramified, it is necessary and sufficient that the canonical homomorphism* (16.4.19)

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is surjective.

Proof. As f is locally of finite presentation (1.4.3, (v)), the proposition follows from (17.2.2). □

Definition (17.3.7). — Let $f : X \to Y$ be a morphism. We say that f is *smooth* (resp. *unramified*, resp. *étale*) at a point $x \in X$, if there exists an open neighbourhood U of x in X such that the restriction $f|U$ is a smooth (resp. unramified, resp. étale) morphism from U to Y.

We then also say that X is *smooth* (resp. *unramified*, resp. *étale*) *over Y at the point x*.

Taking Remark (17.3.2, (iii)) into account, saying that f is smooth (resp. unramified, resp. étale) is equivalent to saying that it is smooth (resp. unramified, resp. étale) at every point of X.

It is clear that the set of points of X at which the morphism $f : X \to Y$ is smooth (resp. unramified, resp. étale) is *open* in X.

Proposition (17.3.8). — *For all preschemes Y and all locally free $\mathscr{O}_Y$-modules $\mathscr{E}$ of finite type, the vector bundle prescheme $\mathbf{V}(\mathscr{E})$ (**II**, 1.7.8) associated to $\mathscr{E}$ is a smooth Y-prescheme.*

Proof. By (17.3.2, (iii)), we may restrict to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{V}(\mathscr{E}) = \operatorname{Spec}(A[T_1, \ldots, T_r])$; since $A[T_1, \ldots, T_r]$ is a formally smooth A-algebra for the discrete topologies (**0**, 19.3.2) and is of finite presentation, this proves the proposition (17.3.2, (ii)). □

Corollary (17.3.9). — *Under the hypotheses of (17.3.8), the projective prescheme $\mathbf{P}(\mathscr{E})$ (**II**, 4.1.1) is a smooth Y-prescheme.*

Proof. We can still restrict to the case where $Y = \operatorname{Spec}(A)$ is affine and $\mathbf{P}(\mathscr{E}) = \mathbf{P}^r_Y$. We then know (**II**, 2.3.14) that we have a finite open cover of $\mathbf{P}^r_A$ by the $D_+(T_i)$ ($0 \leqslant i \leqslant r$) respectively equal to the spectrum of the ring $S_{(f)}$, where we wrote S for $A[T_1, \ldots, T_r]$ and f for T_i; but it follows immediately from the definition of $S_{(f)}$ (**II**, 2.2.1), that this ring, in this case, is isomorphic to $A[T_0, \ldots, T_{i-1}, T_{i+1}, \ldots, T_r]$; hence the corollary follows by (17.3.8). □

17.4. Characterizations of unramified morphisms. IV | 63

Theorem (17.4.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X. The following properties are equivalent:*

(a) f is unramified at the point x.
(b) The diagonal morphism $\Delta_f : X \to X \times_Y X$ is a local isomorphism at the point x.
b') If one sets $\Delta_f = (\psi, \theta)$, $Z = X \times_Y X$ and $z = \psi(x)$, the homomorphism $\theta^\#_z : \mathscr{O}_{Z,z} \to \mathscr{O}_{X,x}$ is bijective.
b") For every morphism $g : Y' \to Y$, every point $y' \in Y'$ above $y = f(x)$, and every Y'-section s' of $X' = X \times_Y Y'$ such that $x' = s'(y')$ lies above x, the section s' is a local isomorphism at y'.
(c) $(\Omega^1_{X/Y})_x = 0$.
(d) The $k(y)$-prescheme $f^{-1}(y)$ is unramified over $k(y)$ at the point x.
*d') The point x is isolated in $X_y = f^{-1}(y)$ (equivalently (**Err$_{\mathbf{III}}$**, 20), the morphism f is quasi-finite at x), and the ring $\mathscr{O}_{X_y,x}$ is a field that is a finite separable extension of $k(y)$.*
d") The ring $\mathscr{O}_{X_y,x} = \mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a field that is a finite separable extension of $k(y)$.
(e) The ring $\mathscr{O}_{X,x}$ is a formally unramified $\mathscr{O}_{Y,y}$-algebra for the discrete topologies.

Proof. Since f is locally of finite type, the $\mathscr{O}_X$-module $\Omega^1_{X/Y}$ is of finite type (16.3.9); hence $(\Omega^1_{X/Y})_x = 0$ is equivalent to the existence of an open neighbourhood U of x such that $\Omega^1_{X/Y}|U = 0$. This proves the equivalence of a) and c). On the other hand, if $A = \mathscr{O}_{Y,y}$ and $B = \mathscr{O}_{X,x}$, then $(\Omega^1_{X/Y})_x = \Omega^1_{B/A}$ (16.4.15), and the equivalence of c) and e) follows from (**0**, 20.7.4).

Since d') involves only properties of the morphism $X_y \to \operatorname{Spec}(k(y))$, the equivalence of a) and d') will imply, *ipso facto*, the equivalence of d) and d'). Moreover, d') and d") are equivalent, since, X_y being a $k(y)$-prescheme locally of finite type, saying that $\mathscr{O}_{X_y,x}$ is a *finite* $k(y)$-algebra is equivalent to saying that x is an isolated point of X_y (**I**, 6.4.4).

We now prove the equivalence of b) and b'). We may restrict to the case where $Y = \operatorname{Spec}(R)$ and $X = \operatorname{Spec}(S)$ are affine and f is of finite presentation. Then $Z = \operatorname{Spec}(S \otimes_R S)$, and Δ_f corresponds to the canonical surjective homomorphism $S \otimes_R S \to S$, whose kernel $\mathfrak{J}$ is an ideal of finite type (**0**, 20.4.4). If $\mathscr{J} = \widetilde{\mathfrak{J}}$, the $\mathscr{O}_Z$-module $\psi_*(\mathscr{O}_X) = \mathscr{O}_Z/\mathscr{J}$ is therefore of finite presentation; the hypothesis that $\theta^\#_z : \mathscr{O}_{Z,z} \to (\psi_*(\mathscr{O}_X))_z = \mathscr{O}_{X,x}$ is bijective then implies that, after replacing X if necessary by a suitable open neighbourhood of x, the homomorphism $\theta : \mathscr{O}_Z \to \psi_*(\mathscr{O}_X)$ is itself bijective (**0**, 5.2.7). This shows that b') implies b); the converse is obvious.

On the other hand, the equivalence of b) and b") follows from (**I**, 5.3.7), without any finiteness hypothesis on f: giving a Y'-section $s' : Y' \to X'$ is equivalent to giving a Y-morphism $h = g' \circ s' :$ IV | 64
$Y' \to X$ (where $g' : X' \to X$ is the canonical projection), so that $s' = (1_{Y'}, h)_X$, and then the diagram

(17.4.1.1)
$$\begin{tikzcd} Y' \arrow[r, "s'"] \arrow[d, "h"'] & X' = Y' \times_Y X \arrow[d, "h \times 1_X"] \\ X \arrow[r, "\Delta_f"'] & X \times_Y X \end{tikzcd}$$

identifies Y' with the $(X \times_Y X)$-prescheme product of X and X'. Hence ((**I**, 4.3.2)) if Δ_f is an isomorphism local at the point x, s' is an isomorphism local at the point y' (since $x = h(y')$), which proves that b) implies b''). The converse is obtained by applying b'') to the case $Y' = X$, $y' = x$, $g = f$, and $s' = \Delta_f$.

To finish the proof of (17.4.1), it suffices to prove the implications d'') $\Rightarrow c$) $\Rightarrow b$) $\Rightarrow d''$).

d'') $\Rightarrow c$): Since $\Omega^1_{X/Y}$ is an $\mathscr{O}_X$-module of finite type, Nakayama's lemma shows that condition c) is equivalent to $(\Omega^1_{X/Y})_x/\mathfrak{m}_y(\Omega^1_{X/Y})_x = 0$, that is, by (16.4.5), to $(\Omega^1_{X_y/\operatorname{Spec}(k(y))})_x = 0$. We are thus reduced to the case where Y is the spectrum of a field k and X is an algebraic k-prescheme. The hypothesis that $\mathscr{O}_{X,x}$ is a field k', finite extension of k, implies first that x is a *closed* point in X ((**I**, 6.4.2)), then that x is a *maximal* point, hence is an isolated point of X. Replacing X by the open $\{x\}$ of X, one may suppose $X = \operatorname{Spec}(k')$; but then the hypothesis that k' is a finite separable extension of k implies $\Omega^1_{k'/k} = 0$ (**0**, 20.6.20), which proves c).

c) $\Rightarrow$ *b*): We saw above that one then has $\Omega^1_{X/Y}|U = 0$ for a neighbourhood open U of x in X; the assertion *b*) follows then from the definition of $\Omega^1_{X/Y}$ (16.3.1) and (16.1.9).

b) $\Rightarrow$ *d''*): Replacing X by an open neighbourhood of x, we may suppose that Δ_f is an open immersion. If $f_y : X_y \to \mathrm{Spec}(k(y))$ denotes the morphism obtained from f by base change, then Δ_{f_y} is also an open immersion (**I**, 5.3.4). Since condition d'') concerns only X_y, we may reduce to the case where Y is the spectrum of a field k and X is the spectrum of a k-algebra A of finite type. What must be proved is that A is a *finite separable* k-algebra, that is, a finite product of finite separable extensions of k. If K is an algebraically closed extension of k, saying this is equivalent to saying that $A \otimes_k K$ is a finite separable K-algebra (4.6.1); we may therefore suppose that k is algebraically closed. We first show that A is finite over k. It is enough to show that every closed point x of X is isolated: the set of such points will then be open and discrete in X, and hence finite because it is quasi-compact (X being Noetherian), which proves the assertion by (**I**, 6.4.4). But then $k(x) = k$, since k is algebraically closed (**I**, 6.4.3); consequently there is a Y-section s of X such that $s(Y) = \{x\}$. By (17.4.1.1), $\{x\}$ is the inverse image of the diagonal $\Delta_X(X)$ under a morphism $X \to X \times_Y X$, and therefore $\{x\}$ is open in X by hypothesis b). Thus we have shown that A is a *finite* k-algebra. Since a **IV | 65**
finite k-algebra is a finite product of finite local k-algebras, to express that Δ_f is an open immersion we may reduce to the case where A is a finite *local* k-algebra. The residue field of A, being a finite extension of k, is necessarily equal to k; hence (**I**, 3.4.9) the underlying space of $X \times_k X$ consists of a single point, and Δ_f is necessarily an *isomorphism*. But since A is a k-algebra, the canonical homomorphism $A \otimes_k A \to A$ can only be bijective if $A = k$. □

Remark (17.4.2). — Suppose only that f is *locally of finite type*. Then $\Omega^1_{X/Y}$ is still an $\mathscr{O}_X$-module of finite type (16.3.9), and Δ_f is an immersion locally of finite presentation (1.4.3.1). The entire proof of (17.4.1) remains valid, provided a) is replaced by: *the restriction of f to a suitable neighbourhood of x is a formally unramified morphism*. One sees moreover that in this case the restriction of f to a suitable neighbourhood of x is a morphism *locally quasi-finite*.

Corollary (17.4.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following properties are equivalent:*

- *(a) f is unramified.*
- *(b) The diagonal morphism $\Delta_f : X \to X \times_Y X$ is an open immersion.*
- *b') For every morphism $Y' \to Y$, every Y'-section of $X' = X \times_Y Y'$ is an open immersion.*
- *(c) $\Omega^1_{X/Y} = 0$.*
- *(d) For every $y \in Y$, the $k(y)$-prescheme $f^{-1}(y)$ is unramified over $k(y)$.*
- *d') For every $y \in Y$, the $k(y)$-prescheme $f^{-1}(y)$ is isomorphic to a prescheme of the form $\coprod_{\lambda \in L} \mathrm{Spec}(K_\lambda)$, where each K_λ is a finite separable extension of $k(y)$.*
- *(e) For every $x \in X$, the ring $\mathscr{O}_{X,x}$ is a formally unramified $\mathscr{O}_{Y,f(x)}$-algebra for the discrete topologies.*

Corollary (17.4.3). — *If $f : X \to Y$ is unramified, then f is locally quasi-finite (**Err$_{\mathbf{III}}$**, 20).*

Proof. Indeed, by ((17.4.1), *d'*)), for every $y \in Y$, the fibre $f^{-1}(y)$ is isomorphic to a prescheme of the form $\coprod_{\lambda \in L} \mathrm{Spec}(K_\lambda)$, where the K_λ are finite separable extensions of $k(y)$; hence $f^{-1}(y)$ is a discrete topological space. □

Proposition (17.4.4). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings, and let k be the residue field of A. Then the conditions equivalent a) to e) of theorem (17.4.1) are also equivalent to each of the following:*

- *f) $\widehat{B} \otimes_{\widehat{A}} k$ is a field, finite separable extension of k (which implies that $\widehat{B}$ is an $\widehat{A}$-algebra finite).*
- *f') $\widehat{B}$ is a formally unramified $\widehat{A}$-algebra for the adic topologies.*

If moreover $k(x) = k(y)$, or if k is separably closed, these conditions are also equivalent to:

- *f'') The homomorphism $\widehat{A} \to \widehat{B}$ is surjective.*

Proof. First note, by the same argument as in (**0**, 19.3.6), that saying that B is a formally unramified A-algebra for the preadic topologies is equivalent to saying that $\widehat{B}$ is a formally unramified $\widehat{A}$-algebra for the adic topologies. On the other hand, the hypothesis that f is locally of finite type implies that $\Omega^1_{B/A}$ is a B-module of finite type (16.3.9), and hence is separated for the $\mathfrak{n}$-preadic topology (where $\mathfrak{n}$ is the maximal ideal of B) (**0**, 7.3.5); consequently, $\Omega^1_{B/A} = 0$ is equivalent to $\widehat{\Omega^1_{B/A}} = 0$. By (**0**, 20.7.4), it is therefore equivalent to say that B is a formally unramified A-algebra for the discrete topologies or for the preadic topologies. This proves the equivalence of e) and f′).

If $\mathfrak{m}$ is the maximal ideal of A, then **IV** | 66

$$k = A/\mathfrak{m} = \widehat{A}/\mathfrak{m}\widehat{A}, \qquad \widehat{B} \otimes_{\widehat{A}} k = \widehat{B}/\mathfrak{m}\widehat{B} = \widehat{B} \otimes_B (B/\mathfrak{m}B).$$

It follows from (**0**, 7.3.5) that $\widehat{B}/\mathfrak{m}\widehat{B}$ is the completion of $B/\mathfrak{m}B = B \otimes_A k$ for the $\mathfrak{n}$-preadic topology; this proves the equivalence of d″) and f). Finally, when $k(x) = k(y)$, or when k is separably closed, condition f) implies that the homomorphism $\widehat{A}/\mathfrak{m}\widehat{A} \to \widehat{B}/\mathfrak{m}\widehat{B}$ is bijective. Condition f) also implies that $\widehat{B}$ is a quasi-finite $\widehat{A}$-algebra (**0**, 7.4.4), hence a finite one, since $\widehat{A}$ is complete and $\widehat{B}$ is separated for the $\mathfrak{m}$-preadic topology, $\mathfrak{m}\widehat{B}$ being an ideal of definition of $\widehat{B}$ (**0**, 7.4.1). Nakayama's lemma therefore shows that $\widehat{A} \to \widehat{B}$ is surjective. Thus f) implies f″), and the converse is immediate. □

(17.4.5). Given an S-prescheme Y and two S-morphisms $f : X \to Y$, $g : X \to Y$, one canonically deduces an S-morphism $(f,g)_S : X \to Y \times_S Y$. We call the *prescheme of coincidences* of f and g the inverse image under $(f,g)_S$ of the diagonal $\Delta_{Y|S}$; it is therefore a subprescheme of X, and is a *closed* subprescheme when Y is an S-scheme (**I**, 5.4.1).

Proposition (17.4.6). — *Let $h : Y \to S$ be an unramified morphism, and let $f : X \to Y$, $g : X \to Y$ be two S-morphisms. Then the prescheme C of coincidences of f and g is a sub-prescheme induced on an open of X; if moreover Y is an S-scheme ((*__I__*, 5.4.1)), C is a closed sub-prescheme of X.*

Proof. Indeed, since $\Delta_h : Y \to Y \times_S Y$ is an open immersion (17.4.2), the inverse image by $(f,g)_S$ of $\Delta_{Y|S}$ is a sub-prescheme induced on an open of X ((**I**, 4.4.1)). The last assertion follows from (17.4.5). □

Corollary (17.4.7). — *Under the hypotheses of (17.4.6), let x be a point of X such that the two composed morphisms* $\operatorname{Spec}(k(x)) \to X \xrightarrow{f} Y$ *and* $\operatorname{Spec}(k(x)) \to X \xrightarrow{g} Y$ *are equal. Then there exists an open neighbourhood U of x such that $f|U = g|U$. If moreover Y is an S-scheme, there exists a closed open neighbourhood X' of x in X such that $f|X' = g|X'$. If finally X is connected, then $f = g$.*

This follows from (17.4.6) and ((**I**, 5.3.17)).

Corollary (17.4.8). — *Under the hypotheses of (17.4.6), suppose that the structural morphism $\varphi = h \circ f = h \circ g$ of X in S is closed. Let s be a point of S; let X_s be the $k(s)$-prescheme $\varphi^{-1}(s)$ and suppose that the two morphisms composed $X_s \to X \xrightarrow{f} Y$ and $X_s \to X \xrightarrow{g} Y$ are equal. Then there exists a neighbourhood open V of s in S such that $f|\varphi^{-1}(V) = g|\varphi^{-1}(V)$. If moreover Y is an S-scheme and φ is open, one can take V open and closed. If finally S is connected, $f = g$.*

Proof. It follows from (17.4.7) that the prescheme C of coincidences of f and g is induced on an open subset of X and contains X_s. Since φ is closed, there exists an open neighbourhood V of s such that $\varphi^{-1}(V) \subset C$. If moreover Y is an S-scheme, then C is closed; hence, when φ is also open, $\varphi(X - C)$ is both open and closed in S, and its complement V in S is therefore an open-and-closed neighbourhood of s such that $\varphi^{-1}(V) \subset C$. If finally S is connected, then $V = S$, and consequently $f = g$. □

Proposition (17.4.9). — *Let Y be a connected prescheme, $f : X \to Y$ an unramified and separated morphism. Then every Y-section g of X is an isomorphism of Y onto a connected component open of X, and the application $g \mapsto g(Y)$ is a bijection of $\Gamma(X/Y)$ onto the set of connected components Z of X (necessarily* open *in X) such that the restriction of f to Z is an isomorphism. In particular, if g' and g'' are two Y-sections of X such that $g'(y) = g''(y)$ for some $y \in Y$, then $g' = g''$.*

Proof. It follows from (17.4.1, b'')) that a Y-section s of X is also an open immersion, and since X is a Y-scheme, s is also a closed immersion ((**I**, 5.4.7)); it follows that s is an isomorphism of Y onto a sub-prescheme of X that is both open and closed, hence a connected component open of X, and since $s(Y)$ is connected, it is necessarily a connected component. The rest of the proposition is immediate. □

Remark (17.4.10). — In view of Remark (17.4.2), in statements (17.4.6) through (17.4.9) the word "unramified" may everywhere be replaced by "formally unramified and locally of finite type".

17.5. Characterizations of smooth morphisms. IV | 67

Theorem (17.5.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$. The following conditions are equivalent:*

(a) f is smooth at the point x.
(b) f is flat at the point x and the $k(y)$-prescheme $f^{-1}(y)$ is smooth on $k(y)$ at the point x (6.8.1).
b') f is regular at the point x (6.8.1).
(c) $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-algebra formally smooth for the discrete topologies.

Proof. One may reduce to the case where $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(C)$, where $C = B/\mathfrak{J}$, $B = A[T_1, \dots, T_n]$ being a polynomial algebra and $\mathfrak{J}$ an ideal *of finite type*. The equivalence of $a)$ and $c)$ follows from (**0**, 22.6.4). On the other hand, applying this result to the morphism locally of finite type $f^{-1}(y) \to \mathrm{Spec}(k(y))$, one sees that the equivalence of $b)$ and $b')$ follows from the equivalence of $a)$ and $b)$ in (6.8.6). It remains to prove the equivalence of $a)$ and $b)$.

Let us first show that $a)$ implies $b)$. Denote by $\mathfrak{p}$ the prime ideal $\mathfrak{i}_x$ of C, and by $\mathfrak{r}$ the prime ideal $\mathfrak{i}_y$ of A; then $\mathfrak{p} = \mathfrak{q}/\mathfrak{J}$, where $\mathfrak{q}$ is a prime ideal of B, and $\mathfrak{r}$ is the inverse image of $\mathfrak{q}$ in A. Hypothesis $a)$ first implies, by (17.3.3), that $f^{-1}(y)$ is smooth over $k(y)$ at x, and it remains to prove that $C_{\mathfrak{p}} = \mathscr{O}_{X,x}$ is a *flat* $A_{\mathfrak{r}}$-module. Since $C_{\mathfrak{p}}$ is a formally smooth $A_{\mathfrak{r}}$-algebra and B is a formally smooth A-algebra (for the discrete topologies), the Jacobian criterion (**0**, 22.6.4), together with (**0**, 19.1.12), implies that there exist in $\mathfrak{J}$ a system of r polynomials u_i ($1 \leqslant i \leqslant r$) and r indices j_i ($1 \leqslant i \leqslant r$) such that the images of the u_i in $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}^2$ generate this $B_{\mathfrak{q}}$-module and

(17.5.1.1) $$\det\left(\frac{\partial u_i}{\partial T_{j_k}}\right) \notin \mathfrak{q}.$$

Now let $g : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be the structural morphism. The fibre $g^{-1}(y)$ is the spectrum of the regular ring $k(y)[T_1, \dots, T_n]$ (**0**, 17.3.7); hence the Noetherian local ring $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ at a point of this fibre is regular. Condition (17.5.1.1) implies that the canonical images v_i of the u_i in the maximal ideal $\mathfrak{m}$ of $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ are *linearly independent* modulo $\mathfrak{m}^2$. Indeed, otherwise there would exist polynomials $w_i \in B$ ($1 \leqslant i \leqslant r$), not all belonging to $\mathfrak{q}$, such that $\sum_{i=1}^r w_i u_i \in \mathfrak{q}^2$. Differentiating with respect to the T_{j_k} would then give $\sum_{i=1}^r w_i(\partial u_i/\partial T_{j_k}) \in \mathfrak{q}$ for $1 \leqslant k \leqslant r$, contradicting (17.5.1.1), since $\mathfrak{q}$ is prime. It follows from (**0**, 17.1.7) that (v_i) is a *regular sequence* in $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$. But since g is locally of finite presentation and B is a flat A-module, (11.3.8) shows that the canonical images u'_i of the u_i in $B_{\mathfrak{q}}$ also form a regular sequence and that $B_{\mathfrak{q}}/(\sum_i u'_i B_{\mathfrak{q}})$ is a flat $A_{\mathfrak{r}}$-module. Since the images of the u'_i in $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}^2$ generate this $B_{\mathfrak{q}}$-module, Nakayama's lemma gives $\sum_i u'_i B_{\mathfrak{q}} = \mathfrak{J}_{\mathfrak{q}}$; hence $C_{\mathfrak{p}} = B_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}$ is indeed a flat $A_{\mathfrak{r}}$-module. IV | 68

Finally, let us prove that $b)$ implies $a)$. With the same notation, the hypothesis that $B_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}} = C_{\mathfrak{p}}$ is a flat $A_{\mathfrak{r}}$-module implies that the canonical homomorphism $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}} \to B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ is injective ($\mathbf{0_I}$, 6.1.2), so that $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}}$ is identified with an ideal of $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$. Since $B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}$ is a formally smooth $A_{\mathfrak{r}}$-algebra for the discrete topologies, the Jacobian criterion (**0**, 22.6.4), applied to

$$C_{\mathfrak{p}} = (B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}})/(\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}}),$$

together with (**0**, 19.1.12), proves, by hypothesis $b)$, the existence of r polynomials $v_i \in k(y)[T_1, \dots, T_n]$ and r indices j_i ($1 \leqslant i \leqslant r$) such that their images in

$$(\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}})/(\mathfrak{J}_{\mathfrak{q}}/\mathfrak{r}\mathfrak{J}_{\mathfrak{q}})^2$$

generate this $(B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}})$-module and

(17.5.1.2) $$\det\left(\frac{\partial v_i}{\partial T_{j_k}}\right) \notin \mathfrak{q}B_{\mathfrak{q}}/\mathfrak{r}B_{\mathfrak{q}}.$$

For each i, choose an element $u_i \in \mathfrak{J}$ whose canonical image is v_i. It follows from (17.5.1.2) that the u_i satisfy (17.5.1.1); on the other hand, Nakayama's lemma shows that the images u'_i of the u_i in $\mathfrak{J}_{\mathfrak{q}}$ generate this $B_{\mathfrak{q}}$-module. The Jacobian criterion (**0**, 22.6.4), together with (**0**, 19.1.12), then proves that $C_{\mathfrak{p}} = B_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}$ is a formally smooth $A_{\mathfrak{r}}$-algebra for the discrete topologies. Q.E.D. □

Corollary (17.5.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be smooth (in the sense of (17.3.1)), it is necessary and sufficient that f be regular (6.8.1), i.e. that f be flat and that for every $y \in Y$, $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular (6.7.6).*

One has thus established the equivalence of the two definitions of "smooth morphism" given in (6.8.1) and (17.3.1).

Proposition (17.5.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings. Then the conditions equivalent a) to c) of (17.5.1) are also equivalent to each of the following:*

IV | 69

d) B is an A-algebra formally smooth for the preadic topologies.
d′) $\widehat{B}$ is an $\widehat{A}$-algebra formally smooth for the adic topologies.

If moreover $k(x) = k(y)$, these conditions are also equivalent to:

d″) $\widehat{B}$ is an $\widehat{A}$-algebra isomorphic to a formal power series algebra $\widehat{A}[[T_1, \dots, T_n]]$.

Proof. The equivalence of condition c) of (17.5.1) and condition d) follows from the equivalence of a) and d) in the Jacobian criterion (**0**, 22.6.4), and the equivalence of d) and d') follows from (**0**, 19.3.6). Moreover, d'') implies d') without any hypothesis on the residue fields (**0**, 19.3.4). Finally, if $\mathfrak{m}$ denotes the maximal ideal of $\widehat{A}$, hypothesis d') implies that $\widehat{B}/\mathfrak{m}\widehat{B}$ is a complete Noetherian local $k(y)$-algebra, formally smooth for its adic topology (**0**, 19.3.5); the hypothesis $k(y) = k(x)$ therefore implies that $\widehat{B}/\mathfrak{m}\widehat{B}$ is $k(y)$-isomorphic to a formal power series algebra $k(y)[[T_1, \dots, T_n]]$ (**0**, 19.6.4). On the other hand, $\widehat{A}[[T_1, \dots, T_n]]$ is a flat $\widehat{A}$-module and a complete Noetherian local $\widehat{A}$-algebra; it follows from (**0**, 19.7.1.5) that this algebra is isomorphic to $\widehat{B}$. Thus d') implies d'') under the additional hypothesis $k(x) = k(y)$. □

Remark (17.5.4). — Suppose that Y is a locally Noetherian prescheme, and $f : X \to Y$ a morphism locally of finite type. The criterion (17.5.3, d)), together with (**0**, 22.1.4) shows that for proving f is smooth, one can apply the definition (17.1.1), *restricting* to the case where the affine scheme Y' is the spectrum of a local Artinian ring.

Proposition (17.5.5). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that Y is reduced at the point y. Then, for f to be smooth at the point x, it is necessary and sufficient that f be universally open in a neighbourhood of x in $f^{-1}(y)$ and that $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular at the point x.*

Proof. By (17.5.1), everything reduces to seeing that, if $f^{-1}(y)$ is a $k(y)$-prescheme geometrically regular at the point x, it is equivalent to say that f is flat or universally open in a neighbourhood of x in $f^{-1}(y)$. But if f is flat at the point x, it is in a neighbourhood of x in X (11.1.1) and hence universally open in this neighbourhood (2.4.6). Reciprocally, the hypothesis that f is universally open in a neighbourhood of x in $f^{-1}(y)$ and that $f^{-1}(y)$ is a $k(y)$-prescheme geometrically regular at the point x implies that f is flat at the point x, since $\mathscr{O}_{Y,y}$ is reduced (15.2.2). □

Corollary (17.5.6). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that Y is reduced and geometrically unibranch (6.15.1) at the point y. Then, for f to be smooth at the point x, it is necessary and sufficient that f be equidimensional at the point x and that $f^{-1}(y)$ be a $k(y)$-prescheme geometrically regular at the point x.*

Observing that the set of points at which f is equidimensional is open (13.3.2), one sees that the corollary follows from (17.5.5) and the Chevalley criterion (14.4.4).

Proposition (17.5.7). — *Let $f : X \to Y$ be a morphism locally of finite presentation, smooth at a point $x \in X$; set $y = f(x)$. Then, for $\mathscr{O}_{X,x}$ to be reduced (resp. integrally closed, resp. geometrically unibranch), it is necessary and sufficient that $\mathscr{O}_{Y,y}$ be so.*

This has been proved in (11.3.13) and (11.3.14) completed by $\mathbf{Err_{IV}}$, 30.

Proposition (17.5.8). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, smooth at a point $x \in X$; set $y = f(x)$. Then:*

(i) $\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y}) + \dim(\mathscr{O}_{X,x} \otimes_{\mathscr{O}_{Y,y}} k(y))$.
(ii) $\operatorname{coprof}(\mathscr{O}_{X,x}) = \operatorname{coprof}(\mathscr{O}_{Y,y})$.
(iii) *For $\mathscr{O}_{X,x}$ to have the property (S_n) (5.7.2) (resp. (R_n) (5.8.2)), it is necessary and sufficient that $\mathscr{O}_{Y,y}$ have it. In particular, for $\mathscr{O}_{X,x}$ to be regular, it is necessary and sufficient that $\mathscr{O}_{Y,y}$ be so.*

These are special cases of (6.1.2), (6.3.2), (6.4.1) and (6.5.3).

17.6. Characterizations of étale morphisms. IV | 70

Theorem (17.6.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$. The following conditions are equivalent:*

a) *f is étale at the point x.*
a′) *f is smooth at the point x and non ramified at the point x.*
b) *f is smooth at the point x and locally quasi-finite (**Err$_{\mathbf{III}}$**, 20).*
c) *f is flat at the point x and non ramified at the point x.*
c′) *f is flat at the point x and the ring $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a field, extension finite separable of $k(y)$.*
d) *$\mathscr{O}_{X,x}$ is an $\mathscr{O}_{Y,y}$-algebra formally étale for the discrete topologies.*

Proof. The equivalence of $a)$ and $a')$ follows immediately from the definitions; that of $a)$ and $d)$ follows from the equivalence of $a)$ and $c)$ in (17.4.1) and the equivalence of $a)$ and $d)$ in (17.5.1). The equivalence of $c)$ and $c')$ follows from the equivalence of $a)$ and $d')$ in (17.4.1). The fact that $a')$ implies $c')$ follows from (17.5.1); conversely, if $c')$ holds, then f is regular (hence smooth by (17.5.1)) at x, because if K is a finite separable extension of a field k, then, for every extension k' of k, $\operatorname{Spec}(K \otimes_k k')$ is regular, being a finite sum of spectra of fields. The fact that $a')$ implies $b)$ follows from (17.4.1, $d')$) and (17.5.1, $b)$). It remains to show that $b)$ implies $c)$; since one already knows from (17.5.1) that f is flat at x, it is enough to prove that $f^{-1}(y)$ is a $k(y)$-prescheme unramified over $k(y)$. In other words, one is reduced to proving that $b)$ implies $c)$ when $Y = \operatorname{Spec}(k)$ is the spectrum of a field k and $X = \operatorname{Spec}(A)$, where A is a finite local k-algebra ($\mathbf{0_I}$, 7.4.1). By hypothesis $b)$, A is a formally smooth k-algebra for the discrete topologies, which here coincide with the preadic topologies; hence ($\mathbf{0}$, 19.6.5) A is a regular local ring.

It is therefore a field since it is Artinian. It then follows from ($\mathbf{0}$, 19.6.5.1) that A is a finite separable extension of k. □ IV | 71

Corollary (17.6.2). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following conditions are equivalent:*

a) *f is étale.*
a′) *f is smooth and non ramified.*
b) *f is smooth and locally quasi-finite (**Err$_{\mathbf{III}}$**, 20).*
c) *f is flat and non ramified.*
c′) *f is flat, and every fibre $f^{-1}(y)$ is a $k(y)$-scheme sum of spectra of fields, extensions finite and separable of $k(y)$.*
c″) *f is flat, and for every $y \in Y$ and every algebraically closed extension k' of $k(y)$, the "geometric fibre" $f^{-1}(y) \otimes_{k(y)} k'$ is a sum of spectra of fields isomorphic to k'.*

Proof. The only point left to prove is the equivalence of $c')$ and $c'')$. It is clear from base change (17.3.3) that $c')$ implies $c'')$. Conversely, since the projection morphism

$$f^{-1}(y) \otimes_{k(y)} k' \longrightarrow f^{-1}(y)$$

is open (2.4.10), hypothesis $c'')$ implies that the space $f^{-1}(y)$ is discrete. Thus, for every point $x \in X_y = f^{-1}(y)$, the local ring $\mathscr{O}_{X_y,x} = A$ is a finite $k(y)$-algebra, hence an Artinian local ring. Moreover, it follows from $c'')$ that $\operatorname{Spec}(A \otimes_{k(y)} k')$ is a sum of spectra of fields isomorphic to k', which is possible only if A is a field that is a finite separable extension of $k(y)$ (4.6.1). □

Proposition (17.6.3). — *Let Y be a locally Noetherian prescheme, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. Set $A = \mathscr{O}_{Y,y}$, $B = \mathscr{O}_{X,x}$, which are local Noetherian rings, and let k be the residue field of A. Then the conditions equivalent $a)$ to $d)$ of (17.6.1) are also equivalent to each of the following:*

e) $\widehat{B}$ *is an* $\widehat{A}$*-algebra formally étale for the adic topologies.*
e′) $\widehat{B}$ *is an* $\widehat{A}$*-module free and* $\widehat{B} \otimes_{\widehat{A}} k$ *is a field, extension finite and separable of* k *(which implies that* $\widehat{B}$ *is an* $\widehat{A}$*-algebra finite).*

If moreover $k(x) = k(y)$*, or if* k *is separably closed, these conditions are also equivalent to:*

e″) The canonical homomorphism $\widehat{A} \to \widehat{B}$ *is bijective.*

Proof. The equivalence of *e*) with each of the conditions of (17.6.1) follows at once from (17.4.4, f')) and (17.5.3, d')). That *e*) implies *e*′) follows from (17.4.4, f)) and (**0**, 19.7.1), since $\widehat{B}$ is then a finite $\widehat{A}$-algebra (17.4.4), and saying that it is a flat $\widehat{A}$-module is equivalent to saying that it is a free $\widehat{A}$-module ($\mathbf{0_{III}}$, 10.1.3). Conversely, the implication *e*′) ⇒ *e*) follows from (17.4.4) and (**0**, 19.7.1). Finally, *e*′) implies that the homomorphism $\widehat{A} \to \widehat{B}$ is injective; if $k(x) = k(y)$, or if k is separably closed, this homomorphism is surjective by (17.4.4). The converse is immediate. □

Proposition (17.6.4). — *Under the hypotheses of (17.6.3), if* f *is étale at the point* x*, then* $\dim(\mathscr{O}_{X,x}) = \dim(\mathscr{O}_{Y,y})$.

This is a particular case of (17.5.8, (i)) since x is isolated in its fibre $f^{-1}(y)$.

17.7. Properties of descent, passage to the limit, and constructibility. IV | 72

Proposition (17.7.1). — *Let* $f : X \to Y$ *be a morphism locally of finite presentation,* $g : Y' \to Y$ *a morphism,* $X' = X \times_Y Y'$*,* $f' = f_{(Y')} : X' \to Y'$ *and* $g' : X' \to X$ *the canonical projections. Let* x' *be a point of* X' *and set* $x = g'(x')$*,* $y' = f'(x')$*.*

(i) If f' *is non ramified at the point* x'*, then* f *is non ramified at the point* x*.*
(ii) Suppose moreover that g *is flat at the point* y'*. Then, if* f' *is smooth (resp. étale) at the point* x'*,* f *is smooth (resp. étale) at the point* x*.*

Proof. Set $y = f(x) = g(y')$, so that $f'^{-1}(y') = f^{-1}(y) \otimes_{k(y)} k(y')$; note that f' is locally of finite presentation.

(i) Since the property for a morphism (locally of finite presentation) of being non ramified at a point only involves the fibre of the morphism at that point (17.4.1, *d*)), one may reduce to the case where Y and Y' are spectra of fields. But then it amounts to the same thing to say that f (resp. f') is non ramified at x (resp. x') or that it is étale at that point (17.6.1, *c*)); thus (i) is a consequence of (ii).

(ii) Since f' is flat at x' (17.5.1), the hypothesis that g is flat at y' implies that f is flat at x: indeed, the projection $g' : X' \to X$ is flat at x', and $f \circ g' = g \circ f'$ is flat at x', whence the conclusion (2.2.11, (iv)). Whether f is smooth (resp. étale) at x then only involves the fibre $f^{-1}(y)$ (17.5.1 and 17.6.1), and one is again reduced to the case where $Y = \operatorname{Spec}(k)$ and $Y' = \operatorname{Spec}(k')$ are spectra of fields. To say that f' is smooth at x' then means (17.5.1) that X' is geometrically regular over k' at x', and this implies (6.7.8) that X is geometrically regular over k at x; hence f is smooth at x. Suppose moreover that f' is étale at x', so that x' is isolated in $f'^{-1}(y')$ (17.6.1). Since the projection $f'^{-1}(y') \to f^{-1}(y)$ is an open morphism (2.4.10), x is isolated in $f^{-1}(y)$; as f is already known to be smooth at x, it is étale at that point (17.6.1). □

Corollary (17.7.2). — *(i) With the notation of (17.7.1), let* U *(resp.* U'*) be the set of points of* X *(resp.* X'*) where* f *(resp.* f'*) is non ramified; then* $U' = g'^{-1}(U)$*.*
(ii) Suppose moreover that g *is flat, and let* V *(resp.* V'*) be the set of points of* X *(resp.* X'*) where* f *(resp.* f'*) is smooth; then* $V' = g'^{-1}(V)$*.*

Corollary (17.7.3). — *(i) Suppose* g *surjective; then, for* f *to be non ramified, it is necessary and sufficient that* f' *be so.*
(ii) Let $f : X \to Y$ *be an* S*-morphism and* $g : S' \to S$ *a faithfully flat morphism. Suppose that* f *is locally of finite presentation, or that* g *is quasi-compact. Then, for* f *to be smooth (resp. non ramified, resp. étale), it is necessary and sufficient that* f' *be so.*

In (ii), the case where f is locally of finite presentation follows from (17.7.1). If g is quasi-compact and f' is smooth (resp. non ramified, resp. étale), then f' is locally of finite presentation by (2.7.1, (iv)), and one is reduced to the first case.

Proposition (17.7.4). — *Let $f : X \to Y$ be a morphism, $g : Y' \to Y$ a morphism flat and locally of finite presentation, $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ and $g' : X' \to X$ the canonical projections.*

Let V (resp. V') be the set of $x \in X$ (resp. $x' \in X'$) where f (resp. f') has one of the following properties: **IV** | 73

(i) *locally of finite type;*
(ii) *locally of finite presentation;*
(iii) *flat;*
(iv) *non ramified;*
(v) *smooth;*
(vi) *étale.*

Then $V' = g'^{-1}(V)$ (in other words, for f' to have the property considered at a point x', it is necessary and sufficient that f have this property at x).

Proof. Property (iii) is immediate and does not even require the hypothesis that g be locally of finite presentation (2.2.11, (iv)), since the projection $g' : X' \to X$ is flat. By (17.7.2), the assertions concerning properties (iv), (v), and (vi) are consequences of the assertion concerning property (ii). It is therefore enough to consider cases (i) and (ii). It is clear that $V' \supset g'^{-1}(V)$; it remains to prove the reverse inclusion. The sets V and V' are open, and $W = g'(V')$ is also open in X by (2.4.6). It is thus enough to prove that the morphism $f|W : W \to Y$ is locally of finite type (resp. locally of finite presentation). By hypothesis, the composite

$$V' \xrightarrow{g''} W \xrightarrow{f|W} Y,$$

where g'' is the restriction of g', is locally of finite type (resp. locally of finite presentation), because it is equal to the composite $V' \xrightarrow{f'} Y' \xrightarrow{g} Y$; moreover, g'' is surjective, hence faithfully flat. The assertion is therefore reduced to the following lemma, which strengthens (11.3.16). □

Lemma (17.7.5). — *Let $f : X \to Y$ be a faithfully flat morphism and locally of finite presentation, $g : Y \to Z$ a morphism such that $g \circ f : X \to Z$ has one of the following properties: being*

(i) *locally of finite type;*
(ii) *locally of finite presentation;*
(iii) *of finite type.*

Then g has the same property. If moreover f is quasi-compact or g quasi-separated, the same conclusion holds for the property:

(iv) *of finite presentation.*

Proof. In cases (i) and (ii), one must show that, for every $y \in Y$, there are an affine open neighbourhood V of y in Y and an affine open neighbourhood W of $z = g(y)$ in Z, containing $g(V)$, such that the morphism $V \to W$ induced by g is of finite type (resp. of finite presentation). By hypothesis, there exist a point $x \in X$ such that $f(x) = y$, an affine open neighbourhood U of x in X, an affine open neighbourhood V' of y in Y containing $f(U)$, and an affine open neighbourhood W of z in Z containing $g(V')$, such that the morphism $f_1 : U \to V'$ induced by f is flat and of finite presentation, while the composite of f_1 with the restriction $g_1 : V' \to W$ of g is of finite type (resp. of finite presentation). Then $f(U)$ is open in Y (2.4.6); if $V \subset f(U)$ is an affine neighbourhood of y, the morphism

$$f_2 : f_1^{-1}(V) \longrightarrow V,$$

obtained by restricting f_1, is still of finite presentation and is moreover faithfully flat. Furthermore, if $g_2 = g_1|V$, then $g_2 \circ f_2$ is of finite type (resp. of finite presentation), while the open set $f_1^{-1}(V)$ is quasi-compact (resp. quasi-compact and quasi-separated). One is thus reduced to the hypotheses of (11.3.16), which proves that g_2 is of finite type (resp. of finite presentation). **IV** | 74

In case (iii), the question is local on Z, so one may suppose Z affine; then X is quasi-compact, and so is $Y = f(X)$. Case (iii) is therefore a consequence of case (i).

In case (iv), under the additional hypotheses on f or g, one may again suppose Z affine, hence X and Y quasi-compact. Moreover, g is already known to be locally of finite presentation and quasi-compact (I.1.3); thus everything reduces to showing that g is quasi-separated, and it is enough to prove this when f is quasi-compact. But since $g \circ f$ is quasi-separated, the same is true of f

(I.2.2, (v)); as f is quasi-compact and locally of finite presentation, it is of finite presentation (I.6.1). It then suffices to repeat the argument of the first paragraph of the proof of (11.3.16). □

Remark (17.7.6). — Regarding the assertion relative to property (iv), one cannot suppress the hypothesis that f is quasi-compact; without this, it would imply that every morphism $g : Y \to Z$ quasi-compact and locally of finite presentation would be of finite presentation, a conclusion known to be erroneous (**I**, 6.4). Indeed, one can only restrict to the case where Z is affine, hence Y quasi-compact; there is then a finite covering of Y by affine opens U_i such that the restrictions $g|U_i$ are of finite presentation; it would then suffice to take for X the prescheme *sum* of the U_i, for $f : X \to Y$ the canonical morphism, which is evidently faithfully flat and locally of finite presentation (**I**, 6.5); $g \circ f$ would be of finite presentation by virtue of the choice of the U_i and of (**I**, 6.5), whence our assertion.

Proposition (17.7.7). — *Let $f : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$. Suppose that g is flat at the point x. Then, if h is smooth (resp. non ramified, resp. étale) at the point x, f is smooth (resp. non ramified, resp. étale) at the point y.*

Proof. Set $s = h(x) = f(y)$. To say that h is non ramified at x (resp. that f is non ramified at y) amounts to saying that $h^{-1}(s)$ is étale over $k(s)$ at x (resp. that $f^{-1}(s)$ is étale over $k(s)$ at y) (17.4.1 and 17.6.1); since the morphism $g_s : h^{-1}(s) \to f^{-1}(s)$ induced by g is flat at x, one may restrict to proving the proposition when h is smooth or étale at x. Moreover, since h is then flat at x (17.5.1), f is flat at y by (2.2.11, (iv)). It therefore remains to say that $f^{-1}(s)$ is smooth (resp. étale) over $k(s)$ at y, and one is thus reduced to the case where $S = \mathrm{Spec}(k)$ is the spectrum of a field.

(i) Case of smooth morphisms. Since g is locally of finite presentation (I.4.3, (v)), there is an open neighbourhood U of x in X on which g is flat (11.3.1) and h is smooth. Moreover, $g(U)$ is an open neighbourhood of y in Y (2.4.6); replacing X by U and Y by $g(U)$, one may therefore suppose that g is faithfully flat and that h is smooth, and one is reduced to proving that f is smooth. If k' is an algebraically closed extension of k, then $X \otimes_k k'$ is smooth over k'; by (17.7.3, (ii)), it is enough to prove that $Y \otimes_k k'$ is smooth over k' (since $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$ is faithfully flat). Thus one may suppose that k is algebraically closed. The set of k-rational points of Y is then very dense in Y (10.4.8), and the locus where Y is smooth over k is open; it is therefore enough to prove that Y is smooth over k at every k-rational point y. At such a point, saying that Y is smooth over k is equivalent to saying that Y is regular at y (17.5.1 and 6.7.8). Now $y = g(x)$ for some $x \in X$, and by hypothesis X is regular at x (17.5.1); since X and Y are locally Noetherian and g is flat, Y is indeed regular at y (6.5.1, (i)).

(ii) Case of étale morphisms. By (i), one already knows that f is smooth at y; by (17.6.1), it is therefore enough to prove that f is quasi-finite at y, or equivalently that $\mathscr{O}_{Y,y}$ is a finite k-algebra. **IV** | 75
Since $\mathscr{O}_{X,x}$ is a faithfully flat $\mathscr{O}_{Y,y}$-module ($\mathbf{0_I}$, 6.6.2), $\mathscr{O}_{Y,y}$ is identified with a k-submodule of $\mathscr{O}_{X,x}$; and since $\mathscr{O}_{X,x}$ is, by hypothesis, a finite k-algebra, the same holds for $\mathscr{O}_{Y,y}$. □

Proposition (17.7.8). — *With the notation of (8.8.1), suppose that X_α and Y_α are locally of finite presentation over S_α. Let $f_\alpha : X_\alpha \to Y_\alpha$ be an S_α-morphism and $f : X \to Y$ the corresponding S-morphism.*

- *(i) Let x be a point of X, x_λ its canonical projection in X_λ. For f to be smooth (resp. non ramified, resp. étale) at x, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ is smooth (resp. non ramified, resp. étale) at x_λ.*
- *(ii) Suppose moreover that X_α is quasi-compact. For f to be smooth (resp. non ramified, resp. étale), it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that f_λ is smooth (resp. non ramified, resp. étale).*

Proof. (i) If $y = f(x)$ and $y_\lambda = f_\lambda(x_\lambda)$ is the canonical projection of y in Y_λ, then

$$f^{-1}(y) = f_\lambda^{-1}(y_\lambda) \otimes_{k(y_\lambda)} k(y).$$

The assertion concerning non-ramified morphisms therefore follows from (17.7.1, (i)), and it is enough to consider smooth morphisms. Since f and f_λ are locally of finite presentation, (17.5.1) says that it amounts to the same thing to require that $f^{-1}(y)$ be geometrically regular at x or that $f_\lambda^{-1}(y_\lambda)$ be geometrically regular at x_λ (6.7.8). The proposition then follows from (17.5.1) and (11.2.6).

(ii) For every λ, let U_λ be the open set of points $x_\lambda \in X_\lambda$ at which f_λ is smooth (resp. non ramified, resp. étale), and let V_λ be its inverse image in X. By (i), for every $x \in X$ there exists a λ such that f_λ has the required property at x_λ; hence X is the union of the V_λ. Moreover, by (17.3.3), $V_\lambda \subset V_\mu$ for $\lambda \leqslant \mu$; since X is quasi-compact, there is therefore an index μ such that $X = V_\mu$. Since the X_λ are quasi-compact, it follows from (8.3.4) that there exists an index $\nu \geqslant \mu$ such that the inverse image of U_μ in X_ν is all of X_ν; this means that f_ν is smooth (resp. non ramified, resp. étale), by (17.3.3). □

Corollary (17.7.9). — *Let $S = \mathrm{Spec}(A)$ be an affine scheme, $f : X \to S$ a morphism. The following conditions are equivalent:*

a) f is a morphism of finite presentation and smooth (resp. non ramified, resp. étale).
b) There exists an affine Noetherian scheme $S_0 = \mathrm{Spec}(A_0)$, a morphism of finite type $f_0 : X_0 \to S_0$ and a morphism $S \to S_0$ such that the S-prescheme $X_0 \otimes_{S_0} S$ is S-isomorphic to X and f_0 is smooth (resp. non ramified, resp. étale).
c) The conditions of b) are satisfied, and moreover A_0 is a sub-$\mathbf{Z}$-algebra of finite type of A, and the morphism $S \to S_0$ corresponds to the canonical injection $A_0 \to A$.

Proof. The proof from (17.7.8) is the same as that of (11.2.7) from (11.2.6). □

Proposition (17.7.10). — *Let $f : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$. Suppose that h is flat at the point x and that f is non ramified at the point y. Then f is étale at the point y and g is flat at the point x.* **IV | 76**

Proof. (One will see further on (18.4.9) that one can in fact dispense with the hypothesis that h is locally of finite presentation.)

The question being local on S, X, and Y, one may suppose that S, X, and Y are affine, that f, g, and h are of finite presentation, that h is flat, and that f is non ramified. In view of (11.2.7) and (17.7.9), one may moreover suppose that S, X, and Y are Noetherian; finally, one may restrict to the case $S = \mathrm{Spec}(\mathscr{O}_{S,s})$, where $s = f(y) = h(x)$. Since $A = \mathscr{O}_{S,s}$ is a Noetherian local ring, there exist a complete Noetherian local ring B with algebraically closed residue field and a local homomorphism $A \to B$ making B a faithfully flat A-module ($\mathbf{0}_{\mathbf{III}}$, 10.3.1). Replacing X and Y by $X \times_S \mathrm{Spec}(B)$ and $Y \times_S \mathrm{Spec}(B)$, one concludes from (2.5.1) and (17.7.1) that one may restrict to the case where A is complete and has algebraically closed residue field. The hypothesis that f is non ramified at y then implies (17.4.4) that $\mathscr{O}_{Y,y}$ is a finite $\mathscr{O}_{S,s}$-algebra and a complete Noetherian local ring ($\mathbf{0}_{\mathbf{I}}$, 7.4.2); since the residue field of $\mathscr{O}_{S,s}$ is algebraically closed, the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is surjective (17.4.4). On the other hand, the hypothesis that h is flat at x implies that the composite homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective ($\mathbf{0}_{\mathbf{I}}$, 6.5.1); consequently $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is bijective, which proves that f is étale at y (17.6.3). Furthermore, $\mathscr{O}_{X,x}$ is a flat $\mathscr{O}_{Y,y}$-module, so g is flat at x. □

Proposition (17.7.11). — *Let S be a prescheme and X, Y two S-preschemes locally of finite presentation over S; let $f : X \to Y$ be an S-morphism. For every $s \in S$, denote X_s, Y_s, f_s the preschemes and morphism deduced from X, Y, f by the change of base $\mathrm{Spec}(k(s)) \to S$. Then:*

(i) The set of $x \in X$ such that, if s is the image of x in S, $f_s : X_s \to Y_s$ is smooth (resp. non ramified, resp. étale, resp. differentially smooth) at x, is locally constructible.
(ii) Suppose moreover that f is of finite presentation. The set of $y \in Y$ such that, if s is the image of y in S, f_s is smooth (resp. non ramified, resp. étale, resp. differentially smooth) at all the points of $f_s^{-1}(y)$, is locally constructible.
(iii) Suppose X and Y of finite presentation on S. The set of $s \in S$ such that f_s is smooth (resp. non ramified, resp. étale, resp. differentially smooth) is locally constructible.

IV | 77

Proof. Let E be the set of $x \in X$ satisfying the property considered in (i). The set F of points $y \in Y$ satisfying the corresponding property in (ii) is none other than $Y - f(X - E)$; hence (ii) follows from (i) and Chevalley's theorem when f is of finite presentation (1.8.4). Likewise, if $h : Y \to S$ is the structural morphism, the set of $s \in S$ satisfying the corresponding property in (iii) is $S - h(Y - F)$;

thus (iii) again follows from (i) and Chevalley's theorem when f and h are of finite presentation. It therefore remains to prove the assertions in (i).

Let us first prove (i) for the property of being smooth. The question is local on X, so one may restrict to the case where $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and $Y = \mathrm{Spec}(C)$ are affine, with B and C algebras of finite presentation. Reasoning as at the beginning of (9.9.1) and using (17.7.2, (ii)), one is reduced to the case where A is Noetherian. By ($\mathbf{0_{III}}$, 9.2.3), it is enough to show that, if $x \in E$ (resp. $x \notin E$), there is a neighbourhood V of x in $\overline{\{x\}}$ contained in E (resp. in $X - E$).

Denote by $g : X \to S$ and $h : Y \to S$ the structural morphisms. One may first replace S by the reduced sub-prescheme S' having $\overline{\{g(x)\}}$ as underlying space, X by $X' = g^{-1}(S')$, and Y by $Y' = h^{-1}(S')$; the fibres of X and X' (resp. of Y and Y') at the points of S' are the same. Thus one may restrict to the case where S is integral and where $\eta = g(x) = h(y)$, with $y = f(x)$, is its generic point.

1° Suppose first that $x \in E$. The local rings $\mathscr{O}_{X_\eta,x}$ and $\mathscr{O}_{Y_\eta,y}$ are respectively equal to $\mathscr{O}_{X,x}$ and $\mathscr{O}_{Y,y}$. Since smoothness of a morphism of finite presentation at a point depends only on the local ring at that point and the local ring at its image (17.5.1), the hypothesis $x \in E$ means that f is smooth at x. It then has this property at every point of an open neighbourhood of x in X, and (17.3.3, (iii)) yields the conclusion.

2° Suppose next that $x \in X - E$ and that f_η is not flat at x. The conclusion follows from the following lemma, which sharpens (11.2.8):

Lemma (17.7.11.1). — *Let $g : X \to S$ and $h : Y \to S$ be two morphisms locally of finite presentation, $f : X \to Y$ an S-morphism, and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. Then the set E of $x \in X$ such that $\mathscr{F}_{g(x)}$ is $f_{g(x)}$-flat at x is locally constructible.*

Proof of Lemma 17.7.11.1. Reasoning again as at the beginning of (9.9.1) and using (2.5.1), one is reduced to the case where S, X, and Y are Noetherian. Then, as above, one reduces to the case where S is integral, where $\eta = g(x) = h(f(x))$ is its generic point, and where it is enough to prove that, if $x \in E$ (resp. $x \notin E$), there is a neighbourhood V of x in $\overline{\{x\}}$ contained in E (resp. in $X - E$). The case $x \in E$ is an immediate consequence of (11.1.1). To treat the case $x \notin E$, one reasons as in (9.4.7.1): after possibly replacing Y and X by neighbourhoods of $f(x)$ and x, respectively, there exist two coherent $\mathscr{O}_Y$-modules $\mathscr{G}$ and $\mathscr{H}$ and an $\mathscr{O}_Y$-homomorphism $u : \mathscr{G} \to \mathscr{H}$ such that, for every $s \in S$, the homomorphism $u_s : \mathscr{G}_s \to \mathscr{H}_s$ is injective, whereas

$$1 \otimes u_\eta : \mathscr{F} \otimes_{\mathscr{O}_{Y_\eta}} \mathscr{G}_\eta \longrightarrow \mathscr{F} \otimes_{\mathscr{O}_{Y_\eta}} \mathscr{H}_\eta$$

is not injective at x; equivalently, $x \in \mathrm{Supp}(\mathrm{Ker}(1 \otimes u_\eta))$. If $T = \overline{\{x\}}$, then $\mathrm{Supp}(\mathrm{Ker}(1 \otimes u_\eta)) \supset T_\eta$. One may suppose that, for every $s \in S$, base change gives

$$\mathrm{Ker}(1 \otimes u_s) = (\mathrm{Ker}(1 \otimes u))_s$$

(9.4.2), and hence

$$\mathrm{Supp}(\mathrm{Ker}(1 \otimes u_s)) = (\mathrm{Supp}(\mathrm{Ker}(1 \otimes u)))_s$$

(I.9.1.13.1). It follows from (9.5.2) that, for s in a neighbourhood of η, one has $\mathrm{Supp}(\mathrm{Ker}(1 \otimes u_s)) \supset T_s$, which proves the lemma. □

3° Suppose now that $x \in X - E$, that f_η is flat at x, but that f_η is not smooth at x. To say that f_η is flat at x is equivalent to saying that f itself is flat at x; replacing X by a neighbourhood of x, one may therefore suppose that f is flat (11.1.1). It follows that f_s is flat for every $s \in S$; and since, for every $y \in Y$, one has $f^{-1}(y) = f_{h(y)}^{-1}(y)$, it amounts to the same thing to say that $f_{g(x')}$ is smooth at x' or that f is smooth at x'. But the locus where f is smooth is open in X (12.1.7); since it does not contain x by hypothesis, it does not contain $\overline{\{x\}}$ either. This completes the proof for the first property considered in (i).

Let us next prove (i) for the property of being étale. For f_s to be étale at x means that it is both smooth and quasi-finite at x (17.6.1). Now it amounts to the same thing to say that $f_{g(x)}$ is quasi-finite at x or that f itself is quasi-finite at that point; it follows from (13.1.4) that the set of points x at which $f_{g(x)}$ is quasi-finite is open in X, hence a fortiori locally constructible. The conclusion therefore follows from the local constructibility of the set of points x at which $f_{g(x)}$ is smooth. **IV** | 78

Let us pass to the proof of (i) for the property of being differentially smooth. Let $p : X \times_Y X \to X$ be the second canonical projection; for every $s \in S$, the second projection $X_s \times_{Y_s} X_s \to X_s$ is just p_s. It follows from (17.12.5.1)[5] that $f_{g(x)}$ is differentially smooth at x if and only if $p_{g(x)}$ is smooth at $\Delta_f(x)$. Since p is locally of finite presentation, the set of points $z \in X \times_Y X$ at which $p_{g(p(z))}$ is smooth is locally constructible; its intersection with the locally closed subset $\Delta_f(X)$ is therefore locally constructible in $\Delta_f(X)$ (1.8.2). As the restriction of p to $\Delta_f(X)$ is an isomorphism onto X, the set of $x \in X$ at which $f_{g(x)}$ is differentially smooth is locally constructible.

Finally, consider the property of being non ramified. The diagonal morphism $\Delta_f : X \to X \times_Y X$ is an immersion locally of finite presentation (I.4.3.1), and for every $s \in S$ the diagonal $\Delta_{f_s} : X_s \to X_s \times_{Y_s} X_s$ is just $(\Delta_f)_s$. To say that $f_{g(x)}$ is non ramified at x amounts to saying that $(\Delta_f)_{g(x)}$ is a local isomorphism at x (17.4.1); since this is an immersion locally of finite presentation, it amounts to the same thing (17.9.1)[6] to say that $(\Delta_f)_{g(x)}$ is étale at x. It is therefore enough to apply what has just been proved for the property of being étale. □

17.8. Criteria for smoothness and non-ramification by fibres. **IV** | 79

Proposition (17.8.1). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation. For an S-morphism $f : X \to Y$ to be non ramified, it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ deduced from f by the change of base* $\operatorname{Spec}(k(s)) \to S$*, be non ramified.*

Proof. This follows immediately from the fact that, for a morphism locally of finite presentation, being non ramified is a property of the *fibres* of this morphism (17.4.1), *d*)), and from the fact that for every $y \in Y$, one has $f^{-1}(y) = f_s^{-1}(y)$ where $s = g(y)$. □

Proposition (17.8.2). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation; suppose moreover that h is flat. For an S-morphism $f : X \to Y$ to be smooth (resp. étale), it is necessary and sufficient that, for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ induced from f by the base change* $\operatorname{Spec}(k(s)) \to S$ *be smooth (resp. étale). When this is the case, the morphism g is flat at all points of $f(X)$.*

Proof. Indeed, one knows from (11.3.10) that f is flat if and only if f_s is flat for every $s \in S$, and that in this case g is flat at the points of $f(X)$. But for a flat morphism locally of finite presentation, smoothness is a property of its fibres (17.5.1, *b*)); and for every $y \in Y$, one has $f^{-1}(y) = f_s^{-1}(y)$ when $s = g(y)$. □

Remark (17.8.3). — The preceding proofs show (taking (11.3.10) into account) that, under the same hypotheses on g and h, for f to be non ramified (resp. smooth, resp. étale) *at a point* $x \in X$, it is enough that, on setting $s = h(x)$, f_s be non ramified (resp. smooth, resp. étale) at x.

17.9. Étale morphisms and open immersions.

Theorem (17.9.1). — *Let $f : X \to Y$ be a morphism. The following properties are equivalent:*

- *a) f is an open immersion.*
- *b) f is a flat monomorphism locally of finite presentation.*
- *c) f is étale and radiciel.*

Proof. It results from (**I**, 4.3), (i)) that *a*) implies *b*). The condition *b*) implies that for every $y \in Y$, the fibre $f^{-1}(y)$ is either empty or isomorphic to $\operatorname{Spec}(k(y))$ (8.11.5.1), hence *b*) implies *c*), by virtue of (17.6.2), *c*)). It remains to show that *c*) implies *a*). The question being local on X and on Y (since f is injective), one can reduce to the case where Y is affine and f of finite presentation. As f is flat, this is an *open* morphism (2.4.6), hence, replacing Y by $f(X)$, one can suppose that f is *surjective*. For every morphism $Y' \to Y$, $f' = f_{(Y')} : X_{(Y')} \to Y'$ is again étale, radiciel, surjective, and of finite presentation; it is therefore open, hence a homeomorphism. In other words, f is a *universal homeomorphism*; and since it is of finite type and separated (I.8.7.1, (ii)), it is proper. The hypothesis that f is radiciel and of finite type implies that f is quasi-finite; hence (8.11.1) f is a finite morphism. To prove that f is an isomorphism, one can reduce to the case where $Y = \operatorname{Spec}(A)$, A being a local ring. As f is of finite presentation, one has $X = \operatorname{Spec}(B)$, where B is a flat A-module of finite

[5] The reader may verify that the result of (17.7.11) is not used in the remainder of §17, so there is no vicious circle.

[6] The reader may verify that the result of (17.7.11) is not used in the remainder of §17, so there is no vicious circle.

presentation (I.4.7), hence free (Bourbaki, *Alg. comm.*, chap. II, §5, no. 2, Cor. 2 to Th. 1). Moreover, if $\mathfrak{m}$ is the maximal ideal of A and k its residue field, $B/\mathfrak{m}B$ is by hypothesis a field, which is both radiciel extension and étale extension of k (17.6.1); hence $B/\mathfrak{m}B$ is isomorphic to k, and as B is a free A-module, B is isomorphic to A. C.Q.F.D. □

Corollary (17.9.2). — *Let X be a connected prescheme; if $f : X \to Y$ is a closed étale immersion, then f is an isomorphism of X onto a connected component of Y.*

Proof. Indeed f is étale and radiciel, hence an isomorphism from X onto a sub-prescheme of Y; but by hypothesis $f(X)$ is closed in Y, hence since open and closed, and since $f(X)$ is connected, it is a connected component of Y. □

IV | 80

Corollary (17.9.3). — *Let $f : X \to Y$ be a morphism étale (resp. étale and separated). Then every Y-section $g : Y \to X$ of X is an open immersion (resp. open and closed); moreover the application $g \mapsto g(Y)$ is a bijection of the set $\Gamma(X/Y)$ of Y-sections of X to the set of open (resp. open and closed) parts Z of X such that the restriction of f to Z is a surjective morphism and radiciel of Z onto Y.*

Proof. Indeed, the fact that g is an open immersion follows already from the fact that f is non ramified (17.4.1), $b'')$), and the restriction of f to the open $g(Y)$ of X is an isomorphism. Conversely, if Z is an open subset of X such that $f|Z$ is surjective and radiciel, then, since f is étale, $f|Z$ is an isomorphism by (17.9.1). If moreover f is separated, then g is a closed immersion (I.5.4.6), which completes the proof. □

Corollary (17.9.4). — *Let Y be a connected prescheme and $f : X \to Y$ an étale and separated morphism. Then every Y-section g of X is an isomorphism of Y onto a connected component of X, and the application $g \mapsto g(Y)$ is a bijection of $\Gamma(X/Y)$ to the set of connected components of X such that the restriction of f to Z is a surjective morphism and radiciel of Z onto Y.*

Corollary (17.9.5). — *Let $g : Y \to S$, $h : X \to S$ be two morphisms locally of finite presentation; suppose moreover that h is flat. For an S-morphism $f : X \to Y$ to be an open immersion (resp. isomorphism), it is necessary and sufficient that for every $s \in S$, the morphism $f_s : h^{-1}(s) \to g^{-1}(s)$ deduced from f by the change of base* $\mathrm{Spec}(k(s)) \to S$*, be an open immersion (resp. isomorphism).*

Proof. Indeed, if f_s is an open immersion for every $s \in S$, it results from (17.8.3) that f is étale, and as for every $y \in Y$, $f^{-1}(y) = f_s^{-1}(y)$ with $s = g(y)$, f is radiciel; hence f is an open immersion by virtue of (17.9.1). If moreover f_s is surjective for every $s \in S$, f is surjective, hence an isomorphism. □

The following proposition specifies (10.4.11):

Proposition (17.9.6). — *Let S be a prescheme and X an S-prescheme of finite presentation. Every S-endomorphism of X which is a monomorphism is an automorphism of X.*

Proof. Let $f : X \to S$ be the structural morphism and g the S-endomorphism under consideration. The question being local on S, one can suppose S affine. Using (8.9.1) and (8.10.5), (i *bis*)), one is reduced to the case where $S = \mathrm{Spec}(A)$, where A is a $\mathbf{Z}$-algebra of finite type, and hence X is a $\mathbf{Z}$-prescheme of finite type. It follows already from (10.4.11) that g is a bijective morphism, since a monomorphism is radiciel ((8.11.5.1)); it will thus suffice to show that g is an open immersion, and since g is radiciel, it will suffice, by virtue of (17.9.1), to prove that g is étale. Moreover, as the set of points where g is étale is open and X is a Jacobson prescheme ((10.4.7)), it suffices to show that g is étale at every *closed* point z of X ((10.3.1)). Set $z' = g(z)$; it follows from (10.4.11), (i) that z' is also a closed point of X, and since g is a monomorphism, the map $k(z') \to k(z)$ deduced from g is an isomorphism. For g to be étale at the point z, it is thus necessary and sufficient that the canonical homomorphism $\widehat{\mathscr{O}}_{X,z'} \to \widehat{\mathscr{O}}_{X,z}$ be bijective ((17.6.3), $e'')$). We will show for this purpose that for every integer n, the canonical homomorphism $\mathscr{O}_{X,z'}/\mathfrak{m}_{z'}^{n+1} \to \mathscr{O}_{X,z}/\mathfrak{m}_z^{n+1}$ is bijective, from which the conclusion will immediately follow. One can suppose that z belongs to the finite set $T_{p,d}$ of closed points t of X such that $k(t)$ is an extension of $\mathbf{F}_p$ whose degree divides d; in which case the same holds for z'. Denote by $\mathscr{J}$ the coherent Ideal of $\mathscr{O}_X$ such that $\mathscr{J}_x = \mathscr{O}_x$ for $x \notin T_{p,d}$, and $\mathscr{J}_x = \mathfrak{m}_x^{n+1}$ for $x \in T_{p,d}$. Let $T_{p,d,n}$ be the closed sub-prescheme of X defined by $\mathscr{J}$ and $j : T_{p,d,n} \to X$ the canonical injection; the composite $g \circ j : T_{p,d,n} \to X$ maps $T_{p,d}$ into itself, and for every $x \in T_{p,d}$ the homomorphism $\mathscr{O}_{X,g(x)} \to \mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}$ factors as

$$\mathscr{O}_{X,g(x)} \longrightarrow \mathscr{O}_{X,g(x)}/\mathfrak{m}_{g(x)}^{n+1} \longrightarrow \mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}.$$

By ((**I**, 4.9)), there therefore exists a unique endomorphism g' of $T_{p,d,n}$ such that $j \circ g' = g \circ j$. As j is a monomorphism, and g is also one by hypothesis, one deduces that g' is also a monomorphism, and the proposition will be proven if one shows that g' is an *automorphism* of $T_{p,d,n}$. But, for every $x \in T_{p,d,n}$, we have seen that $k(x)$ is a finite field, and since $\mathscr{O}_{X,x}$ is Noetherian, each quotient $\mathfrak{m}_x^h/\mathfrak{m}_x^{h+1}$ is a finite-dimensional $k(x)$-vector space. It follows immediately that the local ring $\mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1}$ of $T_{p,d,n}$ at x has finitely many elements and, a fortiori, is a finite-type $\mathbf{Z}$-module. As $T_{p,d,n}$ is the sum of the preschemes $\operatorname{Spec}(\mathscr{O}_{X,x}/\mathfrak{m}_x^{n+1})$ in finite number, it is a finite $\mathbf{Z}$-prescheme; the endomorphism g' is therefore also finite (**II**, 6.1.5, (v)), hence proper. But a proper monomorphism is a closed immersion ((8.11.5)); if $T_{p,d,n} = \operatorname{Spec}(B)$, g' corresponds to a surjective endomorphism $\varphi : B \to B$ of the ring B. But the set B is finite, hence φ is necessarily bijective. □

17.10. Relative dimension of a smooth prescheme over another. **IV** | 81

Definition (17.10.1). — Let $f : X \to Y$ be a morphism locally of finite type. The *relative dimension* of f at a point $x \in X$ (or the *dimension of X relative to Y at the point x*) is the positive integer $\dim_x f = \dim_x(f^{-1}(f(x)))$.

To say that f is *quasi-finite* at the point x (**Err**$_{\mathbf{III}}$, 20) is equivalent to saying that $\dim_x f = 0$. One has seen (13.1.3) that the function $x \mapsto \dim_x f$ is upper semi-continuous. One notes that, even when the morphism f has the property (S_1) (that is to say (6.8.1) is flat and such that its fibres have no embedded associated prime cycle), the function $x \mapsto \dim_x f$ is not necessarily continuous, as shown by the example where $Y = \operatorname{Spec}(k)$, where k is a field, and $X = \operatorname{Spec}(k[\mathrm{U}, \mathrm{V}, \mathrm{W}]/\mathfrak{p}\mathfrak{q})$, where $\mathfrak{p} = (\mathrm{W})$ and $\mathfrak{q} = (\mathrm{U}) + (\mathrm{V} - \mathrm{W})$, the prime ideals of $k[\mathrm{U}, \mathrm{V}, \mathrm{W}]$ (X being thus the reunion, in the affine space of 3 dimensions, of a plane and of a line non parallel to this plane).

To say that a morphism locally of finite presentation $f : X \to Y$ is étale at a point $x \in X$ means that f is smooth at x and that $\dim_x f = 0$ (17.6.1).

Proposition (17.10.2). — *Let $f : X \to Y$ be a smooth morphism. For every $x \in X$, the locally free $\mathscr{O}_X$-module Ω^1_f (17.2.3) has rank at x equal to* $\dim_x f$ *(which implies that $x \mapsto \dim_x f$ is a continuous function on X).*

Proof. Indeed, if $y = f(x)$, then $X_y = f^{-1}(y)$ is smooth over $k(y)$; if $f_y : X_y \to \operatorname{Spec}(k(y))$ is the structural morphism, one has $\Omega^1_{f_y} = \Omega^1_f \otimes_{\mathscr{O}_X} \mathscr{O}_{X_y}$ (16.4.5). Thus one may reduce to the case where $Y = \operatorname{Spec}(k)$ is the spectrum of a field. Moreover, by (16.4.5) and (17.7.1), one may replace k by an algebraically closed extension; in other words, one may suppose that k is algebraically closed. The set of k-rational points of X is then dense in X (10.4.8), so it is enough to consider the case where x is rational over k. Now (16.4.12) identifies $(\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x)$ with $\mathfrak{m}_x/\mathfrak{m}_x^2$; since $\mathscr{O}_{X,x}$ is a regular local ring (17.5.1), one has $\operatorname{rg}_{k(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2) = \dim(\mathscr{O}_{X,x})$ (**0**, 17.1.1). Hence $\operatorname{rg}_{\mathscr{O}_{X,x}}(\Omega^1_{X/k})_x = \dim(\mathscr{O}_{X,x})$. Since $k(x) = k$, one also has $\dim_x f = \dim(\mathscr{O}_{X,x})$ (5.2.3). Q.E.D. □

We will prove a reciprocal of this result further on (17.15.5).

IV | 82

Corollary (17.10.3). — *Let $f : X \to Y$, $g : Y \to Z$ be two smooth morphisms. Then for every $x \in X$, one has*

(17.10.3.1) $$\dim_x(g \circ f) = \dim_x f + \dim_{f(x)} g.$$

Proof. Indeed, $g \circ f$ is smooth (17.3.3), so the three $\mathscr{O}_X$-modules $\Omega^1_{X/Y}$, $\Omega^1_{X/Z}$, and $f^*(\Omega^1_{Y/Z})$ are locally free (17.2.3 and $\mathbf{0_I}$, 5.4.5); moreover, the rank at x of $f^*(\Omega^1_{Y/Z})$ equals the rank at $y = f(x)$ of $\Omega^1_{Y/Z}$. Equality (17.10.3.1) is therefore a consequence of (17.10.2) and the exactness of the sequence (17.2.3.1). □

Corollary (17.10.4). — *Let $f : X \to Y$ be a smooth morphism and X' a sub-prescheme of X such that the composite $X' \xrightarrow{j} X \xrightarrow{f} Y$ (where j is the canonical injection) is smooth. Then the conormal sheaf $\mathscr{N}_{X'/X}$ is a locally free $\mathscr{O}_{X'}$-module, and for every $x \in X'$ one has*

$$\dim_x f = \dim_x(f \circ j) + \operatorname{rg}_{\mathscr{O}_{X',x}}(\mathscr{N}_{X'/X})_x. \tag{17.10.4.1}$$

Proof. Indeed, $\Omega^1_{X/Y} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$ and $\Omega^1_{X'/Y}$ are locally free, and the exact sequence (17.2.5.1) splits locally in a suitable neighbourhood of every point of X'. Thus $\mathscr{N}_{X'/X}$ is locally free, and relation (17.10.4.1) follows at once from the exactness of (17.2.5.1). □

17.11. Smooth morphisms of smooth preschemes.

Theorem (17.11.1). — *Let $f : Y \to S$ and $h : X \to S$ be two morphisms locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X; set $y = g(x)$, $s = f(y) = h(x)$. The following conditions are equivalent:*

- *a) f is smooth at y and g is smooth at x.*
- *b) g and h are smooth at x.*
- *c) h is smooth at x, and the canonical homomorphism (16.4.18)*

$$(g^*(\Omega^1_{Y/S}))_x \longrightarrow (\Omega^1_{X/S})_x \tag{17.11.1.1}$$

is left invertible (in other words is an isomorphism onto a direct factor of $(\Omega^1_{X/S})_x$).

- *c′) h is smooth at x, and the canonical homomorphism*

$$((\Omega^1_{Y/S})_y \otimes_{\mathscr{O}_{Y,y}} k(y)) \otimes_{k(y)} k(x) \longrightarrow (\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x) \tag{17.11.1.2}$$

is injective.

Suppose moreover that the homomorphism $k(y) \to k(x)$ is bijective. Then the preceding conditions are also equivalent to the following:

- *d) h is smooth at x, and the canonical map $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ (16.5.12) from the tangent vector space to X at x to the tangent vector space to Y at y is surjective.*

Proof. That a) implies b) follows at once from (17.3.3), (iii), and b) implies c) by (17.2.3), (ii). To see that c) and c') are equivalent, note that under either condition $\Omega^1_{X/S}$ is locally free of finite type, by (17.2.3), (ii); it is therefore enough to apply (**0**, (19.1.12)) to the local ring $\mathscr{O}_{X,x}$ and to the homomorphism (17.11.1.1) of finite-type $\mathscr{O}_{X,x}$-modules. When $k(x) = k(y)$, the linear tangent map $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ is, by (16.5.12), the transpose of (17.11.1.2); this proves the equivalence of c') and d) in that case.

It remains to prove that c) implies a). We may restrict ourselves to the case in which $S = \operatorname{Spec}(A)$, **IV | 83**
$Y = \operatorname{Spec}(B)$, and $X = \operatorname{Spec}(C)$ are affine. The hypothesis implies, by (17.2.3), (i), that $(\Omega^1_{X/S})_x = (\Omega^1_{C/A})_{\mathfrak{m}}$ is a free $C_{\mathfrak{m}}$-module. By (16.10.6), there are elements t_i $(1 \leqslant i \leqslant r)$ of $B_{\mathfrak{p}} = \mathscr{O}_{Y,y}$ such that their differentials generate $(\Omega^1_{B/A})_{\mathfrak{p}}$ and their images in $(\Omega^1_{C/A})_{\mathfrak{m}}$ are part of a basis of that free $C_{\mathfrak{m}}$-module. Since $\Omega^1_{Y/S}$ and $\Omega^1_{X/S}$ are of finite presentation (16.4.22), after replacing X and Y by suitable affine open neighbourhoods of x and y, we may suppose that $\Omega^1_{C/A}$ is free and that the t_i are the images of elements $s_i \in B$ whose differentials $d_{B/A}(s_i)$ generate $\Omega^1_{B/A}$ and whose images in $\Omega^1_{C/A}$ are part of a basis of that C-module (Bourbaki, *Alg. comm.*, chap. II, § 5, n° 1, prop. 2).

Let $\varphi : B' = A[\mathrm{T}_1, \ldots, \mathrm{T}_r] \to B$ be the A-homomorphism such that $\varphi(\mathrm{T}_i) = s_i$. The corresponding dihomomorphism $\Omega^1_{B'/A} \to \Omega^1_{B/A}$ (**0**, (20.5.2)) sends the basis elements $d_{B'/A}(\mathrm{T}_i)$ (**0**, (20.4.13)) to the $d_{B/A}(s_i)$, and is therefore surjective. If $Y' = \operatorname{Spec}(B')$ and $u : Y \to Y'$ is the S-morphism corresponding to φ, it follows from (17.2.2) that u is unramified. If $u \circ g : X \to Y'$ is smooth at x, then (17.7.10) shows that u is étale at y, and (17.3.5) then shows that g is smooth at x; moreover, since the structural morphism $f' : Y' \to S$ is smooth (17.3.8), $f = f' \circ u$ is smooth at y. The canonical images of the $d_{B'/A}(\mathrm{T}_i)$ in $\Omega^1_{C/A}$ are the images of the $d_{B/A}(s_i)$ and thus form part of a basis. Consequently, in proving that c) implies a), we may restrict ourselves to the case $Y' = Y$, and hence suppose that f is smooth.

By (17.8.2), we may then reduce to the case in which $S = \operatorname{Spec}(k)$ is the spectrum of a field; by (17.7.1), (ii), we may moreover suppose that k is algebraically closed. After replacing X and Y by suitable open neighbourhoods of x and y, we may suppose that f and h are both smooth and that the canonical homomorphism

$$g^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is left invertible (**0**, (19.1.12)). The set of k-rational points of X is then very dense in X (10.4.8); thus, to prove that g is smooth, it is enough to prove smoothness at every k-rational point $x \in X$, or equivalently that $\mathscr{O}_{X,x}$ is flat over $\mathscr{O}_{Y,y}$ and that $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is regular (17.5.1).

Now $y = g(x)$ is a fortiori k-rational. Since $\mathscr{O}_{Y,y}$ is then a formally smooth k-algebra for the discrete topologies and $\mathscr{O}_{Y,y}/\mathfrak{m}_y = k$, it follows from (**0**, (20.5.14)) that the canonical homomorphism

$$\mathfrak{m}_y/\mathfrak{m}_y^2 \longrightarrow (\Omega^1_{Y/S})_y \otimes_{\mathscr{O}_{Y,y}} k(y)$$

is bijective; likewise the canonical homomorphism

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x)$$

is bijective. Condition $c')$, which is equivalent to $c)$, therefore says here that the canonical homomorphism

$$(\mathfrak{m}_y/\mathfrak{m}_y^2) \otimes_{k(y)} k(x) \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2$$

is injective. Since $\mathscr{O}_{X,x}$ is regular, the conclusion follows from (**0**, (17.3.3)). ☐

IV | 84

Corollary (17.11.2). — *Under the general hypotheses of (17.11.1), the following conditions are equivalent:*

a) *f is smooth at y and g is étale at x.*

b) *h is smooth at x and g is étale at x.*

c) *h is smooth at x, and the canonical homomorphism*

$$(g^*(\Omega^1_{Y/S}))_x \longrightarrow (\Omega^1_{X/S})_x$$

is bijective.

c′) *h is smooth at x, and the canonical homomorphism*

$$((\Omega^1_{Y/S})_y \otimes_{\mathscr{O}_{Y,y}} k(y)) \otimes_{k(y)} k(x) \longrightarrow (\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x)$$

is bijective.

Suppose moreover that the homomorphism $k(y) \to k(x)$ is bijective. Then the preceding conditions are also equivalent to the following:

d) *h is smooth at x, and the canonical map $\mathrm{T}_{X/S}(x) \to \mathrm{T}_{Y/S}(y)$ (16.5.12) is bijective.*

Proof. Each of conditions $a)$, $b)$, and $c)$ of (17.11.2) is equivalent to the conjunction of the corresponding condition of (17.11.1) and the assertion that g is unramified at x, taking (17.2.2) into account for condition $c)$; hence $a)$, $b)$, and $c)$ are equivalent. The equivalence of $c)$ and $c')$ follows from the equivalence of the corresponding conditions of (17.11.1) and Nakayama's lemma; when $k(x) = k(y)$, the equivalence of $c')$ and $d)$ follows immediately by transposition (16.5.12). ☐

Corollary (17.11.3). — *Let $h : X \to S$ be a smooth morphism, let s_i $(1 \leqslant i \leqslant r)$ be sections of $\mathscr{O}_X$ over X (and thus also sections of $h_*(\mathscr{O}_X)$ over S), and let $g : X \to S[\mathrm{T}_1, \dots, \mathrm{T}_r] = \mathbf{V}^r_S$ be the S-morphism corresponding to the homomorphism of $\mathscr{O}_S$-modules $\mathscr{O}^r_S \to h_*(\mathscr{O}_X)$ defined by these sections (**II**, (1.2.7)). For g to be smooth (resp. étale) at a point $x \in X$, it is necessary and sufficient that the $(d_{X/S}(s_i))_x$ form part of a basis (resp. form a basis) of the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/S})_x$.*

Proof. It suffices to apply (17.11.1) (resp. (17.11.2)) taking for f the structural morphism $S[\mathrm{T}_1, \dots, \mathrm{T}_r] \to S$. ☐

Corollary (17.11.4). — *For a morphism $h : X \to S$ to be smooth at a point $x \in X$, it is necessary and sufficient that there exist an open neighbourhood U of x, an integer r, and an étale S-morphism $U \to S[\mathrm{T}_1, \dots, \mathrm{T}_r]$.*

Proof. The condition is plainly sufficient, since the structural morphism $S[\mathrm{T}_1, \dots, \mathrm{T}_r] \to S$ is smooth (17.3.8). Conversely, since h is smooth at x, there is an open neighbourhood U of x such that $\Omega^1_{X/S}|U$ is locally free (17.2.3); using (16.10.6) and restricting U if necessary, we obtain sections s_i of $\mathscr{O}_X$ over U such that the $d_{X/S}(s_i)$ form a basis of $\Omega^1_{X/S}|U$. The conclusion follows from (17.11.3). ☐

Proposition (17.11.5). — *Let $f : Y \to S$, $h : X \to S$ be two smooth morphisms. For an S-morphism $g : X \to Y$ to be an open immersion, it is necessary and sufficient that g be a monomorphism of preschemes and that for every $x \in X$, if one sets $y = g(x)$, one has $\dim_y(f) = \dim_x(h)$ (for a generalisation of this proposition, see (18.10.5)).*

Proof. The condition is evidently necessary; let us prove that it is sufficient. By (17.9.5), we are immediately reduced to the case in which $S = \mathrm{Spec}(k)$ is the spectrum of a field, and by (2.7.1), (x), we may suppose that k is algebraically closed. In view of (17.9.1), it is then enough to prove that g is étale; since the locus where g is étale is open, it is enough to prove this at the closed points of X, or equivalently at its k-rational points (10.4.8).

Let x be such a point and set $y = g(x)$; then y too is k-rational. Put $A = \mathscr{O}_{Y,y}$, a regular local ring with residue field k, let $d = \dim(A)$, and let $(t_i)_{1 \leqslant i \leqslant d}$ be a regular system of parameters of A. Put $B = \mathscr{O}_{X,x}$ and $C = B/\mathfrak{m}_y B$. Since g is a monomorphism, so is the morphism $\mathrm{Spec}(C) \to \mathrm{Spec}(k(y)) = \mathrm{Spec}(k)$ obtained from g by base change (**I**, (3.3.12)). This means that the corresponding homomorphism $u : k \to C$ is surjective, and hence bijective: it has a left inverse $v : C \to k$, while $u \circ v$ and the identity of C, after composition with u, give the same homomorphism $u : k \to C$; the monomorphism property therefore gives $u \circ v = 1_C$.

By hypothesis B is a regular ring of dimension d. The images of the t_i in B consequently form a regular system of parameters of B (**0**, (17.1.7)); condition (17.6.3), e''), is therefore satisfied by g at x (**0**, (17.1.1), and Bourbaki, *Alg. comm.*, chap. III, § 2, n° 8, cor. 3 to th. 1), which completes the proof. □

17.12. Smooth subpreschemes of a smooth prescheme. Smooth morphisms and differentially smooth morphisms. **IV** | 85

Theorem (17.12.1). — *Let $f : X \to S$, $h : Y \to S$ be two morphisms locally of finite presentation, $j : Y \to X$ an immersion, y a point of Y, $x = j(y)$. The following conditions are equivalent:*

- *a) h is smooth at y and f is smooth at x.*
- *b) f is smooth at x, and the canonical homomorphism (16.4.21)*

$$(\mathscr{N}_{Y/X})_y \longrightarrow (j^*(\Omega^1_{X/S}))_y$$

  is left invertible.
- *c) h is smooth at y and there exists an open neighbourhood U of y in Y such that $j|U : U \to X$ is a regular immersion (16.9.2).*
- *c′) h is smooth at y and there exists an open neighbourhood U of y in Y such that $j|U : U \to X$ is a quasi-regular immersion; in other words (16.9.8), there is a neighbourhood U of y in Y on which $\mathscr{N}_{Y/X}$ is locally free and the canonical homomorphism*

$$\mathbf{S}_{\mathscr{O}_Y}(\mathscr{N}_{Y/X}) \longrightarrow \mathrm{gr}^\bullet(j)$$

  is bijective.

Proof. To prove the equivalence of a) and b), we may restrict ourselves to the case in which $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, with $Y = \mathrm{Spec}(B/\mathfrak{J})$, where $\mathfrak{J}$ is a finite-type ideal of B, and where B is a smooth A-algebra (17.3.2), (iii). It is then enough to apply the Jacobian criterion (**0**, (22.6.1)) together with (**0**, (19.1.14)), bearing in mind that $\Omega^1_{X/S}$ is locally free in a neighbourhood of x and that $\mathscr{N}_{Y/X}$ is of finite type (16.1.6).

We next prove that a) implies c'). We may again suppose that S, X, and Y are affine, and that B and $B/\mathfrak{J}$ are smooth A-algebras. It follows from (17.7.9) and (8.10.5), (iv), that there exist a Noetherian subring A' of A, a finite-type A'-algebra B', and an ideal $\mathfrak{J}'$ of B' such that

$$B = B' \otimes_{A'} A, qquad \mathfrak{J} = \mathfrak{J}' B,$$

and such that B' and $B'/\mathfrak{J}'$ are smooth A'-algebras. By (**0**, (19.3.8)), B' is formally smooth when A' is given the discrete topology and B' the $\mathfrak{J}'$-preadic topology. Consequently (**0**, (19.5.4)), $\mathfrak{J}'/\mathfrak{J}'^2$ is a projective $B'/\mathfrak{J}'$-module and the canonical homomorphism

$$\mathbf{S}_{B'/\mathfrak{J}'}(\mathfrak{J}'/\mathfrak{J}'^2) \longrightarrow \mathrm{gr}^\bullet_{\mathfrak{J}'}(B')$$

is bijective. Thus the immersion $\mathrm{Spec}(B'/\mathfrak{J}') \to \mathrm{Spec}(B')$ is quasi-regular (16.9.8), and hence regular because B' is Noetherian (16.9.10). Moreover, $B'/\mathfrak{J}'$ is flat over A' (17.5.1) and B' is an A'-algebra of finite presentation; applying (11.3.8) with $\mathfrak{J}'$ in place of $\mathscr{F}$ and then changing base shows that j is a regular immersion, and a fortiori a quasi-regular immersion. The fact that B is an A-algebra of finite presentation and is flat over A also shows, by (11.3.8), that conditions $c')$ and $c)$ are equivalent and are stable under base change.

It remains to prove that $c)$ implies $a)$. Since condition $b)$ of (11.3.8) implies condition $c)$ there, condition $c)$ above already shows that f is flat at x. By (17.5.1), it is therefore enough to prove that the fibre $f^{-1}(s)$, where $s = h(y) = f(x)$, is smooth over $k(s)$ at x. Since condition $c)$ is stable under base change, we reduce to the case in which $S = \mathrm{Spec}(k)$ is the spectrum of a field; in view of (17.7.1), (ii), we may suppose that k is algebraically closed. It is enough to prove that $\mathscr{O}_{X,x}$ is regular (17.5.1). By hypothesis,

$$\mathscr{O}_{Y,y} = \mathscr{O}_{X,x}/\mathfrak{J}_x,$$

where $\mathfrak{J}_x$ is generated by an $\mathscr{O}_{X,x}$-regular sequence (t_i), and $\mathscr{O}_{Y,y}$ is regular. The t_i form part of a system of parameters of $\mathscr{O}_{X,x}$ (**0**, (16.4.1)); the conclusion follows from (**0**, (17.1.7)). □

Corollary (17.12.2). — *With the notation of (17.12.1), suppose that f is smooth at x, and let Y be a closed subprescheme of X, defined by an ideal $\mathscr{J}$ of $\mathscr{O}_X$, which is smooth over S at x. Let $(g_i)_{1\leqslant i\leqslant r}$ be sections of $\mathscr{J}$ over X. The following conditions are equivalent:*

- *a) The $(g_i)_x$ form a system of generators of the $\mathscr{O}_{X,x}$-module $\mathscr{J}_x$ having the smallest possible number of elements.*
- *b) If g'_i denotes the canonical image of g_i in $\Gamma(X, \mathscr{J}/\mathscr{J}^2)$, the $(g'_i)_x$ form a basis of the $(\mathscr{O}_{X,x}/\mathscr{J}_x)$-module $\mathscr{J}_x/\mathscr{J}_x^2$.*
- *c) The images of the $d_{X/S}g_i$ in $(\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} k(x)$ are linearly independent over $k(x)$, and the $(g_i)_x$ generate $\mathscr{J}_x$.*
- *d) There exist an open neighbourhood U of x in X and sections g_j $(r+1 \leqslant j \leqslant n)$ of $\mathscr{O}_X$ over U such that the S-morphism*

$$u : U \longrightarrow S[\mathrm{T}_1, \ldots, \mathrm{T}_n]$$

*corresponding to the homomorphism $\mathscr{O}_S^n \to f_*(\mathscr{O}_U)$ defined by the sections $g_i|U$ $(1 \leqslant i \leqslant r)$ and g_j $(r+1 \leqslant j \leqslant n)$ (**II**, (1.2.7)) is étale, and the inverse image under this morphism of the subprescheme $Y' = S[\mathrm{T}_{r+1}, \ldots, \mathrm{T}_n]$ is the subprescheme induced by $j(Y)$ on the open set $U \cap j(Y)$.*

Proof. Since $(g'_i)_x$ is the canonical image of $(g_i)_x$, the equivalence of $a)$ and $b)$ follows from Nakayama's lemma: $\mathscr{J}_x$ is of finite type and $\mathscr{J}_x/\mathscr{J}_x^2$ is a free $(\mathscr{O}_{X,x}/\mathscr{J}_x)$-module (17.10.4) (Bourbaki, *Alg. comm.*, chap. II, § 3, nº 2, prop. 5). By (17.12.1), $b)$, the module $\mathscr{J}_x/\mathscr{J}_x^2$ is canonically identified with a direct factor of the free $(\mathscr{O}_{X,x}/\mathscr{J}_x)$-module

$$(\Omega^1_{X/S})_x \otimes_{\mathscr{O}_{X,x}} (\mathscr{O}_{X,x}/\mathscr{J}_x),$$

and the equivalence of $b)$ and $c)$ follows from Bourbaki, *loc. cit.*

If $a)$ holds, $(g'_i)_x$ is identified with $(d_{X/S}g_i)_x \otimes 1$ for $1 \leqslant i \leqslant r$. Since $(\Omega^1_{X/S})_x$ is free of rank n, **IV** | 86
Bourbaki, *loc. cit.*, shows that, after replacing X by an open neighbourhood of x, there exist $n-r$ sections g_i $(r+1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ over X such that the $(d_{X/S}g_i)_x$ for $1 \leqslant i \leqslant n$ form a basis of $(\Omega^1_{X/S})_x$. The corresponding morphism u is then étale at x by (17.11.3); after a further restriction, we may suppose it is étale. For the same reason, the restriction $u^{-1}(Y') \to Y'$ is étale at x, and we may suppose both morphisms étale near x. Moreover, Y (identified for simplicity with the closed subprescheme $j(Y)$ of X) is plainly a closed subprescheme of $u^{-1}(Y')$; by the choice of the g_i for $i > r$, together with (17.2.5) and (17.11.2), the restriction of u to Y may also be supposed étale. It follows that for every $y' \in Y'$, the immersion $Y \cap u^{-1}(y') \to u^{-1}(y')$ is open (17.9.6); hence $Y \to u^{-1}(Y')$ is an open immersion. This proves that $a)$ implies $d)$. The converse $d) \Rightarrow a)$ follows at once from (17.11.3). □

Corollary (17.12.3). — *Let $f : X \to S$ be a morphism locally of finite presentation, $u : S \to X$ an S-section of X. For f to be smooth at a point $x \in X$, it is necessary and sufficient that u be a quasi-regular immersion in a neighbourhood of $s = f(x)$.*

Proof. The conclusion follows from (17.12.1), since $f \circ u = 1_S$ is smooth. □

Proposition (17.12.4). — *Every smooth morphism $f : X \to S$ is differentially smooth (in other words (16.10.5) $\Delta_f : X \to X \times_S X$ is a quasi-regular immersion). In particular, the $\mathscr{P}^n_f = \mathscr{P}^n_{X/S}$ and the $\mathrm{gr}_n(\mathscr{P}_{X/S})$ are locally free $\mathscr{O}_X$-modules of finite type.*

Proof. It is enough to note that the structural morphism $p_1 : X \times_S X \to X$ is smooth (17.3.3), (iii), and that Δ_f is an X-section of p_1; the conclusion follows from (17.12.3). □

Proposition (17.12.5). — *Let $f : X \to Y$ be a morphism locally of finite presentation. The following conditions are equivalent:*

a) f is differentially smooth.
b) For every morphism $g : Y' \to Y$, if $X' = X \times_Y Y'$ and $f' = f_{(Y')} : X' \to Y'$, then for every Y'-section s' of X', f' is smooth at all points of $s'(Y')$.
c) The second projection $p_2 : X \times_Y X \to X$ is smooth at all points of the diagonal $\Delta_f(X)$.

IV | 87

Proof. Condition c) is a special case of b): take $Y' = X$ and $g = f$ in b); then f' is the second projection p_2, and Δ_f is an X-section of $X \times_Y X$. Next, c) implies a), since p_2 is locally of finite presentation and, if p_2 is smooth at the points of $\Delta_f(X)$, then Δ_f is a quasi-regular immersion by (17.12.3); hence f is differentially smooth.

Finally, suppose a) and prove b). If Δ_f is quasi-regular, then p_2 is smooth at every point of $\Delta_f(X)$ by (17.12.3). Let $g' : X' \to X$ be the canonical projection and let $v = (g', g')_Y : X' \to X \times_Y X$. Then the diagram

$$\begin{tikzcd} X \times_Y X \arrow[d, "p_2"'] & X' \arrow[l, "v"'] \arrow[d, "f'"] \\ X & Y' \arrow[l, "h = g' \circ s'"] \end{tikzcd}$$

is commutative and identifies X' with the product $(X \times_Y X) \times_X Y'$ (**I**, (3.3.9)). Taking into account that diagram (17.4.1.1) identifies Y' with the product of the $(X \times_Y X)$-preschemes X and X', we conclude from the smoothness of p_2 at every point of $\Delta_f(X)$ that f' is smooth at every point of $s'(Y')$ (17.3.3), (iii). □

Corollary (17.12.6). — *With the notations being those of* (8.8.1), *suppose X_α quasi-compact, and $f_\alpha : X_\alpha \to S_\alpha$ locally of finite presentation. For $f : X \to S$ to be differentially smooth, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $f_\lambda : X_\lambda \to S_\lambda$ is differentially smooth (in which case $f_\mu : X_\mu \to S_\mu$ is differentially smooth for $\mu \geqslant \lambda$).*

Proof. The sufficiency of the condition and the last assertion follow from (16.10.4). To prove necessity, note that $\Delta_f : X \to X \times_S X$ and $p_2 : X \times_S X \to X$ are obtained by base change from $\Delta_{f_\lambda} : X_\lambda \to X_\lambda \times_{S_\lambda} X_\lambda$ and $p_{2,\lambda} : X_\lambda \times_{S_\lambda} X_\lambda \to X_\lambda$. By (17.12.5), for every $z \in \Delta_f(X)$ there is an affine open neighbourhood $U(z)$ of z in $X \times_S X$ on which p_2 is smooth. By (8.2.11) and (17.7.8), there exist an index $\lambda(z)$ and a neighbourhood $U_{\lambda(z)}$ of the projection of z in $X_{\lambda(z)} \times_{S_{\lambda(z)}} X_{\lambda(z)}$ such that $U(z)$ is the inverse image of $U_{\lambda(z)}$ and $p_{2,\lambda(z)}$ is smooth on $U_{\lambda(z)}$. Since X is quasi-compact, $\Delta_f(X)$ is covered by finitely many such neighbourhoods $U(z_i)$. By (8.3.4), there is an index λ greater than all the $\lambda(z_i)$ such that the inverse images of the $U_{\lambda(z_i)}$ in $X_\lambda \times_{S_\lambda} X_\lambda$ cover $\Delta_{f_\lambda}(X_\lambda)$. It follows from (17.12.5) that this λ has the required property. □

Example (17.12.7). — Let S be a prescheme, G an S-*prescheme in groups*, locally of finite presentation on S. For G to be differentially smooth over S, it is necessary and sufficient that G be smooth over S at all points of the "identity section" e of G. The condition is indeed necessary by (17.12.5). Conversely, suppose it holds. For every morphism $S' \to S$, $G' = G \times_S S'$ is an S'-prescheme in groups locally of finite presentation over S'; by (17.12.5), it is therefore enough to prove that for every S-section s of G, the structural morphism is smooth at the points of $s(S)$. If $m : G \times_S G \to G$ is the multiplication, then $m \circ (s \times 1_G) : S \times_S G \to G \times_S G \to G$ is an S-isomorphism of G, the "left translation" τ_s, carrying the points of the identity section of G to the points of $s(S)$. Thus G is smooth over S at the points of $s(S)$.

Remark (17.12.8). — An S-prescheme in groups G, of finite type (and even finite) over a locally Noetherian prescheme S, can be differentially smooth over S without being smooth (or even *flat*) over S. For example, take S to be the spectrum of the dual-number algebra $D = k[\mathrm{T}]/\mathrm{T}^2k[\mathrm{T}]$ over a field k, and take G to be the spectrum of the direct-product D-algebra $E = D \oplus k$. Define an S-group-prescheme structure on G by a "diagonal map" $\delta : E \to E \otimes_D E$ as follows. If e' and e'' are the idempotents given by the canonical images of 1 of D in the two factors D and k of E, then

$$\delta(e') = e' \otimes e' + e'' \otimes e'', \qquad \delta(e'') = e' \otimes e'' + e'' \otimes e'.$$

The identity section of the group scheme G corresponds to the homomorphism $E \to D$ that is the identity on D and zero on k. It is an isomorphism of S onto a connected component of G; hence G is differentially smooth over S (17.12.6), whereas it is clear that G is not flat over S.

17.13. Transversal morphisms.

(17.13.1). Let S be a prescheme, let X, Y, and X' be three S-preschemes, let $i : Y \to X$ be an S-immersion, and let $f : X' \to X$ be an S-morphism. Set $Y' = Y \times_X X'$, and let $g : Y' \to Y$, $j : Y' \to X'$ be the canonical projections, so that one has the commutative diagram

(17.13.1.1)
$$\begin{tikzcd} Y & Y' \arrow[l, "g"'] \arrow[d, "j"] \\ X \arrow[from=u, "i"'] & X' \arrow[l, "f"] \end{tikzcd}$$

and j is an S-immersion. There is then a commutative diagram of quasi-coherent $\mathscr{O}_{Y'}$-modules

(17.13.1.2)
$$\begin{array}{ccccccc} g^*(\mathscr{N}_{Y/X}) & \longrightarrow & f^*(\Omega^1_{X/S}) \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} & \longrightarrow & g^*(\Omega^1_{Y/S}) & \longrightarrow & 0 \\ \big\downarrow{\scriptstyle \mathrm{gr}_1(g)} & & \big\downarrow & & \big\downarrow & & \\ \mathscr{N}_{Y'/X'} & \longrightarrow & \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} & \longrightarrow & \Omega^1_{Y'/S} & \longrightarrow & 0 \end{array}$$

Here the lower row is the exact sequence (16.4.21) applied to j; the upper row is obtained from the same exact sequence for i by applying the right-exact functor g^*, which therefore preserves its exactness. The homomorphism $\mathrm{gr}_1(g)$ is defined in (16.2.1), and the commutativity of the two squares follows from (**0**, (20.5.7.3)) and (**0**, (20.5.11.3)). Moreover, we saw in (16.2.2), (iii), that $\mathrm{gr}_1(g)$ is surjective here; hence diagram (17.13.1.2) gives the exact sequence

(17.13.1.3)
$$g^*(\mathscr{N}_{Y/X}) \xrightarrow{\alpha} \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \longrightarrow \Omega^1_{Y'/S} \longrightarrow 0.$$

Proposition (17.13.2). — *With the notation of (17.13.1), let x' be a point of Y' and let $x = g(x')$ be its image in Y; suppose that X and Y are smooth over S at x, and that X' is smooth over S at x'. Let m and $m - c$ be the relative dimensions of X and Y over S at x (17.10.1), and let n be that of X' over S at x'. The following conditions are equivalent:*

a) Y' is smooth over S at x' and has relative dimension $n - c$.

b) The homomorphism $\alpha \otimes 1 : g^*(\mathscr{N}_{Y/X}) \otimes_{\mathscr{O}_{Y'}} k(x') \to \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} k(x')$ deduced from the homomorphism α of (17.13.1.3) is injective.

Let s be the canonical image of x in S, and suppose that x' is rational over $k(s)$. Then the conditions a) and b) are also equivalent to the following:

b') The transposed homomorphism of $\alpha \otimes 1$

$$\mathrm{T}_{X'/S}(x') \longrightarrow (\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} k(x'))^\vee = \mathrm{T}_{X/S}(x)/\mathrm{T}_{Y/S}(x)$$

(cf. (16.5.12)) is surjective.

Moreover, when conditions a) and b) hold at x', they hold throughout a neighbourhood of x' in Y'; after replacing X' by a neighbourhood of x', the homomorphism

$$\mathrm{gr}_1(g) : g^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Y'/X'}$$

is bijective, and the sequence

$$0 \longrightarrow g^*(\mathscr{N}_{Y/X}) \xrightarrow{\alpha} \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \longrightarrow \Omega^1_{Y'/S} \longrightarrow 0 \tag{17.13.2.1}$$

obtained by adjoining a zero to (17.13.1.3) is exact.

Proof. If the equivalent conditions a) and b) hold at x', they also hold in a neighbourhood of x' in Y': this follows from the openness of the smooth locus (17.3.7) and from (17.10.2).

By (17.12.1), applied to X' and Y', saying that Y' is smooth over S at x' is equivalent to saying that the homomorphism

$$\delta \otimes 1 : \mathscr{N}_{Y'/X'} \otimes_{\mathscr{O}_{Y'}} k(x') \longrightarrow \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} k(x')$$

is injective, taking account of (**0**, (19.1.12)) and of the fact that $\Omega^1_{X'/S}$ is locally free at x' (17.2.3). When this holds, $\Omega^1_{Y'/S}$ is locally free in a neighbourhood of x' and the sequence

$$0 \longrightarrow \mathscr{N}_{Y'/X'} \longrightarrow \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'} \longrightarrow \Omega^1_{Y'/S} \longrightarrow 0 \tag{17.13.2.2}$$

is exact (17.2.5). Thus, by (17.10.2), saying in addition that Y' has relative dimension $n-c$ over S at x' means that $\mathscr{N}_{Y'/X'}$, which is locally free near x', has rank c at x'. By the same reasoning and the hypotheses on X and Y, $\mathscr{N}_{Y/X}$ is locally free of rank c at x; hence $g^*(\mathscr{N}_{Y/X})$ is locally free of rank c at x'. Since $\mathrm{gr}_1(g) : g^*(\mathscr{N}_{Y/X}) \to \mathscr{N}_{Y'/X'}$ is surjective, the preceding conditions are equivalent to its being bijective at x' (Bourbaki, *Alg. comm.*, chap. II, § 3, n$^{\text{o}}$ 2, cor. to prop. 6), and therefore also in a neighbourhood of x' ($\mathbf{0}_{\mathrm{I}}$, (5.2.7)). This plainly implies b) and the final assertion of the statement, by exactness of (17.13.2.2).

Conversely, $\alpha \otimes 1$ factors as

$$g^*(\mathscr{N}_{Y/X}) \otimes k(x') \xrightarrow{\mathrm{gr}_1(g)\otimes 1} \mathscr{N}_{Y'/X'} \otimes k(x') \xrightarrow{\delta\otimes 1} \Omega^1_{X'/S} \otimes k(x').$$

Since $\mathrm{gr}_1(g)$ is surjective, injectivity of $\alpha \otimes 1$ implies both that $\delta \otimes 1$ is injective and that $\mathrm{gr}_1(g) \otimes 1$ is bijective. It follows from (17.12.1) that Y' is smooth over S at x', and $\mathrm{gr}_1(g)$ is bijective in a neighbourhood of x' (Bourbaki, *loc. cit.*). The exact sequence (17.13.2.2) then shows that $\mathscr{N}_{Y'/X'}$, isomorphic to $g^*(\mathscr{N}_{Y/X})$, has rank c at x', while $\Omega^1_{Y'/S}$ has rank $n-c$ there. By (17.10.2), this completes the proof of the equivalence of a) and b).

It remains to prove the equivalence of b) and b') when x' is rational over $k(s)$; this also implies that x is rational over $k(s)$. Then $g^*(\mathscr{N}_{Y/X}) \otimes_{\mathscr{O}_{Y'}} k(x')$ identifies with $\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} k(x')$. Since the sequence

$$0 \longrightarrow \mathscr{N}_{Y/X} \longrightarrow \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_Y \longrightarrow \Omega^1_{Y/S} \longrightarrow 0$$

is exact at x and consists there of locally free $\mathscr{O}_Y$-modules, the dual of $\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} k(x')$ identifies with $\mathrm{T}_{X/S}(x)/\mathrm{T}_{Y/S}(x)$ (16.5.12). Thus injectivity in b) is equivalent to surjectivity of the transposed homomorphism in b'). □

Definition (17.13.3). — With the notation of (17.13.1), we say that the morphism f is *transversal* to Y at x', relative to S, if X and Y are smooth over S at x, X' is smooth over S at x', and the equivalent conditions a) and b) of (17.13.2) hold. When there is no confusion to fear, we suppress the mention of the prescheme S and simply say that f is transversal to Y at the point x'.

Remarks (17.13.4). — (i) Suppose that X, Y and X' are flat and locally of finite presentation on S. For every $s \in S$, denote X_s, Y_s, X'_s, Y'_s the fibres of X, Y, X', Y' at the point s, $f_s : X'_s \to X_s$ the morphism deduced from f by change of base. It follows from (17.5.1) that for f to be transversal to Y at x', it is necessary and sufficient that, if s is the image of x in S, f_s be transversal to Y_s at x', relative to $k(s)$.[7]

(ii) We saw in (17.13.2) that the set of points $x' \in Y'$ at which f is transversal to Y is open in Y'. If Y' is moreover *proper* over S, it follows that the set of $s \in S$ such that f is transversal to Y at all points of Y'_s, relative to S, is open in S.

When X, Y, and X' are flat and locally of finite presentation over S, it follows from (i) that the set of $x' \in Y'$ such that f_s is transversal to Y_s at x' (where s is the image of x' in S), relative to $k(s)$,

[7][Trans.] The original reference (17.5.9) is an erratum in the original; there is no item 17.5.9. The correct reference is (17.5.1).

is open in Y'. If, in addition, Y' is proper over S, the set of $s \in S$ such that f_s is transversal to Y_s at all points of Y'_s (relative to $k(s)$) is open in S.

(iii) The property of being smooth over S at a point, as well as the notion of relative dimension over S at a point, is stable under every base change $S' \to S$ ((17.3.3) and (4.2.7)); the same is therefore true of the property that a morphism $f : X' \to X$ be transversal to a subprescheme of X at a point, relative to S.

(iv) Condition b) of (17.13.2) also says that the homomorphism $\alpha : g^*(\mathscr{N}_{Y/X}) \to \Omega^1_{X'/S} \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{Y'}$ is *universally injective* relative to Y' (11.9.18).

(17.13.5). Consider a prescheme S, three S-preschemes X, Y, and Z, and two S-morphisms $f : Y \to X$, $g : Z \to X$; put $T = Y \times_X Z$. By (**I**, 5.3.5), there is a commutative diagram **IV | 92**

(17.13.5.1)
$$\begin{tikzcd} X \arrow[d, "\Delta"] & T = Y \times_X Z \arrow[l] \arrow[d, "j"] \\ X \times_S X & Y \times_S Z \arrow[l, "u"'] \end{tikzcd}$$

which makes T the product of the $(X \times_S X)$-preschemes X and $Y \times_S Z$, where $u = f \times_S g$. Since Δ is an S-immersion (**I**, 5.3.9), we are in the situation of diagram (17.13.1.1). The object corresponding there to $\Omega^1_{X'/S}$ is, by (16.4.23),

$$(\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_{Y \times_S Z}) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_{Y \times_S Z}).$$

On the other hand, the object corresponding to the $\mathscr{O}_Y$-module $\mathscr{N}_{Y/X}$ in (17.13.1) is here, by definition, $\Omega^1_{X/S}$ (16.3.1). Thus (17.13.1.3) becomes the exact sequence

(17.13.5.2)
$$\Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T \xrightarrow{\rho} (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_T) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_T) \xrightarrow{\sigma} \Omega^1_{T/S} \longrightarrow 0$$

where it remains to describe ρ and σ explicitly. First, taking account of (16.4.23) and (**0**, 20.5.2), if $p : T \to Y$ and $q : T \to Z$ are the canonical projections, then, with the notation of (16.4.19),

(17.13.5.3)
$$\sigma = p_{T/Y/S} + q_{T/Z/S}.$$

To evaluate ρ, use the commutativity of the left square in (17.13.1.2). In the present case, this first requires the canonical homomorphism $\rho' : \Omega^1_{X/S} \to \Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_X$ defined in (16.4.21) applied to the immersion Δ. We may reduce to the affine case $S = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$. Then ρ' corresponds to the homomorphism δ of (**0**, 20.5.11.2), with B there replaced by $B \otimes_A B$ and C by $B = (B \otimes_A B)/\mathfrak{J}_{B/A}$. It sends the class of $x \otimes 1 - 1 \otimes x \pmod{\mathfrak{J}^2_{B/A}}$, for $x \in B$, to the image of

$$(x \otimes 1 - 1 \otimes x) \otimes (1 \otimes 1) - (1 \otimes 1) \otimes (x \otimes 1 - 1 \otimes x)$$

in $\Omega^1_{(B \otimes_A B)/A} \otimes_{B \otimes_A B} B$. The preceding element may be written

$$\begin{aligned} &((x \otimes 1) \otimes (1 \otimes 1) - (1 \otimes 1) \otimes (x \otimes 1)) \\ &\quad - ((1 \otimes x) \otimes (1 \otimes 1) - (1 \otimes 1) \otimes (1 \otimes x)). \end{aligned}$$

Thus ρ' is the difference of the two homomorphisms $\pi^{(1)}$ and $\pi^{(2)}$ from $\Omega^1_{X/S}$ to $\Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_X$ corresponding, by (16.4.3.3), to the first and second projections $X \times_S X \to X$. To obtain ρ, consider the homomorphism $\rho'' : \Omega^1_{(X \times_S X)/S} \otimes_{\mathscr{O}_{X \times_S X}} \mathscr{O}_{Y \times_S Z} \to \Omega^1_{(Y \times_S Z)/S}$ corresponding by (16.4.3.3) to the morphism u of (17.13.5.1); after tensoring with $\mathscr{O}_T$, form the composite $(\rho'' \otimes 1) \circ (\rho' \otimes 1)$. It follows from the preceding computation that, with the notation of (16.4.18),

(17.13.5.4)
$$\rho = (f_{Y/X/S} \otimes 1_{\mathscr{O}_T}) \oplus (-g_{Z/X/S} \otimes 1_{\mathscr{O}_T}).$$

Applying (17.13.2) to diagram (17.13.5.1) (taking account of (17.3.3), (iv), which shows that $X \times_S X$ is smooth over S at (x, x) when X is smooth over S at x) gives the following corollary. **IV | 93**

Corollary (17.13.6). — *Let S be a prescheme, X, Y, Z three S-preschemes, $f : Y \to X$, $g : Z \to X$ two S-morphisms; set $T = Y \times_X Z$, and let $p : T \to Y$, $q : T \to Z$ be the canonical projections. Let t be a point of T, $y = p(t)$, $z = q(t)$, $x = f(y) = g(z)$. Suppose that X is smooth over S at the point x, of relative dimension m, Y (resp. Z) smooth over S at the point y (resp. z), of relative dimension $m + a$ (resp. $m + b$), a and b being positive or negative integers. Then the following conditions are equivalent:*

a) *T is smooth over S at the point t, of relative dimension $m+a+b$.*
b) *The homomorphism*

$$\rho \otimes 1 : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(t) \longrightarrow (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} k(t)) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} k(t))$$

where ρ is given by (17.13.5.4), is injective.
c) *The morphism $u : Y \times_S Z \to X \times_S X$ is transversal to the diagonal of $X \times_S X$ at the point t.*

If t is rational over the residue field of its image s in S, these conditions are also equivalent to the following:

b') *The homomorphism*

(17.13.6.1) $$\mathrm{T}_y(f) - \mathrm{T}_z(g) : \mathrm{T}_{Y/S}(y) \oplus \mathrm{T}_{Z/S}(z) \longrightarrow \mathrm{T}_{X/S}(x)$$

(cf. (16.5.12.5)*) is surjective.*

Moreover, when conditions a) *and* b) *are verified at t, they are so in a neighbourhood of t in T, and restricting T to such a neighbourhood, the sequence*

(17.13.6.2) $$0 \longrightarrow \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T \xrightarrow{\rho} (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_T) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_T) \xrightarrow{\sigma} \Omega^1_{T/S} \longrightarrow 0$$

(where σ is given by (17.13.5.3)) is exact.

The only point in (17.13.6) still requiring proof is b'); it follows from the fact that, when t is rational over $k(s)$, the homomorphism $\rho \otimes 1$ is the transpose *of $\mathrm{T}_y(f) - \mathrm{T}_z(g)$, by (17.13.5.4).*

When the equivalent conditions of (17.13.6) hold, we say that f and g form a pair of transversal morphisms at t, relative to S; *here again, the mention of S is often omitted.*

(17.13.7). Consider in particular the case where Y and Z are *sub-preschemes* of X, f and g being the canonical injections; T is then the sub-prescheme "intersection" $\inf(Y, Z)$ of X (**I**, 4.4.3), and one has $t = y = z = x$. Instead of saying that the couple (f, g) is transversal at the point x, one says then that Y and Z *cut transversally at the point x* (relatively to S). Let X_s, Y_s, Z_s, and T_s denote the fibres **IV** | 94
of X, Y, Z, and T at s, the image of x in S. Taking account of (5.2.3), (5.1.9), and (**0**, 16.5.12), we see that, when X, Y, and Z are smooth over S at x, they cut transversally at x if and only if T is smooth over S at x and

(17.13.7.1) $$\operatorname{codim}_x(T_s, X_s) = \operatorname{codim}_x(Y_s, X_s) + \operatorname{codim}_x(Z_s, X_s).$$

Proposition (17.13.8). — *Let S be a prescheme, X an S-prescheme locally of finite presentation over S, Y, Z two sub-preschemes of X, $T = \inf(Y, Z)$ the sub-prescheme "intersection" of Y and Z, x a point of T. The following conditions are equivalent:*

a) *The canonical injection $i : Y \to X$ is a transversal morphism to Z at the point x, relatively to S (17.13.3).*
a') *The canonical injection $j : Z \to X$ is a transversal morphism to Y at the point x, relatively to S.*
a'') *Y and Z cut transversally at the point x, relatively to S.*
b) *X, Y, and Z are smooth over S at x, and the homomorphism*

$$\rho \otimes 1 : \Omega^1_{X/S} \otimes_{\mathscr{O}_X} k(x) \longrightarrow (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} k(x)) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} k(x))$$

is injective.

When x is rational over the residue field $k(s)$ (s image of x in S), these conditions are also equivalent to the following:

b') *The homomorphism*

$$\mathrm{T}_x(i) - \mathrm{T}_x(j) : \mathrm{T}_{Y/S}(x) \oplus \mathrm{T}_{Z/S}(x) \longrightarrow \mathrm{T}_{X/S}(x)$$

is surjective.

Moreover, when conditions a) *and* b) *are verified at x, they are so in a neighbourhood of x in T, and restricting X to a neighbourhood of x, the sequence (17.13.6.2) is exact, and one has a canonical isomorphism*

(17.13.8.1) $$\mathscr{N}_{T/X} \xrightarrow{\sim} (\mathscr{N}_{Y/X} \otimes_{\mathscr{O}_Y} \mathscr{O}_T) \oplus (\mathscr{N}_{Z/X} \otimes_{\mathscr{O}_Z} \mathscr{O}_T).$$

Proof. Conditions a), a'), and a'') all imply that X, Y, and Z are smooth over S at x. Let m, $m-a$, and **IV** | 95
$m-b$ be their respective relative dimensions over S at x. Applying (17.13.2) with Z in place of X' and $T = Y \times_X Z$ in place of Y', conditions a) and a') are both equivalent to saying that T is smooth over S at x and has relative dimension $m-a-b$ there. By (17.13.7), this is precisely condition

a''); hence a), a'), and a'') are equivalent. The equivalence of a'') and b) (and of b') when x is rational over $k(s)$), the persistence of these conditions in a neighbourhood of x, and the exactness of (17.13.6.2) there were proved in (17.13.6). It remains to define the canonical isomorphism (17.13.8.1). Denote by α and $-\beta$ the two homomorphisms in the middle term of (17.13.5.4), and by γ and δ the two homomorphisms in the middle term of (17.13.5.2). Exactness of (17.13.6.2), that is, of

$$0 \longrightarrow \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T \xrightarrow{\alpha\oplus(-\beta)} (\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_T) \oplus (\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_T) \xrightarrow{\gamma+\delta} \Omega^1_{T/S} \longrightarrow 0,$$

means that, in the category of $\mathscr{O}_T$-modules, $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T$ is canonically the fibre product of $\Omega^1_{Y/S} \otimes_{\mathscr{O}_Y} \mathscr{O}_T$ and $\Omega^1_{Z/S} \otimes_{\mathscr{O}_Z} \mathscr{O}_T$ over $\Omega^1_{T/S}$, for γ and δ. The same reasoning as in (**0**, 18.1.2 and 18.1.3), replacing rings and ideals by modules and submodules, gives the commutative diagram

$$\begin{tikzcd}
 & (\mathscr{N}_{Y/X} \otimes \mathscr{O}_T) \oplus (\mathscr{N}_{Z/X} \otimes \mathscr{O}_T) \arrow[r] \arrow[d] \arrow[dr, "\iota"] & \mathscr{N}_{Z/X} \otimes \mathscr{O}_T \arrow[r, "\sim"] \arrow[d, "\beta"] & \mathscr{N}_{T/Y} \arrow[d] & \\
0 \arrow[r] & \mathscr{N}_{Y/X} \otimes \mathscr{O}_T \arrow[r] \arrow[d] & \Omega^1_{X/S} \otimes \mathscr{O}_T \arrow[r, "\alpha"] \arrow[d, "\beta"] \arrow[dr, "\varepsilon"] & \Omega^1_{Y/S} \otimes \mathscr{O}_T \arrow[r] \arrow[d, "\gamma"] & 0 \\
0 \arrow[r] & \mathscr{N}_{T/Z} \arrow[r] & \Omega^1_{Z/S} \otimes \mathscr{O}_T \arrow[r, "\delta"] \arrow[d] & \Omega^1_{T/S} \arrow[r] \arrow[d] & 0 \\
 & & 0 & 0 &
\end{tikzcd}$$

whose third and fourth rows and columns are exact by the smoothness hypotheses. The identity $\varepsilon = \gamma \circ \alpha = \delta \circ \beta$, and the fact that this is the homomorphism corresponding to the canonical injection $T \to X$, follow from (**0**, 20.5.2.7). Since the diagonal of the preceding diagram is exact, $\ker(\varepsilon)$ identifies canonically with $\mathscr{N}_{T/X}$, taking account of the fact that T is smooth over S (17.2.5). Thus the canonical isomorphism (17.13.8.1) is the inverse of ι. □

(17.13.9). The results of (17.13.5) and (17.13.6) may be generalized to any finite family of S-morphisms $f_i : Y_i \to X$ $(i \in I)$. Let X^I be the product of the family $(X_i)_{i\in I}$ of S-preschemes all identical to X (**I**, 3.3.5), and let $p_i : X^I \to X$ be the canonical projections. The diagonal morphism $\Delta : X \to X^I$ is defined, as in (**I**, 5.3.1), by the identities $p_i \circ \Delta = 1_X$ for every $i \in I$; it is an X-section of X^I for p_1, and hence an immersion (**I**, 5.3.11).

Let Y be the product of the S-preschemes Y_i for the composites $Y_i \xrightarrow{f_i} X \to S$, and let T be the product of the X-preschemes Y_i for the morphisms f_i. As in (**I**, 5.3.5), there is a commutative diagram

(17.13.9.1)
$$\begin{tikzcd}
X \arrow[d, "\Delta"] & T \arrow[l, "v"'] \arrow[d, "j"] \\
X^I & Y \arrow[l, "u"']
\end{tikzcd}$$

(where u is the product over S of the f_i), which makes T the product of the X^I-preschemes X and Y. By induction on the number of elements of I, (16.4.23) shows that $\Omega^1_{X^I/S}$ identifies canonically with the direct sum of the $p_i^*(\Omega^1_{X/S})$; similarly, $\Omega^1_{Y/S}$ identifies with the direct sum of the $q_i^*(\Omega^1_{Y_i/S})$, where $q_i : Y \to Y_i$ are the canonical projections. **IV** | 96

Suppose now that X is smooth over S at a point x, and hence in a neighbourhood of x. Then X^I is smooth at the corresponding point (17.3.3), (iv), and, after restricting X to a neighbourhood of x, (17.2.5) gives an exact sequence of locally free $\mathscr{O}_X$-modules

$$0 \longrightarrow \mathrm{gr}_1(\Delta) \longrightarrow (\Omega^1_{X/S})^I \longrightarrow \Omega^1_{X/S} \longrightarrow 0.$$

Put $\mathscr{J} = \mathrm{gr}_1(\Delta)$. The preceding sequence gives an exact sequence of locally free $\mathscr{O}_T$-modules

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_T \longrightarrow (\Omega^1_{X/S})^I \otimes_{\mathscr{O}_X} \mathscr{O}_T \xrightarrow{\sigma} \Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T \longrightarrow 0$$

which is the first row of diagram (17.13.1.2), and where σ is the canonical homomorphism sending each family of sections $(t_i')_{i\in I}$ of $\Omega^1_{X/S} \otimes_{\mathscr{O}_X} \mathscr{O}_T$ over an open subset of T to its *sum*. On the other hand, the homomorphism corresponding here to the second vertical arrow of diagram (17.13.1.2) is

$$\tau : (\Omega^1_{X/S})^I \otimes_{\mathscr{O}_X} \mathscr{O}_T \longrightarrow \bigoplus_{i\in I}(\Omega^1_{Y_i/S} \otimes_{\mathscr{O}_{Y_i}} \mathscr{O}_T) \tag{17.13.9.2}$$

which sends a family $(t'_i \otimes 1)_{i\in I}$, where the t'_i are sections of $\Omega^1_{X/S}$, to the family of the $((f_i)_{Y_i/X/S}(t'_i)) \otimes 1$, with the notation of (16.4.18). The homomorphism ρ corresponding to α of (17.13.1.3) is therefore the restriction of τ to $\mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_T$. With these hypotheses and notation, Proposition (17.13.2) applied to (17.13.9.1) gives the following corollary.

Corollary (17.13.10). — *Under the general hypotheses of (17.13.9), let t be a point of T, y_i ($i \in I$) its projections in the Y_i, x the point of X equal to each of the $f_i(y_i)$. Suppose that X is smooth over S at the point x and of relative dimension d, and that each of the Y_i is smooth over S at the point y_i; let $d + c_i$ (c_i a positive or negative integer) be the relative dimension of Y_i at the point y_i. Then the following conditions are equivalent:*

a) T is smooth over S at the point t, of relative dimension $d + \sum_{i\in I} c_i$.

b) The homomorphism

$$\rho \otimes 1 : \mathscr{J} \otimes_{\mathscr{O}_X} k(t) \longrightarrow \bigoplus_{i\in I}(\Omega^1_{Y_i/S} \otimes_{\mathscr{O}_{Y_i}} k(t))$$

is injective.

When t is rational over the residue field $k(s)$ (s the image of t in S), these conditions are also equivalent to the following: **IV** | 97

b′) The homomorphism transpose of $\rho \otimes 1$

$$\bigoplus_{i\in I} \mathrm{T}_{Y_i/S}(y_i) \longrightarrow (\mathrm{T}_{X/S}(x))^I/\delta(\mathrm{T}_{X/S}(x))$$

(where δ is the diagonal application) is surjective.

It suffices to note that X^I is smooth over S at the point x and of relative dimension $d \cdot \mathrm{Card}(I)$, and Y smooth over S at the point t and of relative dimension $d \cdot \mathrm{Card}(I) + \sum_{i\in I} c_i$.

When the equivalent conditions of (17.13.10) are verified, one says further that the f_i form a family of transversal morphisms at the point t, relatively to S. *One sees further that the set of points $t \in T$ where this holds is* open *in T. The remark ((17.13.4), (ii)) shows moreover that if X and the Y_i are flat and locally of finite presentation over S, and if moreover T is* proper *over S, the set of $s \in S$ such that the f_i form a family of transversal morphisms at* all *the points of T_s, relatively to S, is* open *in S.*

(17.13.11). Consider in particular the case, generalising (17.13.7), where the Y_i ($i \in I$) are *sub-preschemes* of X, the f_i being the canonical injections, so that $T = \inf_{i\in I}(Y_i)$ is also the sub-prescheme "intersection" of the Y_i, and $t = y_i = x$ for every i; instead of saying that the f_i form a family of transversal morphisms at the point x, one says further that the Y_i *cut transversally at the point x* (relatively to S). The condition a) of (17.13.10) is expressed further by the relation that generalises (17.13.7.1):

$$\mathrm{codim}_x(T_s, X_s) = \sum_{i\in I} \mathrm{codim}_x((Y_i)_s, X_s). \tag{17.13.11.1}$$

Moreover, one has the following property which extends (17.13.8.1), and of which we give another demonstration when I has 2 elements:

Corollary (17.13.12). — *When the Y_i cut transversally at the point x, one has a canonical isomorphism*

$$\mathscr{N}_{T/X} \xrightarrow{\sim} \bigoplus_{i\in I}(\mathscr{N}_{Y_i/X} \otimes_{\mathscr{O}_{Y_i}} \mathscr{O}_T). \tag{17.13.12.1}$$

Proof. One can reduce to the case where the Y_i are closed sub-preschemes of X, defined by quasi-coherent Ideals $\mathscr{J}_i$, so that T is defined by the Ideal $\mathscr{J} = \sum_i \mathscr{J}_i$. By definition of the conormal sheaf of an immersion (16.1.3), the canonical homomorphism $\bigoplus_i \mathscr{J}_i \to \mathscr{J}$ gives, by passage to quotients, a *surjective* homomorphism

$$\bigoplus_{i\in I}(\mathscr{N}_{Y_i/X} \otimes_{\mathscr{O}_{Y_i}} \mathscr{O}_T) \longrightarrow \mathscr{N}_{T/X}. \tag{17.13.12.2}$$

Here the $\mathscr{O}_T$-modules on both sides of (17.13.12.2) are locally free and have the same rank $\sum_i c_i$, where c_i is the rank of $\mathscr{N}_{Y_i/X}$, by (17.2.5) and condition a) of (17.13.10). Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. to prop. 6, therefore shows that (17.13.12.2) is bijective; (17.13.12.1) is its inverse. □ IV | 98

17.14. Local and infinitesimal characterisations of smooth, unramified, and étale morphisms. IV | 98

Proposition (17.14.1). — *Let $f : X \to Y$ be a morphism locally of finite presentation, x a point of X, $y = f(x)$. For f to be smooth (resp. unramified, resp. étale) at x, it is necessary and sufficient that the following condition hold:*

For every local scheme $Y' = \operatorname{Spec}(A')$ with closed point y', every morphism $h : Y' \to Y$ such that $h(y') = y$, every closed subscheme $Y'_0 = \operatorname{Spec}(A'/\mathfrak{J}')$ of Y', where $(\mathfrak{J}')^2 = 0$, and every Y-morphism $g_0 : Y'_0 \to X$ such that $g_0(y') = x$, there exists at least one *(resp.* at most one, *resp.* one and only one*) Y-morphism $g : Y' \to X$ whose restriction to Y'_0 is g_0.*

Proof. In view of Definitions (17.1.1) and (17.3.1), it remains only to show that the stated condition is sufficient in the smooth and unramified cases.

(i) The smooth case. We may restrict ourselves to the affine case $Y = \operatorname{Spec}(A)$ and $X = \operatorname{Spec}(C)$, where $C = B/\mathfrak{R}$, $B = A[\mathrm{T}_1, \dots, \mathrm{T}_n]$, and $\mathfrak{R}$ is a finite-type ideal of B. To prove that f is smooth at x, it is enough to prove that $\mathscr{O}_{X,x}$ is formally smooth over $\mathscr{O}_{Y,y}$ for the discrete topologies (17.5.1). Now $\mathscr{O}_{X,x} = B_{\mathfrak{q}}/\mathfrak{R}B_{\mathfrak{q}}$, where $\mathfrak{q}$ is a prime ideal of B; if $\mathfrak{p}$ is its inverse image in A, then $B_{\mathfrak{q}}$ is formally smooth over $A_{\mathfrak{p}}$ for the discrete topologies (**0**, 19.3.2 and 19.3.5). By the Jacobian criterion (**0**, 22.6.1 and 20.5.12), it is therefore enough to show that $B_{\mathfrak{q}}/\mathfrak{R}^2B_{\mathfrak{q}}$ is an $A_{\mathfrak{p}}$-trivial extension of $B_{\mathfrak{q}}/\mathfrak{R}B_{\mathfrak{q}}$ by $\mathfrak{R}B_{\mathfrak{q}}/\mathfrak{R}^2B_{\mathfrak{q}}$. But this is precisely the hypothesis applied to

$$A' = B_{\mathfrak{q}}/\mathfrak{R}^2B_{\mathfrak{q}}, \qquad \mathfrak{J}' = \mathfrak{R}B_{\mathfrak{q}}/\mathfrak{R}^2B_{\mathfrak{q}},$$

and to the morphism g_0 corresponding to the identity automorphism of $A'/\mathfrak{J}' = \mathscr{O}_{X,x}$.

(ii) The unramified case. Put $A = \mathscr{O}_{Y,y}$ and $B = \mathscr{O}_{X,x}$. By (17.4.1), c'), it is enough to prove that $\Omega^1_{B/A} = 0$. With the notation of (**0**, 20.4.1), the ring $C = \mathscr{P}^1_{B/A} = (B \otimes_A B)/\mathfrak{J}^2$ is local, since $C/(\mathfrak{J}/\mathfrak{J}^2)$ is isomorphic to B. Applying the hypothesis to $A' = C$ and $\mathfrak{J}' = \mathfrak{J}/\mathfrak{J}^2$ shows, by definition, that the map $d_{B/A} : B \to \Omega^1_{B/A}$ is zero (**0**, 20.4.6); hence $\Omega^1_{B/A} = 0$ by (**0**, 20.4.7). □

Proposition (17.14.2). — *Let Y be a prescheme locally Noetherian, $f : X \to Y$ a morphism locally of finite type, x a point of X, $y = f(x)$. For f to be smooth (resp. unramified, resp. étale) at x, it is necessary and sufficient that the condition of (17.14.1) hold when one additionally requires that A' be Artinian, that $\mathfrak{m}'\mathfrak{J}' = 0$, where $\mathfrak{m}'$ is the maximal ideal of A', and that the residue field $A'/\mathfrak{m}'$ be equal to $k(x)$.*

IV | 99

Proof. It remains again only to prove sufficiency in the smooth and unramified cases.

(i) The smooth case. Put $A = \mathscr{O}_{Y,y}$ and $B = \mathscr{O}_{X,x}$. In view of (17.5.3), the point is to prove that B is a formally smooth A-algebra for the preadic topologies. By (**0**, 22.1.4), this follows from the hypothesis if, in it, the condition $\mathfrak{m}'\mathfrak{J}' = 0$ is replaced by $\mathfrak{J}' \subset \mathfrak{m}'$. Since $\mathfrak{m}'$ is nilpotent, however, the condition in the statement is already sufficient: consider the rings $A'/\mathfrak{m}'^j\mathfrak{J}'$ and argue by induction on j, as in (**0**, 19.4.3).

(ii) The unramified case. Resume the notation of the proof of (17.14.1), (ii). To prove that the ideal $\mathfrak{J}/\mathfrak{J}^2$ of C is zero, note first that C is a Noetherian local ring: it is the local ring of an affine open subscheme of $X \times_Y X$, which is locally of finite type over X, and therefore also over Y. If $\mathfrak{r}$ is the maximal ideal of C, it is consequently enough to verify that

$$\mathfrak{J}/\mathfrak{J}^2 \subset \mathfrak{r}^n \quad \text{for every } n \qquad (\mathbf{0}_{\mathrm{I}}, 7.3.5),$$

or, equivalently, that $\mathfrak{r}^n = \mathfrak{r}^n + (\mathfrak{J}/\mathfrak{J}^2)$. With the notation of (**0**, 20.4.6), this says that for every n the two composite maps

$$B \xrightarrow{p_1} C \longrightarrow C/\mathfrak{r}^n, \qquad B \xrightarrow{p_2} C \longrightarrow C/\mathfrak{r}^n$$

are identical. But $C/\mathfrak{r}^n$ is Artinian, and this identity follows by induction on n from the hypothesis applied to

$$A' = C/\mathfrak{r}^n, \qquad \mathfrak{J}' = \big(\mathfrak{r}^n + (\mathfrak{J}/\mathfrak{J}^2)\big)/\mathfrak{r}^n.$$

□

Remark (17.14.3). — Under the hypotheses of (17.14.2), if in addition the residue field $k(x)$ is a *finite* extension of $k(y)$, then the A'-modules $\mathfrak{m}'^j/\mathfrak{m}'^{j+1}$ are finite-dimensional $k(y)$-vector spaces; hence, *a fortiori*, A' is a finite $\mathscr{O}_{Y,y}$-algebra.

17.15. The case of preschemes over a base field. We first recall (6.7.7), (6.7.8), and (6.8.1):

Proposition (17.15.1). — *Let k be a field, X a prescheme locally of finite type over k, and x a point of X. For X to be smooth at x, it is necessary and sufficient that, for every radicial extension k' of k (or only for every finite radicial extension k' of k), the local ring $\mathscr{O}_{X,x} \otimes_k k'$ be regular. If $k(x)$ is a separable extension of k (in particular if k is perfect), this amounts to saying that $\mathscr{O}_{X,x}$ is regular.*

Corollary (17.15.2). — *Under the conditions of (17.15.1), for X to be smooth over k, it is necessary and sufficient that X be geometrically regular over k* (6.7.6)*; this implies that X is reduced. If k is perfect, X is smooth over k if and only if X is regular.*

Proposition (17.15.3). — *Let k be a field, X a prescheme locally of finite type over k, x a point of X, and $n = \dim_x(X)$. Let $(g_i)_{1\leqslant i\leqslant n}$ be a family of n sections of $\mathscr{O}_X$ over X, and let $f : X \to Y = \operatorname{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ be the morphism corresponding to the k-homomorphism $k[\mathrm{T}_1, \dots, \mathrm{T}_n] \to \Gamma(X, \mathscr{O}_X)$ sending T_i to g_i* (**I**, 2.2.4)*. The following conditions are equivalent (and imply that X is smooth over k at x, and therefore regular at x):*

a) f is étale at the point x.

b) The images $d_{X/k}g_i$ form a basis of the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/k})_x$.

c) The images $d_{X/k}g_i$ generate the $\mathscr{O}_{X,x}$-module $(\Omega^1_{X/k})_x$.

If f is étale at x, then X is smooth over k at x, since $Y = \operatorname{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ is smooth over k (17.3.8); the implication $a) \Rightarrow b)$ is therefore a special case of (17.11.3). As $b)$ trivially implies $c)$, it remains to see that $c)$ implies $a)$.

Proof. First note that hypothesis $c)$ makes the homomorphism $(f^*\Omega^1_{Y/k})_x \to (\Omega^1_{X/k})_x$ surjective. After replacing X by an open neighbourhood of x, we may therefore suppose that $f^*\Omega^1_{Y/k} \to \Omega^1_{X/k}$ is surjective (Bourbaki, *Alg. comm.*, chap. II, § 5, n° 1, prop. 2); hence f is unramified (17.2.2). Let us first show that one can reduce to the case where x is rational over k. Indeed, put $k' = k(x)$, $X' = X \otimes_k k'$, and $Y' = Y \otimes_k k' = \operatorname{Spec}(k'[\mathrm{T}_1, \dots, \mathrm{T}_n])$. There exists a point $x' \in X'$ above x such that $k(x') = k'$. To prove that f is étale at x, it suffices to show that $f' = f_{(k')} : X' \to Y'$ is étale at x' (17.7.1), (ii); moreover, f' is unramified (17.3.3), (iii), and $\dim_{x'}(X') = n$ (4.2.7). In the same way, replacing k' by an algebraically closed extension of k', one can suppose that k is algebraically closed. Now put $y = f(x)$, $A = \mathscr{O}_{Y,y}$, and $B = \mathscr{O}_{X,x}$. Since the residue field of B is k, the same is true of the residue field of A; thus x (resp. y) is a closed point of X (resp. Y) (**I**, 6.4.2), and consequently $\dim A = \dim B = n$. Since f is unramified, the homomorphism $\widehat{A} \to \widehat{B}$ is surjective (17.4.4), f''). But $\dim \widehat{A} = \dim \widehat{B} = n$, and $\widehat{A}$, being regular, is an integral domain; hence $\widehat{A} \to \widehat{B}$ is also injective (**0**, 16.3.10). It is therefore bijective, which completes the proof that f is étale at x (17.6.3), e''). □

IV | 100

Corollary (17.15.4). — *Under the general hypotheses of (17.15.3), suppose moreover that $k(x)$ is a finite and separable extension of k and that the germs $(g_i)_x$ belong to $\mathfrak{m}_x$. Then the conditions $a)$, $b)$, $c)$ of (17.15.3) are also equivalent to each of the following:*

d) The n germs $(g_i)_x$ generate the maximal ideal $\mathfrak{m}_x$ of $\mathscr{O}_{X,x}$.

d') The ring $\mathscr{O}_{X,x}$ is regular and the $(g_i)_x$ form a regular system of parameters for this ring (**0**, 17.1.6)*.*

Proof. Clearly $d')$ implies $d)$. Since $k(x)$ is a finite separable extension of k, one has $\Omega^1_{k(x)/k} = 0$ (**0**, 20.6.20), and $k(x) = \mathscr{O}_{X,x}/\mathfrak{m}_x$ is a formally smooth k-algebra for the discrete topologies (**0**, 19.6.1). The exact sequence (**0**, 20.5.14.1), applied with $A = k$, $B = \mathscr{O}_{X,x}$, and $\mathfrak{R} = \mathfrak{m}_x$, therefore gives the canonical isomorphism

$$\delta : \mathfrak{m}_x/\mathfrak{m}_x^2 \xrightarrow{\sim} (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x).$$

It follows first, by Nakayama's lemma, that $c)$ and $d)$ are equivalent. If f is étale at x, the ring $\mathscr{O}_{X,x}$ is regular of dimension n, since x is closed (**I**, 6.4.2); and n elements which generate its maximal ideal necessarily form a regular system of parameters (**0**, 17.1.6). Thus $a)$ implies $d')$. □

Proposition (17.15.5). — *Let k be a field, X a prescheme locally of finite type over k, x a point of X, and $n = \dim_x(X)$. The following conditions are equivalent:*

a) X is smooth over k at x.

b) X is differentially smooth over k at x.

c) $(\Omega^1_{X/k})_x$ is a free $\mathscr{O}_{X,x}$-module of rank n.

d) $(\Omega^1_{X/k})_x$ admits a system of n generators as an $\mathscr{O}_{X,x}$-module. **IV** | 101

e) There exist a perfect extension k' of k and an open neighbourhood U of x in X such that the prescheme $U \otimes_k k'$ is regular.

Proof. If X is smooth over k at x, there is an open neighbourhood U of x which is smooth over k; consequently $U \otimes_k k'$ is regular for every extension k' of k (17.15.2). This proves $a) \Rightarrow e)$. Conversely, $e)$ implies $a)$, because $U \otimes_k k'$ is then smooth over the perfect field k' (17.15.2), and hence U is smooth over k (17.7.1), (ii).

We have already proved $a) \Rightarrow c)$ (17.10.2); clearly $c) \Rightarrow d)$, and $d) \Rightarrow a)$ follows from (**0**, 20.4.7) and (17.15.3), $c)$.

Finally, $a) \Rightarrow b)$ was proved in (17.12.4). Conversely, to prove $b) \Rightarrow a)$, we may reduce to the case where x is rational over k: as in the proof of (17.15.3), take a point x' of $X' = X \otimes_k k'$, above x, with $k(x') = k' = k(x)$, using again (17.7.1), (ii), and the fact that differential smoothness at x implies differential smoothness of X' at x' (16.10.4). When x is rational over k, the smoothness of f at x follows from $b)$ and (17.12.5), applied to the k-section $u : \mathrm{Spec}(k) \to X$ such that $u(\mathrm{Spec}(k)) = \{x\}$. □

Corollary (17.15.6). — *Let X be a prescheme of finite type over a field k. For X to be smooth over k, it is necessary and sufficient that the $\mathscr{O}_X$-module $\Omega^1_{X/k}$ be locally free and that the local rings of X at its maximal points be fields which are separable extensions of k (the latter condition is automatically satisfied if k is perfect and X is* reduced*).*

Proof. The conditions are necessary. Indeed, if X is smooth over k, then $\Omega^1_{X/k}$ is locally free by (17.10.2). Moreover, X is regular and hence, *a fortiori*, reduced; at every maximal point x of X, therefore, $\mathscr{O}_{X,x}$ is a field. This field must be a formally smooth k-algebra for the discrete topologies (17.5.1), and is consequently a separable extension of k (**0**, 19.6.1).

The conditions are sufficient. We may reduce to the case where X is connected; then $\Omega^1_{X/k}$ is locally free of constant rank n. At every maximal point x of X one has $\mathrm{rank}_{k(x)} \Omega^1_{k(x)/k} = n$, since $\mathscr{O}_{X,x} = k(x)$. By hypothesis $k(x)$ is separable over k, so $\Upsilon_{k(x)/k} = 0$ (**0**, 20.6.3), and Cartier's equality gives $\deg.\mathrm{tr}_k\, k(x) = n$ (**0**, 21.7.1). All irreducible components of X therefore have the same dimension n (5.2.1), and the implication $c) \Rightarrow a)$ of (17.15.5) shows that X is smooth over k at every point. □

Corollary (17.15.7). — *If k has characteristic 0, then X is smooth over k at x if and only if $(\Omega^1_{X/k})_x$ is a free $\mathscr{O}_{X,x}$-module.*

Proof. This results from (16.12.2) and the equivalence of $b)$ and $a)$ in (17.15.5). □

Proposition (17.15.8). — *Let k be a field, X a prescheme locally of finite type over k, and x a point of X; put $n = \dim_x(X)$ and $r = \dim(\mathscr{O}_{X,x})$, so that $n - r = \deg.\mathrm{tr}_k\, k(x)$ (5.2.3). Let $(g_i)_{1 \leqslant i \leqslant n}$ be a family of n sections of $\mathscr{O}_X$ over X, with $(g_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant r$; let $f : X \to Y = \mathrm{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ be the morphism corresponding to the k-homomorphism which sends T_i to g_i. The following conditions are equivalent (and imply that X is smooth over k at x and that $k(x)$ is a separable extension of k):*

a) f is étale at the point x. **IV** | 102

b) The germs $(g_i)_x$ for $1 \leqslant i \leqslant r$ generate $\mathfrak{m}_x$ (and hence form a regular system of parameters for $\mathscr{O}_{X,x}$), and the images in $\Omega^1_{k(x)/k}$ of the elements $(d_{X/k} g_i)_x$ for $r+1 \leqslant i \leqslant n$ generate $\Omega^1_{k(x)/k}$.

Proof. There is an exact sequence of $k(x)$-modules (**0**, 20.5.12.1)

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x) \longrightarrow \Omega^1_{k(x)/k} \longrightarrow 0. \tag{17.15.8.1}$$

Condition $b)$ and Nakayama's lemma therefore imply that the $(d_{X/k} g_i)_x$ generate $(\Omega^1_{X/k})_x$; hence $b) \Rightarrow a)$ by (17.15.3).

Conversely, suppose $a)$ holds. By (17.15.3), the $(d_{X/k}g_i)_x$, $1 \leqslant i \leqslant n$, form a basis of $(\Omega^1_{X/k})_x$. If t_i is the image of $(g_i)_x$ in $k(x)$, it follows that the dt_i generate $\Omega^1_{k(x)/k}$. Since $t_i = 0$ for $1 \leqslant i \leqslant r$, the dt_i for $r+1 \leqslant i \leqslant n$ already generate $\Omega^1_{k(x)/k}$. As $\deg.\mathrm{tr}_k\, k(x) = n - r$, Cartier's equality gives $\Upsilon_{k(x)/k} = 0$; thus $k(x)$ is separable over k (**0**, 20.6.3), and the sequence

$$0 \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x) \longrightarrow \Omega^1_{k(x)/k} \longrightarrow 0 \tag{17.15.8.2}$$

is exact ((**0**, 20.5.14) and (**0**, 19.6.1)). Moreover the dt_i, $r+1 \leqslant i \leqslant n$, form a basis of $\Omega^1_{k(x)/k}$, so none of the corresponding t_i is zero. It follows that the images of the $(g_i)_x$, $1 \leqslant i \leqslant r$, generate $\mathfrak{m}_x/\mathfrak{m}_x^2$; by Nakayama's lemma the $(g_i)_x$, $1 \leqslant i \leqslant r$, generate $\mathfrak{m}_x$. This proves $a) \Rightarrow b)$. □

Corollary (17.15.9). — *Let X be a prescheme locally of finite type over a field k, and x a point of X; put $n = \dim_x(X)$ and $r = \dim(\mathscr{O}_{X,x})$. The following conditions are equivalent:*

a) The ring $\mathscr{O}_{X,x}$ is regular and $k(x)$ is a separable extension of k.

b) X is smooth over k at x, and the canonical homomorphism

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow (\Omega^1_{X/k})_x \otimes_{\mathscr{O}_{X,x}} k(x)$$

is injective.

c) There exist n sections g_i of $\mathscr{O}_X$ over an open neighbourhood U of x, with $(g_i)_x \in \mathfrak{m}_x$ for $1 \leqslant i \leqslant r$, such that the morphism $U \to \mathrm{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n])$ corresponding to the g_i (cf. (17.15.8)) is étale at x.

d) There exist n sections g_i $(1 \leqslant i \leqslant n)$ of $\mathscr{O}_X$ over an open neighbourhood U of x, such that the $(g_i)_x$, $1 \leqslant i \leqslant r$, generate $\mathfrak{m}_x$, and the images in $\Omega^1_{k(x)/k}$ of the $(d_{U/k}g_i)_x$, $r+1 \leqslant i \leqslant n$, generate $\Omega^1_{k(x)/k}$.

Proof. The equivalence of $c)$ and $d)$ follows from (17.15.8). Its proof also shows that $c)$ implies that X is smooth over k at x and that the sequence (17.15.8.2) is exact; hence $c) \Rightarrow b)$.

Condition $b)$ implies that $\mathscr{O}_{X,x}$ is regular, so $\mathfrak{m}_x/\mathfrak{m}_x^2$ has rank r; moreover $(\Omega^1_{X/k})_x$ is then free of rank n (17.10.2). Since the sequence (17.15.8.1) is exact by hypothesis, $\Omega^1_{k(x)/k}$ has rank $n - r = \deg.\mathrm{tr}_k\, k(x)$. Cartier's equality therefore gives $\Upsilon_{k(x)/k} = 0$, and $k(x)$ is separable over k (**0**, 20.6.3). Thus $b) \Rightarrow a)$.

Finally, if $a)$ holds, Cartier's equality again shows that $\Omega^1_{k(x)/k}$ has rank $n - r$. Since $\mathscr{O}_{X,x}$ is **IV** | 103
regular, the existence of sections g_i satisfying $d)$ is immediate. □

Remarks (17.15.10). — (i) For a prescheme X of finite type over k to be smooth over k, it is not enough that $\Omega^1_{X/k}$ be locally free. This is shown by $X = \mathrm{Spec}(K)$, where K is a finite inseparable extension of k.

(ii) When k is not perfect, X may be smooth over k even though $k(x)$ is not separable over k. For example, take $X = \mathrm{Spec}(k[\mathrm{T}])$ and let x correspond to the principal prime ideal $(\mathrm{T}^p - \lambda)$, where $p > 0$ is the characteristic of k and $\lambda \in k - k^p$.

(iii) Nevertheless, if X is smooth over k, the set of closed points x of X for which $k(x)$ is a separable (and finite) extension of k is *dense* in X. Indeed, let $f : X \to \mathrm{Spec}(k)$ be the structural morphism. For every $x_0 \in X$, there is an open neighbourhood U of x_0 and a factorisation

$$f|U : U \xrightarrow{g} \mathrm{Spec}(k[\mathrm{T}_1, \dots, \mathrm{T}_n]) \xrightarrow{h} \mathrm{Spec}(k),$$

in which g is étale (17.11.4). We may restrict to the case where k is not perfect, and hence is infinite. The open set $g(U)$ then contains a point y rational over k; if $x \in U$ lies above y, then x is closed in X and $k(x)$ is separable over $k(y) = k$ (17.6.2).

Proposition (17.15.11). — *Let X be a prescheme of finite type over a field k. The following conditions are equivalent:*

a) X is étale over k.

b) X is unramified over k.

c) X is isomorphic to $\mathrm{Spec}(A)$, where A is a finite and separable k-algebra.

Proof. This follows from (17.6.2) and (17.4.2), noting that condition $b)$ here implies that X is finite over k. □

Proposition (17.15.12). — *Let k be a field, X a prescheme locally of finite type over k. There exists an everywhere dense open subset U of X which is smooth over k if and only if, at every maximal point x of X, the prescheme X is reduced and $k(x)$ is a separable extension of k. There exists an everywhere dense open subset U of X such that U_{red} is smooth over k if and only if $k(x)$ is a separable extension of k for every maximal point x of X.*

Proof. The second assertion follows immediately from the first. To say that there is an everywhere dense open subset U smooth over k means that X is smooth over k at each of its maximal points x. Necessarily $\mathscr{O}_{X,x}$ is regular (17.15.1), and *a fortiori* reduced; it is therefore the field $k(x)$. Moreover, (17.15.1) requires $k(x) \otimes_k k'$ to be regular for every radicial extension k' of k, so $k(x)$ must be separable over k (4.3.5). The converse follows at once from (17.15.1). □

Corollary (17.15.13). — *Let X be an algebraic prescheme over a field k. Then there exist an everywhere dense open subset U of X and a finite radicial extension k' of k such that $(U_{(k')})_{\mathrm{red}}$ is smooth over k'.*

Proof. There is such an extension k' for which $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k' (4.6.6); **IV** | 104
equivalently (4.6.1), $k(x')$ is separable over k' at every maximal point x' of $X_{(k')}$. It follows from (17.15.12) that there is an everywhere dense open subset U' of $X_{(k')}$ such that U'_{red} is smooth over k'. The morphism $X_{(k')} \to X$ is a homeomorphism (2.4.5), so $U' = U_{(k')}$ for an everywhere dense open subset U of X. □

Proposition (17.15.14). — *Let X be an algebraic prescheme over a field k, of dimension $\leqslant 1$. Then there exists a finite radicial extension k' of k such that the normalization* (**II**, *6.3.8) of* $(X_{(k')})_{\mathrm{red}}$ *is smooth over k'.*

We first prove the following two lemmas.

Lemma (17.15.14.1). — *Let X be a reduced prescheme whose set of irreducible components is locally finite, and let $f : Y \to X$ be a morphism. Then Y is X-isomorphic to the normalization X' of X* (**II**, *6.3.8) if and only if the following conditions hold:*

(i) Y is normal;

(ii) f is integral and birational.

If X contains a dense normal open subset (as is always the case when X is excellent (7.8.3)*, in particular when X is locally of finite type over a field), condition (ii) may be replaced by:*

(ii bis) f is integral, and there exists a dense open subset U of X such that $f^{-1}(U)$ is dense in Y and the restriction $f^{-1}(U) \to U$ is an isomorphism.

Proof. The question is local on X, so we may suppose that X has only finitely many irreducible components. Let X_i $(1 \leqslant i \leqslant m)$ be the reduced subpreschemes of X having these components as their underlying spaces. The normalization X' is the sum of the normalizations X'_i of the X_i (**II**, 6.3.6). Each structural morphism $f_i : X'_i \to X_i$ is integral and birational, so f is integral and birational. If X has a dense normal open subset V, condition (ii) plainly implies (ii bis), with $U = V$.

Conversely, suppose (i) and (ii) hold. The prescheme Y has only finitely many irreducible components Y_i $(1 \leqslant i \leqslant m)$, with Y_i dominating X_i for each i (6.15.4). Since Y is normal it is the sum of the Y_i, and the morphism $g_i : Y_i \to X_i$ through which the restriction $Y_i \to X$ of f factors is birational (**I**, 5.2.2). As X_i and Y_i are integral, g_i is integral, and Y_i is normal, for every affine open V of X_i the ring $\Gamma(g_i^{-1}(V), \mathscr{O}_{Y_i})$ is the integral closure of $\Gamma(V, \mathscr{O}_{X_i})$. Thus Y identifies canonically with X' (**II**, 6.3.4).

Finally, suppose (ii bis). We may take U to be a disjoint union of irreducible open subsets U_i $(1 \leqslant i \leqslant m)$; then $f^{-1}(U)$ is the disjoint union of the $V_i = f^{-1}(U_i)$. Since the V_i are irreducible and $f^{-1}(U)$ is dense in Y, they are the irreducible components of Y, and f is therefore birational. This proves the lemma. □

Lemma (17.15.14.2). — *Let k be a field and X an algebraic prescheme over k. There exists a finite radicial extension k' of k such that $(X_{(k')})_{\mathrm{red}}$ is geometrically reduced over k' and its normalization is geometrically normal over k'.*

Proof. In view of (4.6.6) and (**I**, 5.1.8), we may reduce to the case where X is already geometrically reduced over k. Let p be the characteristic exponent of k, and let $k_1 = k^{p^{-\infty}}$ be the smallest perfect extension of k; thus k_1 is the filtered direct limit of the finite radicial extensions of k. Put $X_1 = X_{(k_1)}$, which is reduced by hypothesis, and let Y_1 be its normalization. If $f_1 : Y_1 \to X_1$ is the structural morphism, then f_1 is finite (7.8.3), (vi), and surjective.

There is a dense open subset U_1 of X_1 such that $V_1 = f_1^{-1}(U_1)$ is dense in Y_1 and $V_1 \to U_1$ is an isomorphism (17.15.14.1). **IV** | 105

Applying (8.9.1) and (8.10.5), (vi) and (x), we first obtain a finite radicial extension k'' of k and a finite surjective morphism $Y'' \to X'' = X_{(k'')}$ such that

$$Y_1 = Y'' \times_{X''} X_1 = Y'' \otimes_{k''} k_1.$$

Since Y_1 is normal and k_1 is perfect, (6.7.7) shows that Y'' is geometrically normal over k''. The projections $X_1 \to X''$ and $Y_1 \to Y''$ are integral, surjective, and radicial, hence homeomorphisms (2.4.5). If U'' and V'' are the respective images of U_1 and V_1, they are dense open subsets and $V'' = f''^{-1}(U'')$, where $f'' : Y'' \to X''$ is the structural morphism. Since $U_1 = U'' \otimes_{k''} k_1$, (8.10.5), (ii), gives a finite radicial extension k' of k'' such that, after putting

$$X' = X \otimes_k k' = X'' \otimes_{k''} k', \qquad Y' = Y'' \otimes_{k''} k',$$

and writing $f' : Y' \to X'$ for the resulting morphism, the images U' and V' of U'' and V'' satisfy that $V' \to U'$ is an isomorphism. Since Y' is normal and f' is integral and birational, (17.15.14.1) identifies Y' with the normalization of X'. This proves the lemma, because Y' is geometrically normal. □

Proof of Proposition 17.15.14. Apply (17.15.14.2) to k and X. Since $\dim X \leqslant 1$, one also has $\dim X_{(k')} \leqslant 1$ (4.1.4), and the normalization Y' of $X_{(k')}$ has dimension at most 1 ((5.4.2) and **II**, 7.4.6). By the definitions and (**II**, 7.4.5), geometric normality of Y' is then equivalent to geometric regularity over k', and hence to smoothness over k' (17.5.1). □

Proposition (17.15.15). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be smooth at a point $x \in X$, it is necessary and sufficient that f be flat at x and that Ω^1_f be a locally free $\mathscr{O}_X$-module in a neighbourhood of x, of rank at x equal to $\dim_x f$.*

Proof. Necessity follows from (17.5.1) and (17.10.2). Conversely, suppose the conditions hold and put $y = f(x)$. By (17.5.1), it is enough to show that $f^{-1}(y)$ is smooth over $k(y)$ at x; this follows from the definition of $\dim_x f$ (17.10.1), from (16.4.5), and from (17.15.5). □

17.16. Quasi-sections of flat or smooth morphisms. The statements of this number complement those of (14.5), by means of hypotheses of flatness.

Proposition (17.16.1). — *Let $f : X \to S$ be a flat morphism locally of finite presentation. Let $s \in S$, and let x be a closed point of X_s such that $\mathscr{O}_{X_s,x}$ is a Cohen–Macaulay ring. Then there exist an open neighbourhood U of s in S and a subprescheme $X' \subset f^{-1}(U)$ of X, containing x, such that the restriction $X' \to U$ of f is flat, quasi-finite, and of finite presentation.*

Proof. The question is local on X and S, so we may suppose that f is of finite presentation. Let $(t_i)_{1 \leqslant i \leqslant m}$ be a system of parameters of the local ring $\mathscr{O}_{X_s,x}$. There exist an affine open neighbourhood $V = \mathrm{Spec}(A)$ of x in X and m sections g_i of $\mathscr{O}_X$ over V such that
the images of the germs $(g_i)_x \in \mathscr{O}_{X,x}$ in **IV** | 106

$$\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$$

are the t_i. Let X' be the closed subprescheme of V defined by the ideal of A generated by the g_i. The sequence (t_i) is regular by hypothesis (**0**, 16.5.7); it therefore follows from (11.3.8), after replacing V if necessary by a smaller open neighbourhood, that the restriction $X' \to S$ of f is flat and of finite presentation. Since the t_i form a system of parameters of $\mathscr{O}_{X_s,x}$, the ring $\mathscr{O}_{X'_s,x}$ is Artinian. As x is closed in X'_s, it is consequently isolated in X'_s. Replacing V once more by a smaller neighbourhood of x, (13.1.4) shows that $X' \to S$ is quasi-finite. □

Corollary (17.16.2). — *Let $f : X \to S$ be faithfully flat and locally of finite presentation. Then there exists a faithfully flat, locally finitely presented, and locally quasi-finite morphism $g : S' \to S$ such that there is an S-morphism $S' \to X$ (equivalently, an S'-section of $X' = X \times_S S'$;* **I***, 3.3.14). If S is quasi-compact (resp. quasi-compact and quasi-separated), S' may be chosen affine (resp. S' may be chosen affine and g of finite presentation and quasi-finite).*

Proof. For every $s \in S$, the fibre X_s is nonempty and is a prescheme locally of finite type over $k(s)$. The set U_s of points x of X_s for which $\mathscr{O}_{X_s,x}$ is Cohen–Macaulay is open in X_s (6.11.3) and nonempty, since it contains the maximal points of X_s (**0**, 16.5.1). It therefore contains a point x_s closed in X_s (10.4.7). Let $X'(s)$ be an affine subprescheme of X, containing x_s, and satisfying the conclusion of (17.16.1).

To obtain a prescheme S' satisfying the assertion, it is enough to take the sum of the $X'(s)$ as s runs through S. Indeed, since $X'(s) \to S$ is flat and locally of finite presentation, its image $U(s)$ is open in S (2.4.6). If S is quasi-compact, it can therefore be covered by finitely many of the $U(s_i)$, and the sum of the corresponding $X'(s_i)$ is still suitable and is affine. If, in addition, S is quasi-separated, the open immersions $U(s_i) \to S$ may be supposed quasi-compact (**I**, 2.7), hence of finite presentation (**I**, 6.2). The morphisms $X'(s_i) \to S$ are then of finite presentation, and so is $S' \to S$. □

Corollary (17.16.3). — *(i) Let $f : X \to S$ be smooth. Let $s \in S$, and let x be a closed point of X_s such that the residue field $k(x)$ is separable over $k(s)$. In the conclusion of (17.16.1), the restriction $X' \to U$ of f may then be chosen étale.*

(ii) Let $f : X \to S$ be smooth and surjective. In the conclusion of (17.16.2), one may moreover suppose that $g : S' \to S$ is étale.

Proof. Assertion (ii) follows from (i) just as (17.16.2) follows from (17.16.1): for every $s \in S$, the nonempty smooth $k(s)$-prescheme X_s has a closed point x for which $k(x)$ is separable over $k(s)$ (17.15.10), (iii). It remains to prove (i). The ring $\mathscr{O}_{X_s,x}$ is regular. Repeat the construction of (17.16.1), taking for (t_i) a regular system of parameters of $\mathscr{O}_{X_s,x}$. The ring $\mathscr{O}_{X'_s,x}$ is then a field isomorphic to $k(x)$, and hence separable over $k(s)$ by hypothesis. The conclusion follows from (17.6.1), c'). □

Proposition (17.16.4). — *Let S be a quasi-compact and quasi-separated prescheme, and let $f : X \to S$ be a surjective morphism locally of finite presentation. Then there exists a finite family $(S_i)_{i \in I}$ of affine subpreschemes of S, of finite presentation over S, pairwise disjoint, and with union S, having the following property: for every $i \in I$, there exist two finite, finitely presented, and surjective morphisms*

$$S''_i \xrightarrow{g_i} S'_i \xrightarrow{h_i} S_i,$$

where h_i is étale and g_i is flat and radicial, and an S-morphism $S''_i \to X$ (in other words, an S''_i-section of $X''_i = X \times_S S''_i$).

IV | 107

Proof. There is a finite covering $(U_i)_{1 \leqslant i \leqslant n}$ of S by affine open subsets. For every i, the intersection

$$W_i = U_i \cap \Big(\bigcup_{j<i} U_j \Big)$$

is quasi-compact (**I**, 2.7); hence there exists a closed subprescheme V_i of U_i whose underlying space is $U_i - W_i$ and which is defined by an ideal of finite type in the ring of U_i. Thus V_i is an affine subprescheme of S of finite presentation over S (**I**, 6.2). It is clear that it is enough to prove the proposition after replacing S by each V_i and X by $X \times_S V_i$. In other words, we may already suppose that S is affine.

On the other hand, X is the union of affine open subsets W_α, and the $f(W_\alpha)$ are constructible in S (**I**, 8.4) and cover S. Consequently (**I**, 9.9), there is a finite subfamily $(W_{\alpha_j})_{1 \leqslant j \leqslant m}$ whose images already cover S. Let X' be the sum prescheme of the subpreschemes induced on the open subsets W_{α_j} of X. It is immediate that it is enough to prove the proposition after replacing X by X', since an S-morphism $S''_i \to X'$ gives, by composition, an S-morphism $S''_i \to X' \to X$. We may therefore suppose that X is affine and that f is of finite presentation.

We may then write (8.9.1)

$$X = X_0 \times_{S_0} S,$$

where S_0 is Noetherian and $X_0 \to S_0$ is a morphism of finite type which may be supposed surjective (8.10.5), (vi). If the proposition is proved for this morphism, it follows at once for X by the base change $S \to S_0$. We may thus suppose in addition that S is Noetherian and that f is of finite type.

Let s_h $(1 \leqslant h \leqslant r)$ be the maximal points of S. We show that it is enough to prove the assertion after replacing S by a sufficiently small open neighbourhood U_h of each s_h, and X by the inverse image of U_h in X. Indeed, suppose the proposition established in this case, and argue by Noetherian induction ($\mathbf{0_I}$, 2.2.2), assuming the assertion established for every closed subprescheme of S whose underlying space is not S. We may suppose that the U_h are pairwise disjoint. If T is a closed subprescheme whose underlying space is

$$S - \bigcup_h U_h,$$

the induction hypothesis implies that the assertion is true for T; since it is also true for each U_h, it is plainly true for S.

Since we may clearly suppose S reduced (by replacing S by S_{red} and X by $X \times_S S_{\mathrm{red}}$), each $\mathscr{O}_{S,s_h}$ is a field $k(s_h)$. Suppose that the proposition has been proved when S is the spectrum of a field and f is of finite type. The existence of the U_h then follows from the method of (8.1.2), a), together with (8.8.2), (i) and (ii), (8.10.5), (vi), (vii), and (x), (11.2.6), (ii), and (17.7.8), (ii).

We may therefore suppose that $S = \mathrm{Spec}(k)$, where k is a field, and that X is a nonempty prescheme of finite type over k. Since X is Noetherian, it has a closed point x ($\mathbf{0_I}$, 2.1.3), and $k(x)$ is then a finite extension of k (**I**, 6.4.2). There is consequently a finite separable extension k' of k such that $k'' = k(x)$ is a finite radicial extension of k'. The assertion follows by taking I to have one element, $S'_i = \mathrm{Spec}(k')$, and $S''_i = \mathrm{Spec}(k'')$. □

Corollary (17.16.5). — *Let $f : X \to S$ be a surjective morphism locally of finite presentation. Then there exists a surjective morphism $g : S' \to S$, locally of finite presentation and locally quasi-finite, such that there is an S-morphism $S' \to X$ (in other words, an S'-section of $X' = X \times_S S'$). If S is quasi-compact (resp. quasi-compact and quasi-separated), S' may be supposed affine (resp. S' may be supposed affine and g of finite presentation and quasi-finite).*

IV | 108

Proof. It is enough to prove the corollary when S is affine. Indeed, S is the union of a family (S_α) of affine open subsets. If, for every α, S'_α is affine and a morphism $g_\alpha : S'_\alpha \to S_\alpha$ satisfies the assertion and is of finite presentation and quasi-finite, take for S' the sum prescheme of the S'_α, with $g : S' \to S$ agreeing with g_α on each S'_α. This morphism has the required properties. Moreover, if S is quasi-compact, the family (S_α) may be supposed finite, so S' is affine; if S is also quasi-separated, the immersions $S_\alpha \to S$ are of finite presentation (**I**, 6.2), and consequently so is g.

Now suppose that S is affine. Form the finite morphisms of finite presentation $S''_i \to S_i$ supplied by (17.16.4). Since the immersions $S_i \to S$ are of finite presentation (the S_i being affine), the composite morphisms

$$g''_i : S''_i \longrightarrow S_i \longrightarrow S$$

are of finite presentation and quasi-finite. The assertion follows by taking S' to be the sum of the S''_i and g to agree with g''_i on each S''_i. □

Proposition (17.16.6). — *Let S be a quasi-compact and quasi-separated prescheme, and let $f : X \to S$ be a morphism of finite presentation such that, for every $s \in S$, the fibre $X_s = f^{-1}(s)$ is a prescheme proper over $k(s)$. Then there exists a finite family $(S_i)_{i \in I}$ of affine subpreschemes of S, of finite presentation over S, pairwise disjoint and with union S, such that, for every i, the morphism $X_i = X \times_S S_i \to S_i$ is proper and flat. If, moreover, X_s is finite over $k(s)$ for every $s \in S$, the S_i may be chosen so that each morphism $X_i \to S_i$ factors as*

$$X_i \xrightarrow{u_i} X'_i \xrightarrow{h_i} S_i,$$

where u_i and h_i are finite and locally free (18.2.7), h_i is étale, and u_i is radicial and surjective.

Proof. We follow a course analogous to that of (17.16.4). First reduce to the case in which S is affine and f is flat, using the generic-flatness theorem (8.9.5); note that the subpreschemes S'_i defined in the proof of (8.9.5) are affine. Since f is of finite presentation, we may then write

$$X = X_0 \times_{S_0} S,$$

where S_0 is Noetherian and $X_0 \to S_0$ is a morphism of finite type and flat (11.2.7). Moreover, every fibre $(X_0)_{s_0}$ is still proper over $k(s_0)$, by (2.7.1), (vii), and we are therefore reduced to the case in which S is Noetherian. By Noetherian induction and application of the method of (8.1.2), a) (using also (8.10.5), (xii) on this occasion), the proof of the first assertion is finally reduced to the case in which S is the spectrum of a field; it then follows trivially from the hypothesis, with $S_i = S$.

The case in which X_s is supposed finite over $k(s)$ for every $s \in S$ is treated in the same way, now using (2.7.1), (xv), (8.10.5), (x), (iv), (vi), and (vii), (17.7.8), and (2.1.12). We are reduced to proving the last assertion when $S = \mathrm{Spec}(k)$ is the spectrum of a field. Then $X = \mathrm{Spec}(A)$, where A is a finite k-algebra (**I**, 6.4.4); thus A is a direct product of finite k-algebras A_j which are local rings $(1 \leqslant j \leqslant m)$. Since A_j is an Artinian ring and a k-algebra, it contains a subfield K'_j canonically isomorphic to the residue field of A_j (**0**, 19.6.2).

Take $X_i = X$ and take X'_i to be the sum of the $\mathrm{Spec}(K'_j)$. This answers the question, since for every j there are homomorphisms **IV** | 109

$$K'_j \longrightarrow A_j \longrightarrow k(A_j)$$

whose composite is an isomorphism. □

§18. Supplement on étale morphisms. Henselian local rings and strictly local rings

In the present section we study various properties peculiar to étale morphisms. Moreover, the notion of an étale morphism makes it possible to develop, in a very natural way, Nagata's theory of Henselian rings, as well as that of strictly local rings. These rings play an important role in many recent developments, appearing whenever one needs a process of "localisation" finer than that furnished by the Zariski topology (cf. for example [43]), pending the appearance of the chapter of our Treatise devoted to the study of the "étale topology."

18.1. A remarkable equivalence of categories.

Proposition (18.1.1). — *Let S be a prescheme, S_0 a closed subprescheme of S, X_0 an S_0-prescheme smooth (resp. étale) over S_0, and x_0 a point of X_0. Then there exist an open neighbourhood U_0 of x_0 in X_0, an S-prescheme U smooth (resp. étale) over S, and an S_0-isomorphism*

$$U \times_S S_0 \xrightarrow{\sim} U_0.$$

Proof. Observe that if X_0 is étale over S_0 at x_0, it is *a fortiori* unramified over S_0 at that point. If an S-prescheme U smooth over S has been constructed such that $U \times_S S_0$ is isomorphic to U_0, then the fibres of the morphisms $U_0 \to S_0$ and $U \to S$ containing x_0 are isomorphic. It follows that U is unramified over S at x_0 (17.4.1), d), hence also in a neighbourhood of x_0; replacing U by this neighbourhood, we conclude that U is étale over S. It is therefore enough to prove the proposition when X_0 is merely supposed smooth over S_0.

The question being local on S and on X_0, we may suppose that $S = \mathrm{Spec}(A)$ and $X_0 = \mathrm{Spec}(C_0)$ are affine, so that $S_0 = \mathrm{Spec}(A_0)$, where A_0 is a quotient ring of A, and

$$C_0 = B_0/\mathfrak{J}_0, qquad B_0 = A_0[T_1, \ldots, T_n],$$

where $\mathfrak{J}_0$ is an ideal of finite type in B_0; finally, C_0 is a formally smooth A_0-algebra for the discrete topologies. Let $\mathfrak{p}_0$ be the ideal $\mathfrak{j}_{x_0}$ in C_0; then $\mathfrak{p}_0 = \mathfrak{q}_0/\mathfrak{J}_0$, where $\mathfrak{q}_0$ is a prime ideal of B_0. The Jacobian criterion (**0**, 22.6.4), together with (**0**, 19.1.12), shows that there exist in $\mathfrak{J}_0$ a family of r polynomials u_i $(1 \leqslant i \leqslant r)$ and indices j_h $(1 \leqslant h \leqslant r)$ such that the images of the u_i in

$$(\mathfrak{J}_0)_{\mathfrak{q}_0}/(\mathfrak{J}_0^2)_{\mathfrak{q}_0}$$

generate this $(B_0)_{\mathfrak{q}_0}$-module and

$$\det\left(\frac{\partial u_i}{\partial T_{j_h}}\right) \notin \mathfrak{q}_0. \tag{18.1.1.1}$$

Since $(B_0)_{\mathfrak{q}_0}$ is a local ring, Nakayama's lemma shows that we may suppose that the images of the u_i in $(\mathfrak{J}_0)_{\mathfrak{q}_0}$ generate this $(B_0)_{\mathfrak{q}_0}$-module and then, replacing X_0 if necessary by an affine open neighbourhood of x_0, that the u_i generate $\mathfrak{J}_0$ ($\mathbf{0_I}$, 5.2.2). Put $B = A[T_1, \dots, T_n]$; then B_0 is a quotient ring of B, $\mathfrak{q}_0$ is the image of a prime ideal $\mathfrak{q}$ of B, and $\mathfrak{q}$ is the inverse image of $\mathfrak{q}_0$. For every i, let $v_i \in B$ be an element whose image in B_0 is u_i, and let $\mathfrak{J}$ be the ideal of B generated by the v_i, so that $\mathfrak{J}_0$ is the image of $\mathfrak{J}$ in B_0.

The proposition will be proved by taking for U an open neighbourhood of the point of $\mathrm{Spec}(B/\mathfrak{J})$ **IV** | 110
corresponding to the prime ideal $\mathfrak{p} = \mathfrak{q}/\mathfrak{J}$, provided that we show that $(B/\mathfrak{J})_{\mathfrak{q}/\mathfrak{J}}$ is a formally smooth A-algebra for the discrete topologies. This follows from the Jacobian criterion: the images of the v_i in $\mathfrak{J}_{\mathfrak{q}}/\mathfrak{J}_{\mathfrak{q}}^2$ generate this $B_{\mathfrak{q}}$-module, and (18.1.1.1) gives

$$\det\left(\frac{\partial v_i}{\partial T_{j_h}}\right) \notin \mathfrak{q}.$$

□

Theorem (18.1.2). — *Let S be a prescheme, S_0 a closed sub-prescheme of S whose underlying space is identical to that of S. Then the functor*

$$X \longmapsto X \times_S S_0$$

from the category of S-preschemes étale over S, to the category of S_0-preschemes étale over S_0, is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let X, Y be two S-preschemes étale over S, and set $X_0 = X \times_S S_0$, $Y_0 = Y \times_S S_0$. If $Z = X \times_S Y$, the set $\mathrm{Hom}_S(X, Y)$ is in canonical bijective correspondence with the set of X-sections $\Gamma(Z/X)$, and likewise $\mathrm{Hom}_{S_0}(X_0, Y_0)$ is in canonical bijective correspondence with $\Gamma(Z_0/X_0)$, where $Z_0 = Z \times_S S_0 = X_0 \times_{S_0} Y_0$. Now Z is étale over X, Z_0 is étale over X_0, and X_0 (resp. Z_0) is a closed subprescheme of X (resp. Z) having the same underlying space. The open subsets of Z on which the restriction of $Z \to X$ is surjective and radicial are therefore the *same* as the open subsets of Z_0 having the corresponding properties, and our assertion follows from (17.9.3).

To complete the proof, it suffices to show that for every S_0-prescheme X_0 étale over S_0, there exists an S-prescheme X étale over S and an S_0-isomorphism $X_0 \xrightarrow{\sim} X \times_S S_0$. By virtue of Proposition (18.1.1), there exists an open covering (U_α) of X_0 and for each α, an S-prescheme V_α which is étale over S, and finally an S_0-isomorphism $\theta_\alpha : U_\alpha \xrightarrow{\sim} V_\alpha \times_S S_0$. Furthermore, by virtue of the first part of the proof, there exists a unique S_0-isomorphism $\varphi_{\alpha\beta}$ from $\theta_\alpha(U_\alpha \cap U_\beta)$ to $\theta_\beta(U_\alpha \cap U_\beta)$, corresponding to the identity automorphism of $U_\alpha \cap U_\beta$, and it is immediate, for the same reason, that these isomorphisms satisfy the glueing condition (($\mathbf{0}$, 4.1.7)). Hence there exists an S-prescheme X such that the V_α identify canonically with sub-preschemes induced on the open subsets of X, the θ_α identify with S_0-isomorphisms which coincide on the intersections $U_\alpha \cap U_\beta$, and define an S_0-isomorphism $X_0 \xrightarrow{\sim} X \times_S S_0$. It is clear that X is étale over S (17.3.2), which completes the proof. □

Corollary (18.1.3). — *Let S be a prescheme, X an S-prescheme étale over S, S' an S-prescheme, S'_0 a closed sub-prescheme of S' having the same underlying space. Then the canonical map $X(S')_S \to X(S'_0)_S$* (**I**, 3.4.3) *is bijective.*

Proof. Indeed, set $X' = X \times_S S'$, $X'_0 = X \times_S S'_0$, so that $X(S')_S = \Gamma(X'/S')$ and $X(S'_0)_S = \Gamma(X'_0/S'_0)$; the corollary follows from the fact that the functor defined in (18.1.2) (with S and S_0 replaced by S' and S'_0 respectively) is fully faithful (or directly from (17.9.3)). □

18.2. Étale coverings. IV | 111

(18.2.1). Given a ring A and a commutative A-algebra B which is *finite* and is a *free* A-module, recall (Bourbaki, *Alg.*, ch. VIII, § 12, no. 2) that one defines on B an *A-linear form* $\mathrm{Tr}_{B/A}$, the "trace form"; from it one obtains a *symmetric A-bilinear form* (also called the "trace form")

$$(x, y) \longmapsto \mathrm{Tr}_{B/A}(xy) \tag{18.2.1.1}$$

whose datum is equivalent to that of the associated A-linear map $\mathrm{astr}_{B/A} : B \to \check{B}$ from the A-module B into its *dual* $\check{B}$; this map is equal to its transpose. When A is a *field*, saying that this bilinear form is *non-degenerate* is equivalent to saying that B is a *separable* A-algebra (Bourbaki, *Alg.*, ch. IX, § 2, prop. 5).

Let $f : A \to A'$ be a ring homomorphism; set $B' = B \otimes_A A'$, and let $g : B \to B'$ be the canonical homomorphism; the image under g of a basis of the A-module B is then a basis of the A'-module B', and it follows from the definitions that, for every $x \in B$,

$$\mathrm{Tr}_{B'/A'}(g(x)) = f(\mathrm{Tr}_{B/A}(x)). \tag{18.2.1.2}$$

(18.2.2). Consider now a *ringed space* $(X, \mathscr{O}_X)$ and let $\mathscr{B}$ be an $\mathscr{O}_X$-algebra which, as an $\mathscr{O}_X$-module, is *locally free of finite rank*. For every open subset $U \subset X$ such that $\mathscr{B}|_U$ is, as an $\mathscr{O}_U$-module, isomorphic to $\mathscr{O}_U^n$ (for some n depending on U), $\Gamma(U, \mathscr{B})$ is a $\Gamma(U, \mathscr{O}_X)$-algebra which, as a $\Gamma(U, \mathscr{O}_X)$-module, is free of finite rank. It therefore defines a $\Gamma(U, \mathscr{O}_X)$-linear form $\mathrm{Tr}_{\Gamma(U,\mathscr{B})/\Gamma(U,\mathscr{O}_X)}$, which we also denote by $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X, U}$, and hence an associated linear map

$$\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X, U} : \Gamma(U, \mathscr{B}) \longrightarrow \Gamma(U, \mathscr{B})^\vee = \Gamma(U, \check{\mathscr{B}}).$$

Furthermore, it follows from (18.2.1.2) that these linear maps are compatible with restriction from U to a smaller open subset and therefore define, on the one hand, an *$\mathscr{O}_X$-module homomorphism*, again called the *trace homomorphism*,

$$\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X} : \mathscr{B} \longrightarrow \mathscr{O}_X \tag{18.2.2.1}$$

and, on the other hand, an *$\mathscr{O}_X$-module homomorphism*

$$\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X} : \mathscr{B} \longrightarrow \check{\mathscr{B}} \tag{18.2.2.2}$$

called the *homomorphism associated to the trace*, and equal to its transpose. It also follows from (18.2.1.2) that for every $x \in X$, we have

$$(\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X})_x = \mathrm{Tr}_{\mathscr{B}_x|\mathscr{O}_{X,x}} \tag{18.2.2.3}$$

$$(\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X})_x = \mathrm{astr}_{\mathscr{B}_x|\mathscr{O}_{X,x}}\,. \tag{18.2.2.4}$$

Finally, under the conditions of (18.2.1), if we set $X = \mathrm{Spec}(A)$ and $\mathscr{B} = \widetilde{B}$, the form $\mathrm{Tr}_{\mathscr{B}|\mathscr{O}_X}$ (resp. the homomorphism $\mathrm{astr}_{\mathscr{B}|\mathscr{O}_X}$) corresponds to the form $\mathrm{Tr}_{B/A}$ (resp. the A-module homomorphism $\mathrm{astr}_{B/A}$), as also follows from (18.2.1.2).

Proposition (18.2.3). — *Let $f : X \to Y$ be a finite morphism of preschemes and set $\mathscr{B} = f_*(\mathscr{O}_X)$. The following conditions are equivalent:*

- a) *f is étale.*
- a') *f is a flat morphism of finite presentation, and for every $x \in X$, if $y = f(x)$, the ring $\mathscr{O}_{X,x}/\mathfrak{m}_y\mathscr{O}_{X,x}$ is a field which is a finite separable extension of $\kappa(y)$.*
- b) *$\mathscr{B}$ is a locally free $\mathscr{O}_Y$-module and, for every $y \in Y$, $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a finite separable $\kappa(y)$-algebra (hence a direct product of finitely many fields which are finite separable extensions of $\kappa(y)$).*
- c) *$\mathscr{B}$ is a locally free $\mathscr{O}_Y$-module and the homomorphism $\mathrm{astr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \check{\mathscr{B}}$ (18.2.2) is bijective.*

Proof. Since f is quasi-compact, the equivalence of a) and a') has already been shown (17.6.2). To prove the rest of the proposition, we may restrict to the case where $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ are affine, B being a finite A-algebra and $\mathscr{B} = \widetilde{B}$. To say that f is a morphism of finite presentation is equivalent to saying that B is an A-module of finite presentation (1.4.7). If, in addition, f is flat, then B is a flat A-module; it is known (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 2, cor. 2 to th. 1) that B is then a *projective* A-module. Thus $\mathscr{B}$ is a locally free $\mathscr{O}_Y$-module (*loc. cit.*, no. 2, th. 1), and the converse is immediate. On the other hand, $f^{-1}(y)$ is nothing but the spectrum of the $\kappa(y)$-algebra IV | 112

$\mathscr{B}(y) = \mathscr{B}_y/\mathfrak{m}_y\mathscr{B}_y$, which completes the proof of the equivalence of $a')$ and $b)$. To see that $b)$ is equivalent to $c)$, note that the second assertion of $b)$ is equivalent to the bijectivity of

$$\operatorname{astr}_{\mathscr{B}(y)|\kappa(y)} : \mathscr{B}(y) \longrightarrow \check{\mathscr{B}(y)}.$$

Since

$$\mathscr{B}(y) = \mathscr{B}_y/\mathfrak{m}_y\mathscr{B}_y, \qquad \check{\mathscr{B}(y)} = \check{\mathscr{B}}_y/\mathfrak{m}_y\check{\mathscr{B}}_y,$$

and $\mathscr{B}_y$ and $\check{\mathscr{B}}_y$ are free $\mathscr{O}_{Y,y}$-modules, it follows from (18.2.2.4) and Bourbaki, *Alg. comm.*, ch. II, § 3, no. 3, cor. to prop. 6, that the homomorphism $\operatorname{astr}_{\mathscr{B}_y|\mathscr{O}_{Y,y}} : \mathscr{B}_y \to \check{\mathscr{B}}_y$ is itself bijective; the converse being evident, this completes the proof.

When an $\mathscr{O}_X$-Algebra $\mathscr{B}$ satisfies the equivalent conditions $b)$ and $c)$ of (18.2.3), we say that $\mathscr{B}$ is a *finite étale $\mathscr{O}_X$-Algebra*. When $X = \operatorname{Spec}(A)$ is affine and $\mathscr{B} = \widetilde{B}$, this amounts, by virtue of (18.2.3), to saying that B is a finite étale A-algebra, or that B is a finite A-algebra that is étale (in the sense of (17.3.2)). □

Corollary (18.2.4). — *Let $f : X \to Y$ be a finite morphism of finite presentation, and set $\mathscr{B} = f_*(\mathscr{O}_X)$. Let y be a point of Y. The following conditions are equivalent:*

- a) *There exists an open neighbourhood U of y in Y such that the restriction $f^{-1}(U) \to U$ of f is an étale morphism.*
- b) *$\mathscr{B}_y$ is a free $\mathscr{O}_{Y,y}$-module of finite type and $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a separable $\kappa(y)$-algebra.*

Proof. It is clear that $a)$ implies $b)$ by virtue of (18.2.3). On the other hand, $\mathscr{B}$ is an $\mathscr{O}_Y$-module of finite presentation (1.6.3) and (1.4.7); hence, if $\mathscr{B}_y$ is a free $\mathscr{O}_{Y,y}$-module, there exists an open neighbourhood U of y in Y such that $\mathscr{B}|_U$ is a locally free $\mathscr{O}_U$-module ($\mathbf{0_I}$, 5.2.7). Moreover, since by hypothesis the homomorphism $\operatorname{astr}_{\mathscr{B}_y|\mathscr{O}_{Y,y}} : \mathscr{B}_y \to \check{\mathscr{B}}_y$ is bijective, the same result shows that U may be chosen so that

the homomorphism $\operatorname{astr}_{\mathscr{B}|_U/\mathscr{O}_U}$ is bijective. The fact that $b)$ implies $a)$ then follows from (18.2.3). **IV** | 113 □

Corollary (18.2.5). — *Let Y be a quasi-compact or locally Noetherian prescheme, and let $f : X \to Y$ be a finite morphism of finite presentation; set $\mathscr{B} = f_*(\mathscr{O}_X)$. Suppose that for every* closed *point y of Y, $\mathscr{B}_y$ is a free $\mathscr{O}_{Y,y}$-module and $\mathscr{B}_y \otimes_{\mathscr{O}_{Y,y}} \kappa(y)$ is a separable $\kappa(y)$-algebra. Then f is étale.*

Proof. Indeed, it follows from (18.2.4) that every closed point y of Y has an open neighbourhood U such that the restriction $f^{-1}(U) \to U$ of f is étale; the conclusion follows from the fact that in the two cases considered, every non-empty closed subset of Y contains a closed point ((5.1.11) and ($\mathbf{0}$, 2.1.3)). □

Corollary (18.2.6). — *If $f : X \to Y$ is a finite étale morphism, and if we set $\mathscr{B} = f_*(\mathscr{O}_X)$, then the $\mathscr{O}_Y$-homomorphism $\operatorname{Tr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \mathscr{O}_Y$ (18.2.2) (also denoted Tr_f) is surjective.*

Proof. The question being local, we may, by virtue of (18.2.3), suppose that $Y = \operatorname{Spec}(A)$, $X = \operatorname{Spec}(B)$, with B a free A-module; as by virtue of (18.2.3) the bilinear form (18.2.1.1) is non-degenerate, this implies in particular that the linear form $\operatorname{Tr}_{B/A}$ is surjective. □

Remarks (18.2.7). — (i) When $f : X \to Y$ is a finite morphism such that $f_*(\mathscr{O}_X)$ is a locally free $\mathscr{O}_Y$-Module (resp. locally free of rank n), we say further that f is a *finite locally free* morphism (resp. *locally free of rank n*). This condition, by virtue of (18.2.3), is satisfied if f is a finite étale morphism, but does not imply by itself that f is étale, as shown by the example where $X = \operatorname{Spec}(K)$ and $Y = \operatorname{Spec}(k)$ are spectra of fields, K being a finite non-separable extension of k. When $f : X \to Y$ is a finite étale morphism, we say further that X is an *étale covering* of Y. One notes that in this case, f is universally open and universally closed, and in particular $f(X)$ is a subset that is both open and closed of Y.

We say that an étale covering X of Y is *trivial* if X is a *sum* of a finite number of preschemes *isomorphic* to Y. We say that an étale covering X of Y is *locally trivial* if the morphism $f : X \to Y$ is such that for every point $y \in Y$ there is an open neighbourhood U for which the covering $f^{-1}(U)$ of U is trivial.

(ii) Let $f : X \to Y$ be a finite morphism, locally free of rank n; set $\mathscr{B} = f_*(\mathscr{O}_X)$ and let $u = \mathrm{astr}_{\mathscr{B}|\mathscr{O}_Y} : \mathscr{B} \to \check{\mathscr{B}}$; we deduce from this an n-th exterior power homomorphism $\bigwedge^n u : \bigwedge^n \mathscr{B} \to \bigwedge^n \check{\mathscr{B}} = (\bigwedge^n \mathscr{B})^\vee$ of *invertible* $\mathscr{O}_Y$-Modules, and by (**0**, 5.4.2) an element

$$d_{X/Y} \in \Gamma(Y, (\bigwedge^n \check{\mathscr{B}}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \check{\mathscr{B}})) \tag{18.2.7.1}$$

which we call the *discriminant* of X over Y. Furthermore, since $(\bigwedge^n \check{\mathscr{B}}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \check{\mathscr{B}})$ is the dual of $(\bigwedge^n \mathscr{B}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \mathscr{B})$, $d_{X/Y}$ can also be identified with a homomorphism

$$(\bigwedge^n \mathscr{B}) \otimes_{\mathscr{O}_Y} (\bigwedge^n \mathscr{B}) \longrightarrow \mathscr{O}_Y \tag{18.2.7.2}$$

and we denote by $\mathscr{D}_{X/Y}$ the quasi-coherent ideal of $\mathscr{O}_Y$ of finite type, *image* of the homomorphism (18.2.7.2), also called the *discriminant ideal* of X over Y.

For the homomorphism u to be bijective, it is necessary and sufficient that $\bigwedge^n u$ be bijective, or again that the section $d_{X/Y}$ have an *invertible* germ at every point $y \in Y$, which is also written $d_{X/Y}(y) \neq 0$ for every y ((**0**, 5.5.2)). It amounts to the same to say that the discriminant ideal $\mathscr{D}_{X/Y}$ is *equal* to $\mathscr{O}_Y$. **IV** | 114

The terminology "covering" introduced in (18.2.7), (i), is justified by the following proposition:

Proposition (18.2.8). — *Let $f : X \to Y$ be a morphism étale, separated and of finite type, and for every $y \in Y$, let $n(y)$ be the geometric number of points of $f^{-1}(y)$. Then the function $y \mapsto n(y)$ is lower semi-continuous on Y. For it to be continuous at a point y (and hence constant in a neighbourhood of y), it is necessary and sufficient that there exists an open neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is a* finite *(étale) morphism.*

Proof. Since f is quasi-finite (17.6.1) and locally of finite presentation, it amounts to saying that f is finite or that f is proper (8.11.1): moreover, each fibre $f^{-1}(y)$ is geometrically reduced over $\kappa(y)$. The conclusions therefore follow from (15.5.9), (i) and (ii) and from the fact that f is flat. □

Corollary (18.2.9). — *Let Y be a connected prescheme, $f : X \to Y$ a morphism étale, separated and of finite type. For f to be finite (in other words, for X to be an étale covering of Y), it is necessary and sufficient that all the fibres of f have the same geometric number of points.*

Remarks (18.2.10). — (i) The example of the "affine line with a doubled point" ((**I**, 5.5.11)) shows that a morphism étale of finite type of Noetherian preschemes may not be separated; for this example, the first assertion of (18.2.8) is no longer exact.

(ii) For a separated morphism of finite type and étale $f : X \to Y$ to make X a *locally trivial* covering, it is necessary and sufficient that for every $x \in X$ there exist an open neighbourhood U of $y = f(x)$ and a U-section g of $f^{-1}(U)$ such that $g(y) = x$. Indeed, the condition is obviously necessary; the fact that it is also sufficient follows from the fact that every fibre $f^{-1}(y)$ is finite ((17.6.1)) and (**I**, 6.2.2), from the characterisation of sections of a Y-scheme étale (17.9.3) and from Proposition (18.2.8).

18.3. Finite algebras and étale algebras.

Proposition (18.3.1). — *Let A be a ring, B an A-algebra of finite presentation.*

(i) For B to be an unramified A-algebra, it is necessary and sufficient that B be an A-module of finite presentation and be a projective *$(B \otimes_A B)$-module.*

(ii) Suppose moreover that B is a finite A-algebra. For B to be an étale A-algebra, it is necessary and sufficient that B be a projective A-module and a projective $(B \otimes_A B)$-module.

Proof. The $(B \otimes_A B)$-module structure on B is that coming from the canonical ring homomorphism $B \otimes_A B \to B$, which is surjective and has kernel $\mathfrak{J}_{B/A}$ ((**0**, 20.4.1)). **IV** | 115

(i) To say that the morphism $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is locally of finite presentation is equivalent to saying that B is an A-algebra of finite presentation (1.4.6). To say that B is an unramified A-algebra means (17.4.2) that

$$\mathrm{Spec}\big((B \otimes_A B)/\mathfrak{J}_{B/A}\big)$$

is the subprescheme induced on an open and closed subset of $\mathrm{Spec}(B \otimes_A B)$. This holds if and only if $\mathfrak{J}_{B/A}$ is a *direct-factor ideal* of $B \otimes_A B$ (Bourbaki, *Alg. comm.*, ch. II, § 4, no. 3, prop. 15); but this amounts to saying that the $(B \otimes_A B)$-module quotient $(B \otimes_A B)/\mathfrak{J}_{B/A}$ is projective (Bourbaki, *Alg.*, ch. II, 3[e] éd., § 2, no. 2, prop. 4).

(ii) Recall that a flat A-module of finite presentation is projective, and conversely (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 2, cor. 2 du th. 1), so the assertion (ii) follows from that of (i) and from (17.6.2). □

Proposition (18.3.2). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$-preadic topology, A is separated and complete; set $A_0 = A/\mathfrak{J}$. Then the functor*

$$B \longmapsto B \otimes_A A_0$$

is an equivalence of the category of finite étale A-algebras, to the category of finite étale A_0-algebras.

Proof. We will first prove the following lemma:

Lemma (18.3.2.1). — *Let A be a ring, $\mathfrak{J}$ an ideal of A such that, for the $\mathfrak{J}$-preadic topology, A is separated and complete.*

(i) Every projective A-module of finite type M is separated and complete for the $\mathfrak{J}$-preadic topology, and hence is the projective limit of the projective $(A/\mathfrak{J}^{n+1})$-modules $M/\mathfrak{J}^{n+1}M$.

(ii) Conversely, set $A_n = A/\mathfrak{J}^{n+1}$, and let (M_n) be a projective system of A_n-modules such that, for every n, the homomorphism $M_{n+1} \otimes_{A_{n+1}} A_n \to M_n$ induced by the transition homomorphism $M_{n+1} \to M_n$ is bijective. Suppose moreover that the M_n are projective and M_0 of finite type. Then $M = \varprojlim M_n$ is a projective A-module of finite type such that the canonical homomorphism $M \otimes_A A_0 \to M_0$ is bijective.

Proof. (i) There is a free A-module of finite type L such that M is isomorphic to a direct factor of L; as L is separated for the $\mathfrak{J}$-preadic topology, so is every sub-module N of L since $\mathfrak{J}^{n+1}N \subset \mathfrak{J}^{n+1}L$; in particular M is separated for this topology, and since the surjective homomorphism $f : L \to M$ is continuous for the $\mathfrak{J}$-preadic topology, its kernel N is closed for the topology induced by that of L; since L is complete and f a strict morphism, we conclude that M is complete (Bourbaki, *Top. gén.*, ch. IX, 2[e] éd., § 3, no. 1, prop. 4).

(ii) It follows from Nakayama's lemma that, if M_0 is generated by a finite family (x_{i0}) of r elements and if, for every n, x_{in} is an element of M_n whose image in M_{n-1} is $x_{i,n-1}$, then (x_{in}) $(1 \leqslant i \leqslant r)$ is a system of generators of M_n (Bourbaki, *Alg. comm.*, ch. II, § 3, cor. 2 to prop. 4). For every n, put $L_n = A_n^r$. If $(e_{in})_{1 \leqslant i \leqslant r}$ is the canonical basis of L_n, let $u_n : L_n \to M_n$ be the A_n-linear map such that $u_n(e_{in}) = x_{in}$ for every i. By hypothesis, we have a split exact sequence

$$0 \longrightarrow N_n \xrightarrow{v_n} L_n \xrightarrow{u_n} M_n \longrightarrow 0$$

and since $L_n = L_{n+1}/\mathfrak{J}^{n+1}L_{n+1}$ and $M_n = M_{n+1}/\mathfrak{J}^{n+1}M_{n+1}$, the vertical arrows in the commutative diagram **IV** | 116

$$\begin{tikzcd}
0 \arrow[r] & N_{n+1} \arrow[r, "v_{n+1}"] \arrow[d] & L_{n+1} \arrow[r, "u_{n+1}"] \arrow[d] & M_{n+1} \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & N_n \arrow[r, "v_n"] & L_n \arrow[r, "u_n"] & M_n \arrow[r] & 0
\end{tikzcd}$$

are all three surjective. Now we have $M = \varprojlim M_n$ and $L = A^r = \varprojlim L_n$; setting $N = \varprojlim N_n$, $u = \varprojlim u_n$, $v = \varprojlim v_n$, by (EGA 0_{III}, 13.2.2) the exact sequence

$$0 \longrightarrow N \xrightarrow{v} L \xrightarrow{u} M \longrightarrow 0 \qquad (18.3.2.2)$$

is exact. Moreover, since for every n, v_n is left-invertible and M_n is a projective A_n-module, by virtue of ((**0**, 19.1.8)) the exact sequence (18.3.2.2) is split, which proves the lemma. □

This being established, let us first show that the functor of (18.3.2) is *fully faithful*. Set, as in the lemma, $A_n = A/\mathfrak{J}^{n+1}$; let B, C be two finite étale A-algebras, and set, for every n, $B_n = B \otimes_A A_n$, $C_n = C \otimes_A A_n$; by virtue of (18.3.1) and (18.3.2.1), B and C are separated and complete for the $\mathfrak{J}$-preadic topology, and we have $B = \varprojlim B_n$, $C = \varprojlim C_n$; moreover, every A-algebra homomorphism

$u : B \to C$ is continuous for the $\mathfrak{J}$-preadic topologies, and hence gives a projective system of A_n-algebra homomorphisms $u_n = u \otimes 1 : B_n \to C_n$, of which it is the projective limit; the converse being evident, we have a canonical bijection

$$\mathrm{Hom}_{A\text{-alg.}}(B, C) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n).$$

But since B and C are finite étale A-algebras, it follows also from (18.1.2) that the canonical map

$$\mathrm{Hom}_{A_{n+1}\text{-alg.}}(B_{n+1}, C_{n+1}) \longrightarrow \mathrm{Hom}_{A_n\text{-alg.}}(B_n, C_n)$$

is *bijective* for $n \geqslant 0$, which proves that the canonical map $\mathrm{Hom}_{A\text{-alg.}}(B, C) \to \mathrm{Hom}_{A_0\text{-alg.}}(B_0, C_0)$ is bijective.

To complete the proof of (18.3.2), it suffices to show that for every finite étale A_0-algebra B_0, there exists a finite étale A-algebra B and an A_0-isomorphism $B_0 \xrightarrow{\sim} B \otimes_A A_0$. Now, by (18.1.2), there is a projective system (B_n) such that B_n is an étale A_n-algebra and that the homomorphisms $B_{n+1} \otimes_{A_{n+1}} A_n \to B_n$ are bijective. It follows from (18.3.1) and (18.3.2.1) that the A-algebra $B = \varprojlim B_n$ is a projective A-module of finite type and that B_0 is isomorphic to $B \otimes_A A_0$. To prove that B is a finite étale A-algebra, it suffices, by (18.2.5), to show that for every maximal ideal $\mathfrak{m}$ of A, $B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}}$ is a separable $(A/\mathfrak{m})$-algebra. Now, since $\mathfrak{J}$ is contained in the radical of A ((**0**, 7.1.10)), we have $\mathfrak{J} \subset \mathfrak{m}$, so $\mathfrak{m}_0 = \mathfrak{m}/\mathfrak{J}$, we have $A_0/\mathfrak{m}_0 = A/\mathfrak{m}$ and $B_{\mathfrak{m}}/\mathfrak{m}B_{\mathfrak{m}} = (B_0)_{\mathfrak{m}_0}/\mathfrak{m}_0(B_0)_{\mathfrak{m}_0}$; the **IV** | 117
conclusion therefore follows from the fact that B_0 is a finite étale A_0-algebra (18.2.5). □

(18.3.3). *Example.* — Proposition (18.3.2) applies in particular when A is a *local* ring, separated and complete, $\mathfrak{J}$ being the maximal ideal of A, so that A_0 is a field and the finite étale A_0-algebras are precisely the finite separable A_0-algebras, hence the direct products of finite separable field extensions of A_0. In particular, if the field A_0 is *separably closed*, these extensions are all identical to A_0, and hence every étale covering of $\mathrm{Spec}(A)$ is *trivial* (18.2.7) by virtue of (18.3.2).

Theorem (18.3.4). — *Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is separated and complete for the $\mathfrak{J}$-preadic topology, $A_0 = A/\mathfrak{J}$. Let $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(A_0)$. Let X be a proper S-scheme, and set $X_0 = X \times_S S_0$. Then the functor*

$$Z \longmapsto Z \times_X X_0$$

from the category of finite étale X-schemes over X, to the category of finite étale X_0-schemes over X_0, is an equivalence of categories.

Proof. Let us first show that this functor is *fully faithful*. Let Z' and Z'' be two finite étale X-schemes over X. Set $S_n = \mathrm{Spec}(A/\mathfrak{J}^{n+1})$, $X_n = X \times_S S_n$, $Z'_n = Z' \times_S S_n$, $Z''_n = Z'' \times_S S_n$. It follows from ((**III**, 5.4.1)) that we have a canonical bijection $\mathrm{Hom}_X(Z', Z'') \xrightarrow{\sim} \varprojlim \mathrm{Hom}_{X_n}(Z'_n, Z''_n)$. Now, by virtue of (18.1.2), the canonical map $\mathrm{Hom}_{X_{n+1}}(Z'_{n+1}, Z''_{n+1}) \to \mathrm{Hom}_{X_n}(Z'_n, Z''_n)$ is bijective, which proves our assertion.

It remains to show that if $\mathscr{B}_0$ is a finite étale $\mathscr{O}_{X_0}$-algebra, there exists a finite étale $\mathscr{O}_X$-algebra $\mathscr{B}$ and an isomorphism $\mathscr{B}_0 \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_0}$. It follows from (18.1.2) that there is a projective system $(\mathscr{B}_n)$, where $\mathscr{B}_n$ is a finite étale $\mathscr{O}_{X_n}$-algebra, and the second comparison theorem ((**III**, 5.1.4)) proves the existence of a coherent $\mathscr{O}_X$-module $\mathscr{B}$ and of a projective system of isomorphisms $\mathscr{B}_n \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_n}$. The datum of an algebra structure on an $\mathscr{O}_X$-module $\mathscr{F}$ is equivalent to that of a multiplication homomorphism $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{F} \to \mathscr{F}$ making commute certain diagrams in which only tensor powers of $\mathscr{F}$ occur; by (**III**, 5.1.3), (**I**, 10.11.4) and (**I**, 10.11.7), $\mathscr{B}$ is therefore endowed in a natural way with a structure of $\mathscr{O}_X$-algebra for which the isomorphism $\mathscr{B}_0 \xrightarrow{\sim} \mathscr{B} \otimes_{\mathscr{O}_X} \mathscr{O}_{X_0}$ is an isomorphism of algebras. Moreover, $\mathscr{B}$ is a *locally free* $\mathscr{O}_X$-module; this also follows from (**III**, 5.1.3), (**I**, 10.11.4), (**I**, 10.11.7) and from the fact that, in the category of coherent $\mathscr{O}_X$-modules, the locally free $\mathscr{O}_X$-modules $\mathscr{F}$ can be characterized as those for which the functors $\mathscr{G} \mapsto \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ are *exact*. Finally, to see that $\mathscr{B}$ is a finite étale $\mathscr{O}_X$-Algebra, it suffices (18.2.5) to show that for every *closed* point $x \in X$, $\mathscr{B}_x \otimes_{\mathscr{O}_{X,x}} \kappa(x)$ is a separable $\kappa(x)$-algebra. But since the structural morphism $f : X \to S$ is *proper*, $f(x)$ is a closed point of S, so it belongs to S_0, since $\mathfrak{J}$ is contained in the radical of A ((**0**, 7.1.10)); the conclusion therefore follows from the fact that $X_0 = f^{-1}(S_0)$ and that $\mathscr{B}_0$ is a finite étale $\mathscr{O}_{X_0}$-algebra. □

18.4. Local structure of unramified morphisms and of étale morphisms. IV | 118

Lemma (18.4.1). — *Let A be a ring, B a finite monogenic A-algebra, u a generator of the A-algebra B, $F \in A[\mathrm{T}]$ a polynomial such that $F(u) = 0$, F' the derivative polynomial; set $u' = F'(u)$. Then the annihilator ideal of $\Omega^1_{B/A}$ in B contains $u'B$; it is equal to $u'B$ if the ideal $\mathfrak{J}$ of polynomials $G \in A[\mathrm{T}]$ such that $G(u) = 0$ is generated by F, in other words if the canonical surjection $\varphi : A[\mathrm{T}]/FA[\mathrm{T}] \to B$, which sends the class of* T *to u, is bijective.*

Proof. Set $C = A[\mathrm{T}]$, so that $B = C/\mathfrak{J}$. We have the exact sequence ((**0**, 20.5.12.1))

$$\mathfrak{J}/\mathfrak{J}^2 \longrightarrow \Omega^1_{C/A} \otimes_C B \longrightarrow \Omega^1_{B/A} \longrightarrow 0$$

and $\Omega^1_{B/A}$ therefore identifies with the quotient $B/\mathfrak{J}'$, where $\mathfrak{J}'$ is the ideal generated by the elements $G'(u)$, where G ranges over a system of generators of the ideal $\mathfrak{J}$ ((**0**, 20.5.13)); whence the lemma immediately follows. □

Proposition (18.4.2). — *Let the notation be those of (18.4.1), let $\mathfrak{q}$ be a prime ideal of B. Then:*

- *(i) If $\mathfrak{q}$ does not contain u', $B_\mathfrak{q}$ is a formally unramified $A_\mathfrak{p}$-algebra ($\mathfrak{p}$ being the inverse image of $\mathfrak{q}$ in A); in other terms,* $\mathrm{Spec}(B_{u'})$ *is formally unramified over* $\mathrm{Spec}(A)$.
- *(ii) Suppose moreover that F is monic and generates $\mathfrak{J}$. Then, for* $\mathrm{Spec}(B)$ *to be étale over* $\mathrm{Spec}(A)$ *at the point $\mathfrak{q}$, it is necessary and sufficient that $u' \notin \mathfrak{q}$.*

Proof. The hypothesis that $u' \notin \mathfrak{q}$ implies that $\Omega^1_{B_\mathfrak{q}/A_\mathfrak{p}} = 0$ ((**0**, 20.5.9)), hence (i) follows from (17.2.1). Moreover, under the hypotheses of (ii), B is a free A-module by virtue of the Euclidean division; as the annihilator $\mathfrak{J}'$ of $\Omega^1_{B/A}$ is then equal to $u'B$ by virtue of (18.4.1) and $\Omega^1_{B/A}$ is a B-module of finite presentation (16.4.22), the annihilator of $\Omega^1_{B_\mathfrak{q}/A_\mathfrak{p}}$ is equal to $u'B_\mathfrak{q}$ (Bourbaki, *Alg. comm.*, ch. II, § 2, no. 4, formule (g)), and (ii) follows from (i) and from the implication $c) \Rightarrow a)$ in (17.6.1). □

Corollary (18.4.3). — *With the notation of (18.4.2), suppose that F is monic and generates $\mathfrak{J}$. Then, for B to be an étale A-algebra, it is necessary and sufficient that u' be invertible in B (or, what amounts to the same, that the ideal of $A[\mathrm{T}]$ generated by F and F' be equal to $A[\mathrm{T}]$).*

Proof. In view of (18.4.2), (ii), saying that $\mathrm{Spec}(B)$ is étale over $\mathrm{Spec}(A)$ means that u' does not belong to any prime ideal of B, i.e. that it is invertible in B. □

We say that a monic polynomial $F \in A[\mathrm{T}]$ such that the ideal of $A[\mathrm{T}]$ generated by F and F' is equal to $A[\mathrm{T}]$ is *separable*; it is immediate that this definition coincides with the usual definition (Bourbaki, *Alg.*, ch. V, § 7, no. 6) when A is a field.

Lemma (18.4.4). — *Let A be a local ring, B a finite monogenic A-algebra, u a generator of the A-algebra B. Let $\mathfrak{n}_i$ ($1 \leqslant i \leqslant r$) be the maximal ideals of B such that* $\mathrm{Spec}(B)$ *is formally unramified over* $\mathrm{Spec}(A)$ *at the points $\mathfrak{n}_i$. Then there exists a monic polynomial $F \in A[\mathrm{T}]$ such that $F(u) = 0$ and $F'(u) \notin \mathfrak{n}_i$ for every i ($1 \leqslant i \leqslant r$). Moreover, if k is the residue field of A and $f \in k[\mathrm{T}]$ is the minimal polynomial of the image of* IV | 119 *u in $B \otimes_A k$, one can choose F so that its canonical image in $k[\mathrm{T}]$ is f.*

Proof. The maximal ideal $\mathfrak{m}$ of A is the inverse image of each $\mathfrak{n}_i$ ((**II**, 6.1.10)). Set $L = B \otimes_A k$, a finite k-algebra, and let ξ be the image of u in L; if n is the rank of L over k, the minimal polynomial $f \in k[\mathrm{T}]$ of ξ over k has degree n, and L is isomorphic to $k[\mathrm{T}]/fk[\mathrm{T}]$. If $\mathfrak{n}'_i = \mathfrak{n}_i/\mathfrak{m}B$, the hypothesis implies that $\mathrm{Spec}(L)$ is étale over k at the points $\mathfrak{n}'_i$ ($1 \leqslant i \leqslant r$), by (17.6.1), c); hence $f'(\xi) \notin \mathfrak{n}'_i$ by (18.4.2), (ii). Now $\mathfrak{m}B$ is contained in the radical of B; since $1, \xi, \dots, \xi^{n-1}$ form a basis of L over k, Nakayama's lemma shows that $1, u, \dots, u^{n-1}$ generate the A-module B. Consequently there exists a monic polynomial $F \in A[\mathrm{T}]$ of degree n such that $F(u) = 0$; since ξ is a root of the canonical image of F in $k[\mathrm{T}]$, this image is necessarily f. The image of $F'(u)$ in L is then $f'(\xi)$, and therefore $F'(u) \notin \mathfrak{n}_i$ for every i. □

Proposition (18.4.5). — *Let A be a local ring, k its residue field, and B a finite A-algebra (respectively, a finite A-algebra of finite presentation). Suppose moreover that the residue field k is* infinite, *or else that B is a local ring. Let n be the rank of $L = B \otimes_A k$ over k. For B to be a formally unramified A-algebra (resp. étale), it is necessary and sufficient that there exist a monic separable polynomial $F \in A[\mathrm{T}]$ such that B is* *isomorphic to a quotient of $A[\mathrm{T}]/F.A[\mathrm{T}]$ (resp. isomorphic to $A[\mathrm{T}]/F.A[\mathrm{T}]$). Moreover, one can suppose that F has degree n (respectively, F necessarily has degree n).*

Proof. The conditions are sufficient by virtue of (18.4.2), without assuming either that k is infinite or that B is local. To see that the conditions are necessary, note that if B is a formally unramified A-algebra, L is a finite separable k-algebra, hence a direct product of finitely many finite separable field extensions k_j of k ($1 \leqslant j \leqslant r$), each generated by an element $\bar{\xi}_j$ with minimal polynomial f_j (Bourbaki, *Alg.*, ch. V, § 11, no. 4, prop. 4). Let us show that by virtue of the hypotheses made on k or B, there exists an element $\bar{\xi}$ of L generating the k-algebra L. This is immediate if B is local, since then $r = 1$. Otherwise, k being assumed infinite, one can suppose that the irreducible polynomials $f_j \in k[\mathrm{T}]$ are all *distinct* (by replacing each $\bar{\xi}_j$ by $\bar{\xi}_j + a_j$, for a suitable element $a_j \in k$); if we set $f = f_1 f_2 \ldots f_r$, it is clear that L is isomorphic to $k[\mathrm{T}]/fk[\mathrm{T}]$ in both cases, hence generated by an element $\bar{\xi}$ of minimal polynomial $f \in k[\mathrm{T}]$ of degree n. Choose $u \in B$ whose image in L is $\bar{\xi}$. Since $\mathfrak{m}B$ is contained in the radical of B and $1, \bar{\xi}, \ldots, \bar{\xi}^{n-1}$ form a basis of L over k, Nakayama's lemma shows that $1, u, \ldots, u^{n-1}$ generate the A-module B; hence there exists a monic polynomial $F \in A[\mathrm{T}]$ of degree n such that $F(u) = 0$; moreover, since $\bar{\xi}$ is a root of the canonical image of F in $k[\mathrm{T}]$, this image is necessarily equal to f. Thus u generates the A-algebra B, which is consequently a quotient of $A[\mathrm{T}]/FA[\mathrm{T}]$. Moreover, B is semilocal and, at each of its maximal ideals $\mathfrak{n}_i$, (18.4.4) gives $F'(u) \notin \mathfrak{n}_i$; hence $F'(u)$ is invertible in B, and F is separable. Finally, if B is an étale A-algebra, then B, being a flat A-module of finite presentation ((1.4.7)), is free, and $1, u, \ldots, u^{n-1}$ form a basis of the A-module B (Bourbaki, *Alg. comm.*, ch. II, § 3, no. 2, prop. 5); in other words, the A-algebra B is **IV** | 120
isomorphic to $A[\mathrm{T}]/FA[\mathrm{T}]$; and any other monic polynomial $G \in A[\mathrm{T}]$ such that $B \simeq A[\mathrm{T}]/GA[\mathrm{T}]$ necessarily has degree n. □

Theorem (18.4.6). — *(Chevalley). —*

- *(i) Let $f : X \to Y$ be a morphism locally of finite type, x a point of X, $y = f(x)$, and set $A = \mathscr{O}_{Y,y}$. For $\mathscr{O}_{X,x}$ to be a formally étale (respectively, formally unramified) A-algebra, it is necessary and sufficient that there exist a monic polynomial $F \in A[\mathrm{T}]$ and a maximal ideal $\mathfrak{n}$ of $B = A[\mathrm{T}]/FA[\mathrm{T}]$ (respectively, of a quotient algebra B of $A[\mathrm{T}]/FA[\mathrm{T}]$) such that $\mathscr{O}_{X,x}$ is A-isomorphic to $B_{\mathfrak{n}}$ and, if u is the image of T in B, one has $F'(u) \notin \mathfrak{n}$.*
- *(ii) Suppose in addition that f is locally of finite presentation. Then, for f to be étale at the point x, it is necessary and sufficient that we can moreover take $B = A[\mathrm{T}]/F.A[\mathrm{T}]$.*

Proof. The conditions are sufficient by virtue of (18.4.2). To see that they are necessary, we may plainly reduce to the case where X and Y are affine and, in view of (17.4.1.2), to the case where f is formally unramified and quasi-finite. It then follows from (8.12.8) that there exist a finite A-algebra C and a maximal ideal $\mathfrak{r}$ of C (necessarily lying above the maximal ideal $\mathfrak{m}$ of A) such that $\mathscr{O}_{X,x}$ is A-isomorphic to $C_{\mathfrak{r}}$. Moreover, by (17.4.1.2), the residue field $C/\mathfrak{r} = C_{\mathfrak{r}}/\mathfrak{r}C_{\mathfrak{r}}$ is a finite separable extension of $k = A/\mathfrak{m}$, and hence is of the form $k[v]$, where v is separable over k. Let $\mathfrak{r}_i$ ($1 \leqslant i \leqslant h$) be the maximal ideals of the semilocal ring C other than $\mathfrak{r}$. There exists an element $u \in C$ belonging to every $\mathfrak{r}_i$ and whose image in $C/\mathfrak{r}$ is v (Bourbaki, *Alg. comm.*, ch. II, § 1, no. 2, prop. 5). We shall show that the A-subalgebra $B = A[u]$ of C and the ideal $\mathfrak{n} = \mathfrak{r} \cap B$ (necessarily maximal, since C is finite over B) answer the question.

For the case where $\mathscr{O}_{X,x}$ is a formally unramified A-algebra, it suffices to prove that $B_{\mathfrak{n}}$ is isomorphic to $C_{\mathfrak{r}}$; indeed, $B_{\mathfrak{n}}$ will then be formally unramified over A, and the existence of a polynomial $F \in A[\mathrm{T}]$ having the asserted properties will follow from (18.4.4). Since f is formally unramified, $C/\mathfrak{r}$ is isomorphic to the A-algebra $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$ ((17.4.1.2)). We are thus reduced to proving the following lemma.

Lemma (18.4.6.1). — *Let A be a local ring with maximal ideal $\mathfrak{m}$, let C be a finite A-algebra, and let $\mathfrak{r}$ be a maximal ideal of C. Let $u \in C$ belong to every maximal ideal of C distinct from $\mathfrak{r}$, but not to $\mathfrak{r}$, and suppose that $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$ is a monogenic algebra over $k = A/\mathfrak{m}$, generated by the image of u. Set $B = A[u]$ and $\mathfrak{n} = \mathfrak{r} \cap B$. Then the canonical homomorphism $B_{\mathfrak{n}} \to C_{\mathfrak{r}}$ is an isomorphism.*

Proof. Set $R = B \setminus \mathfrak{n}$ and $S = C \setminus \mathfrak{r}$, so that $B_{\mathfrak{n}} = R^{-1}B$ and $C_{\mathfrak{r}} = S^{-1}C$; the canonical homomorphism $B_{\mathfrak{n}} \to C_{\mathfrak{r}}$ factors as

$$R^{-1}B \xrightarrow{g} R^{-1}C \xrightarrow{h} S^{-1}C.$$

It suffices to prove that both homomorphisms are bijective. First, $h : R^{-1}C \to S^{-1}C = C_{\mathfrak{r}}$ is bijective. Indeed, it suffices to see that the images in $R^{-1}C$ of the elements of S are invertible, or equivalently that every maximal ideal $\mathfrak{p}$ of $R^{-1}C$ has inverse image in C disjoint from S, and hence equal to $\mathfrak{r}$. Since $R^{-1}C$ is finite over the local ring $R^{-1}B = B_{\mathfrak{n}}$, the inverse image of $\mathfrak{p}$ in $R^{-1}B$ is its **IV** | 121
unique maximal ideal $\mathfrak{n}B_{\mathfrak{n}}$; if $\mathfrak{q}$ is the inverse image of $\mathfrak{p}$ in C, then $\mathfrak{q} \cap B = \mathfrak{n}$ and, since $\mathfrak{n}$ is maximal in B and C is finite over B, $\mathfrak{q}$ is one of the maximal ideals of C. Furthermore $u \notin \mathfrak{q}$, since $u \notin \mathfrak{r}$ and $u \in B$; the hypothesis therefore forces $\mathfrak{q} = \mathfrak{r}$.

Since $B \subset C$, the homomorphism $g : R^{-1}B \to R^{-1}C$ is injective (EGA 0_{I}, 1.3.2). To prove that it is surjective, observe that $R^{-1}C$ is an $R^{-1}B$-module of finite type and that $\mathfrak{m}R^{-1}B$ is contained in the maximal ideal of the local ring $R^{-1}B$. By Nakayama's lemma, it suffices to prove that

$$R^{-1}B/\mathfrak{m}R^{-1}B \longrightarrow R^{-1}C/\mathfrak{m}R^{-1}C$$

is surjective. But, by the first part of the proof, $R^{-1}C/\mathfrak{m}R^{-1}C$ identifies with $C_{\mathfrak{r}}/\mathfrak{m}C_{\mathfrak{r}}$; by hypothesis this k-algebra is generated by the image of u, and hence it is the image of $R^{-1}B/\mathfrak{m}R^{-1}B$. □

Consider next the case where f is étale at x. Replacing X by a neighbourhood of x, we may suppose that X is a neighbourhood of $\mathfrak{n}$ in $\operatorname{Spec}(B)$ ((**I**, 7.2)). Set $B' = A[\mathrm{T}]/FA[\mathrm{T}]$ and let $\mathfrak{n}'$ be the inverse image of $\mathfrak{n}$ in B'. Since the image of $F'(\mathrm{T})$ in B' does not belong to $\mathfrak{n}'$, the morphism $\operatorname{Spec}(B') \to \operatorname{Spec}(A)$ is étale at $\mathfrak{n}'$ by (18.4.2), (ii). As $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is étale at $\mathfrak{n}$, (17.3.4) shows that $\operatorname{Spec}(B) \to \operatorname{Spec}(B')$ is étale at $\mathfrak{n}$; but this morphism is an immersion, so it can be étale at a point only if it is a local isomorphism there ((17.9.1)). Thus $B_{\mathfrak{n}}$ and $B'_{\mathfrak{n}'}$ are isomorphic.

Finally, suppose that $\mathscr{O}_{X,x}$ is a formally étale A-algebra. With the preceding notation, (17.1.5) shows that $B_{\mathfrak{n}}$ is a formally étale $B'_{\mathfrak{n}'}$-algebra; since the homomorphism $B'_{\mathfrak{n}'} \to B_{\mathfrak{n}}$ is surjective, this is possible only if it is *bijective* ((**0**, 19.10.3), (i)). This completes the proof of (18.4.6). □

Corollary (18.4.7). — *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X. For f to be formally unramified at the point x, it is necessary and sufficient that there exist an open neighbourhood U of x such that $f|U$ factorises as $U \xrightarrow{j} X' \xrightarrow{h} Y$, where h is an étale morphism and j is a closed immersion.*

Proof. We can obviously reduce to the case where $Y = \operatorname{Spec}(R)$ is affine and f is of finite type. If $A = \mathscr{O}_{Y,y}$, with $y = f(x)$, the condition for f to be formally unramified at the point x is equivalent, by virtue of (17.4.1.2), to saying that $\mathscr{O}_{X,x}$ is a formally unramified A-algebra. If it is so, we can apply (18.4.6), (i); replacing Y as needed by an affine neighbourhood of y, we may suppose (with the notation of (18.4.6)) that the polynomial F is the image in $A[\mathrm{T}]$ of a monic unitaire polynomial $G \in R[\mathrm{T}]$. We then set $X' = \operatorname{Spec}(R[\mathrm{T}]/G.R[\mathrm{T}])$; let x' be the image of the point $\mathfrak{n}$ of $\operatorname{Spec}(B)$ by the morphism corresponding to the composite homomorphism $R[\mathrm{T}]/G.R[\mathrm{T}] \to A[\mathrm{T}]/F.A[\mathrm{T}] \to B$. It follows from (18.4.2) that the canonical morphism $h : X' \to Y$ is étale at the point x', hence, restricting X' and Y to open neighbourhoods of x' and y if necessary, we may suppose that h is étale. On the other hand, by virtue of (**I**, 6.5.1), (ii) and (**I**, 7.2), the local homomorphism $\varphi : \mathscr{O}_{X',x'} \to \mathscr{O}_{X,x}$ corresponds to a morphism $j : U \to X'$ from an open neighbourhood U of x in X; in applying (**I**, 6.5.1), **IV** | 122
(i)), suppose that $h \circ j = f|U$, hence j is of finite type ((**I**, 6.3.4), (v)); finally, the homomorphism φ is surjective by (18.4.6), (i), hence it follows from ((**I**, 6.5.4), (i)) that we can, restricting further U and X', suppose that j is a closed immersion. This proves the necessity of the condition stated; the sufficiency is immediate (17.1.3), (i) and (ii). □

Corollary (18.4.8). — *Let $f : X \to Y$ be a morphism locally of finite presentation. For f to be unramified (respectively, étale), it is necessary and sufficient that there exist a family of flat morphisms $g_\alpha : Y'_\alpha \to Y$ and, for each α, an open subset U_α of $X'_\alpha = X \times_Y Y'_\alpha$, such that, if $g'_\alpha : X'_\alpha \to X$ and $f'_\alpha : X'_\alpha \to Y'_\alpha$ are the canonical projections, the $g'_\alpha(U_\alpha)$ cover X, and each composite $U_\alpha \to X'_\alpha \xrightarrow{f'_\alpha} Y'_\alpha$ is a closed immersion (respectively, an open immersion). One can moreover take the g_α to be étale.*

Proof. The necessity of the condition for étale morphisms is immediate: take a single $Y'_\alpha = X$, let $g_\alpha = f$, and take for $U_\alpha \subset X \times_Y X$ the diagonal. When f is unramified, necessity follows from (18.4.7): choose an open covering (V_α) of X such that, for each α, $f|V_\alpha$ factors as $V_\alpha \xrightarrow{j_\alpha} Y'_\alpha \xrightarrow{g_\alpha} Y$, where j_α is a closed immersion and g_α is étale. Then $j_\alpha : V_\alpha \to Y'_\alpha$ factors as $V_\alpha \xrightarrow{s_\alpha} X'_\alpha \xrightarrow{f'_\alpha} Y'_\alpha$, where

s_α is a V_α-section of X'_α; since $g'_\alpha : X'_\alpha \to X$ is étale, s_α is an open immersion ((17.4.1)), and it suffices to take $U_\alpha = s_\alpha(V_\alpha)$.

The sufficiency of the conditions follows from (17.7.1); one concludes at each point x of $g'(U_\alpha)$, hence in all of X since the $g'_\alpha(U_\alpha)$ cover X. □

Proposition (18.4.9). — *Let S be a prescheme, $f : X \to S$ a morphism, $h : Y \to S$ a morphism locally of finite presentation, $g : X \to Y$ an S-morphism, x a point of X, and $y = g(x)$. The following conditions are equivalent:*

a) *h is étale at the point y and g is flat at the point x.*
b) *h is unramified at the point y and f is flat at the point x.*

Proof. Since $f = h \circ g$, condition $a)$ implies that f is flat at x ((2.1.6)); moreover, an étale morphism is unramified, so $a)$ implies $b)$.

To prove that $b)$ implies $a)$, we may first suppose that h is unramified, after replacing Y by an open neighbourhood of y. Then, after replacing S by an open neighbourhood of $s = h(y) = f(x)$, we may suppose that there exist an étale morphism $u : S' \to S$, a point y' of $Y' = Y_{(S')}$ above y, and an open neighbourhood V of y' in Y' such that, if $h' = h_{(S')} : Y' \to S'$, the restriction $h'|V$ is a closed immersion ((18.4.8)). If one proves that h' is étale at the point y', it will follow that h is étale at the point y (17.7.1), (ii); moreover, $f' = f_{(S')} : X_{(S')} \to S'$ is flat at every point x' above x. As the projections $v : Y' \to Y$, $w : X' \to X$ are étale morphisms (hence flat), if one proves that $g' = g_{(S')} : X_{(S')} \to Y_{(S')}$ is flat at the point x', it will follow from (2.2.11), (iv) that g is flat at the point x. We may therefore reduce to the case where h is a *closed immersion* of finite presentation, with f flat at x. Let $\mathscr{J}$ be the quasi-coherent ideal of finite type in $\mathscr{O}_S$ defining the closed subprescheme Y of S. The hypothesis that f is flat at x implies that the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{X,x}$ is injective (EGA 0_{I}, 6.5.1); *a fortiori* the homomorphism $\mathscr{O}_{S,s} \to \mathscr{O}_{Y,y}$ is injective, which means that $\mathscr{J}_s = 0$, since it is the kernel of the preceding homomorphism. Since $\mathscr{J}$ is of finite type, there exists an open neighbourhood U of s in S such that $\mathscr{J}|U = 0$ ((**0**, 5.2.2)). We can hence suppose that h is an open immersion, and it is clear that g is flat at the point x. □ **IV | 123**

Proposition (18.4.10). — *Let us denote by $\mathbf{P}(f, x)$ a property satisfying the following conditions:*

1° *For every morphism $f : X \to Y$ and every local isomorphism $h : Y \to Z$, $\mathbf{P}(f, x)$ is equivalent to $\mathbf{P}(h \circ f, x)$ for $x \in X$.*
2° *For every morphism $f : X \to Y$, every étale morphism $g : Y' \to Y$, every point $x \in X$, if we set $X' = X \times_Y Y'$, $f' = f_{(Y')} : X' \to Y'$ and if $x' \in X'$ is above x, the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(f', x')$ are equivalent ("invariance by étale base change").*

Let then S be a prescheme, $f : X \to S$ and $h : Y \to S$ two morphisms, $g : X \to Y$ an S-morphism, x a point of X, $y = g(x)$, and suppose h étale at the point y. Then the properties $\mathbf{P}(f, x)$ and $\mathbf{P}(g, x)$ are equivalent.

Proof. With the same notation as in the proof of (18.4.9), and after replacing S (respectively, Y) by a neighbourhood of $s = h(y)$ (respectively, of y), (18.4.8) gives an étale morphism $u : S' \to S$ such that h' is an open immersion. If $x' \in X'$ lies above x, the hypothesis makes $\mathbf{P}(f', x')$ (respectively, $\mathbf{P}(g', x')$) equivalent to $\mathbf{P}(f, x)$ (respectively, $\mathbf{P}(g, x)$). We may therefore reduce to the case where h is an open immersion; condition 1° then makes $\mathbf{P}(g, x)$ equivalent to $\mathbf{P}(h \circ g, x) = \mathbf{P}(f, x)$. □

(18.4.11). *Examples.* — One can take for the property $\mathbf{P}(f, x)$ any one of the following, taking into account (17.7.4), (ii):

(i) f is flat at the point x (2.2.11), (iv);
(ii) f is locally of finite presentation and has codepth $\leqslant n$ at the point x ((6.8.1) and (6.7.8));
(iii) f is locally of finite presentation and Cohen–Macaulay at the point x ((6.8.1) and (6.7.8));
(iv) f is locally of finite presentation and possesses the property (S_n) at the point x ((6.8.1) and (6.7.8));
(v) f is locally of finite presentation and possesses the property (R_n) at the point x ((6.8.1) and (6.7.8));
(vi) f is locally of finite presentation and normal at the point x ((6.8.1) and (6.7.8));
(vii) f is locally of finite presentation and reduced at the point x ((6.8.1) and (6.7.8));
(viii) f is unramified at the point x (17.7.4);

(ix) f is smooth at the point x (17.7.4);
(x) f is étale at the point x (17.7.4).

IV | 124

Corollary (18.4.12). — *(i) Let $f : X \to Y$ be a morphism locally of finite type, x a point of X. If f is flat and formally unramified at the point x, then, in every factorisation $f|U : U \xrightarrow{j} X' \xrightarrow{h} Y$, where U is an open neighbourhood of x, j a closed immersion and h an étale morphism, the homomorphism $\mathscr{O}_{X',j(x)} \to \mathscr{O}_{X,x}$ corresponding to j is bijective (in particular $\mathscr{O}_{X,x}$ is an essentially étale $\mathscr{O}_{Y,f(x)}$-algebra (18.6.1)).*

(ii) For a morphism $f : X \to Y$ to be étale, it is necessary and sufficient that it be locally of finite type, formally unramified and flat.

Proof. (i) Since the homomorphism $\mathscr{O}_{X',j(x)} \to \mathscr{O}_{X,x}$ is surjective, it suffices to prove that it is injective, and for that it suffices to prove that $\mathscr{O}_{X,x}$ is an $\mathscr{O}_{X',j(x)}$-module *faithfully flat*, or merely *flat* ((**0**, 6.5.1) and (6.6.2)); in other words, it amounts to showing that j is a flat morphism at the point x; but since $h \circ j = f$ is flat at the point x by hypothesis and h is étale, this follows from (18.4.10) and (18.4.11), (i).

(ii) It remains only to prove the sufficiency of the stated conditions. For every $x \in X$, there is an open neighbourhood U of x on which $f|U$ admits a factorization having the properties in (i). Since f is flat and formally unramified at *every* point of U, part (i) applies not only at x but at every $z \in U$. Thus, if $\mathscr{J}$ is the quasi-coherent ideal of $\mathscr{O}_{X'}$ defining the closed subprescheme U associated with j, then $\mathscr{J}_{j(z)} = 0$ for every $z \in U$; consequently j is an open immersion, and f is étale at every point of U, hence at every point of X. □

Corollary (18.4.13). — *Let $f : X \to Y$ be a morphism locally of finite type, x a point of X, $y = f(x)$. Suppose that y admits an open neighbourhood which is a reduced prescheme and has only a finite number of irreducible components. Then, for f to be étale at the point x, it is necessary and sufficient that f be flat and formally non ramified at the point x.*

Proof. There is only the sufficiency of the condition to prove. The question being local on X and Y, one can suppose (18.4.7) that f factorises as $X \xrightarrow{j} X' \xrightarrow{h} Y$ where h is étale and j a closed immersion. Then X' is reduced (17.5.7) and, replacing X' by an open neighbourhood of $j(x)$, one can suppose that X' has only a finite number of irreducible components. Then the maximal points of X' are above the maximal points of Y (2.3.4) and since they are finite, their number is finite. It all amounts to showing, with the notations of the proof of (18.4.12), that $\mathscr{J}_{x'} = 0$ for all points x' in a neighbourhood of $j(x)$ in X', knowing that $\mathscr{J}_{j(x)} = 0$. Now, replacing X' by an affine neighbourhood of $j(x)$, we can suppose that all the irreducible components of X' contain $j(x)$; if $X' = \operatorname{Spec}(A')$, and if $\mathfrak{p}'$ is the prime ideal of A' corresponding to the point $j(x)$, the morphism $\operatorname{Spec}(A'_{\mathfrak{p}'}) \to \operatorname{Spec}(A')$ is dominant, hence the corresponding homomorphism $A' \to A'_{\mathfrak{p}'}$ is *injective* since A' is reduced (**I**, 1.2.7). If $\mathscr{J} = \widetilde{\mathfrak{J}}$, where $\mathfrak{J}$ is an ideal of A', then $\mathfrak{J}$ identifies with a submodule of $A'_{\mathfrak{p}'}$; since $\mathfrak{J}_{\mathfrak{p}'} = 0$ by hypothesis, it follows that $\mathfrak{J} = 0$. □

Corollary (18.4.14). — *Let A be a local ring with maximal ideal $\mathfrak{m}$, residue field k, B a finite A-algebra.* **IV | 125**

(i) For B to be a formally unramified A-algebra, it is necessary and sufficient that $B \otimes_A k$ be a k-algebra étale.

*(ii) For B to be a finite étale A-algebra, it is necessary and sufficient that $B \otimes_A k$ be an étale k-algebra and that B be a flat A-module (which is equivalent to saying ((**0**, 6.3.3)) that for every maximal ideal $\mathfrak{n}$ of B (necessarily above $\mathfrak{m}$), $B_{\mathfrak{n}}$ is a flat A-module).*

Proof. Only sufficiency needs proof. It is clear that (ii) follows from (i) and (18.4.12), (ii), taking into account (17.1.2), (i). To prove (i), let us note that if $B \otimes_A k$ is an étale k-algebra, we have $\Omega^1_{B\otimes_A k/k} = 0$ (17.2.1). Since $\Omega^1_{B\otimes_A k/k} = \Omega^1_{B/A} \otimes_B k$ ((**0**, 20.5.5)) and since B is an A-algebra of finite type, $\Omega^1_{B/A}$ is a B-module of finite type ((**0**, 20.4.7)). But since B is a finite A-algebra, $\mathfrak{m}B$ is contained in the radical of B (Bourbaki, *Alg. comm.*, ch. V, § 2, no. 1, prop. 1), hence the lemma of Nakayama proves that $\Omega^1_{B/A} = 0$, and hence B is a formally unramified A-algebra (17.2.1). □

18.5. Henselian local rings. [8]

(18.5.1). Let X be a prescheme and $\mathscr{E}$ a locally free $\mathscr{O}_X$-module of finite rank. Its dual $\check{\mathscr{E}} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{E}, \mathscr{O}_X)$ is therefore a locally free $\mathscr{O}_X$-module whose rank at every point equals that of $\mathscr{E}$ there, and the canonical homomorphism $\mathscr{E} \to \mathscr{H}om_{\mathscr{O}_X}(\check{\mathscr{E}}, \mathscr{O}_X) = (\check{\mathscr{E}})^\vee$ is *bijective*. For every morphism $X' \to X$, set $\mathscr{E}_{(X')} = \mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'}$, which is a locally free $\mathscr{O}_{X'}$-module, and consider the *set* $\Gamma(X', \mathscr{E}_{(X')})$ of its sections over X'. We shall see that this defines a *contravariant representable functor*

(18.5.1.1) $$^tV_{\mathscr{E}} : X' \longmapsto \Gamma(X', \mathscr{E}_{(X')})$$

from the category of X-preschemes to the category of *sets* (EGA 0_{III}, 8.1.8).

First of all, this is indeed a well-defined functor, for if $f : X'' \to X'$ is an X-morphism, we have $\mathscr{E}_{(X'')} = f^*(\mathscr{E}_{(X')})$, whence ((**0**, 4.3.2)) a map $\Gamma(X', \mathscr{E}_{(X')}) \to \Gamma(X'', \mathscr{E}_{(X'')})$ which completes the definition of the functor ^tV. Let us then show that the X-prescheme $\mathbf{V}(\check{\mathscr{E}})$ ((**II**, 1.7.8)) represents the functor ^tV. It is immediate that $(\mathscr{E}_{(X')})^\vee = (\check{\mathscr{E}})_{(X')}$, hence $\mathbf{V}(\check{\mathscr{E}}) \times_X X' = \mathbf{V}((\check{\mathscr{E}})_{(X')})$. Taking (**I**, 3.3.14) into account, it suffices to define a bijection

$$\Gamma(\mathbf{V}(\check{\mathscr{E}})/X) \xrightarrow{\sim} \Gamma(X, \mathscr{E})$$

and to verify that the corresponding bijection $\Gamma(\mathbf{V}((\check{\mathscr{E}})_{(X')})/X') \xrightarrow{\sim} \Gamma(X', \mathscr{E}_{(X')})$ is functorial in X'. There is a canonical bijection from $\Gamma(\mathbf{V}(\check{\mathscr{E}})/X)$ to $\operatorname{Hom}_{\mathscr{O}_X}(\check{\mathscr{E}}, \mathscr{O}_X)$ ((**II**, 1.7.8)), and transposition $u \mapsto {}^tu$ gives a canonical bijection from $\operatorname{Hom}_{\mathscr{O}_X}(\check{\mathscr{E}}, \mathscr{O}_X)$ to $\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{E}) = \Gamma(X, \mathscr{E})$, because $(\check{\mathscr{E}})^\vee$ is identified with $\mathscr{E}$. The functoriality is immediate.

Observe also that, in accordance with the general theory (EGA 0_{III}, 8.1.6), the identity automorphism of $\mathbf{V}(\check{\mathscr{E}})$ corresponds canonically to a section c of $\mathscr{E}_{(\mathbf{V}(\check{\mathscr{E}}))}$ over $\mathbf{V}(\check{\mathscr{E}})$. Equivalently ((**II**, 1.4.1)), it corresponds to a homomorphism of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$-modules **IV** | 126

$$u : \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \longrightarrow \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \otimes_{\mathscr{O}_X} \mathscr{E}.$$

For every affine open $W \subset X$, if $\Gamma(W, \mathscr{O}_X) = A$ and $\Gamma(W, \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}))$ is identified with $C = A[\mathrm{T}_1, \dots, \mathrm{T}_n]$ so that the T_i form a basis of $\Gamma(W, \check{\mathscr{E}})$, and if (e_i) denotes the dual basis of $\Gamma(W, \mathscr{E})$, then u corresponds to the C-module homomorphism satisfying

$$u(1) = \sum_{i=1}^{n} \mathrm{T}_i \otimes e_i.$$

If $X' = \operatorname{Spec}(A)$ is affine and $\mathscr{E}_{(X')} \simeq \mathscr{O}_{X'}^n$, then $\Gamma(X', \mathscr{E}_{(X')})$ is a free A-module of rank n; figuratively, $\mathbf{V}(\check{\mathscr{E}})$ represents "the set of points of the twisted affine space over X defined by $\mathscr{E}$".

(18.5.2). Recall ((**II**, 1.7.8)) that by definition $\mathbf{V}(\check{\mathscr{E}}) = \operatorname{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}))$. Let $\mathscr{J}$ be a quasi-coherent ideal of $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, so that $\operatorname{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ is a closed subprescheme of $\mathbf{V}(\check{\mathscr{E}})$; we shall interpret it as *representing a functor* from the category of X-preschemes to that of sets. A section $u \in \Gamma(X, \mathscr{E})$ identifies canonically with an $\mathscr{O}_X$-homomorphism $u : \mathscr{O}_X \to \mathscr{E}$, to which correspond by transposition an $\mathscr{O}_X$-homomorphism ${}^tu : \check{\mathscr{E}} \to \mathscr{O}_X$ and hence an $\mathscr{O}_X$-algebra homomorphism $v : \mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}}) \to \mathscr{O}_X$. Let $\operatorname{Al}(X, \mathscr{E}, \mathscr{J})$ denote the set of $u \in \Gamma(X, \mathscr{E})$ such that $\mathscr{J}$ is *contained in the kernel* of v; it follows immediately that

$$X' \longmapsto \operatorname{Al}(X', \mathscr{E}_{(X')}, \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'})$$

is represented by $\operatorname{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$. If $X' = \operatorname{Spec}(A)$ is affine, $\mathscr{E}_{(X')} \simeq \mathscr{O}_{X'}^n$, and $\mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{O}_{X'} = \widetilde{\mathfrak{J}}$ for an ideal $\mathfrak{J}$ of $A[\mathrm{T}_1, \dots, \mathrm{T}_n]$, then this set identifies with the subset of A^n consisting of the points $(t_1, \dots, t_n)$ such that $P(t_1, \dots, t_n) = 0$ for every $P \in \mathfrak{J}$. Thus $\operatorname{Spec}(\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})/\mathscr{J})$ represents "the algebraic subset of the twisted affine space defined by $\mathscr{E}$ consisting of the points annihilating $\mathscr{J}$". We also denote this X-prescheme by $\mathbf{Al}(\mathscr{E}, \mathscr{J})$. If $\mathscr{J}$ is an ideal of finite type in $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$, then $\mathbf{Al}(\mathscr{E}, \mathscr{J})$ is an X-prescheme of finite presentation, since $\mathbf{S}_{\mathscr{O}_X}(\check{\mathscr{E}})$ is an $\mathscr{O}_X$-algebra of finite presentation.

[8] The notion of a Henselian local ring is due to Azumaya, and that of Henselization to Nagata, to whom the principal results of this theory are also due.

Lemma (18.5.3). — *Let S be a prescheme, $f : X \to S$ a finite and locally free morphism ((18.2.7)). Consider the contravariant functor from the category of S-preschemes to the category of sets*

(18.5.3.1) $$S' \longmapsto \mathrm{Of}(X \times_S S')$$

where $\mathrm{Of}(X \times_S S')$ denotes the set of subsets that are both open and closed of the underlying space of $X \times_S S'$. Then this functor is representable by an S-prescheme $\mathbf{Of}(X)$, which is affine, étale and of finite presentation over S.

Proof. By hypothesis $X = \mathrm{Spec}(\mathscr{B})$, where $\mathscr{B} = f_*(\mathscr{O}_X)$ is a finite and locally free $\mathscr{O}_S$-algebra. Set $X' = X \times_S S'$, $f' = f_{(S')}$, $\mathscr{B}' = \mathscr{B} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'} = f'_*(\mathscr{O}_{X'})$, so that $X' = \mathrm{Spec}(\mathscr{B}')$; there is then a canonical bijection, functorial in S', from the set $\mathrm{Of}(X')$ to the set $\mathrm{Id}(\mathscr{B}')$ of *idempotents* of the ring $\Gamma(S', \mathscr{B}') = \Gamma(X', \mathscr{O}_{X'})$. Indeed, by virtue of the equivalence of the category of affine S'-schemes and the opposite category of quasi-coherent $\mathscr{O}_{S'}$-algebras ((**II**, 1.2.7) and (1.3.1)), there is a canonical bijective correspondence between decompositions of X' as a *disjoint union* $X'_1 \sqcup X'_2$ of two induced open subpreschemes and decompositions of $\mathscr{B}'$ as a *direct sum* of two ideals $\mathscr{B}'_1$, $\mathscr{B}'_2$; these latter in turn form a set in canonical bijective correspondence with $\mathrm{Id}(\mathscr{B}')$. It suffices therefore to prove the lemma for the functor $S' \mapsto \mathrm{Id}(\mathscr{B}')$. IV | 127

For this, we shall show that there exists an ideal $\mathscr{J}$ of finite type in $\mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}})$ such that $\mathrm{Id}(\mathscr{B}')$ is of the form $\mathrm{Al}(S', \mathscr{B}', \mathscr{J} \otimes_{\mathscr{O}_S} \mathscr{O}_{S'})$ ((18.5.2)). Since $\mathscr{B}$ is an $\mathscr{O}_S$-algebra, its inverse image $\mathscr{B}_{\mathbf{V}(\check{\mathscr{B}})}$ is an $\mathscr{O}_{\mathbf{V}(\check{\mathscr{B}})}$-algebra, and we may form in this algebra the square c^2 of the canonical section c ((18.5.1)); it corresponds canonically to a homomorphism $u^{(2)} : \mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}}) \to \mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}}) \otimes_{\mathscr{O}_S} \mathscr{B}$, and one verifies immediately that for every affine open subset W of S, with the notations of (18.5.1), $u^{(2)}$ corresponds to the C-module homomorphism such that $u^{(2)}(1) = \sum_k (\sum_{i,j} c_{ijk} \mathrm{T}_i \mathrm{T}_j) \otimes e_k$, where (c_{ijk}) is the table of multiplication of the algebra $\Gamma(W, \mathscr{B})$. Let $\mathscr{J}$ be the ideal of $\mathbf{S}_{\mathscr{O}_S}(\check{\mathscr{B}})$ locally generated by the coefficients of $(u - u^{(2)})(1)$ with respect to a basis of $\mathscr{B}$; it answers the question. Indeed, $\Gamma(W, \mathscr{J})$ is generated by the polynomials $\mathrm{P}_k(\mathrm{T}_1, \ldots, \mathrm{T}_n) = \mathrm{T}_k - \sum_{i,j} c_{ijk} \mathrm{T}_i \mathrm{T}_j$, and the idempotents of $\Gamma(W, \mathscr{B})$ are precisely the elements $\sum_i t_i e_i$ such that $(t_1, \ldots, t_n)$ annihilates every P_k $(1 \leqslant k \leqslant n)$. We deduce from (18.5.2) that the S-prescheme affine $\mathbf{Of}(X) = \mathbf{Al}(\mathscr{B}, \mathscr{J})$ does represent the functor (18.5.3.1). It remains to show that $\mathbf{Of}(X)$ is étale over S, or equivalently, that it is *formally étale* over S. But if S' is an S-prescheme, S'_0 a closed sub-prescheme of S' defined by a locally nilpotent Ideal (and having the same underlying space as S'), it is clear that $X' = X \times_S S'$ and $X'_0 = X \times_S S'_0$ have the same underlying space, hence the canonical map $\mathrm{Of}(X') \to \mathrm{Of}(X'_0)$ is *bijective*, which completes the proof (17.1.1). □

Proposition (18.5.4). — *Let S be a prescheme, S_0 a closed sub-prescheme of S; consider the following properties:*

a) *For every finite morphism $g : S' \to S$, the canonical map*

(18.5.4.1) $$\mathrm{Of}(S') \longrightarrow \mathrm{Of}(S' \times_S S_0)$$

is bijective (*cf.* (18.5.3)).

a′) *For every finite and locally free morphism $g : S' \to S$, the map* (18.5.4.1) *is bijective.* IV | 128

b) *For every étale and separated morphism $g : S' \to S$, the canonical map*

(18.5.4.2) $$\Gamma(S'/S) \longrightarrow \Gamma(S' \times_S S_0/S_0)$$

is bijective.

Condition *b*) implies a′); if moreover S is quasi-compact and quasi-separated, condition *a*) implies *b*).

Proof. Let us first prove that *b*) implies *a′*). Suppose therefore *b*) satisfied, and let $g : S' \to S$ be a finite and locally free morphism; set $S'_0 = S' \times_S S_0$, so that $g_0 = g_{(S_0)} : S'_0 \to S_0$ is finite and locally free. Then it follows from (18.5.3) that $\mathrm{P} = \mathbf{Of}(S')$ is an S-prescheme *étale and separated*; moreover, by the definition of the functor Of, we see immediately that if we set $\mathrm{P}_0 = \mathbf{Of}(S'_0)$ (for the category of S_0-preschemes), we have $\mathrm{P}_0 = \mathrm{P} \times_S S_0$. This being so, by definition we have the commutative diagram

$$\begin{array}{ccc} \Gamma(\mathrm{P}/S) & \longrightarrow & \Gamma(\mathrm{P}_0/S_0) \\ {\scriptstyle\sim}\big\downarrow & & {\scriptstyle\sim}\big\downarrow \\ \mathrm{Of}(S') & \longrightarrow & \mathrm{Of}(S'_0) \end{array}$$

where the vertical arrows are the bijections. Since hypothesis b), applied to the morphism $\mathrm{P} \to S$, implies that the first row is a bijection, the same holds for the second, which establishes our assertion.

Before proving that a) implies b) when S is quasi-compact and quasi-separated, we will establish the following lemma.

Lemma (18.5.4.3). — *If S and S_0 satisfy condition a) of (18.5.4), then, for every finite morphism $g : S' \to S$, S' is the unique open and closed neighbourhood of $S'_0 = g^{-1}(S_0) = S' \times_S S_0$ in S'.*

Proof. Indeed, it amounts to saying that if T' is a closed subset of S' such that $T' \cap S'_0 = \varnothing$, then $T' = \varnothing$. Now, if we also denote by T' the closed sub-prescheme of S' having T' as underlying space, the composite morphism $h : T' \to S' \xrightarrow{g} S$ is finite and $h^{-1}(S_0)$ is empty; condition a), applied to the morphism h, implies that T' is necessarily empty. □

This lemma being established, let us first prove that, under hypothesis a), the map (18.5.4.2) is *injective*. Indeed, if u', u'' are two S-sections of S', the fact that the morphism $S' \to S$ is unramified implies that the prescheme of coincidences of u' and u'' is induced on an open subset U of S (17.4.6). If the restrictions to S_0 of u' and u'' are equal, then the fact that u' and u'' are open immersions (17.4.1) implies that U contains S_0, and hence is equal to S by virtue of Lemma (18.5.4.3), applied to the case $S' = S$.

It remains to show that, under hypothesis a), the map (18.5.4.2) is *surjective* (when S is quasi-compact and quasi-separated). Let therefore $u_0 : S_0 \to S'_0$ be an S_0-section of S'_0; $u_0(S_0)$ being quasi-compact in S', can be covered by a finite number of open affine subsets V_i such that the restriction $g|V_i$ is a morphism of finite type; if V is the union of the V_i, we conclude that $g|V : V \to S$ is a morphism *of finite presentation*, being separated and locally of finite presentation (1.6.1). Replacing S' by V, one can hence suppose that g is of finite presentation. Since S is quasi-compact and quasi-separated, and g is quasi-finite and separated (17.6.1), it follows from the "Main theorem" (8.12.6) that g factorises as $S' \xrightarrow{j} S'' \xrightarrow{f} S$, where j is an open immersion and f a *finite* morphism. Set $S''_0 = S'' \times_S S_0$, $j_0 = j_{(S_0)} : S'_0 \to S''_0$, which is an open immersion, and $f_0 = f_{(S_0)} : S''_0 \to S_0$, which is a finite morphism. Then u_0 is also an S_0-section of S''_0. Since $g_0 : S'_0 \to S_0$ is étale, u_0 is an open immersion of S_0 into S'_0 (17.4.1), so $u_0(S_0)$ is open in S'_0, and *a fortiori* in S''_0; on the other hand, since f_0 is finite and hence separated, u_0 is a closed immersion of S_0 into S''_0 (**I**, 5.4.6). Thus $X_0 = u_0(S_0)$ is *both open and closed* in S''_0. In virtue of condition a), there exists an open and closed subset X of S'' such that $X \cap S''_0 = X_0$. Let us show that the morphism $f : S'' \to S$ is *étale at the points of* X: indeed, the set of points of X where $f|X$ is étale is open and contains $X_0 \subset S'_0$. But the lemma (18.5.4.3) applied to the finite morphism $f|X$ proves that $U = X$. On the other hand, $S' \cap X$ is open in X and contains X_0, hence by the same reasoning proves that $S' \cap X = X$, i.e. $X \subset S'$. It remains to show that, for every $s \in S$, the geometric number $n(s)$ of points of $X \cap f^{-1}(s)$ is equal to 1; it will then follow that $f|X$ is radicial and surjective, and, since $f|X$ is étale, we shall have proved (17.9.1) that $f|X$ is an isomorphism from the open subset $X \subset S'$ onto S, whose inverse will be the desired S-section extending u_0. But since $f|X$ is étale and finite, $s \mapsto n(s)$ is continuous on S (18.2.8), and since $X \cap S''_0 = X_0$, we have $n(s) = 1$ on S_0; the set of points $s \in S$ such that $n(s) = 1$ is therefore open and closed in S and contains S_0, so it is equal to S by (18.5.4.3). □ **IV** | 129

Remark (18.5.4.4). — One can show that the statement (18.5.4) remains valid when, in condition b), one supposes merely the morphism g étale (but not necessarily separated) [**?**, exp. XII].

Definition (18.5.5). — We say that a prescheme S and a closed sub-prescheme S_0 of S form a *Henselian couple* if they satisfy condition a) of (18.5.4).

Taking into account (**I**, 5.1.8), it amounts to the same to say that (S, S_0) is a Henselian couple or that $(S_{\mathrm{red}}, (S_0)_{\mathrm{red}})$ is one.

Proposition (18.5.6). — *(i) If (S, S_0) is a Henselian couple, then, for every finite morphism $f : S' \to S$, if S'_0 is the closed sub-prescheme $f^{-1}(S_0)$ of S', the couple (S', S'_0) is Henselian.*

(ii) Let $S = \coprod_\alpha S^{(\alpha)}$ be a sum of preschemes, S_0 a closed sub-prescheme of S, the sum of sub-preschemes $S_0^{(\alpha)}$ of the $S^{(\alpha)}$. For the couple (S, S_0) to be Henselian, it is necessary and sufficient that each of the couples $(S^{(\alpha)}, S_0^{(\alpha)})$ be Henselian.

Proof. Assertion (i) is an immediate consequence of the definition, since for every finite morphism $g : S'' \to S'$, $f \circ g : S'' \to S$ is a finite morphism. Similarly, under the conditions of (ii), for a morphism $g : S' \to S$ to be finite, it is necessary and sufficient that each of its restrictions $g^{(\alpha)} :$ **IV** | 130
$S'^{(\alpha)} = g^{-1}(S^{(\alpha)}) \to S^{(\alpha)}$ be finite. If we set $S'_0 = g^{-1}(S_0)$ and $S_0'^{(\alpha)} = (g^{(\alpha)})^{-1}(S_0^{(\alpha)})$, the open-and-closed subsets U (resp. U_0) of S' (resp. S'_0) correspond bijectively to the families $(U^{(\alpha)})$ (resp. $(U_0^{(\alpha)})$), where $U^{(\alpha)}$ (resp. $U_0^{(\alpha)}$) is an open-and-closed subset of $S'^{(\alpha)}$ (resp. $S_0'^{(\alpha)}$); assertion (ii) follows. □

Remark (18.5.7). — Let $S = \mathrm{Spec}(A)$ be an affine scheme, S_0 a closed sub-prescheme defined by an ideal $\mathfrak{J}$ of A. Then, if the couple (S, S_0) is Henselian, the ideal $\mathfrak{J}$ is necessarily contained in the *radical* of A. Indeed, if $\mathfrak{m}$ is a maximal ideal of A, $\mathfrak{m}$ must belong to $V(\mathfrak{J}) = S_0$, by virtue of (18.5.4.3), whence the conclusion. In particular, suppose that S_0 is *reduced to a point*, then $\mathfrak{J}$ must be the radical of A, so A must be a *local* ring, and S_0 the unique closed point of $\mathrm{Spec}(A)$.

Definition (18.5.8). — We say that a ring A is *Henselian* if it is semi-local and if, denoting by $\mathfrak{r}$ the radical of A, the couple $(\mathrm{Spec}(A), \mathrm{Spec}(A/\mathfrak{r}))$ is Henselian. We call a *Henselian local scheme* a scheme isomorphic to the spectrum of a Henselian local ring.

Proposition (18.5.9). — *(i) For a semi-local ring A to be Henselian, it is necessary and sufficient that it be a direct product of Henselian local rings.*

(ii) For a local ring A to be Henselian, it is necessary and sufficient that every finite A-algebra B be isomorphic to a product of local rings.

Proof. (i) Indeed, the definition, applied to $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/\mathfrak{r})$, shows that S_0 is a finite discrete set, closed in S, and S is the union of a finite number of open and closed subsets S_i $(1 \leqslant i \leqslant n)$, pairwise disjoint, each containing exactly one of the maximal ideals $\mathfrak{m}_i$ of A; the conclusion follows from (18.5.6), (ii) and the remark (18.5.7).

(ii) Every finite morphism $S' \to \mathrm{Spec}(A)$ is of the form $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$, where B is a finite A-algebra. If k is the residue field of A, $\mathrm{Spec}(B \otimes_A k)$ is a finite Artinian spectrum, hence finite and discrete. To say that the couple $(\mathrm{Spec}(A), \mathrm{Spec}(k))$ is Henselian means that B is a direct product of rings A_i such that $\mathrm{Spec}(A_i \otimes_A k)$ is reduced to a point, i.e. that each A_i (which is a finite A-algebra) must have only a single maximal ideal (Bourbaki, *Alg. comm.*, ch. V, § 2, no. 1, prop. 1). □

The study of Henselian rings is thus essentially reduced to that of Henselian *local* rings.

Proposition (18.5.10). — *If A is a Henselian ring, every finite A-algebra B is a Henselian ring (hence a direct product of Henselian local rings (18.5.9)).*

Proof. Indeed, if $\mathfrak{r}'$ is the radical of B, the inverse image of $\mathfrak{r}'$ in A is the radical $\mathfrak{r}$ of A, and every prime ideal of B lying over a maximal ideal of A is a maximal ideal of B; hence the set $V(\mathfrak{r}')$ in $\mathrm{Spec}(B)$ is the inverse image of the set $V(\mathfrak{r})$ in $\mathrm{Spec}(A)$. The proposition is then a consequence of (18.5.6), (i). □

Theorem (18.5.11). — *Let A be a local ring, $\mathfrak{m}$ its maximal ideal. The following conditions are equivalent:*

a) *A is Henselian, in other words, every finite A-algebra B is isomorphic to a product of local rings.* **IV** | 131

a′) Condition a) is satisfied for all A-algebras B of the form $A[\mathrm{T}]/F.A[\mathrm{T}]$, where $F \in A[\mathrm{T}]$ is a monic polynomial.

b) *Let $S = \mathrm{Spec}(A)$, $S_0 = \mathrm{Spec}(k)$. For every étale morphism $g : S' \to S$, if we set $S'_0 = S' \otimes_A k$, every S_0-section u_0 of S'_0 is the restriction of an S-section u of S'.*

c) *For every morphism locally of finite type $f : X \to S$, separated, and every point $x \in X$ such that $f(x)$ is equal to the closed point s of S and that f is quasi-finite at the point x ((**III**, Err.20)), X is the sum of two preschemes X', X'' such that $X' = \operatorname{Spec}(\mathscr{O}_{X,x})$ and that $f|X' : X' \to S$ is a finite morphism.*

c') *For every morphism locally of finite type $f : X \to S$, and every point $x \in X$ such that f be quasi-finite at the point x and $f(x)$ equal to the closed point s of S, $\mathscr{O}_{X,x}$ is a finite algebra over $\mathscr{O}_{S,s} = A$.*

c'') *For every morphism locally of finite presentation $f : X \to S$, and every point $x \in X$ such that f be quasi-finite at the point x and $f(x)$ equal to the closed point s of S, $\mathscr{O}_{X,x}$ is a finite algebra and of finite presentation over $\mathscr{O}_{S,s} = A$.*

Proof. Let us first note that c') (resp. c'') is equivalent to the same condition where moreover f is supposed *separated*, the question being local on X. Similarly, condition b) is equivalent to the same condition where moreover g is supposed *separated*: indeed, it suffices to apply the latter to the restriction of g to an open neighbourhood affine of the point $u_0(S_0)$ in S'. Let us henceforth restrict to the case where, in b), c') and c'', the morphisms are separated.

The fact that a) implies b) and that b) implies a') is a particular case of (18.5.4). Let us moreover show that a') implies a). It is a matter of proving that if e_0 is an idempotent of $C = B \otimes_A k$, there is an idempotent $e \in B$ whose image is e_0 in C. If b is an element of B whose image in C is e_0, the sub-A-algebra $A[b] = B'$ of B is finite, and the canonical image C' of $B' \otimes_A k$ in C contains e_0. Now, $B' \otimes_A k$ is a finite k-algebra, hence a direct product of finite local k-algebras, and consequently e_0 is the image in C' of an idempotent e'_0 of $B' \otimes_A k$. We are thus reduced to the case where B is monogenic, and hence by a'), a quotient of an algebra of the form $A[\mathrm{T}]/F.A[\mathrm{T}]$; now, by a'), $A[\mathrm{T}]/F.A[\mathrm{T}]$ is a direct product of local rings, hence so is each of its quotient algebras, which proves the existence of the idempotent e.

It is immediate that c) implies a), as one sees by induction on the number of maximal ideals of the semi-local ring B. To see that a) implies c), we may, by (13.1.4), restrict to the case where f is affine and quasi-finite. Then, by the "Main theorem" (8.12.8), f can be written as the composite

$$X \xrightarrow{j} Y \xrightarrow{g} S,$$

where g is finite and j is an open immersion. Since $Y = \operatorname{Spec}(B)$, where B is a finite A-algebra, condition a) implies that B is a direct product of local rings, which are evidently finite A-algebras, and $\mathscr{O}_{X,x}$ identifies with one of these local rings because $f(x) = s$; moreover, every open subset of Y containing x necessarily contains $\operatorname{Spec}(\mathscr{O}_{X,x})$.

It is trivial that c) implies c'), by virtue of the remark at the beginning. Let us prove that c') implies c''). Suppose c') satisfied and let us prove that, under the hypotheses of c''), the subset **IV** | 132 $Z = \operatorname{Spec}(\mathscr{O}_{X,x})$ identifies with an *open and closed* subset of X, which will establish c'') (**I**, 2.4.2). First, the composite morphism $Z \xrightarrow{j} X \xrightarrow{f} S$, where j is the canonical morphism (**I**, 2.4.1), is finite and of finite presentation by hypothesis; and since f is separated and locally of finite presentation, j is also finite and of finite presentation ((**II**, 6.1.5) and (**I**, 4.3)), hence j is a closed morphism (**II**, 6.1.10), which proves that Z is closed in X. It then follows from (**I**, 2.4.2) and (**I**, 4.2.2) that j is a closed immersion. If $\mathscr{J}$ is the ideal of $\mathscr{O}_X$ defining Z, then by hypothesis $\mathscr{J}_x = 0$, and hence $\mathscr{J}_z = 0$ at every point z of some open neighbourhood V of x in X, since $\mathscr{J}$ is a quasi-coherent $\mathscr{O}_X$-module of finite type ((1.4.7) and (**0**, 5.2.2)). Such a neighbourhood contains Z, so Z is open in X, since $\mathscr{J}|V = 0$.

Finally, c'') implies a'): indeed, if $B = A[\mathrm{T}]/F.A[\mathrm{T}]$, the morphism $X = \operatorname{Spec}(B) \to \operatorname{Spec}(A) = Y$ is finite and of finite presentation, and the preceding argument shows that B is a direct product of finite A-algebras $\mathscr{O}_{X,x_i}$, where the x_i are the points of the fibre of the closed point of Y. ☐

Corollary (18.5.12). — *Let A be a semi-local ring, $\mathfrak{r}$ its radical; set $S = \operatorname{Spec}(A)$, $S_0 = \operatorname{Spec}(A/\mathfrak{r})$. For A to be Henselian, it is necessary and sufficient that, for every finite morphism $f : X \to S$ and every étale and separated morphism $g : Y \to S$, if we set $X_0 = X \times_S S_0$ and $Y_0 = Y \times_S S_0$, the canonical map*

$$\operatorname{Hom}_S(X, Y) \longrightarrow \operatorname{Hom}_{S_0}(X_0, Y_0)$$

is bijective.

Proof. The sufficiency of the condition follows from the equivalence of $a)$ and $b)$ in (18.5.11), by applying this condition to the case where $f = 1_S$. To see that the condition is necessary, note that if A is Henselian, the couple (X, X_0) is Henselian by (18.5.6), (i); moreover $\mathrm{Hom}_S(X, Y) = \Gamma(X \times_S Y/X)$ and $\mathrm{Hom}_{S_0}(X_0, Y_0) = \Gamma(X_0 \times_{S_0} Y_0/X_0)$; since $X \times_S Y$ is étale and separated over X, the conclusion follows from the fact that $a)$ implies $b)$ in (18.5.4). □

Remark (18.5.13). — The equivalent conditions $a)$, $a')$, $b)$ of Theorem (18.5.11) are also equivalent to the following ("Hensel's lemma"):

a$''$) For every monic polynomial $F \in A[\mathrm{T}]$, with canonical image $F_0 \in k[\mathrm{T}]$, and every decomposition $F_0 = G_0 H_0$ of F_0 into a product of two monic polynomials that are coprime G_0, H_0 in $k[\mathrm{T}]$, there exists a unique couple (G, H) of monic polynomials of $A[\mathrm{T}]$ possessing the following properties: G_0 and H_0 are the canonical images respectively of G and H, we have $F = GH$, and the ideal of $A[\mathrm{T}]$ generated by G and H is equal to $A[\mathrm{T}]$.

Lemma (18.5.13.1). — *Let A be a local ring with residue field k, $F \in A[\mathrm{T}]$ a monic polynomial, B the A-algebra $A[\mathrm{T}]/F.A[\mathrm{T}]$. There exists a canonical bijective correspondence between the decompositions of B into the direct product of two quotient A-algebras B', B'' and the decompositions $F = GH$ of F into a product of two monic polynomials G, H of $A[\mathrm{T}]$, such that the ideal generated by G and H is equal to $A[\mathrm{T}]$; the quotient algebras B', B'' corresponding to such a couple of polynomials G, H are respectively $A[\mathrm{T}]/H.A[\mathrm{T}]$ and $A[\mathrm{T}]/G.A[\mathrm{T}]$.*

Proof. If $F = GH$ and G and H generate the ideal $A[\mathrm{T}]$, there are two polynomials P, Q of $A[\mathrm{T}]$ such that $1 = PG + QH$. We deduce that the intersection of the principal ideals $\mathfrak{a} = G.A[\mathrm{T}]$ and $\mathfrak{b} = H.A[\mathrm{T}]$ is equal to $\mathfrak{c} = F.A[\mathrm{T}]$: indeed, if $R \in G.A[\mathrm{T}] \cap H.A[\mathrm{T}]$, we can write $R = PRG + QRH$; since RG (resp. RH) is a multiple of F since R is a multiple of H (resp. G), therefore $R \in F.A[\mathrm{T}]$. Since $A[\mathrm{T}] = \mathfrak{a} + \mathfrak{b}$, $A[\mathrm{T}]/\mathfrak{c}$ is the direct sum of ideals $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{b})$ and $\mathfrak{b}/(\mathfrak{a} \cap \mathfrak{b})$, canonically isomorphic respectively to $A[\mathrm{T}]/\mathfrak{b}$ and $A[\mathrm{T}]/\mathfrak{a}$. **IV** | 133

Conversely, suppose given a decomposition of B into a direct product of two A-algebras B', B'', which identify canonically with two ideals $e'B$, $e''B$ of B, corresponding to a decomposition $1 = e' + e''$ into orthogonal idempotents e', e'' of B. Set further $B_0 = B \otimes_A k = k[\mathrm{T}]/F_0.k[\mathrm{T}]$, where F_0 is the canonical image of F in $k[\mathrm{T}]$, of the same degree n as F; if e'_0, e''_0 are the canonical images of e', e'' in B_0, these are two orthogonal idempotents such that $1 = e'_0 + e''_0$, and B_0 is the direct product of $B'_0 = e'_0 B_0$ and $B''_0 = e''_0 B_0$. Let t and t_0 be the canonical images of T in B and B_0; B'_0 (resp. B''_0) is a k-algebra generated by $t'_0 = e'_0 t_0$ (resp. $t''_0 = e''_0 t_0$), and it admits a base of the form $\{e'_0, t'_0, t'^2_0, \dots, t'^{s-1}_0\}$ (resp. $\{e''_0, t''_0, t''^2_0, \dots, t''^{r-1}_0\}$) with $r + s = n$. On the other hand, B being a free A-module (of base $\{1, t, \dots, t^{n-1}\}$), B' and B'' are projective A-modules, hence free since A is a local ring (Bourbaki, *Alg. comm.*, ch. II, § 5, no. 3, cor. de la prop. 5); it follows from the above and from Bourbaki, *Alg. comm.*, ch. II, § 3, no. 3, prop. 5, that if we set $t' = e't$, $t'' = e''t$, $\{e', t', \dots, t'^{s-1}\}$ (resp. $\{e'', t'', \dots, t''^{r-1}\}$) is a base of the A-module B' (resp. B''). There is hence a monic polynomial H (resp. G) of degree s (resp. r) in $A[\mathrm{T}]$ such that $e'H(t') = 0$ and $e''G(t'') = 0$; as $t^h = t'^h + t''^h$ for every integer $h \geqslant 1$, with $t'^h = e't'^h$ and $t''^h = e''t''^h$, we also have $G(t) = e'G(t')$ and $H(t) = e''H(t'')$, whence $G(t)H(t) = 0$; we conclude that the polynomial $G(T)H(T)$ is divisible by $F(T)$; but since these two monic polynomials have the same degree, we have $GH = F$. Moreover, B' (resp. B'') is isomorphic to $A[\mathrm{T}]/H.A[\mathrm{T}]$ (resp. $A[\mathrm{T}]/G.A[\mathrm{T}]$). Finally, there are two polynomials R, S in $A[\mathrm{T}]$ such that $e' = R(t)$ and $e'' = S(t)$; as $R(t) = e'R(t') + e''R(t'')$, we necessarily have $e''R(t'') = 0$, and similarly $e'S(t') = 0$, so that, by definition of G and H, $R = QH$ and $S = PG$, where P, Q belong to $A[\mathrm{T}]$; the relation $1 = R(t) + S(t)$ in B gives $PG + QH = 1 + LF$ for some polynomial $L \in A[\mathrm{T}]$, and since $F = GH$, this proves that the ideal generated by G and H is equal to $A[\mathrm{T}]$, and completes the proof of the lemma. □

This lemma being established, it suffices to apply it, to the ring A on the one hand, to the field k on the other, to see immediately that conditions $a')$ and $a'')$ are equivalent.

Proposition (18.5.14). — *Every semi-local ring, separated and complete for the $\mathfrak{r}$-preadic topology (where $\mathfrak{r}$ is the radical of A) is Henselian.*

Proof. Indeed, A is a direct product of separated and complete local rings (Bourbaki, *Alg. comm.*, ch. III, § 2, no. 13, cor. of prop. 19), so we are reduced to the case where A is a local ring. Let us verify criterion a') of (18.5.11). Since B is a free A-module of finite type, B is separated and complete for the $\mathfrak{r}$-adic topology, which is also the $\mathfrak{s}$-adic topology, where $\mathfrak{s}$ is the radical of the semi-local ring B, because $B/\mathfrak{r}B$ is an Artinian ring with radical $\mathfrak{s}/\mathfrak{r}B$. It is then known (Bourbaki, *Alg. comm.*, ch. III, § 2, no. 13, cor. of prop. 19) that B is a direct product of local rings. □ **IV | 134**

Proposition (18.5.15). — *Let A be a Henselian ring, $\mathfrak{r}$ its radical. Then the functor $B \mapsto B/\mathfrak{r}B$ is an equivalence of the category of finite étale A-algebras with the category of finite étale $(A/\mathfrak{r})$-algebras.*

Proof. The fact that the functor is fully faithful is a particular case of (18.5.12). To show that this functor is an equivalence of categories, we may reduce to the case where A is local; it suffices then to apply (18.1.1) (for étale morphisms), with $S = \mathrm{Spec}(A)$ and $S_0 = \mathrm{Spec}(A/\mathfrak{r})$, reduced to a single point. □

Remarks (18.5.16). —

(i) We do not know if Proposition (18.5.15) generalises to a Henselian couple (S, S_0), even when S is affine and Noetherian.

(ii) Let A be a Noetherian ring, $\mathfrak{J}$ an ideal of A such that A is *separated and complete* for the $\mathfrak{J}$-preadic topology. Then the couple $(\mathrm{Spec}(A), \mathrm{Spec}(A/\mathfrak{J}))$ is Henselian: indeed, for every finite A-algebra B, B is separated and complete for the $\mathfrak{J}$-preadic topology ((**0**, 7.3.6)). Replacing B by A and $\mathfrak{J}B$ by $\mathfrak{J}$, everything amounts to seeing that the map that to every idempotent of A associates its class mod $\mathfrak{J}$, is *bijective*. Now $A = \varprojlim A/\mathfrak{J}^n$. Write $\mathrm{Idem}(A)$ for the set of idempotents of A, and, for every ring homomorphism $\varphi : A \to B$, let $\mathrm{Idem}(\varphi)$ denote the map $\mathrm{Idem}(A) \to \mathrm{Idem}(B)$ induced by φ. It follows from the definition of the projective limit that

$$\mathrm{Idem}(A) = \varprojlim_n \mathrm{Idem}(A/\mathfrak{J}^n),$$

for the transition maps $\psi_{nm} : \mathrm{Idem}(A/\mathfrak{J}^m) \to \mathrm{Idem}(A/\mathfrak{J}^n)$ induced by the canonical maps $A/\mathfrak{J}^m \to A/\mathfrak{J}^n$. But since $\mathrm{Spec}(A/\mathfrak{J}^n) \to \mathrm{Spec}(A/\mathfrak{J}^m)$ is a homeomorphism, the ψ_{nm} are bijections (as seen in the proof of (18.5.3)); this proves our assertion.

Theorem (18.5.17). — *Let A be a Henselian local ring, $S = \mathrm{Spec}(A)$, s the closed point. For every smooth morphism $f : X \to S$, the canonical map*

$$\Gamma(X/S) \longrightarrow \Gamma(X_s/\kappa(s))$$

(where $X_s = f^{-1}(s)$) is surjective.

Proof. The datum of a $\kappa(s)$-section of X_s is equivalent to that of a point $x \in X$ above s whose residue field $\kappa(x)$ is isomorphic to $\kappa(s)$, and it is a matter of proving that there exists an S-section $u : S \to X$ such that $u(s) = x$. Taking into account (17.16.3), (i), we can suppose that f is étale. Then the conclusion follows from the criterion (18.5.11), b). (The reader will note that by virtue of this criterion, the validity of (18.5.17) for a local ring is necessary and sufficient for A to be Henselian.) □

Remark (18.5.18). — The conditions of (18.5.11) are also equivalent to the following:

d) For every morphism locally of finite type $f : X \to S$ and every point $x \in X$ such that $f(x)$ be equal to the closed point s of S and that $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ be a field canonically isomorphic to $k = \kappa(s)$, there exists an open neighbourhood U of x in X such that $f|U$ is a closed immersion.

Proof. It is immediate that d) implies condition b) of (18.5.11) since a closed immersion étale is an open immersion (18.9.1). Conversely, suppose the conditions of (18.5.11) satisfied, and let us prove d). The hypothesis of d) implies that f is quasi-finite at the point x ((**III**, Err.20)), by virtue of (13.1.4); hence, by condition c) of (18.5.11), $\mathscr{O}_{X,x}$ is a finite A-algebra and $\mathrm{Spec}(\mathscr{O}_{X,x})$ is an open neighbourhood U of x in X. Moreover, since $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ is isomorphic to $k = A/\mathfrak{m}_s$, the Nakayama lemma proves that the homomorphism $A \to \mathscr{O}_{X,x}$ is surjective, hence $f|U$ is a closed immersion. □ **IV | 135**

Proposition (18.5.19). — *Let A be a Noetherian Henselian local ring, $S = \mathrm{Spec}(A)$, s the closed point of S, and $f : X \to S$ a proper morphism; set $X_s = f^{-1}(s)$. Then the map $Y \to Y \cap X_s$ is a bijection from the set of connected components of X to the set of connected components of X_s.*

Proof. By considering a closed sub-prescheme of X having for underlying space a connected component of X, we are reduced to showing that if X is *connected and non-empty*, then X_s is *connected and non-empty*. The fact that X_s is non-empty follows from ((**II**, 7.2.1)); to show that X_s is connected, we argue by contradiction, considering the Stein factorisation $f : X \xrightarrow{f'} S' \xrightarrow{g} S$ of the proper morphism f ((**III**, 4.3.3)); by hypothesis the finite discrete set $g^{-1}(s)$ would contain at least two points. Since A is Henselian and g separated and of finite type, it would then follow from (18.5.11), c) that S' would be the sum of two non-empty preschemes S'_1, S'_2 (since the intersection of any one of them with $g^{-1}(s)$ is reduced to a single point). Since f' is surjective ((**III**, 4.3.1)), we would conclude that X is a sum of two non-empty preschemes, contradicting the hypothesis. □

18.6. Henselization.

(18.6.1). Given a local ring A, we shall say that a local A-algebra B is *essentially étale* if there exist an étale A-algebra C and a prime ideal $\mathfrak{n}$ of C such that B is A-isomorphic to $C_{\mathfrak{n}}$ and the composite homomorphism $A \to C \to C_{\mathfrak{n}}$ is local (in other words, $\mathfrak{n}$ lies above the maximal ideal $\mathfrak{m}$ of A). This implies (17.6.1) that $\mathfrak{n}$ is the only prime ideal of B lying above $\mathfrak{m}$, since B is a local ring; if B, B' are two local essentially étale A-algebras, every A-homomorphism $B \to B'$ is therefore local.

If A' is a local essentially étale A-algebra and A'' a local essentially étale A'-algebra, then A'' is a local essentially étale A-algebra. Indeed, by hypothesis $A' = B_{\mathfrak{n}}$, where B is an étale A-algebra and $\mathfrak{n}$ a prime ideal of B above the maximal ideal of A, and $A'' = B'_{\mathfrak{n}'}$, where B' is an étale A'-algebra and $\mathfrak{n}'$ a prime ideal of B' above $\mathfrak{n}B_{\mathfrak{n}}$. If we set $S = B - \mathfrak{n}$, so that $A' = S^{-1}B$, then B' is of the form $S^{-1}C$, where C is a B-algebra of finite presentation. Consequently C is an A-algebra of finite presentation and A'' is of the form $C_{\mathfrak{r}}$, where $\mathfrak{r}$ is a prime ideal of C above the maximal ideal of A. Since A' is formally étale over A and A'' is formally étale over A', A'' is formally étale over A (for **IV** | 136
the discrete topologies (17.1.3)); the morphism $\operatorname{Spec}(C) \to \operatorname{Spec}(A)$ is therefore étale at the point $\mathfrak{r}$ (17.6.1), and hence there exists an element $g \in C$ such that C_g is an étale A-algebra, which proves that A'' is an essentially étale A-algebra.

Given a local ring A, there exists a *set* $\mathfrak{E}$ of local essentially étale A-algebras such that every local essentially étale A-algebra is A-isomorphic to an algebra belonging to $\mathfrak{E}$. It suffices indeed to observe that there exists a set $\mathfrak{F}$ of A-algebras of finite type such that every A-algebra of finite type is isomorphic to an algebra belonging to $\mathfrak{F}$; one may take $\mathfrak{F}$ to be the set of quotients of the polynomial algebras $A[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ ($n \in \mathbf{N}$).

In what follows, if A and B are two local rings, we denote by $\operatorname{Hom.loc}(A, B)$ the set of *local* homomorphisms from A to B.

Lemma (18.6.2). — *Let* A, A′ *be two local rings, k, k' their residue fields, $\phi : \mathrm{A} \to \mathrm{A}'$ a local homomorphism making* A′ *an A-algebra essentially étale* (18.6.1) *and such that the corresponding homomorphism $k \to k'$ is bijective. Then, for every local Henselian ring* B, *the canonical map*

$$\operatorname{Hom}(\phi, 1_{\mathrm{B}}) : \operatorname{Hom.loc}(\mathrm{A}', \mathrm{B}) \longrightarrow \operatorname{Hom.loc}(\mathrm{A}, \mathrm{B})$$

is bijective.

Proof. Let $\mathrm{S} = \operatorname{Spec}(\mathrm{A})$, $\mathrm{Y} = \operatorname{Spec}(\mathrm{B})$, and let s and y be the closed points of S and Y respectively. By hypothesis, A′ is isomorphic to a local ring $\mathscr{O}_{\mathrm{X},x}$ of an S-scheme X étale over S, at a point $x \in \mathrm{X}$ above s. Suppose given a local homomorphism $\psi : \mathrm{A} \to \mathrm{B}$, making Y an S-scheme; it amounts to showing that there exists *one and only one* S-morphism $f : \mathrm{Y} \to \mathrm{X}$ such that $f(y) = x$. Let $\mathrm{X}' = \mathrm{X} \times_{\mathrm{S}} \mathrm{Y}$, and note that since $k(x) = k(s)$, there exists a unique point $x' \in \mathrm{X}'$ above x and y, and that $k(x') = k(y)$. It remains to show that there exists a unique Y-section f' of X′ such that $f'(y) = x'$. Now, the morphism $g : \mathrm{X}' \to \mathrm{Y}$ is étale and separated, and the fibre $\mathrm{X}'_0 = g^{-1}(y)$ at the point x' has the residue field $k(x') = k(y)$. If we set $\mathrm{Y}_0 = \operatorname{Spec}(k(y))$, there exists a unique Y_0-section f'_0 of X'_0 such that $f'_0(y) = x'$, and the conclusion follows from the fact that B is assumed Henselian and from (18.5.11), b). □

We say that a local A-algebra A′ satisfying the conditions of (18.6.2) is *strictly essentially étale*. Note that the criterion (18.5.11), b) means that, *for A to be Henselian, it is necessary and sufficient that every strictly essentially étale A-algebra be A-isomorphic to* A.

Lemma (18.6.3). — *Let* A *be a local ring,* A_1, A_2 *two A-algebras that are local and strictly essentially étale.*

i) *There exists at most one A-homomorphism (necessarily local) from* A_1 *to* A_2*.*

ii) *There exists a local A-algebra* A_3 *that is strictly essentially étale, and two A-homomorphisms* $A_1 \to A_3$, $A_2 \to A_3$.

Proof. Let $S = \operatorname{Spec}(A)$; by hypothesis there are two S-schemes étale over S, X_1, X_2, and two points $x_1 \in X_1$, $x_2 \in X_2$ above the closed point s of S and such that $A_1 = \mathscr{O}_{X_1,x_1}$, $A_2 = \mathscr{O}_{X_2,x_2}$; Let $X_3 = X_1 \times_S X_2$; the hypotheses $k(x_1) = k(x_2) = k(s)$ (**I**, 3.4.9) imply that there exists a unique point $x_3 \in X_3$ above x_1 and x_2, and that $k(x_3) = k(s)$. Furthermore, X_3 is étale over S (17.3.3), so $A_3 = \mathscr{O}_{X_3,x_3}$ satisfies the conditions of (ii). On the other hand, we have seen that every A-homomorphism from A_1 to A_2 is necessarily local; it corresponds to an S-morphism f from $X_2' = \operatorname{Spec}(A_2)$ to X_1 such that $f(x_2) = x_1$, or, equivalently, on setting $X_3' = X_1 \times_S X_2'$, to an X_2'-section f' of X_3' such that $f'(x_2) = x_3$. Since $k(x_3) = k(x_2)$ and X_2' is connected, the uniqueness of f' follows from (17.4.9), whence (i). ☐ **IV | 137**

(18.6.4). Let us denote by $\mathfrak{S}$ the subset of the set $\mathfrak{E}$ defined in (18.6.1) formed by the A-algebras that are strictly essentially étale belonging to $\mathfrak{E}$. It follows from (18.6.3) that the relation "there exists an A-homomorphism from A_1 to A_2" is a *preorder* relation on $\mathfrak{S}$, and makes $\mathfrak{S}$ a *filtered increasing* set. Indexing $\mathfrak{S}$ by itself by means of the identity map, it follows from (18.6.3), (i) that if $\lambda \leqslant \mu$ in $\mathfrak{S}$, there exists a *unique* $\phi_{\mu\lambda} : A_\lambda \to A_\mu$ and $(A_\lambda, \phi_{\mu\lambda})$ is clearly an *inductive system* of local A-algebras, the $\phi_{\mu\lambda}$ being local homomorphisms. Furthermore, it follows from (17.3.5) that for $\lambda \leqslant \mu$, A_μ is an A_λ-algebra that is strictly essentially étale.

Definition (18.6.5). — The *Henselization* of a local ring A is the inductive limit of the inductive system $(A_\lambda, \phi_{\mu\lambda})$, denoted ${}^hA$, defined in (18.6.4).

This definition does not depend on the choice of $\mathfrak{E}$; if $\mathfrak{E}'$ is another set having the same property as $\mathfrak{E}$, and $\mathfrak{S}'$ the subset of $\mathfrak{E}'$ consisting of the strictly essentially étale A-algebras, it follows from (18.6.3), (ii) that, for the preorder relations considered, $\mathfrak{S}$ and $\mathfrak{S}'$ are *cofinal* in $\mathfrak{S}'' = \mathfrak{S} \cup \mathfrak{S}'$, so they give the same inductive limit up to an isomorphism. We shall moreover see below (18.6.6), (i) and (ii) that ${}^hA$ and the canonical homomorphism $A \to {}^hA$ are the solution of a universal problem, and are therefore determined up to a unique isomorphism.

One will observe that if A is a field, then evidently ${}^hA = A$. In the general case, ${}^hA$ is also the Henselization of all the A-algebras that are strictly essentially étale A_λ ($\lambda \in \mathfrak{S}$), since if B is an A_λ-algebra that is strictly essentially étale, B is also an A-algebra that is strictly essentially étale (18.6.1), so (up to an A-isomorphism) one of the A_μ for $\mu \geqslant \lambda$, and the A_μ such that $\mu \geqslant \lambda$ and that A_μ is an A_λ-algebra that is strictly essentially étale form a cofinal set in the preordered set of the A_λ, by virtue of (18.6.1) and (18.6.3).

Theorem (18.6.6). — *Let* A *be a local ring,* ${}^hA$ *its Henselization.*

i) ${}^hA$ *is a Henselian local ring and the structural homomorphism* $A \to {}^hA$ *is local.*

ii) *For every Henselian local ring* B*, the canonical map*

$$\operatorname{Hom.loc}({}^hA, B) \longrightarrow \operatorname{Hom.loc}(A, B)$$

is bijective.

iii) ${}^hA$ *is a faithfully flat A-module, and if* $\mathfrak{m}$ *is the maximal ideal of* A, $\mathfrak{m}.{}^hA$ *is the maximal ideal of* ${}^hA$, *and the homomorphism* $A/\mathfrak{m} \to {}^hA/\mathfrak{m}.{}^hA$ *of residue fields is bijective.*

iv) *If* $\widehat{A}$ *and* $({}^hA)^\wedge$ *are the separated completions of the local rings* A *and* ${}^hA$, *the homomorphism* $\widehat{A} \to ({}^hA)^\wedge$ *deduced from the structural homomorphism* $A \to {}^hA$ *by completion is bijective.*

v) *For* ${}^hA$ *to be Noetherian, it is necessary and sufficient that A be so.*

vi) *If A is Henselian, the canonical homomorphism* $A \to {}^hA$ *is bijective.*

Proof. Let $\mathfrak{m}_\lambda$ be the maximal ideal of A_λ; it follows from the fact that, in (17.6.1), a) implies c'), that for $\lambda \leqslant \mu$ one has $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$, that the homomorphism $A_\lambda/\mathfrak{m}_\lambda \to A_\mu/\mathfrak{m}_\mu$ is bijective, and that A_μ is a flat A_λ-module. The fact that ${}^hA$ is local, that the homomorphism $A \to {}^hA$ is local, assertion (iii) and the sufficiency of (v) follow therefore from (**0**, 10.3.1.3); the necessity of (v) follows from the fact that ${}^hA$ is a flat A-module (6.5.2). **IV | 138**

To prove that ${}^h\mathrm{A}$ is Henselian, let us apply the criterion (18.5.11), *b*). Let $\mathrm{S} = \mathrm{Spec}({}^h\mathrm{A})$, $\mathrm{S}_0 = \mathrm{Spec}({}^h\mathrm{A}/\mathfrak{m}.{}^h\mathrm{A})$, let $g : \mathrm{S}' \to \mathrm{S}$ be an étale morphism, set $\mathrm{S}'_0 = g^{-1}(\mathrm{S}_0)$, and let $f_0 : \mathrm{S}_0 \to \mathrm{S}'_0$ be an S_0-section. reasoning as in (18.5.4), one can suppose that g is of finite presentation. It follows then from (8.8.2) and (17.7.5) that there exist an index $\lambda \in \mathfrak{S}$, an étale morphism $g_\lambda : \mathrm{S}'^{(\lambda)} \to \mathrm{S}^{(\lambda)} = \mathrm{Spec}(\mathrm{A}_\lambda)$ and, on setting $\mathrm{S}_0^{(\lambda)} = \mathrm{Spec}(\mathrm{A}_\lambda/\mathfrak{m}\mathrm{A}_\lambda)$ and $\mathrm{S}_0'^{(\lambda)} = g_\lambda^{-1}(\mathrm{S}_0^{(\lambda)})$, an $\mathrm{S}_0^{(\lambda)}$-section $f_0^{(\lambda)} : \mathrm{S}_0^{(\lambda)} \to \mathrm{S}_0'^{(\lambda)}$, such that

$$\mathrm{S}' = \mathrm{S}'^{(\lambda)} \times_{\mathrm{S}^{(\lambda)}} \mathrm{S}, \qquad g = g_\lambda \times 1, \qquad f_0 = f_0^{(\lambda)} \times 1.$$

Let s be the closed point of S, $x = f_0(s)$, let x_λ be the projection of x in $\mathrm{S}'^{(\lambda)}$, and let s_λ be the closed point of $\mathrm{S}^{(\lambda)}$. Since x_λ lies above s_λ, the local ring C_λ of $\mathrm{S}'^{(\lambda)}$ at x_λ is an essentially étale A_λ-algebra; moreover, since $f_0^{(\lambda)}(s_\lambda) = x_\lambda$, we have $k(x_\lambda) = k(s_\lambda)$, in other words C_λ is a local strictly essentially étale A_λ-algebra, and consequently (18.6.1) is A_λ-isomorphic to an A-algebra A_μ with $\mu \geqslant \lambda$. There is therefore an A_λ-homomorphism $\mathrm{C}_\lambda \to {}^h\mathrm{A}$, that is, an $\mathrm{S}^{(\lambda)}$-morphism $h : \mathrm{S} \to \mathrm{S}'^{(\lambda)}$ such that $h(s) = x_\lambda$, and hence an S-section f of S' such that $f(s) = x$; this completes the proof of (i).

To prove (ii), it suffices to note that $\mathrm{Hom\,.\,loc}({}^h\mathrm{A}, \mathrm{B}) \cong \varprojlim \mathrm{Hom\,.\,loc}(\mathrm{A}_\lambda, \mathrm{B})$ and that by (18.6.2) the canonical homomorphisms $\mathrm{Hom\,.\,loc}(\mathrm{A}_\lambda, \mathrm{B}) \leftarrow \mathrm{Hom\,.\,loc}(\mathrm{A}, \mathrm{B})$ are bijective.

To prove (iv), note that for every λ and every integer $n > 0$, $\mathfrak{m}_\lambda^n = \mathfrak{m}^n \mathrm{A}_\lambda$, and

$$(\mathfrak{m}.{}^h\mathrm{A})^n = \mathfrak{m}^n.{}^h\mathrm{A} = \varinjlim_\lambda \mathfrak{m}_\lambda^n.$$

By exactness of the functor $\varinjlim$ in the category of A-modules,

$${}^h\mathrm{A}/(\mathfrak{m}.{}^h\mathrm{A})^n = \varinjlim_\lambda (\mathrm{A}_\lambda/\mathfrak{m}_\lambda^n),$$

and it therefore suffices to show that, for every n and every index λ, the homomorphism $\mathrm{A}/\mathfrak{m}^n \to \mathrm{A}_\lambda/\mathfrak{m}_\lambda^n$ is bijective. On the other hand, since A_λ is a flat A-module, one has

$$\mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1} = (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_\mathrm{A} \mathrm{A}_\lambda = (\mathfrak{m}^n/\mathfrak{m}^{n+1}) \otimes_{\mathrm{A}/\mathfrak{m}} (\mathrm{A}_\lambda/\mathfrak{m}\mathrm{A}_\lambda)$$

and as $\mathrm{A}/\mathfrak{m} \to \mathrm{A}_\lambda/\mathfrak{m}\mathrm{A}_\lambda$ is bijective, the homomorphism $\mathfrak{m}^n/\mathfrak{m}^{n+1} \to \mathfrak{m}_\lambda^n/\mathfrak{m}_\lambda^{n+1}$ is also bijective; the conclusion then follows from Bourbaki, *Alg. comm.*, chap. III, § 2, no 8, cor. 3 du th. 1.

Finally, if A is Henselian, it follows from the remark preceding (18.6.3) that the canonical homomorphisms $\mathrm{A} \to \mathrm{A}_\lambda$ are bijective, which proves (vi) by definition of ${}^h\mathrm{A}$. □

(18.6.7). Let A be a *semi-local* ring, $\mathfrak{m}_i$ $(1 \leqslant i \leqslant r)$ its maximal ideals. The *Henselization* of A is the ring denoted again ${}^h\mathrm{A}$, defined as the *product* $\prod_i {}^h(\mathrm{A}_{\mathfrak{m}_i})$ of the Henselizations of the local rings $\mathrm{A}_{\mathfrak{m}_i}$. It is a faithfully flat A-module and a semi-local A-algebra, whose maximal ideals are the $\mathfrak{m}_i.{}^h\mathrm{A}$ by virtue of (18.6.6), (iii); furthermore, if $\mathfrak{r} = \bigcap \mathfrak{m}_i$ is the radical of A, it follows from the above that $\mathfrak{r}.{}^h\mathrm{A}$ is the **IV | 139**
radical of ${}^h\mathrm{A}$ and that the canonical map $\mathrm{A}/\mathfrak{r} \to {}^h\mathrm{A}/\mathfrak{r}.{}^h\mathrm{A}$ is bijective. As the separated completion $\widehat{\mathrm{A}}$ of A for the $\mathfrak{r}$-adic topology is the product of the separated completions $\widehat{\mathrm{A}}_{\mathfrak{m}_i}$ (Bourbaki, *Alg. comm.*, chap. III, § 2, no 13, prop. 18), the canonical homomorphism $\widehat{\mathrm{A}} \to ({}^h\mathrm{A})^\wedge$ is bijective by (18.6.6), (iv), and it is clear by (18.6.6), (v) that for ${}^h\mathrm{A}$ to be Noetherian, it is necessary and sufficient that A be so.

To obtain the analogue of the universal property (18.6.6), (ii), when A and B are two *semi-local* rings of radicals $\mathfrak{r}$, $\mathfrak{s}$, let us agree to call a *semi-local homomorphism* ϕ from A to B any homomorphism such that $\phi(\mathfrak{r}) \subset \mathfrak{s}$. In the particular case where B is a *product* of local rings B_j $(1 \leqslant j \leqslant n)$, the data of such a homomorphism amounts to the data of its projections $\phi_j : \mathrm{A} \to \mathrm{B}_j$, which are homomorphisms subject only to the condition that $\phi_j(\mathfrak{r}) \subset \mathfrak{n}_j$, where $\mathfrak{n}_j$ is the maximal ideal of B_j. Moreover, since $\mathfrak{m}_i$ $(1 \leqslant i \leqslant m)$ are the maximal ideals of A, the prime ideal $\phi_j^{-1}(\mathfrak{n}_j)$ must contain one of the $\mathfrak{m}_i$, since it must contain their intersection $\mathfrak{r}$, hence it is equal to one of the $\mathfrak{m}_i$ and ϕ_j factorises as $\mathrm{A} \to \mathrm{A}_{\mathfrak{m}_i} \xrightarrow{\psi_{ij}} \mathrm{B}_j$, where ψ_{ij} is a *local* homomorphism. If $\mathrm{Hom\,.\,sloc}(\mathrm{A}, \mathrm{B})$ denotes the set of semi-local homomorphisms from A to B, one can thus identify this set with $\prod_j \left(\bigcup_i \mathrm{Hom\,.\,loc}(\mathrm{A}_{\mathfrak{m}_i}, \mathrm{B}_j)\right)$; in the case considered, since a semi-local Henselian ring is a direct product of Henselian local rings, one sees that the universal property (18.6.6), (ii) is still valid for a semi-local ring A, provided one replaces "local homomorphisms" by "semi-local homomorphisms" (18.6.7).

Proposition (18.6.8). — *Let* A *be a semi-local ring, and let* B *be a semi-local A-algebra integral over* A*. Then* $B \otimes_A ({}^hA)$ *is a semi-local B-algebra isomorphic to* ${}^hB$*.*

Proof. Since $B \otimes_A ({}^hA)$ is integral over ${}^hA$, each of its maximal ideals lies above a maximal ideal of ${}^hA$, hence above a maximal ideal of A. Since B is integral over A, the prime ideals of B lying above a maximal ideal of A are maximal ideals of B. Thus the projections in $\mathrm{Spec}({}^hA)$ and $\mathrm{Spec}(B)$ of a maximal ideal of $B \otimes_A ({}^hA)$ are both maximal ideals. Taking account of (18.6.6), (iii), and (**I**, 3.4.9), one concludes that $B \otimes_A ({}^hA)$ is semi-local.

Furthermore, for every Henselian local ring C, $\mathrm{Hom.sloc}(B \otimes_A ({}^hA), C)$ is in bijection with the set of pairs (ϕ, ψ) of semi-local homomorphisms $\phi : B \to C$ and $\psi : {}^hA \to C$ such that the composites

$$A \longrightarrow B \xrightarrow{\phi} C, \qquad A \longrightarrow {}^hA \xrightarrow{\psi} C$$

are equal. By the bijection between $\mathrm{Hom.sloc}(A, C)$ and $\mathrm{Hom.sloc}({}^hA, C)$, for every $\phi \in \mathrm{Hom.sloc}(B, C)$ there exists a unique ψ having this property. Hence the map

$$\mathrm{Hom.sloc}(B \otimes_A ({}^hA), C) \longrightarrow \mathrm{Hom.sloc}(B, C)$$

is bijective, which proves the proposition by the uniqueness of the solution of a universal problem. □

Theorem (18.6.9). — *Let* A *be a semi-local ring.*

i) *For* ${}^hA$ *to be reduced (resp. normal), it is necessary and sufficient that A be so.*
ii) *Suppose A Noetherian. Then, for every prime ideal* $\mathfrak{p}$ *of A, the ring* $({}^hA)_{\mathfrak{p}}/\mathfrak{p}.({}^hA)_{\mathfrak{p}}$ *is a finite direct product of separable algebraic extensions of* $k(\mathfrak{p})$ *(which implies that the fibres of the canonical morphism* $\mathrm{Spec}({}^hA) \to \mathrm{Spec}(A)$ *are* geometrically regular*).*

Proof. (i) As ${}^hA$ is a faithfully flat A-module, it follows from (2.1.13) that if ${}^hA$ is reduced (resp. normal), so is A. To prove the converse, one can restrict to the case where A is local, so that, with the notations of (18.6.4), one has ${}^hA = \varinjlim A_\alpha$. Now, it follows from the definition of the A_α and from (17.5.7) that if A is reduced (resp. integral and integrally closed), so is each of the A_α; furthermore, the homomorphisms $A_\alpha \to A_\beta$ for $\alpha \leqslant \beta$ are injective by virtue of (**0**, 6.5.1) since A_β is a flat A_α-module; hence the morphisms $\mathrm{Spec}(A_\beta) \to \mathrm{Spec}(A_\alpha)$ are dominant, and one concludes from (5.13.2) (resp. (5.13.4)) that ${}^hA$ is reduced (resp. integral and integrally closed). **IV** | 140

(ii) One can restrict to the case where A is local. By virtue of the fact that the functor $\varinjlim$ commutes with tensor products, the fibre of the morphism $\mathrm{Spec}({}^hA) \to \mathrm{Spec}(A)$ at a point $\mathfrak{p}$ is the projective limit of the fibres of the morphisms $\mathrm{Spec}(A_\alpha) \to \mathrm{Spec}(A)$ at this point. As ${}^hA$ is Noetherian, one is reduced to proving the following lemma: □

Lemma (18.6.9.1). — *Let* (B_α) *be an inductive system of Artinian rings,* $B = \varinjlim B_\alpha$ *its inductive limit. If* B *is Noetherian, then* B *is Artinian; if moreover, for* $\alpha \leqslant \beta$*, the homomorphisms* $\phi_{\beta\alpha} : B_\alpha \to B_\beta$ *are injective, then there exists* λ *such that, for* $\alpha \geqslant \lambda$*, the number of local components* $B_\alpha^{(i)}$ *of* B_α *is constant, that* $\phi_{\beta\alpha}(B_\alpha^{(i)}) \subset B_\beta^{(i)}$ *for every i and that the* $B^{(i)} = \varinjlim B_\alpha^{(i)}$ *are the local components of* B*.*

Proof. Indeed $Y_\alpha = \mathrm{Spec}(B_\alpha)$, $Y = \mathrm{Spec}(B)$, which, as a topological space, is equal to $\varprojlim Y_\alpha$ (8.2.10). The spaces Y_α are finite and discrete; furthermore, if the $\phi_{\beta\alpha}$ are injective, the canonical morphisms $Y_\beta \to Y_\alpha$ are dominant, hence surjective. The lemma then follows from the following purely topological lemma: □

Lemma (18.6.9.2). — *Let* $(Y_\alpha, \psi_{\alpha\beta})$ *be a projective system of finite discrete topological spaces. If* $Y = \varprojlim Y_\alpha$ *is Noetherian, Y is finite and discrete. If moreover the* $\psi_{\alpha\beta}$ *are surjective, then there exists* λ *such that for* $\alpha \geqslant \lambda$*, the number of elements of* Y_α *is constant and equal to the number of elements of* Y*.*

Proof. Indeed, since Y is compact and Noetherian, every subset of Y is compact, hence closed; thus Y is discrete, and therefore finite since it is compact. If moreover the $\psi_{\alpha\beta}$ are surjective, the same holds for $\psi_\alpha : Y \to Y_\alpha$, so $\mathrm{Card}(Y) \geqslant \mathrm{Card}(Y_\alpha)$, and as $\mathrm{Card}(Y_\alpha)$ is an increasing function of α, there exists λ such that for $\alpha \geqslant \lambda$, $\mathrm{Card}(Y_\alpha) = \mathrm{Card}(Y_\lambda)$; the $\psi_{\alpha\beta}$ are then bijective for $\alpha \geqslant \lambda$ and one therefore has $\mathrm{Card}(Y) = \mathrm{Card}(Y_\lambda)$. □

Corollary (18.6.10). — *Let* A *be a local Noetherian ring. For* ^{h}A *to possess one of the following properties:*

a) *being a Cohen-Macaulay ring* (**0**, 16.5.3)*;*
b) *satisfying the property* (S_n) (5.7.3)*;*
c) *being regular;*
d) *satisfying the property* (R_n) (5.8.2)*;*

it is necessary and sufficient that A possess the same property.

Proof. It follows immediately from the fact that the fibres of the morphism $\mathrm{Spec}({}^h\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ are geometrically regular (18.6.9) and from (6.4.1), (6.5.1) and (6.5.3). ☐ **IV | 141**

Corollary (18.6.11). — *Let* A *be a semi-local Noetherian ring. For a prime ideal* $\mathfrak{p}$ *of* ^{h}A *to belong to* $\mathrm{Ass}({}^h\mathrm{A})$*, it is necessary and sufficient that* $\mathfrak{p} \cap \mathrm{A}$ *belong to* $\mathrm{Ass}(\mathrm{A})$*.*

Proof. Taking account of the description of the fibres of the morphism $\mathrm{Spec}({}^h\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$, this also follows immediately from (3.3.1). ☐

Proposition (18.6.12). — *Let* A *be a local ring; the following properties are equivalent:*

a) *A is unibranch* (**0**, 23.2.1) *(resp. reduced and unibranch).*
b) *For every A-algebra* A′ *that is local and strictly essentially étale,* $\mathrm{Spec}(\mathrm{A}')$ *is irreducible (resp. integral).*
c) $\mathrm{Spec}({}^h\mathrm{A})$ *is irreducible (resp. integral).*

Proof. First note that ${}^h(\mathrm{A}_{\mathrm{red}}) = ({}^h\mathrm{A})_{\mathrm{red}}$: indeed, ${}^h(\mathrm{A}_{\mathrm{red}}) = ({}^h\mathrm{A}) \otimes_{\mathrm{A}} \mathrm{A}_{\mathrm{red}}$ by (18.6.8), and since ${}^h(\mathrm{A}_{\mathrm{red}})$ is reduced by (18.6.9), it is equal to $({}^h\mathrm{A})_{\mathrm{red}}$. This allows us, in the proof, to consider only the case where A is reduced; then so are A′ (17.5.7) and ^{h}A (18.6.9).

With the notations of (18.6.4), condition *b*) means that all the A_α are integral; one therefore concludes that ${}^h\mathrm{A} = \varinjlim \mathrm{A}_\alpha$ is integral (5.13.3), so *b*) implies *c*).

It is clear that if ^{h}A is integral, so are each of the $\mathrm{A}_\alpha \subset {}^h\mathrm{A}$, hence *c*) implies *b*).

Let us show that *c*) implies *a*). Note first that $\mathrm{A} \subset {}^h\mathrm{A}$ is integral; let K and L be the fields of fractions of A and ^{h}A respectively. It suffices to show that for every *finite* A-algebra B contained in K, the ring B is local. Now B is semi-local, and its Henselization is $\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A})$ by (18.6.8). By flatness, $\mathrm{B} \otimes_{\mathrm{A}} ({}^h\mathrm{A})$ identifies with a subring of $\mathrm{K} \otimes_{\mathrm{A}} ({}^h\mathrm{A})$, and the latter with a subring of L; hence ^{h}B is integral. But since ^{h}B is a semi-local Henselian ring, it is a finite direct product of local rings (18.5.9); it can therefore be integral only if it is local, which indeed implies that B is local.

Let us finally prove that *a*) implies *c*). Let C be the integral closure of the integral ring A. As C is a local ring by hypothesis, $\mathrm{C} \otimes_{\mathrm{A}} ({}^h\mathrm{A})$ is the Henselization of C (18.6.8), hence is integral and integrally closed (18.6.9). Since A is a subring of C, flatness identifies ^{h}A with a subring of ^{h}C; it is therefore integral as well, which completes the proof. ☐

Corollary (18.6.13). — *Let* A *be a local Henselian ring. Then, for A to be unibranch, it is necessary and sufficient that* $\mathrm{A}_{\mathrm{red}}$ *be integral.*

Proposition (18.6.14). — i) *Every filtered inductive limit of Henselian local rings (where the transition homomorphisms are local) is a Henselian local ring.*
ii) *The functor* $\mathrm{A} \rightsquigarrow {}^h\mathrm{A}$ *in the category of local rings (where the morphisms are local homomorphisms) commutes with filtered inductive limits.*
iii) *Every Henselian local ring A is an inductive limit of an inductive filtered system of Henselian and Noetherian local rings* A_λ *(the transition homomorphisms* $\mathrm{A}_\lambda \to \mathrm{A}_\mu$ *being local).*

Proof. (i) Let (A_λ) be a filtered inductive system of Henselian local rings, where the transition homomorphisms are local. Let us show that $\mathrm{A}' = \varinjlim \mathrm{A}_\lambda$, which is local (**0**, 10.3.1.3), is Henselian. Now, let C be an A′-algebra étale, $\mathfrak{n}$ a prime ideal of C above the maximal ideal $\mathfrak{m}'$ of A′, such that **IV | 142** $\mathrm{C}_{\mathfrak{n}}$ is strictly essentially étale over A′ (18.6.1), and in virtue of (8.8.2), (ii) and (17.7.8), there exist an index λ and an A_λ-algebra C_λ étale such that $\mathrm{C} = \mathrm{C}_\lambda \otimes_{\mathrm{A}_\lambda} \mathrm{A}'$ up to an isomorphism. Furthermore, if $\mathfrak{n}_\lambda$ is the inverse image of $\mathfrak{n}$ in C_λ, and $\mathfrak{m}_\lambda$ the maximal ideal of A_λ, one can, by virtue of (8.8.2.4) and of the transitivity of fibres (**I**, 3.6.4), suppose that $(\mathrm{C}_\lambda)_{\mathfrak{n}_\lambda}$ is strictly essentially étale over A_λ; since A_λ is Henselian, $(\mathrm{C}_\lambda)_{\mathfrak{n}_\lambda}$ is necessarily isomorphic to A_λ (18.5.11), *b*); there is therefore a neighbourhood of $\mathfrak{n}_\lambda$ in $\mathrm{Spec}(C_\lambda)$ isomorphic to $\mathrm{Spec}(A_\lambda)$ (**I**, 6.5.4); one concludes that there is a neighbourhood of $\mathfrak{n}$ in $\mathrm{Spec}(C)$ isomorphic to $\mathrm{Spec}(A')$, hence that $C_\mathfrak{n}$ is isomorphic to A'. This proves that A' is Henselian (18.6.2) and completes the proof.

(ii) Suppose that $B = \varinjlim B_\lambda$, where the B_λ are local rings, the transition morphisms $B_\lambda \to B_\mu$ ($\lambda \leqslant \mu$) being local. Set $A_\lambda = {}^hB_\lambda$; the A_λ are Henselian local rings (18.6.6), (vi), and by virtue of (18.6.6), (ii), the transition morphisms $B_\lambda \to B_\mu$ uniquely determine local homomorphisms $A_\lambda \to A_\mu$, so that (A_λ) is an inductive system. It amounts to showing that $A' = \varinjlim A_\lambda$ is canonically isomorphic to $A = {}^hB$. By virtue of (18.6.6), (ii) and of the definition of inductive limits, one has, for every Henselian local ring E,

$$\mathrm{Hom.loc}(A', E) = \varprojlim \mathrm{Hom.loc}(A_\lambda, E) = \varprojlim \mathrm{Hom.loc}(B_\lambda, E) = \mathrm{Hom.loc}(B, E).$$

But as A' is Henselian by (i), this proves our assertion.

(iii) The ring A is a filtered inductive limit of its sub-local Noetherian rings B_λ, the transition homomorphisms being local (5.13.3), (iii). As the ${}^hB_\lambda$ are Henselian and Noetherian local rings (18.6.6), (v) and that $A = {}^hA$ (18.6.6), (vi), it suffices to apply (ii) to the inductive system (B_λ). ☐

Corollary (18.6.15). — *Let* A *be a Henselian local ring,* X *an A-prescheme of finite presentation over* A*. Then there exists a Noetherian Henselian local ring* A_0*, a local homomorphism* $A_0 \to A$*, an* A_0*-prescheme* X_0 *of finite type over* A_0 *and an A-isomorphism* $X_0 \otimes_{A_0} A \xrightarrow{\sim} X$.

Proof. This follows from (18.6.14) and (8.8.2), (ii). ☐

18.7. Henselization and excellent rings.

(18.7.1). In this paragraph we denote by $\boldsymbol{P}(Z, k)$ a property of the form considered in (7.3.1), where we further suppose that the property $\boldsymbol{Q}(A, k)$ satisfies the following condition: **IV | 142**

For every separable algebraic extension k' of k, and every k'-local algebra that is Noetherian, the property $\boldsymbol{Q}(A, k)$ is *equivalent* to $\boldsymbol{Q}(A, k')$.

It is immediate that all the properties $\boldsymbol{P}$ considered in (7.3.8) satisfy the preceding condition, since if K is a finite extension of k, $k' \otimes_k K$ is a finite direct product of separable algebraic extensions of K.

Proposition (18.7.2). — *Let* A *be a local Noetherian ring. For* ${}^hA$ *to be a* $\boldsymbol{P}$*-ring* (7.3.13)*, it is necessary and sufficient that A be so.*

Proof. Indeed, by virtue of (18.6.6), (iv), the completion $\widehat{A}$ of A is also the completion of ${}^hA$ and one therefore has $A \subset {}^hA \subset \widehat{A}$. By (18.6.9), for every $x \in \mathrm{Spec}(A)$, the fibre at x of the morphism $\mathrm{Spec}({}^hA) \to \mathrm{Spec}(A)$ is discrete, finite, and each of the points y_i of this fibre is a separable algebraic extension of $k(x)$. The formal fibre of A at x is therefore the prescheme *sum* of the formal fibres of ${}^hA$ at the points y_i. The conclusion then follows from the hypothesis on $\boldsymbol{P}$ in (18.7.1). ☐ **IV | 143**

Corollary (18.7.3). — *Let* A *be a local Noetherian ring. For* ${}^hA$ *to be universally Japanese, it is necessary and sufficient that A be so.*

Proof. Indeed, for a Noetherian local ring B, it amounts to the same to say that B is universally Japanese, or that the formal fibres of B are geometrically reduced (7.6.4) and (7.7.1); the conclusion therefore follows from (18.7.2). ☐

Corollary (18.7.4). — *Let* A *be a local Noetherian ring. For the formal fibres of* ${}^hA$ *to be geometrically regular, it is necessary and sufficient that those of A be so.*

Proof. This is a particular case of (18.7.2). ☐

Proposition (18.7.5). — *Let* A *be a local Noetherian ring. If A is universally catenary* (5.6.2) *(resp. formally catenary* (7.1.9)*, resp. strictly formally catenary* (7.2.6)*), so is* ${}^hA$.

Proof. With the notations of (18.6.4), if A is universally catenary, the same holds for the A_α, which are A-algebras essentially of finite type (5.6.3). To prove that in this case ${}^hA$ is universally catenary, it will therefore suffice to establish the following lemma: ☐

Lemma (18.7.5.1). — *Let* $(\mathrm{B}_\alpha, \phi_{\beta\alpha})$ *be an inductive system of Noetherian rings and suppose that, for* $\alpha \leqslant \beta$, $\phi_{\beta\alpha}$ *makes* B_β *a flat* B_α*-module and that* $\mathrm{B} = \varinjlim \mathrm{B}_\alpha$ *is Noetherian. Then, if the* B_α *are catenary (resp. universally catenary), the same holds for* B.

Proof. To say that B is universally catenary is equivalent to saying that $\mathrm{B}[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is catenary for all n (5.6.2), and it is immediate that the inductive system $(\mathrm{B}_\alpha[\mathrm{T}_1, \ldots, \mathrm{T}_n])$ satisfies the same conditions as (B_α); it therefore suffices to prove that if the B_α are catenary, then B is catenary. If $\mathfrak{p}_0 \supset \mathfrak{p}_1 \supset \cdots \supset \mathfrak{p}_r$ is a saturated chain of prime ideals, the inverse images $\mathfrak{p}_{0\alpha} \supset \mathfrak{p}_{1\alpha} \supset \cdots \supset \mathfrak{p}_{r\alpha}$ of these ideals in B_α form a chain of prime ideals for each α sufficiently large, and it suffices to show that for α sufficiently large, this chain is saturated. Now, each $\mathfrak{p}_i$ is an ideal of finite type, hence there exists λ such that, for $\alpha \geqslant \lambda$, one has $\mathfrak{p}_i = \mathfrak{p}_{i\alpha}.\mathrm{B}$. One therefore has $1 = \operatorname{codim}(\mathrm{V}(\mathfrak{p}_i), \mathrm{V}(\mathfrak{p}_{i+1})) = \operatorname{codim}(\mathrm{V}(\mathfrak{p}_{i\alpha}), \mathrm{V}(\mathfrak{p}_{i+1,\alpha}))$ since B is a flat B_α-module (6.1.4) and (**0**, 6.2.3), which completes the proof of the lemma.

Suppose now A formally catenary. Let $\mathfrak{p}$ be a prime ideal of ${}^h\mathrm{A}$, and let $\mathfrak{q}$ be its inverse image in A, so that $\mathfrak{q}.{}^h\mathrm{A} \subset \mathfrak{p}$ and consequently ${}^h\mathrm{A}/\mathfrak{p}$ is a quotient ring of ${}^h\mathrm{A}/\mathfrak{q}.{}^h\mathrm{A}$. It therefore suffices to prove that ${}^h\mathrm{A}/\mathfrak{q}.{}^h\mathrm{A}$ is formally catenary for every prime ideal $\mathfrak{q}$ of A (7.1.9). Since ${}^h\mathrm{A}/\mathfrak{q}.{}^h\mathrm{A} = {}^h(\mathrm{A}/\mathfrak{q})$ by (18.6.8), it suffices (7.1.11) to show that if A is *integral* and formally catenary, ${}^h\mathrm{A}$ is formally *equidimensional*. But as the completion of ${}^h\mathrm{A}$ is equal to $\widehat{\mathrm{A}}$, this follows from the hypothesis that A is formally catenary.

Finally, suppose that A is strictly formally catenary. We have just seen that ${}^h\mathrm{A}$ is formally catenary, and it remains to prove that the fibres of the morphism $\operatorname{Spec}(\widehat{\mathrm{A}}) \to \operatorname{Spec}({}^h\mathrm{A})$ satisfy property (S_1) (7.2.5), b). This follows from the hypothesis on A and from (18.7.2), applied to the case where $\boldsymbol{P}(\mathrm{Z}, k)$ is property (S_1) for Z. □

Corollary (18.7.6). — *If a local Noetherian ring A is excellent* (7.8.2), *so is* ${}^h\mathrm{A}$.

Remark (18.7.7). — It can happen that ${}^h\mathrm{A}$ is an excellent local ring without A being universally catenary (nor *a fortiori* excellent). To see this, let us take the example of A from (5.6.11), with the same notations. The ring E constructed in (5.6.11) is excellent when k_0 is of characteristic 0 (7.8.3), (ii) and (iii); consider two rings E_1, E_2 isomorphic to E, and "glue" $\operatorname{Spec}(\mathrm{E}_1)$ and $\operatorname{Spec}(\mathrm{E}_2)$ such that if $\mathfrak{m}_i$ and $\mathfrak{m}'_i$ are the maximal ideals of E_i ($i = 1, 2$), corresponding to $\mathfrak{m}$ and $\mathfrak{m}'$, the point $\mathfrak{m}_1$ is "glued" to $\mathfrak{m}'_2$ and the point $\mathfrak{m}_2$ to $\mathfrak{m}'_1$; to be precise, if ε_i and ε'_i are the canonical homomorphisms from E_i onto $k(\mathfrak{m}_i)$ and $k(\mathfrak{m}'_i)$ ($i = 1, 2$), the scheme obtained is $\operatorname{Spec}(\mathrm{R})$, where R is the subring of $\mathrm{E}_1 \times \mathrm{E}_2$ formed by the pairs (x_1, x_2) such that $\varepsilon'_2(x_2) = \sigma(\varepsilon_1(x_1))$ and $\varepsilon'_1(x_1) = \sigma(\varepsilon_2(x_2))$. One verifies easily (for example with the help of (17.6.3)) that R is a finite étale C-algebra, and that there are two maximal ideals $\mathfrak{r}_1$, $\mathfrak{r}_2$ of R above the maximal ideal $\mathfrak{n}$ of C, the completions of $\mathrm{R}_{\mathfrak{r}_1}$ and $\mathrm{R}_{\mathfrak{r}_2}$ being canonically isomorphic to that of $\mathrm{A} = \mathrm{C}_\mathfrak{n}$. Thus $\mathrm{R}_{\mathfrak{r}_1}$ is an A-algebra that is strictly essentially étale, and consequently its Henselization is equal to ${}^h\mathrm{A}$. Furthermore, we saw in (7.8.4), (ii), that the formal fibres of A are geometrically regular; the same therefore holds for those of ${}^h\mathrm{A}$ by (18.7.2). Finally, R is universally catenary, since its quotients by its two minimal prime ideals are isomorphic to E, the conclusion following from (5.6.3), (iii). Hence $\mathrm{R}_{\mathfrak{r}_1}$ is excellent, and consequently so is ${}^h\mathrm{A}$ by (18.7.6), whereas A is not universally catenary.

18.8. Strictly local rings and strict Henselization. IV | 144

Proposition (18.8.1). — *Let* A *be a local ring. The following conditions are equivalent:*

a) *A is a Henselian ring and its residue field k is separably closed.*
b) *A is a Henselian ring and every étale covering of* $\mathrm{S} = \operatorname{Spec}(\mathrm{A})$ *is trivial.*
c) *For every étale morphism* $f : \mathrm{X} \to \mathrm{S}$ *and every point* $x \in \mathrm{X}$ *such that* $f(x)$ *is the closed point s of S, there exists an S-section* $u : \mathrm{S} \to \mathrm{X}$ *of X such that* $u(s) = x$.

Proof. The hypothesis c) implies that for every étale morphism $f : \mathrm{X} \to \mathrm{S}$, the residue fields at the points of $f^{-1}(s)$ are all isomorphic to k, and moreover the condition b) of (18.5.11) is satisfied. Hence A is Henselian, and the fact that every étale covering of S is trivial follows from (18.2.10), (ii); hence c) implies b). As étale coverings of $\operatorname{Spec}(k)$ are trivial if and only if k is separably closed, b) implies a) by virtue of (18.5.11), b). Finally, it follows also from (18.5.11), b) that a) implies c). □

Definition (18.8.2). — A local ring is said to be *strictly local* if it satisfies the equivalent conditions of (18.8.1). A scheme isomorphic to the spectrum of a strictly local ring is called a *strictly local scheme*.

Remark (18.8.3). — The conditions of (18.8.1) are also equivalent to the following:

d) For every morphism locally of finite type $f : X \to S$ and every point $x \in X$ such that $f(x)$ is the closed point s of S and that $\mathscr{O}_{X,x}/\mathfrak{m}_s\mathscr{O}_{X,x}$ is a field, a finite separable extension of $k(s) = k$, there exists an open neighbourhood U of x in X such that $f|U$ is a closed immersion.

(One will note that the hypothesis of d) is satisfied if f is *formally unramified* at the point x (17.4.1.2).)

We leave the reader the proof, which is substantially the same as that of (18.5.18).

Lemma (18.8.4). — *Let* A, A' *be two local rings,* $\phi : A \to A'$ *a local homomorphism making* A' *an A-algebra essentially étale (18.6.1). Then, for every strictly local ring* B*, every local homomorphism* $\psi : A \to B$ *and every homomorphism of* $k(A)$*-algebras* $\alpha : k(A') \to k(B)$*, there exists one and only one local homomorphism* $\psi' : A' \to B$ *such that* $\psi = \psi' \circ \phi$ *and that* α *is equal to the homomorphism* $\overline{\psi}'$ *deduced from* ψ' *by passage to quotients.* IV | 145

Proof. With the notations of (18.6.2), the homomorphism α corresponds to a $k(s)$-morphism $\mathrm{Spec}(k(y)) \to \mathrm{Spec}(k(x))$, hence to a point x' well determined in X' above y. The existence of a Y-section of $f' : X' \to Y$ taking y to x' follows from (18.8.1), c), since f' is étale and B is strictly local; its uniqueness follows from the fact that f' is separated and from (17.4.9). □

Lemma (18.8.5). — *Let* A *be a local ring,* A_1, A_2 *two A-algebras that are local and essentially étale, K a field that is an extension of* $k(A)$*.*

i) *For every* $k(A)$*-homomorphism* $\gamma : k(A_1) \to k(A_2)$*, there exists at most one A-homomorphism (necessarily local)* $\psi : A_1 \to A_2$ *such that* γ *is the homomorphism* $\overline{\psi}$ *deduced from* ψ *by passage to quotients.*
ii) *For every pair of local A-homomorphisms* $\beta_1 : A_1 \to K$, $\beta_2 : A_2 \to K$ *there exists a local A-algebra* A_3 *that is essentially étale, two A-homomorphisms* $\phi_1 : A_1 \to A_3$, $\phi_2 : A_2 \to A_3$ *and an A-homomorphism* $\beta : A_3 \to K$ *such that* $\beta_1 = \beta \circ \phi_1$ *and* $\beta_2 = \beta \circ \phi_2$*.*

Proof. With the notations of (18.6.3), the homomorphisms β_1, β_2 in (ii) correspond to two S-morphisms $\mathrm{Spec}(K) \to X_1$, $\mathrm{Spec}(K) \to X_2$, with respective images x_1, x_2; one thus deduces a well-determined $\mathrm{Spec}(K) \to X_3$ (**I**, 3.2.1) whose image x_3 lies above x_1 and x_2; the A-algebra $A_3 = \mathscr{O}_{X_3,x_3}$ and the homomorphism $A_3 \to K$ corresponding to these data then satisfy the conditions of (ii). On the other hand, the data of γ in (i) corresponds to an S-morphism $\mathrm{Spec}(k(x_2)) \to \mathrm{Spec}(k(x_1))$ or also to a point x_3' of X_3 above x_2; the uniqueness of ψ follows from this and from (17.4.9). □

(18.8.6). Let A be a local ring, $\mathfrak{m}$ its maximal ideal, k its residue field, Ω a separable closure of k. Consider the set $\mathfrak{E}$ of essentially étale A-algebras defined in (18.6.1), and denote by L the set of *local A-homomorphisms* $A' \to \Omega$, where A' ranges over $\mathfrak{E}$; one will note that such a homomorphism factorises as $\lambda : A' \to k(A') \xrightarrow{\omega_\lambda} \Omega$, where $\omega_\lambda : k(A') \to \Omega$ is a k-homomorphism (since $\mathfrak{m}A'$ is the maximal ideal of A'); conversely the data of a homomorphism ω_λ uniquely determines a homomorphism $\lambda \in L$. For every $\lambda \in L$, we denote by A_λ the A-algebra essentially étale belonging to $\mathfrak{E}$ on which λ is defined. The set L is *preordered* by the relation "there exists a homomorphism $\phi_{\mu\lambda} : A_\lambda \to A_\mu$ such that $\lambda = \mu \circ \phi_{\mu\lambda}$", and it follows from (18.8.5), (i) that this homomorphism $\phi_{\mu\lambda}$ is *unique*. Furthermore, L is *filtered increasing* for the preceding preorder relation (which one also denotes $\lambda \leqslant \mu$), by virtue of (18.8.5), (ii). It is clear that $(A_\lambda, \phi_{\mu\lambda})$ is an *inductive system* of local A-algebras, the $\phi_{\mu\lambda}$ being local homomorphisms; furthermore, it follows from (17.3.5) that for $\lambda \leqslant \mu$, A_μ is an A_λ-algebra that is essentially étale. If $\overline{\phi}_{\mu\lambda} : k(A_\lambda) \to k(A_\mu)$ is the k-homomorphism deduced from $\phi_{\mu\lambda}$ by passage to quotients, $(k(A_\lambda), \overline{\phi}_{\mu\lambda})$ is an inductive system of separable algebraic extensions of k, and the $\omega_\lambda : k(A_\lambda) \to \Omega$ form an inductive system of homomorphisms of k. Furthermore, the inductive limit IV | 146

(18.8.6.1) $$\omega : \varinjlim k(A_\lambda) \longrightarrow \Omega$$

is a *k-isomorphism*. It suffices indeed to prove that for every finite separable extension k' of k, there exists an A-algebra A' such that k' is the residue field of A' and that the homomorphism $k \to k'$ is deduced from $A \to A'$ by passage to quotients. But this follows from (18.1.1) applied to étale morphisms, since $\mathrm{Spec}(A)$ is the unique neighbourhood of its closed point.

Note now that if $\mathfrak{m}_\lambda$ is the maximal ideal of A_λ, it follows that $\mathfrak{m}_\mu = \mathfrak{m}_\lambda A_\mu$ and that A_μ is a flat A_λ-module. Hence it follows from (**0**, 10.3.1.3) that the ring $\varinjlim A_\lambda$ is *local*, that the homomorphism $w : A \to \varinjlim A_\lambda$ is local, and that one has a k-isomorphism

$$\varinjlim k(A_\lambda) \xrightarrow{\sim} k(\varinjlim A_\lambda); \tag{18.8.6.2}$$

identifying these two fields, one deduces a k-isomorphism

$$\omega^{-1} : \Omega \xrightarrow{\sim} k(\varinjlim A_\lambda).$$

Definition (18.8.7). — Let A be a local ring, $k = k(A)$ its residue field, $i : k \to \Omega$ a homomorphism from k into a separable closure Ω of k; the *strict Henselization* of A relative to i, denoted ${}^{hs}A_{(i)}$ (or ${}^{hs}A$ when this causes no confusion), is the inductive limit of the inductive system $(A_\lambda, \phi_{\mu\lambda})$ defined in (18.8.6).

As one has a k-isomorphism $\omega^{-1} : \Omega \xrightarrow{\sim} k({}^{hs}A_{(i)})$, Ω is endowed canonically with the structure of a ${}^{hs}A_{(i)}$-algebra by the local homomorphism

$${}^{hs}A_{(i)} \longrightarrow k({}^{hs}A_{(i)}) \xrightarrow{\omega} \Omega.$$

As in (18.6.5) one sees that the definition given in (18.8.7) does not depend on the choice of $\mathfrak{E}$; we shall moreover see below (18.8.8), (i) and (ii) that ${}^{hs}A_{(i)}$ is an object representing a functor entirely defined by the data of A and i (**0**, 8.1.8), and is hence defined as such up to an isomorphism.

Proposition (18.8.8). — *Let* A *be a local ring, k its residue field, $i : k \to \Omega$ a homomorphism from k into a separable closure Ω of k, ${}^{hs}A_{(i)}$ the strict Henselization of* A *corresponding to i.*

i) *${}^{hs}A_{(i)}$ is a strictly local ring* (18.8.2) *and the structural homomorphism $w : A \to {}^{hs}A_{(i)}$ is local.*
ii) *For every local homomorphism $u : A \to B$, where* B *is a strictly local ring, and every k-homomorphism $\psi : \Omega \to k(B)$, there exists one and only one A-homomorphism $v : {}^{hs}A_{(i)} \to B$ such that ψ factorises as $\Omega \xrightarrow{\omega^{-1}} k({}^{hs}A_{(i)}) \xrightarrow{\overline{v}} k(B)$, where $\overline{v}$ is deduced from v by passage to quotients.*
iii) *${}^{hs}A_{(i)}$ is a faithfully flat A-module, and if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}.{}^{hs}A_{(i)}$ is the maximal ideal of ${}^{hs}A_{(i)}$, and $k({}^{hs}A_{(i)})$ is a separable closure of k.*
iv) *For ${}^{hs}A_{(i)}$ to be Noetherian, it is necessary and sufficient that A be so.*
v) *If A is strictly local (hence $\Omega = k$), one has ${}^{hs}A \cong A$.*
vi) *If $i' : k \to \Omega'$ is a second homomorphism from k into a separable closure of k, for every k-isomorphism $\sigma : \Omega \xrightarrow{\sim} \Omega'$, the corresponding A-homomorphism (by (ii)) $v_\sigma : {}^{hs}A_{(i)} \to {}^{hs}A_{(i')}$ is an isomorphism.*

Proof. We have already seen above that ${}^{hs}A_{(i)}$ is a local ring and that w is a local homomorphism; the fact that ${}^{hs}A_{(i)}$ is Henselian (and hence strictly local) is demonstrated as in (18.6.6). The assertions of (iii), as well as the sufficiency of condition (iv), also follow from (**0**, 10.3.1.3); the necessity of condition (iv) follows from the fact that ${}^{hs}A_{(i)}$ is a faithfully flat A-module (**0**, 6.5.2). **IV** | 147

To prove (ii), remark, with the notations of (18.8.6), that for every $\lambda \in L$, there exists one and only one local homomorphism $v_\lambda : A_\lambda \to B$, such that the composite $k(A_\lambda) \xrightarrow{\omega_\lambda} \Omega \xrightarrow{\psi} k(B)$ is deduced from v_λ by passage to quotients, by virtue of (18.8.4) and the hypothesis that B is strictly local. The uniqueness of v_λ implies moreover that the v_λ form an inductive system, whence the existence of the homomorphism v satisfying the conditions of (ii); its uniqueness follows from (17.4.9). The uniqueness of v immediately implies assertion (vi), by considering the composites $v_\sigma \circ v_{\sigma^{-1}}$ and $v_{\sigma^{-1}} \circ v_\sigma$. Finally, if A is strictly local, and $A' = B_\mathfrak{n}$ is an A-algebra local and essentially étale, where B is an A-algebra étale, it follows from (18.8.1) that there exists a $\mathrm{Spec}(A)$-section of $\mathrm{Spec}(B)$ taking the value $\mathfrak{n}$ at the closed point of $\mathrm{Spec}(A)$, hence by (17.4.1) that the homomorphism $A \to A'$ is bijective, which proves (v). □

(18.8.8.1). Let $\mathscr{C}$ be the category of strictly local rings, with local homomorphisms as morphisms; for every $\mathrm{B} \in \mathscr{C}$, denote by $\mathrm{F}(\mathrm{B})$ the set of pairs (u, ψ) consisting of a local homomorphism $u : \mathrm{A} \to \mathrm{B}$ and a k-homomorphism $\psi : \Omega \to k(\mathrm{B})$ such that the composite $k(\mathrm{A}) = k \xrightarrow{i} \Omega \xrightarrow{\psi} k(\mathrm{B})$ is deduced from u by passage to quotients. One can say that the object ${}^{hs}\mathrm{A}_{(i)}$ *represents* the covariant functor $\mathrm{F} : \mathscr{C} \to \mathtt{Ens}$ ($\mathbf{0}$, 8.1.11).

One will remark that by virtue of (18.8.8), (vi), the strict Henselizations ${}^{hs}\mathrm{A}_{(i)}$ are all A-isomorphic (for the various k-homomorphisms i of k into separable closures of k), but for given i and i', there is in general an infinity of A-isomorphisms ${}^{hs}\mathrm{A}_{(i)} \xrightarrow{\sim} {}^{hs}\mathrm{A}_{(i')}$. To be precise, the group of A-automorphisms of ${}^{hs}\mathrm{A}_{(i)}$ is isomorphic to the Galois group of Ω over k.

(18.8.9). Let now A be a *semi-local* ring, $\mathfrak{m}_j$ ($1 \leqslant j \leqslant r$) its maximal ideals, and for every j, consider a homomorphism $i_j : k(\mathrm{A}_{\mathfrak{m}_j}) \to \Omega_j$ of the residue field into a separable closure. The *strict Henselization* of A relative to the i_j is the product $\prod_{j=1}^{r}({}^{hs}(\mathrm{A}_{\mathfrak{m}_j}))$, denoted ${}^{hs}\mathrm{A}$ (when there is no risk of confusion). It is therefore a faithfully flat A-module and a semi-local A-algebra, whose maximal IV | 148
ideals are the $\mathfrak{m}_j.{}^{hs}\mathrm{A}$ and the radical $\mathfrak{r}.{}^{hs}\mathrm{A}$ (if $\mathfrak{r}$ denotes the radical of A). Finally, the universal property (18.8.8), (ii) still holds when replacing "local homomorphism" by "semi-local homomorphism" (18.6.7), and replacing Ω by $\prod_j \Omega_j$.

Proposition (18.8.10). — *Let* A *be a semi-local ring,* B *a finite A-algebra. Let* $\mathfrak{n}_j$ *($1 \leqslant j \leqslant r$) be the maximal ideals of* B*. Then there is, for every j, a maximal ideal* $\mathfrak{q}_j$ *of* $\mathrm{B} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$ *above* $\mathfrak{n}_j$*, such that* ${}^{hs}\mathrm{B}$ *is isomorphic to the direct product of the local rings* $(\mathrm{B} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A}))_{\mathfrak{q}_j}$ *($1 \leqslant j \leqslant r$).*

Proof. One can evidently restrict to the case where A is local, replacing A by $\mathrm{A}_{\mathfrak{m}_j}$. On the other hand, let $\mathrm{C} = \mathrm{B}_{\mathfrak{n}_j}$; by virtue of the definition of ${}^{hs}\mathrm{B}$ (18.8.9), everything amounts to seeing that ${}^{hs}\mathrm{C}$ is C-isomorphic to $(\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A}))_{\mathfrak{q}}$, where $\mathfrak{q}$ is one of the maximal ideals of $\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$. Let k and K be the residue fields of A and C, Ω the separable closure of k, and let us suppose K and Ω contained in the same closure of k; as K is a finite extension of k, it is immediate that $\mathrm{K} \otimes_k \Omega$ is a finite direct product of fields, all isomorphic to the separable closure $\mathrm{K}\Omega$ of K. On the other hand, since ${}^{hs}\mathrm{A}$ is Henselian, $\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$ is a direct product of the local rings $(\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A}))_{\mathfrak{q}_i}$, where $\mathfrak{q}_i$ ($1 \leqslant i \leqslant s$) ranges over the maximal ideals of $\mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$; these local rings are Henselian (18.5.10) and have $\mathrm{K}\Omega$ as residue field, hence they are strictly local (18.8.1). Let $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$ and $\mathrm{C}'_i = \mathrm{C}'_{\mathfrak{q}_i}$, which is therefore a quotient ring of C'; let $f : \mathrm{C} \to \mathrm{C}'$, $g : {}^{hs}\mathrm{A} \to \mathrm{C}'$, $\phi_i : \mathrm{C}' \to \mathrm{C}'_i$ be the canonical maps.

$$
\begin{tikzcd}
 & & {}^{hs}\mathrm{C} \arrow[dl, bend left, "v"'] \\
\mathrm{C} \arrow[r, "f"'] & \mathrm{C}' \arrow[r, "\phi_i"'] & \mathrm{C}'_i \arrow[u, bend left, "w"] \\
\mathrm{A} \arrow[u] \arrow[r] & {}^{hs}\mathrm{A} \arrow[u, "g"] \arrow[ur, "u"'] &
\end{tikzcd}
$$

Let $v : \mathrm{C} \to {}^{hs}\mathrm{C}$ be the canonical local homomorphism; there exists a unique $u : {}^{hs}\mathrm{A} \to {}^{hs}\mathrm{C}$ which, by passage to quotients, gives the canonical injection $\Omega \to \mathrm{K}\Omega$ and such that the composite $\mathrm{A} \to \mathrm{C} \xrightarrow{v} {}^{hs}\mathrm{C}$ is equal to the composite $\mathrm{A} \to {}^{hs}\mathrm{A} \xrightarrow{u} {}^{hs}\mathrm{C}$ (18.8.8); hence there exists a unique local homomorphism $w_0 : \mathrm{C}' \to {}^{hs}\mathrm{C}$ such that $w_0 \circ g = u$, $w_0 \circ f = v$, and since ${}^{hs}\mathrm{C}$ is a local ring, this homomorphism factorises as $\mathrm{C}' \to \mathrm{C}'_i \xrightarrow{w} {}^{hs}\mathrm{C}$ for a well-determined index i, w being a local homomorphism. On the other hand, the ring C'_i being strictly local, there exists a local homomorphism $w' : {}^{hs}\mathrm{C} \to \mathrm{C}'_i$ which, by passage to quotients, gives the identity automorphism of $\mathrm{K}\Omega$ and which is such that $w' \circ v = \phi_i \circ f$ (18.8.8). One deduces first that $w \circ w'$ is an endomorphism of ${}^{hs}\mathrm{C}$ which, by passage to quotients, gives the identity automorphism of $\mathrm{K}\Omega$, hence is the identity IV | 149
(18.8.8). On the other hand, $w' \circ w \circ \phi_i \circ f = w' \circ v = \phi_i \circ f$ and $w' \circ w \circ \phi_i \circ g = w' \circ u = \phi_i \circ g$; whence $w' \circ w \circ \phi_i = \phi_i$, which is to say $w' \circ w$ is the identity automorphism of C'_i. □

Remark (18.8.11). — The proof of (18.8.10) uses the fact that B is integral over A, but uses the fact that B is a *finite* A-algebra only to establish that $\mathrm{K} \otimes_k \Omega$ is a finite direct product of fields. This latter property still holds when B is assumed to be an *integral semi-local* A-algebra whose maximal ideals $\mathfrak{n}_j$ have finite separable residue-field degrees $[k(\mathfrak{n}_j) : k]_s$.

Proposition (18.8.12). — *Let* A *be a semi-local ring.*

i) *For* ${}^{hs}\mathrm{A}$ *to be reduced (resp. normal), it is necessary and sufficient that A be so.*
ii) *Suppose A Noetherian. Then, for every prime ideal* $\mathfrak{p}$ *of A, the ring* $({}^{hs}\mathrm{A})_{\mathfrak{p}}/\mathfrak{p}.({}^{hs}\mathrm{A})_{\mathfrak{p}}$ *is a finite direct product of separable algebraic extensions of* $k(\mathfrak{p})$ *(which implies that the fibres of the canonical morphism* $\mathrm{Spec}({}^{hs}\mathrm{A}) \to \mathrm{Spec}(\mathrm{A})$ *are* geometrically regular *and are* discrete *spaces).*

Corollary (18.8.13). — *Let* A *be a local Noetherian ring. For* ${}^{hs}\mathrm{A}$ *to possess one of the following properties:*

a) *being a Cohen-Macaulay ring* (**0**, 16.5.3)*;*
b) *satisfying the condition* (S_n) (5.7.3)*;*
c) *being regular;*
d) *satisfying the condition* (R_n) (5.8.2)*;*

it is necessary and sufficient that A possess the same property.

Corollary (18.8.14). — *Let* A *be a semi-local Noetherian ring. For a prime ideal* $\mathfrak{p}$ *of* ${}^{hs}\mathrm{A}$ *to belong to* $\mathrm{Ass}({}^{hs}\mathrm{A})$*, it is necessary and sufficient that* $\mathfrak{p} \cap \mathrm{A}$ *belong to* $\mathrm{Ass}(\mathrm{A})$*.*

Proof. The proofs are the same as those of (18.6.9), (18.6.10) and (18.6.11) respectively. □

Proposition (18.8.15). — *Let* A *be a local ring; the following properties are equivalent:*

a) *A is geometrically unibranch (resp. reduced and geometrically unibranch).*
b) *For every A-algebra* A' *that is local and essentially étale,* $\mathrm{Spec}(\mathrm{A}')$ *is irreducible (resp. integral).*
c) $\mathrm{Spec}({}^{hs}\mathrm{A})$ *is irreducible (resp. integral).*

Proof. As in (18.6.12), one reduces to the case where A is reduced, using the relation ${}^{hs}(\mathrm{A}_{\mathrm{red}}) = ({}^{hs}\mathrm{A}) \otimes_{\mathrm{A}} \mathrm{A}_{\mathrm{red}}$ (18.8.11); the equivalence of *b*) and *c*) is proved as in (18.6.12). The same is true of the implication *a*) $\Rightarrow$ *c*), taking account of (18.8.11) and of the definition of a geometrically unibranch integral local ring. Finally, to prove that *c*) implies *a*), using the same notations as in (18.6.12), it suffices to see that ${}^{hs}\mathrm{B}$ is integral. But, by (18.8.10), ${}^{hs}\mathrm{B}$ is a localization of the semi-local ring $\mathrm{B} \otimes_{\mathrm{A}} ({}^{hs}\mathrm{A})$. By flatness, the latter again identifies with a subring of the field L and is therefore integral; consequently ${}^{hs}\mathrm{B}$ is integral as well, which completes the proof. □

Corollary (18.8.16). — *Let* A *be a local Henselian ring (resp. strictly local). For A to be unibranch (resp. geometrically unibranch), it is necessary and sufficient that* $\mathrm{Spec}(\mathrm{A})$ *be irreducible.*

Proposition (18.8.17). — *Let* A *be a local Noetherian ring. If A is universally catenary, the same holds for* ${}^{hs}\mathrm{A}$*.*

Proof. The proof is the same as that of (18.7.5). □

Proposition (18.8.18). — i) *Every filtered inductive limit of strictly local rings (where the transition homomorphisms are local) is a strictly local ring.*
ii) *The functor* $\mathrm{A} \rightsquigarrow {}^{hs}\mathrm{A}$ *in the category of local rings (where the morphisms are local homomorphisms) commutes with filtered inductive limits.*
iii) *Every strictly local ring A is an inductive limit of a filtered inductive system of Noetherian strictly local rings (the transition homomorphisms being local).*

Proof. The proofs are modelled on those of (18.6.14). □

IV | 150

18.9. Formal fibres of Noetherian Henselian rings.

Theorem (18.9.1). — *Let* A *be a local Noetherian Henselian ring, whose formal fibres are geometrically normal. Then the formal fibres of* A *are geometrically integral (hence geometrically connected).*

Proof. The two assertions stated are in fact equivalent, a prescheme that is locally Noetherian connected and normal being integral. As every finite integral A-algebra is a Henselian local ring (18.5.9) and (18.5.10), it follows from (7.3.16.2) that one is reduced to proving that if one further supposes A *integral*, then the fibre of the morphism $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ at the generic point of $\mathrm{Spec}(\mathrm{A})$ is *integral*, which follows from the following corollary. ☐

Corollary (18.9.2). — *Let* A *be a local Noetherian Henselian integral ring, whose formal fibres are geometrically normal. Then* $\widehat{\mathrm{A}}$ *is integral.*

Proof. Indeed, it follows from (18.8.16) that A is unibranch, hence the theorem follows from the hypothesis made on the formal fibres of A and from (7.6.3). ☐

Corollary (18.9.3). — *Under the hypotheses of* (18.9.2)*, the field of fractions L of* $\widehat{\mathrm{A}}$ *is a separable extension of the field of fractions K of A, and K is algebraically closed in L.*

Proof. This follows from the fact that the fibre of the morphism $\mathrm{Spec}(\widehat{\mathrm{A}}) \to \mathrm{Spec}(\mathrm{A})$ at the generic point is geometrically integral and from (4.3.2) and (4.3.5). ☐

Corollary (18.9.4). — *Let* A *be a local Noetherian ring, Henselian, and whose formal fibres are geometrically normal (for example a Noetherian Henselian local ring that is excellent), and let* $\mathrm{A}' = \widehat{\mathrm{A}}$*. Let* X *be an A-prescheme,* $\mathrm{X}' = \mathrm{X} \otimes_{\mathrm{A}} \mathrm{A}'$*,* $g : \mathrm{X}' \to \mathrm{X}$ *the canonical projection. Then the map* $\mathrm{U} \mapsto g^{-1}(\mathrm{U})$ *is a bijection from the set of subsets of* X *that are both open and closed to the set of subsets of* X' *that are both open and closed.*

Proof. As g is a faithfully flat and quasi-compact morphism (2.3.12), one knows that the topology of X is the *quotient* of that of X' by the equivalence relation defined by g. It therefore amounts to seeing that an open and closed subset U' of X' is *saturated* for this relation. Now, as the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ has its fibres geometrically connected (18.9.1), the fibres of g are connected, whence our assertion immediately follows.

In particular, if X is locally connected (which will be the case if X is locally Noetherian), then, for every connected component U of X (which is open and closed), $g^{-1}(\mathrm{U})$ is connected (Bourbaki, *Top. gén.*, chap. I^er^, 3^e^ éd., § 11, no 3, prop. 7). If one denotes by $\pi_0(\mathrm{X})$ the set of connected components of X, the map $\pi_0(\mathrm{X}') \to \pi_0(\mathrm{X})$ canonically deduced from g is therefore *bijective*. ☐

Corollary (18.9.5). — *Under the hypotheses of* (18.9.4)*, the functor*

(18.9.5.1) $$\mathrm{Z} \longmapsto \mathrm{Z} \otimes_{\mathrm{A}} \mathrm{A}'$$

from the category of étale preschemes over X*, to that of étale preschemes over* X'*, is fully faithful.*

Proof. Let Z_1, Z_2 be two étale preschemes over X, and set $\mathrm{Z}'_i = \mathrm{Z}_i \otimes_{\mathrm{A}} \mathrm{A}'$ ($i = 1, 2$); it is a matter **IV | 151** of proving that every X'-morphism $\mathrm{Z}'_1 \to \mathrm{Z}'_2$ comes, by base change, from a unique X-morphism $\mathrm{Z}_1 \to \mathrm{Z}_2$. Suppose first that Z_2 is *separated* over X; then, as $\mathrm{Hom}_{\mathrm{X}}(\mathrm{Z}_1, \mathrm{Z}_2)$ identifies with $\Gamma((\mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2)/\mathrm{Z}_1)$, and that $\mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2$ is étale and separated over Z_1, by virtue of (17.9.3), to the set of open and closed subsets U of $\mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2$ such that the restriction to U of the projection $p : \mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2 \to \mathrm{Z}_1$ is a surjective and radicial morphism. The assertion thus follows from (18.9.4) and the fact that $\mathrm{Z}'_1 \times_{\mathrm{X}'} \mathrm{Z}'_2 = (\mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2) \otimes_{\mathrm{A}} \mathrm{A}'$.

Let us pass to the general case. It will suffice to prove that the graph Γ' of an X'-morphism $\mathrm{Z}'_1 \to \mathrm{Z}'_2$, which is open in $\mathrm{Z}'_1 \times_{\mathrm{X}'} \mathrm{Z}'_2$ (17.9.3), is of the form $\Gamma \otimes_{\mathrm{A}} \mathrm{A}'$, where Γ is induced by an *open* subset of $\mathrm{Z} = \mathrm{Z}_1 \times_{\mathrm{X}} \mathrm{Z}_2$. Indeed, the restriction to Γ of the projection $p : \mathrm{Z} \to \mathrm{Z}_1$ is then an isomorphism: after the base change $\mathrm{A} \to \mathrm{A}'$, it becomes the restriction of $p' : \mathrm{Z}' \to \mathrm{Z}'_1$ to Γ', which is an isomorphism, and it suffices to apply (2.7.1), (viii). The conclusion then follows from the characterisation of graphs (**I**, 5.3).

If $q : \mathrm{Z}' = \mathrm{Z} \otimes_{\mathrm{A}} \mathrm{A}' \to \mathrm{Z}$ is the canonical projection, it remains to prove that $\Gamma' = q^{-1}(\Gamma)$ for an open subset Γ of Z. Since q is faithfully flat and quasi-compact, it suffices (2.3.12) to prove that $\Gamma' = q^{-1}(\mathrm{U})$ for an open subset $\mathrm{U} \subset \mathrm{Z}$. Let $\mathrm{Z}'' = \mathrm{Z}' \times_{\mathrm{Z}} \mathrm{Z}' = \mathrm{Z} \times_{\mathrm{X}} \mathrm{X}''$ where $\mathrm{X}'' = \mathrm{X}' \times_{\mathrm{X}} \mathrm{X}'$, and let $q_1 : \mathrm{Z}'' \to \mathrm{Z}'$, $q_2 : \mathrm{Z}'' \to \mathrm{Z}'$ be the canonical projections. Applying (4.5.19.1), it suffices to show that $q_1^{-1}(\Gamma') = q_2^{-1}(\Gamma')$. But this is a property which is true if and only if it is true after each base change $\mathrm{Spec}(k(x)) \to \mathrm{X}$, where x ranges over X. In other words, it suffices to prove the corollary when X is the *spectrum of a field*; but as every X-prescheme that is étale is automatically *separated* over X (17.6.2), c'), one is reduced to the case considered at the beginning of the proof. □

Remarks (18.9.6). — *i)* It is possible that, if the residue field of A is of characteristic 0, the functor (18.9.5.1) induces even an *equivalence* of the category of étale coverings of X and the category of étale coverings of X′. One can indeed prove this when A is *excellent*, using the resolution of singularities of Hironaka (M. Artin). It is plausible that an analogous statement is still true without restriction on the characteristic of the residue field, provided one restricts to the étale coverings whose "principal Galois groups" have order prime to the residual characteristic (cf. [41]).

ii) One does not know whether the formal fibres of A are geometrically connected (or even geometrically irreducible) when A is any Henselian local ring. It would suffice that, for every Noetherian Henselian local *integral* ring A, $\mathrm{Spec}(\widehat{\mathrm{A}})$ be irreducible. One does not know if this is always the case.

iii) One does not know whether, when A is an excellent local ring, its strict Henselization ${}^{hs}\mathrm{A}$ has geometrically regular formal fibres. To elucidate this question, one can reduce to the case where $\mathrm{A} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$, k being a field of characteristic $p > 0$. Taking account of (18.8.17), the question is whether the formal fibres of ${}^{hs}\mathrm{A}$ are geometrically regular. The answer is affirmative when $n = 1$, but is not known for $n = 2$. One can show that the answer is affirmative whenever, for every scheme Y_1 finite over $\mathrm{Y} = \mathrm{Spec}(\widehat{\mathrm{A}})$, the singularities of Y_1 can be resolved (7.9.1).

The connectedness assertion made in (18.9.1) generalises in the following way:

Theorem (18.9.7). — *Let* A, B *be two local Noetherian rings,* $\rho : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* f *:* **IV | 152**
$\mathrm{Spec}(\mathrm{B}) = \mathrm{X} \to \mathrm{Spec}(\mathrm{A}) = \mathrm{Y}$ *the corresponding morphism. Suppose the following conditions satisfied:*

i) f *is flat and all the fibres* $\mathrm{X}_y = f^{-1}(y)$ $(y \in \mathrm{Y})$ *are geometrically reduced* (4.6.2).

ii) *Either the ring* A *is geometrically unibranch* (**0**, 23.2.1)*, or else the ring A is unibranch* (**0**, 23.2.1) *and* $k(\mathrm{B})$ *is a primary extension* (4.3.1) *of* $k(\mathrm{A})$*.*

Then, if η *is the generic point of* Y, X_η *is connected, and for every closed subset* T *of X such that* $f(\mathrm{T})$ *is rare in* Y, $\mathrm{X} - \mathrm{T}$ *is connected.*

Proof. The two conclusions stated are in fact equivalent; indeed, the hypothesis that $f(\mathrm{T})$ is rare implies that $f(\mathrm{T})$ does not contain η. On the other hand, since $\{\eta\}$ is proconstructible (9.6) and f is flat, X_η is dense in X (2.3.10); consequently, if X_η is connected, then so is $\mathrm{X} - \mathrm{T}$, which contains it. Conversely, as U ranges over the affine open neighbourhoods of η in Y, X_η is the projective limit of the subschemes induced on the open subsets $f^{-1}(\mathrm{U})$ of X (8.1.2), a). If one supposes that $\mathrm{X} - \mathrm{T}$ is connected for every closed subset T of X such that $f(\mathrm{T})$ is rare in Y, the $f^{-1}(\mathrm{U})$ are connected, and therefore so is X_η (8.4.1).

Let us first show that one can reduce to the case where A is integral and $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ is geometrically unibranch (6.15.1) (which implies that A is geometrically unibranch, but is not equivalent to this condition (6.15.2)). It is clear that one can first suppose Y reduced, by considering the morphism $\mathrm{X} \times_\mathrm{Y} \mathrm{Y}_{\mathrm{red}} \to \mathrm{Y}_{\mathrm{red}}$ obtained from f by base change, which is flat and has the same fibres as f. One can therefore suppose A integral and unibranch. If K is the field of fractions of A, there then exists a finite A-subalgebra A″ of K such that, if A′ is the integral closure of A, the morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A}'')$ is radicial (**0**, 23.2.5); it follows (6.15.5) that $\mathrm{Spec}(\mathrm{A}'')$ is geometrically unibranch. Since A is unibranch, A′ is a local ring, and hence so is A″. Let us show that the ring $\mathrm{B}'' = \mathrm{B} \otimes_\mathrm{A} \mathrm{A}''$ is also local. Indeed, B″ is a finite B-algebra, hence a Noetherian semilocal ring. If k'' is the residue field of A″, the maximal ideals of B″ are the points of $\mathrm{Spec}(\mathrm{B}'')$ lying over the closed point of Spec(B), hence the points of $\mathrm{Spec}(k(\mathrm{B}) \otimes_{k(\mathrm{A})} k'')$ (**I**, 3.4.9), since the closed point of $\mathrm{Spec}(\mathrm{A}'')$ is the only point lying over the closed point of Spec(A). Now, when A is geometrically unibranch, k'' is a radicial extension of $k(\mathrm{A})$, so $\mathrm{Spec}(k(\mathrm{B}) \otimes_{k(\mathrm{A})} k'')$ is radicial over $\mathrm{Spec}(k(\mathrm{B}))$

and consequently consists of a single point. When A is unibranch and $k(\mathrm{B})$ is a primary extension of $k(\mathrm{A})$, $\mathrm{Spec}(k(\mathrm{B}) \otimes_{k(\mathrm{A})} k'')$ is irreducible (4.3.2); since it is a finite discrete space (**I**, 6.4.4), it again consists of a single point. Thus B'' is local in both cases considered. If $\mathrm{Y}'' = \mathrm{Spec}(\mathrm{A}'')$, $\mathrm{X}'' = \mathrm{Spec}(\mathrm{B}'') = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}''$, the morphism $f'' = f_{(\mathrm{Y}'')} : \mathrm{X}'' \to \mathrm{Y}''$ is flat and has geometrically reduced fibres. Moreover, if η'' is the generic point of Y'', then $k(\eta'') = k(\eta) = \mathrm{K}$ by definition, so the fibres X_η and $\mathrm{X}''_{\eta''}$ are isomorphic. It therefore suffices to prove that $\mathrm{X}''_{\eta''}$ is connected. □

Remark (18.9.7.1). — When the ring A is *Japanese*, one can, in the case preceding (18.9.7.2), take $\mathrm{A}'' = \mathrm{A}'$ by definition (**0**, 23.1.1); one sees therefore that in this case one would even be able to prove the theorem when A is *integrally closed*.

(18.9.7.2). Suppose therefore now that Y is integral and geometrically unibranch. Note that if T is a closed subset of X such that $f(\mathrm{T})$ is rare in Y, T contains no maximal point of X since f is flat (2.3.4), and is therefore rare in X. Since X is a local scheme, hence connected, to prove that $\mathrm{X} - \mathrm{T}$ is connected it suffices, by virtue of (15.5.6.1), to show that for every $x \in \mathrm{X}$ such that $f(x) \neq \eta$, $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) - \{x\}$ is connected. Set $y = f(x)$, $\mathrm{A}_1 = \mathscr{O}_{\mathrm{Y},y}$, $\mathrm{B}_1 = \mathscr{O}_{\mathrm{X},x}$, $\mathrm{Y}_1 = \mathrm{Spec}(\mathrm{A}_1)$, $\mathrm{X}_1 = \mathrm{Spec}(\mathrm{B}_1)$, and let $f_1 : \mathrm{X}_1 \to \mathrm{Y}_1$ be the morphism corresponding to the local homomorphism $\mathrm{A}_1 \to \mathrm{B}_1$ induced by ρ. It is clear that f_1 is flat, and it follows from (4.6.1) that its fibres are geometrically reduced. Moreover, A_1 is integral and geometrically unibranch (6.15.1), and $\dim(\mathrm{A}_1) \geqslant 1$. One is thus reduced to proving the following lemma.

Lemma (18.9.7.3). — *Let* A, B *be two local Noetherian rings,* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X} = \mathrm{Spec}(\mathrm{B})$, $\rho : \mathrm{A} \to \mathrm{B}$ *a local homomorphism,* $f : \mathrm{X} \to \mathrm{Y}$ *the corresponding morphism. Suppose the following conditions satisfied:*

i) *f is flat and all the fibres X_y ($y \in \mathrm{Y}$) are geometrically reduced.*
ii) *A is integral, geometrically unibranch and* $\dim(\mathrm{A}) \geqslant 1$.

Then, if b is the closed point of X, $\mathrm{X} - \{b\}$ *is connected.*

Proof. Note first that by virtue of the theorem of Hartshorne (5.10.7), $\mathrm{X} - \{b\}$ is connected if $\mathrm{prof}(\mathrm{B}) \geqslant 2$. Now one has (6.3.1) **IV** | 153

$$\mathrm{prof}(\mathrm{B}) = \mathrm{prof}(\mathrm{A}) + \mathrm{prof}(\mathrm{B} \otimes_{\mathrm{A}} k) \tag{18.9.7.4}$$

where k denotes the residue field of A. On the other hand, since A is integral and $\dim(\mathrm{A}) \geqslant 1$, one has $\mathrm{prof}(\mathrm{A}) \geqslant 1$, hence $\mathrm{prof}(\mathrm{B}) \geqslant 2$ *unless* $\mathrm{prof}(\mathrm{B} \otimes_{\mathrm{A}} k) = 0$; moreover, $\mathrm{B} \otimes_{\mathrm{A}} k$ is reduced by (i), hence the relation $\mathrm{prof}(\mathrm{B} \otimes_{\mathrm{A}} k) = 0$ means that $\mathrm{B} \otimes_{\mathrm{A}} k$ is a *field* (**0**, 16.4.7). We shall henceforth suppose that this last condition is satisfied. We shall begin by treating a case where the proof is very simple.

A) *Case where A is integrally closed.* Then, if $\dim(\mathrm{A}) \geqslant 2$, one has $\mathrm{prof}(\mathrm{A}) \geqslant 2$ (**0**, 16.5.1), hence $\mathrm{prof}(\mathrm{B}) \geqslant 2$. One therefore only has to consider the case where $\dim(\mathrm{A}) = 1$ and where $\mathrm{B} \otimes_{\mathrm{A}} k$ is a field; A is then a discrete valuation ring, hence regular, and since $\mathrm{B} \otimes_{\mathrm{A}} k$ is a field and f is flat, B is **IV** | 154
regular (**0**, 17.3.3). Moreover (6.1.1.1), one has $\dim(\mathrm{B}) = \dim(\mathrm{A}) + \dim(\mathrm{B} \otimes_{\mathrm{A}} k) = \dim(\mathrm{A}) = 1$, so B is also a discrete valuation ring. Thus $\mathrm{X} - \{b\}$ consists of a single point, which proves the lemma (18.9.7.3) in this case.

One will observe that this proves the statement of the theorem (18.9.7) when one supposes in addition that, in that statement, the ring A is *Japanese*, for by virtue of the remark (18.9.7.1), one is reduced to the case where A is *integrally closed*, and then, in the preceding reduction (18.9.7.3), the ring $\mathrm{A}_1 = \mathscr{O}_{\mathrm{Y},y}$ is also integrally closed and one can apply the result just proved.

B) *Case where* $\dim(\mathrm{A}) \geqslant 2$. The result of (18.9.7.3) will follow from the two following lemmas. □

Lemma (18.9.7.5). — *Let* A *be a semi-local Noetherian ring, reduced,* A' *its integral closure in its total ring of fractions,* $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, $\mathrm{X}' = \mathrm{Spec}(\mathrm{A}')$; *let S be the set of closed points of* X *and* $\mathrm{U} = \mathrm{X} - \mathrm{S}$. *Suppose that for every point $x' \in \mathrm{X}'$ above a point of S,* $\dim(\mathscr{O}_{\mathrm{X}',x'}) \geqslant 2$; *then* $\mathrm{A}^{(1)} = \Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}})$ *is integral over* A *and is an A-algebra isomorphic to a subalgebra of* A'.

Proof. Recall that if the $\mathfrak{p}_i$ ($1 \leqslant i \leqslant m$) are the minimal prime ideals of A, then A' is the direct product of the integral closures A'_i of the integral rings $\mathrm{A}_i = \mathrm{A}/\mathfrak{p}_i$, so that X' is the sum of the schemes $\mathrm{X}'_i = \mathrm{Spec}(\mathrm{A}'_i)$ ($1 \leqslant i \leqslant m$). If X_0 is the sum of the schemes $\mathrm{X}_i = \mathrm{Spec}(\mathrm{A}_i)$ ($1 \leqslant i \leqslant m$) and U_0 the inverse image of U in X_0, the canonical homomorphism $\Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}}) \to \Gamma(\mathrm{U}_0, \mathscr{O}_{\mathrm{X}_0})$ is injective since X is reduced and the morphism $X_0 \to X$ (and hence also $U_0 \to U$) is surjective. As $\Gamma(U_0, \mathscr{O}_{X_0})$ is the direct product of the $\Gamma(U_i, \mathscr{O}_{X_i})$ (where U_i is the inverse image of U in X_i), it suffices to prove that for each i, $\Gamma(U_i, \mathscr{O}_{X_i})$ is isomorphic to a sub-A_i-algebra of A'_i. In other words, one is reduced to the case where A is a Noetherian integral semilocal ring. Since X is reduced and the morphism $X' \to X$ is surjective, the homomorphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U', \mathscr{O}_{X'})$ (where U' is the inverse image of U in X') is injective, and it suffices to verify that the canonical homomorphism $A' = \Gamma(X', \mathscr{O}_{X'}) \to \Gamma(U', \mathscr{O}_{X'})$ is *bijective*. Now (**I**, 8.2.1.1), $\Gamma(U', \mathscr{O}_{X'})$ is the intersection of the local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over U', and by virtue of the hypothesis, among these local rings are included all those of height 1. But the reasoning of (**0**, 23.2.7) applies also to a Noetherian integral semilocal ring, so A' is a *semilocal Krull ring*, and is therefore the intersection of the local rings $A'_{\mathfrak{p}'}$, where $\mathfrak{p}'$ ranges over the set of prime ideals of height 1 of A' (Bourbaki, *Alg. comm.*, chap. VII, § 1, no 6, th. 4); *a fortiori*, A' is the intersection of the $A'_{\mathfrak{p}'}$ for $\mathfrak{p}' \in U'$, which completes the proof of the lemma. □

Remark (18.9.7.6). — We know (**0**, 23.2.5) that there exists a finite sub-A-algebra A'' of K, the field of fractions of A, such that the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A'')$ is *radicial*; as this morphism is also integral and dominant, hence surjective (**II**, 6.1.10), it is a *homeomorphism* (2.4.5). By virtue of the geometric interpretation of $\dim(\mathscr{O}_{X',x'})$ (5.1.2), one has therefore, for every point x'' of $X'' = \mathrm{Spec}(A'')$ above a point $x \in X$, $\dim(\mathscr{O}_{X'',x''}) = \dim(\mathscr{O}_{X',x'}) \geqslant 2$, where x' is the unique point of X' above x. This being so, it follows first from (**0**, 16.1.5) and (**0**, 16.1.4.1) that the relation $\dim(\mathscr{O}_{X',x'}) \geqslant 2$ **IV** | 155
implies $\dim(\mathscr{O}_{X,x}) \geqslant 2$. Conversely, suppose that $\dim(\mathscr{O}_{X,x}) \geqslant 2$ and that A is universally catenary (5.6.2); then so is $\mathscr{O}_{X,x}$ (5.6.3), hence $\dim(\mathscr{O}_{X'',x''}) = \dim(\mathscr{O}_{X,x}) \geqslant 2$ by virtue of (5.6.10), and consequently $\dim(\mathscr{O}_{X',x'}) \geqslant 2$. The same conclusion holds if $\mathscr{O}_{X,x}$ is *geometrically unibranch*, for the fibre of the morphism $X' \to X$ at the point x is then reduced to a single point, and $\mathscr{O}_{X',x'}$ is an *integral* $\mathscr{O}_{X,x}$-algebra; it suffices then to apply (**0**, 16.1.5).

Lemma (18.9.7.7). — *Let* A *be a local Noetherian ring, of dimension* $\geqslant 2$, *integral and geometrically unibranch,* $Y = \mathrm{Spec}(A)$, y *the closed point of* Y. *Let* X *be a locally Noetherian and connected prescheme and* $f : X \to Y$ *a flat morphism. Then, for every closed subset* $T \subset f^{-1}(y)$, $X - T$ *is connected.*

Proof. The same reasoning as at the beginning of (18.9.7.2) reduces to proving that, for every point $x \in f^{-1}(y)$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. One can therefore restrict to the case where $X = \mathrm{Spec}(B)$, B being a local Noetherian ring and the homomorphism $A \to B$ being local; let $U = Y - \{y\}$, and note that $V = f^{-1}(U)$ is dense in X (2.3.10); it suffices therefore to prove that V is *connected*. Keeping the notations of (18.9.7.5), and setting $A^{(1)} = \Gamma(U, \mathscr{O}_Y)$, $Y_1 = \mathrm{Spec}(A^{(1)})$, $X_1 = X \times_Y Y_1 = \mathrm{Spec}(B \otimes_A A^{(1)})$, we have the commutative diagram

$$\begin{tikzcd} X \arrow[d, "f"'] & X_1 \arrow[l, "g_1"'] \arrow[d, "f_1"] \\ Y & Y_1 \arrow[l, "g"'] \end{tikzcd}$$

By virtue of (2.3.1) and of the definition of $A^{(1)}$, one has a canonical isomorphism $\Gamma(V, \mathscr{O}_X) \xrightarrow{\sim} B \otimes_A A^{(1)}$, and it suffices therefore to prove that this last ring is *local*, since a local ring cannot be a product of two nonzero rings (**III**, 7.8.6.1). But we have seen (18.9.7.5) that $A^{(1)}$ is a sub-A-algebra of the integral closure A' of A, since $\dim(A) \geqslant 2$. The hypothesis that A is geometrically unibranch implies that the morphism $\mathrm{Spec}(A') \to \mathrm{Spec}(A)$ is radicial at the point y (6.15.3); *a fortiori* g_1 is radicial at the closed point of X; furthermore g_1 is an integral morphism, hence the closed points of X_1 are above x, hence there is only one closed point of X_1, which completes the proof of (18.9.7.7).

It is clear that (18.9.7.7) proves (18.9.7.3) when $\dim(A) \geqslant 2$ (even without supposing the fibres geometrically reduced).

C) *Case where* $\dim(A) = 1$. One has seen at the beginning of the proof of (18.9.7.3) that one can suppose $\dim(B \otimes_A k) = 0$, and by flatness (6.1.1.1), one therefore has

$$\dim(B) = \dim(A) + \dim(B \otimes_A k) = 1.$$

Since A is integral, Y consists of two points, the generic point η and the closed point a. Furthermore, since f is flat, every irreducible component of X dominates Y (2.3.4), hence $X - \{b\}$, which is the set of maximal points ξ_i of X, is equal to the underlying set of $f^{-1}(\eta)$, and the fibre $f^{-1}(\eta)$ is the sum of the $\mathrm{Spec}(k(\xi_i))$. The hypothesis on the X_y and the fact that these fibres are of dimension 0, show that they are *geometrically regular* (6.7.6), and *a fortiori* geometrically normal, in other words f is a *normal morphism*. But since A is integral and geometrically unibranch, it follows then from (6.15.10) that B is also integral and geometrically unibranch, hence $f^{-1}(\eta) = X - \{b\}$ is reduced to a single point, and *a fortiori* connected; this concludes the proof of (18.9.7.3) and of (18.9.7). □ **IV | 156**

Remark (18.9.7.8). — In case C) of the proof of (18.9.7.3), one can avoid making appeal to the delicate result (6.15.10) by reasoning as follows: since A is integral and unibranch of dimension 1, it follows from the theorem of Krull-Akizuki (Bourbaki, *Alg. comm.*, chap. VII, § 2, no 5, prop. 5) that its integral closure A' is a *Noetherian* ring. The same reasoning as at the beginning of the proof of (18.9.7) shows that $B' = B \otimes_A A'$ is a local ring; furthermore, by flatness, B' is contained in the total ring of fractions R of the reduced B (3.3.5). Now, the theorem of Krull-Akizuki (Bourbaki, *loc. cit.*) also applies to a local Noetherian *reduced* ring of dimension 1, and shows that for every such ring B, everything contained between B and its total ring of fractions is *Noetherian*. The ring B' being a local Noetherian ring and the morphism $\mathrm{Spec}(B') \to \mathrm{Spec}(A')$ being normal, B' is an integral ring and integrally closed (6.5.4), and one concludes as at the beginning of the proof of (18.9.7).

Corollary (18.9.8). — *With the notations of* (18.9.7), *suppose verified the condition (i) of* (18.9.7) *and one of the two conditions:*

(ii′) *A is a strictly local ring* (18.8.2).
(ii″) *A is a Henselian local ring and* $k(B)$ *is a primary extension of* $k(A)$.

Then all the fibres X_y ($y \in Y$) *are geometrically connected.*

Proof. Let $\mathfrak{p}$ be the prime ideal of A corresponding to the point y, and set $A' = A/\mathfrak{p}$ and $B' = B \otimes_A A' = B/\mathfrak{p}B$, which are evidently local Noetherian rings; it is clear that the morphism $f' : \mathrm{Spec}(B') \to \mathrm{Spec}(A')$ satisfies condition (i) of (18.9.7); and as y is the generic point of $\mathrm{Spec}(A')$ and $k(A') = k(A)$, $k(B') = k(B)$, one sees that it suffices to verify that, in the case (ii′) (resp. (ii″)), A' is geometrically unibranch (resp. unibranch). But A' is integral, and in the case (ii′) (resp. (ii″)) it is strictly local (resp. Henselian) by virtue of (18.5.10). One therefore concludes that A' is geometrically unibranch (resp. unibranch) by virtue of (18.8.15) (resp. (18.6.12)). □

Corollary (18.9.9). — *Let* A *be a local Noetherian ring, Henselian, universally Japanese (i.e.* (7.6.4) *and* (7.7.2)*), whose formal fibres* (7.3.13) *are geometrically reduced; then all the formal fibres of A are geometrically connected.*

Proof. It suffices to apply (18.9.8) in the case where $B = \widehat{A}$, since B is a flat A-module and $k(B) = k(A)$. □

Remark (18.9.10). — The proofs of (18.9.4) and (18.9.5) do not use the fact that the formal fibres of A are geometrically connected. The conclusions of these two propositions are therefore still valid when one supposes the local ring A Noetherian, Henselian and universally Japanese. **IV-4 | 157**

Corollary (18.9.11). — *Let* X, Y *be two locally Noetherian preschemes,* Y *being supposed geometrically unibranch and X connected. Let* $f : X \to Y$ *be a reduced morphism (i.e.* (6.8.1)*), flat and with geometrically reduced fibres. Then, for every closed subset* T *of* X *such that* $f(T)$ *is rare in* Y, $X - T$ *is connected.*

Proof. As f is flat and $f(T)$ is rare, T cannot contain any maximal point of X (2.3.4), hence is rare in X. Utilising (15.5.6.1), it suffices to show that, for every $x \in T$, $\mathrm{Spec}(\mathscr{O}_{X,x}) - \{x\}$ is connected. Setting $y = f(x)$, as the ring $\mathscr{O}_{Y,y}$ is geometrically unibranch, one can apply (18.9.7) to the morphism

$$f_1 : \mathrm{Spec}(\mathscr{O}_{X,x}) \longrightarrow \mathrm{Spec}(\mathscr{O}_{Y,y}),$$

which satisfies conditions (i) and (ii) of this theorem, and to the closed subset $\{x\}$ of $\mathrm{Spec}(\mathscr{O}_{X,x})$, since $f_1(\{x\}) = \{y\}$ is rare in $\mathrm{Spec}(\mathscr{O}_{Y,y})$. □

18.10. Étale preschemes over a geometrically unibranch or normal prescheme

Theorem (18.10.1). — *Let $f : X \to Y$ be a locally of finite presentation morphism, x a point of X, $y = f(x)$. Suppose that Y is integral and geometrically unibranch at the point y. Then, for f to be étale at the point x, it is necessary and sufficient that the following two conditions be satisfied:*

i) *f is formally unramified at the point x;*
ii) *the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective.*

Moreover, if this is the case, X is integral and geometrically unibranch at the point x.
Finally, if $\mathscr{O}_{Y,y}$ is Noetherian, one can, in the preceding criterion, replace the condition (ii) *by the condition* (ii bis) $\dim \mathscr{O}_{Y,y} \leqslant \dim \mathscr{O}_{X,x}$.

Proof. The fact that, if f is étale, X is integral and geometrically unibranch at the point x is a particular case of (17.5.7). It is clear on the other hand that the conditions (i) and (ii) are necessary for f to be étale at the point x, since $\mathscr{O}_{X,x}$ is then an $\mathscr{O}_{Y,y}$-module faithfully flat (17.6.1). Let us prove that these conditions are sufficient. **IV-4** | 157

Set $\mathrm{A} = \mathscr{O}_{Y,y}$, $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$ (18.8.7), $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$. Let y' be the closed point of Y', x' a point of X' above x and y'; since the morphism $g : \mathrm{Y}' \to \mathrm{Y}$ is *flat* (18.8.8), it amounts to showing that f' is étale at the point x' (17.7.1),(ii)). The hypothesis (ii) implies that the morphism $\mathrm{Spec}(\mathscr{O}_{X,x}) \to \mathrm{Spec}(\mathscr{O}_{Y,y})$ is dominant (**I**, 1.2.7); as $\mathscr{O}_{Y,y}$ is integral, $\mathrm{Spec}(\mathscr{O}_{Y,y})$ has a single generic point y_1, and there is a generization x_1 of x in X above y_1. The projection morphism $\mathrm{X}' \to \mathrm{X}$ being flat, there exists a generization x_1' of x' above x_1 (2.3.4); set $y_1' = f'(x_1')$. Now, the hypothesis on y implies that A' is *integral* (18.8.15), hence Y' has a single generic point η, necessarily above y_1 since g is flat (2.3.4); on the other hand, η is also the generic point of $g^{-1}(y_1)$ (**0**, 2.1.8), and one knows on the other hand that the fibres of g are discrete spaces (18.8.12), hence $g^{-1}(y_1) = \{\eta\}$, and consequently $y_1' = \eta$. The morphism $\mathrm{Spec}(\mathscr{O}_{X',x'}) \to \mathrm{Spec}(\mathscr{O}_{Y',y'})$, restriction of f', is therefore dominant, and since $\mathscr{O}_{Y',y'} = \mathrm{A}'$ is integral, the corresponding homomorphism $\mathscr{O}_{Y',y'} \to \mathscr{O}_{X',x'}$ is injective (**I**, 1.2.7). On the other hand, f' is not ramified at the point x', (hence (18.8.3), *d*)), there exists a neighbourhood U of x' in X' such that $f'|\mathrm{U}$ is a *closed immersion* (18.8.3); *a fortiori* the homomorphism $\mathscr{O}_{Y',y'} \to \mathscr{O}_{X',x'}$ is surjective, and since it is injective, it is therefore *bijective*. But then $\mathscr{O}_{X,x'}$ is an $\mathscr{O}_{Y',y'}$-module *flat*, and f' is therefore étale at the point x' (17.6.1). **IV-4** | 158

Finally, when $\mathscr{O}_{Y,y}$ is Noetherian, we know that if f is étale at the point x, we have $\dim \mathscr{O}_{Y,y} = \dim \mathscr{O}_{X,x}$ (17.6.4). Conversely, suppose the conditions (i) and (ii bis) satisfied, and let us show that they imply conditions (i) and (ii). Let $\mathfrak{J}$ be the kernel of the canonical homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$; restricting Y as needed to a neighbourhood of y, we may suppose that $\mathfrak{J} = \mathscr{J}_y$, where $\mathscr{J}$ is an Ideal of $\mathscr{O}_Y$; if Y_1 is the closed sub-prescheme of Y defined by $\mathscr{J}$, one can, restricting further X and Y to neighbourhoods of x and y respectively, suppose that f factorises as $\mathrm{X} \xrightarrow{h} \mathrm{Y}_1 \to \mathrm{Y}$ (**I**, 6.5.1). It is clear that f_1 is still not ramified at the point x, hence (17.4.1) quasi-finite at this point; the demonstration of (5.4.1), (i)) then shows that $\dim(\mathscr{O}_{X,x}) \leqslant \dim(\mathscr{O}_{Y,y}/\mathfrak{J})$. But if one had $\mathfrak{J} \neq 0$, one would conclude that $\dim(\mathscr{O}_{Y,y}/\mathfrak{J}) < \dim(\mathscr{O}_{Y,y})$ since Y is integral (**0**, 16.1.2.2); one would then have $\dim \mathscr{O}_{X,x} < \dim \mathscr{O}_{Y,y}$, contradicting the hypothesis, which proves (ii). ☐

Remark (18.10.2). —

(i) When $\mathrm{A} = \mathscr{O}_{Y,y}$ is Noetherian, and one knows that its completion $\widehat{\mathrm{A}}$ is integral (for example if A is regular), one can, in the preceding proof, replace A' by $\widehat{\mathrm{A}}$.

(ii) When Y is locally Noetherian, one can give a faster proof of (18.10.1), without using strict henselisation, but invoking the rather delicate results of §§ 14 and 15. As f is quasi-finite at the point x by hypothesis (17.4.1), one can, replacing X by an open neighbourhood of x, suppose that $f^{-1}(y) = \{x\}$. The hypothesis (ii) implies, as we have seen, the existence of an irreducible component Z of X containing x and dominating the unique irreducible component Y_0 of Y containing y. The morphism $f|\mathrm{Z}$ being quasi-finite at the point x, hence equidimensional at this point (13.2.2), it follows from the hypothesis on Y and from the Chevalley criterion (14.4.4) that $g = f|\mathrm{Z}$ is universally open at the point x. Furthermore since f (and hence also g) is not ramified at the point x, $g^{-1}(y)$ is geometrically reduced on

$k(y)$ (17.4.1); as $\mathscr{O}_{Y,y}$ is integral, one concludes from (15.2.3) that g is *flat* at the point x, hence f is *étale* at this point (18.4.9).

(iii) With the same notations and hypotheses on Y as in (18.10.1), suppose only that f is *locally of finite type* (and not necessarily locally of finite presentation). Then, for $\mathscr{O}_{X,x}$ to be an $\mathscr{O}_{Y,y}$-algebra *essentially étale* (18.6.1), it is necessary and sufficient that f satisfy the two following conditions:

IV-4 | 159

1° f is formally unramified at the point x;
2° the homomorphism $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective.

There is nothing more than proving the sufficiency of these conditions. Taking into account (18.4.12), it suffices to show that f is *flat* at the point x. Now, by going back to the proof of (18.10.1) and the notations used in this proof, it suffices to show that f' is flat at the point x' (2.5.1); by (17.1.3) furthermore f' is formally unramified at the point x'. One can therefore restrict to carrying out the proof when $Y = \mathrm{Spec}(A)$, where A is a *strictly local* local ring and $y = f(x)$ the closed point of Y. Using then (18.8.3) we see that $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is surjective, hence *bijective* by virtue of the hypothesis, and this suffices to show the flatness of f at the point x.

Suppose furthermore that Y is *locally integral* at the point y (**I**, 2.1.8). Then the conditions 1° and 2° above (together with the fact that Y is geometrically unibranch at the point y and f locally of finite type) already imply that f is *étale at the point* x. Indeed, this follows from what precedes and from (18.4.13).

Corollary (18.10.3). — *Let* Y *be an integral prescheme, integral and geometrically unibranch, η its generic point,* X *a connected prescheme, $f : X \to Y$ a locally of finite type morphism, and such that $f^{-1}(\eta)$ is non-empty. Then, if f is formally unramified, f is étale, and* X *is integral and geometrically unibranch.*

Proof. It follows from the hypothesis and from (18.4.13) that f is étale at all the points where it is flat, and in particular at the points of $f^{-1}(\eta)$, since $\mathscr{O}_{Y,\eta} = k(\eta)$ is a field. The open set U of points of X where f is étale is therefore *non-empty*. Let us show on the other hand that for all $x \in U$, the homomorphism $\mathscr{O}_{Y,f(x)} \to \mathscr{O}_{X,x}$ is *injective*. Indeed let s be a section of $\mathscr{O}_Y$ above an open neighbourhood of $f(x)$, that one can always suppose to be X and Y themselves (the question being local on X and Y), and suppose that its image $t \in \Gamma(Y, f_*(\mathscr{O}_X)) = \Gamma(X, \mathscr{O}_X)$ has zero germ at the point x, hence vanishes in a neighbourhood V of x; then the restriction of t to the open set $U \cap V \neq \varnothing$ is zero; but the restriction $f|(U \cap V)$ is étale, hence $W = f(U \cap V)$ is open in Y and contains by construction the generic point η; the homomorphism $\mathscr{O}_{Y,\eta} \to \mathscr{O}_{X,z}$ being injective for all points $z \in f^{-1}(\eta)$, the hypothesis that $t|(U \cap V) = 0$ implies that $s_\eta = 0$, and as Y is integral, this implies $s = 0$, whence our assertion. Applying now (18.10.2), (iii)), we conclude that f is *flat* at the point x, in other words $x \in U$, hence U is both open and closed in X; as it is non-empty and X is connected, we have $U = X$.

Note now that since f is étale, hence locally quasi-finite and Y is irreducible, the maximal points of X belong to $f^{-1}(\eta)$ (2.3.4) and for all $x \in X$, there is a neighbourhood of x that contains only a finite number of these maximal points; in other words, the set of irreducible components of X is locally finite. But by (18.10.1) X is integral and geometrically unibranch at every point, hence every point of X belongs to only a single irreducible component, and since these latter form a locally finite set of open (and closed) subsets in X; since X is connected, it is integral. □

IV-4 | 160

Remark (18.10.3.1). — If $f : X \to Y$ is locally of finite presentation, dominant and quasi-compact, the fibre $f^{-1}(\eta)$ is non-empty by virtue of (**I**, 1.1.5). On the other hand, if one does not suppose f quasi-compact, f *can be non-ramified and dominant without f being étale* (always assuming the hypotheses of (18.10.3) verified for X and Y). This is what the following example shows: one takes for Y the affine plane $\mathrm{Spec}(\mathbf{C}[T, U])$; one considers on the other hand a family $(X_i)_{i \in \mathbf{Z}}$ of preschemes isomorphic to the affine line $\mathrm{Spec}(\mathbf{C}[T])$, and one forms the prescheme obtained by "gluing together" X_{2j} and X_{2j+1} at the point -1 and X_{2j+1} and X_{2j+2} at the point $+1$. Let us define on the other hand $f : X \to Y$ as being equal on X_{2j} to the closed immersion whose image is the line of equation $y = 2j$, transforming the point -1 to $(2j-1, 2j)$ and the point 1 to $(2j+1, 2j)$; on X_{2j+1}, f_j is the closed immersion whose image is the line of equation $x = 2j+1$, transforming the point -1 to $(2j+1, 2j)$ and the point 1 to $(2j+1, 2j+2)$. It is clear that f is a local immersion (hence is not ramified) and is dominant but the fibre of the generic point of Y is empty and f is not étale.

Corollary (18.10.4). — *Let $g : \mathrm{X} \to \mathrm{S}$, $h : \mathrm{X} \to \mathrm{S}$ be two locally of finite presentation morphisms, $f : \mathrm{X} \to \mathrm{Y}$ an* S*-morphism, x a point of* X*, $y = f(x)$, $s = g(y) = h(x)$. Suppose that h is flat at the point x and $g^{-1}(s)$ normal at the point y. Then, for f to be étale at the point x, it is necessary and sufficient that f be formally unramified at the point x and that $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$.*

Proof. The conditions are necessary by virtue of the fact that if f is étale at the point x, it follows from this that f_s is also flat and quasi-finite at the point x (6.1.2). Conversely, if the conditions of the statement are satisfied, in order to see that f is étale at the point x, it suffices, by virtue of (17.8.3), to see that f_s is so. One can therefore restrict to the case where S is the spectrum of a field. It then follows from the hypothesis $\dim_x \mathrm{X} = \dim_y \mathrm{Y}$ and the fact that f is not ramified at the point x that one has (by (5.2.3) and (17.4.1)) $\dim \mathscr{O}_{\mathrm{X},x} = \dim \mathscr{O}_{\mathrm{Y},y}$ integral, and we deduce from (**0**, 7.4.4) and (**0**, 16.3.10) that the homomorphism $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{X},x}$ is injective; it then suffices to apply (18.10.1). ☐

Corollary (18.10.5). — *With the notations of (18.10.4), suppose that h is flat and that $g^{-1}(s)$ is normal for all $s \in \mathrm{S}$ (which will be the case for example if g is smooth). For f to be an open immersion, it is necessary and sufficient that f be a monomorphism and that, for all $x \in \mathrm{X}$, $\dim_x h^{-1}(s) = \dim_y g^{-1}(s)$, where $y = f(x)$ and $s = g(y) = h(x)$.*

Proof. One needs only to prove the sufficiency of the conditions; now, it follows from (17.1.3) that a monomorphism locally of finite presentation is formally unramified, hence the hypotheses imply that f is étale (18.10.4); on the other hand, by (17.9.1), we conclude from these and from (18.10.4). ☐

Lemma (18.10.6). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a flat and locally quasi-finite morphism.*

(i) The set $\mathrm{Max}(\mathrm{X})$ of maximal points of X *is in canonical bijection with the set $\coprod_{y \in \mathrm{Max}(\mathrm{Y})} f^{-1}(y)$.*
(ii) If Y *is irreducible with generic point η, the set of irreducible components of* X *is in canonical bijection with $f^{-1}(\eta)$, and is in particular finite if and only if $f^{-1}(\eta)$ is finite.* **IV-4** | 161
(iii) If the set of irreducible components of Y *is locally finite, so is the set of irreducible components of* X*, and in particular* X *is locally connected.*

Proof. The assertion (i) follows immediately from (2.3.4) and (**0**, 2.1.6) and from the fact that the fibres $f^{-1}(y)$ are discrete sets. It is clear that (ii) is a particular case of (i). Finally, to prove (iii), one can, by virtue of (i), restrict to the case where f is quasi-finite and where Y is irreducible (**0**, 2.1.6) and the conclusion follows from (ii) and from the fact that the fibres of f are finite sets (**II**, 6.2.2). ☐

Proposition (18.10.7). — *Let* Y *be a geometrically unibranch and irreducible prescheme (resp. integral), $f : \mathrm{X} \to \mathrm{Y}$ an étale morphism. Then* X *is isomorphic to a sum of preschemes irreducible (resp. integral) X_λ, where λ ranges over the fibre $f^{-1}(\eta)$ of the generic point η of* Y*. The X_λ are geometrically unibranch; if* Y *is normal, the X_λ are normal.*

Proof. Indeed, X is geometrically unibranch (17.5.7), hence every point of X belongs to only a single irreducible component; moreover, by virtue of (18.10.6) and (17.6.1) the set of irreducible components of X is locally finite, hence (**0**, 2.1.6) these are the connected components X_λ of X ($\lambda \in f^{-1}(\eta)$), and X is the sum of the X_λ; by virtue of (17.5.7), if Y is integral (resp. normal), the X_λ are integral (resp. normal). ☐

Proposition (18.10.8). — *Let* Y *be a normal and integral prescheme, of generic point η, $\mathrm{K} = k(\eta)$ its field of rational functions, $f : \mathrm{X} \to \mathrm{Y}$ an étale morphism. Suppose that $f^{-1}(\eta)$ is finite and non-empty, so that the* K*-scheme $f^{-1}(\eta)$ is the spectrum of a finite separable* K*-algebra* L*, composed as a direct product of a finite number of fields K_i ($1 \leqslant i \leqslant n$), finite separable extensions of* K *(17.6.2). Let Y' be the integral closure of* Y *in* L*, isomorphic to the sum of the integral closures Y_i of* Y *in K_i (**II**, 6.3.6). Then the morphism f factorises uniquely as $\mathrm{X} \xrightarrow{f'} \mathrm{Y}' \xrightarrow{g} \mathrm{Y}$, such that the restriction $f'_\eta : f^{-1}(\eta) \to g^{-1}(\eta)$ is the canonical isomorphism. Moreover f' is a local isomorphism; for f' to be an open immersion, it is necessary and sufficient that f be separated.*

Proof. By virtue of (18.10.7), one can restrict to the case where X is integral and normal. The existence and uniqueness of the factorisation considered for f follow from (**II**, 6.3.9). As f' is birational and locally quasi-finite, the final assertions follow from (8.12.10). □

Corollary (18.10.9). — *Under the hypotheses of (18.10.8), for f to be a finite morphism (in other words, for* X *to be an étale covering of* Y*), it is necessary and sufficient that* X *be isomorphic to the integral closure* Y′ *of* Y *in* L.

Proof. If f is finite, the canonical injection $j : \mathrm{X} \to \mathrm{Y}'$ (which is an open immersion) is also a finite morphism (**II**, 6.1.5), (v)), hence a closed morphism (**II**, 6.1.10), and as $j(\mathrm{X})$ is dense in Y′ by (18.10.8), we have $j(\mathrm{X}) = \mathrm{Y}'$. Conversely, as Y is normal and L is a composed direct product of finite separable extensions of K, Y′ is finite over Y (Bourbaki, *Alg. comm.*, chap. V, § 1, n° 6, cor. 1 of prop. 18), whence the corollary. □

(18.10.10). Let Y be a normal and integral prescheme, $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ its field of rational functions. We say that a K-algebra of finite rank L is *non-ramified over* Y if: 1° L is a separable K-algebra, hence composed as a direct product of finite separable extensions K_i of K ($1 \leqslant i \leqslant n$); 2° the integral closure Y′ of Y in L (sum of the preschemes that are integral closures of Y in the K_i) is *non-ramified over* Y (which, by (18.10.3), is equivalent to saying that Y′ is étale over Y). When $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, **IV-4** | 162
where A is an integral ring, integrally closed, K its field of fractions, one says also (by a slight abuse of language, and when there is no confusion to be feared with the terminology of (17.3.2), (ii)) that L is non-ramified over A instead of "non-ramified over Y".

Remark (18.10.11). — Instead of saying that L is non-ramified over Y, certain authors say also that L is "non-ramified over K"; we shall not follow this usage, which lends itself to confusions.

One can then express (18.10.9) in the following form:

Corollary (18.10.12). — *Let* Y *be a normal and integral prescheme,* $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ *its field of rational functions. Then the functor* $\mathrm{X} \mapsto \mathrm{R}(\mathrm{X})$ *is an equivalence of the category of étale coverings of* Y *and of the category of finite* K*-algebras that are non-ramified over* Y*. One obtains a quasi-inverse functor by making correspond to every finite étale* K*-algebra* L*, non-ramified over* Y*, the integral closure* Y′ *of* Y *in* L.

Proposition (18.10.13). — *Let* Y *be a normal and integral prescheme,* $\mathrm{K} = \mathrm{R}(\mathrm{Y})$ *its field of rational functions.*

(i) The field K *is a* K*-algebra non-ramified over* Y.

(ii) Let L *be a finite extension of* K*, non-ramified over* Y*, and let* Z *be the integral closure of* Y *in* L*. Then, if* M *is an* L*-algebra non-ramified over* Z*,* M *is also a* K*-algebra non-ramified over* Y *("transitivity" of non-ramification).*

(iii) Let Y′ *be a normal and integral prescheme,* K′ *its field of rational functions,* $g : \mathrm{Y}' \to \mathrm{Y}$ *a dominant morphism. If* L *is a* K*-algebra non-ramified over* Y*, then* $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ *is a* K′*-algebra non-ramified over* Y′ *("translation" property). Furthermore if* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$ *and* $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ *are affine, and if* C *is the integral closure of* A *in* L*, then* $\mathrm{C}' = \mathrm{C} \otimes_{\mathrm{A}} \mathrm{A}'$ *is the integral closure of* A′ *in* $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$.

Proof. The assertion (i) is immediate, Y being its own normalisation in K. To prove (ii), let Z′ be the integral closure of Z in M, which is by hypothesis an étale covering of Z. As Z is étale and separated over Y, so is Z′ (17.3.3), and it is clearly the integral closure of Y in M. As M is a K-algebra, finite and separable, it follows from (18.10.10) that M is non-ramified over Y. Finally, to prove (iii), let us note that if Z is the integral closure of Y in L, $\mathrm{Z} \times_{\mathrm{Y}} \mathrm{Y}'$ is étale and separated over Y′ (17.3.3), finite over Y′ since Z is finite over Y, and admits for ring of rational functions $\mathrm{L} \otimes_{\mathrm{K}} \mathrm{K}'$ (**I**, 3.4.9); as Y′ is normal, it follows from (17.5.7), which finishes the proof. □

Corollary (18.10.14). — *Let* Y *and* Y′ *be two normal preschemes and integral,* K, K′ *their respective fields of rational functions,* $g : \mathrm{Y}' \to \mathrm{Y}$ *a dominant morphism, so that* K′ *is an extension of* K*. Let* L *be a finite extension of* K *non-ramified over* Y*,* L_1 *a composite extension (Bourbaki,* Alg.*, chap. VIII, § 8, def. 1) of* L *and* K′*. Then* L_1 *is an extension of* K′ *non-ramified over* Y′*. Furthermore if* $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$*,* $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$ *are affine, and if* C *is the integral closure of* A *in* L*, then the subring* $\mathrm{C}_1 = \mathrm{A}[\mathrm{C}, \mathrm{A}']$ *of* L_1 *is the integral closure of* A′ *in* L_1.

Proof. Indeed, $L \otimes_K K'$ is composed as a direct product of finite separable extensions of K', and L_1 is one of these extensions; hence the integral closure of Y in L is the sum of preschemes, one of which is the integral closure of Y in L_1. The corollary therefore follows from (18.10.14). **IV-4** | 163

When for example A is the ring of integers of an algebraic number field K, K' and L algebraic extensions of K, there are classical examples showing that the relation $C_1 = A[C, A']$ does not hold when L is not ramified over $A[\quad]$. □

(18.10.15). Suppose that $Y = \mathrm{Spec}(A)$ is affine, A being integral and integrally closed; let K be its field of fractions, L a finite separable extension of K, and we suppose that the integral closure C of A in L is a *projective* A-module of finite type (which will be the case for example if A is a Dedekind ring (Bourbaki, *Alg. comm.*, chap. VII, § 4, n° 10, prop. 22)). To say that L is non-ramified over A means that C is étale (18.10.10) and by virtue of (18.2.7), (ii)), this is equivalent to saying that the *discriminant* $d_{C/A}$ of $\mathrm{Spec}(C)$ over $\mathrm{Spec}(A)$ is *invertible* in A. In the particular case where C is a *free* A-module and $(x_i)_{1\leqslant i\leqslant n}$ a basis of C over A, this means that $\det(\mathrm{Tr}_{C/A}(x_i x_j))$ is invertible in A.

Theorem (18.10.16). — *Let Y be an integral prescheme, η its generic point, $f : X \to Y$ a separated and locally of finite type morphism. Suppose that every irreducible component of X dominates Y and that the generic fibre $X_\eta = f^{-1}(\eta)$ is a finite prescheme over $K = k(\eta)$, hence $(X_\eta)_{\mathrm{red}}$ is the spectrum of a finite separable K-algebra L, where $L = \prod_i L_i$ is a product of finite extensions L_i of K. Set*

$$n = [L : K] = \sum_i [L_i : K], \quad n_s = \sum_i [L_i : K]_s \quad \textit{(sum of separable degrees of the } L_i\textit{)}.$$

Let y be a geometrically unibranch point of Y, and denote by $n(y)$ the sum of the separable degrees over $k(y)$ of the residue fields of the isolated points of $f^{-1}(y)$. Then:

(i) One has $n(y) \leqslant n_s \leqslant n$.

(ii) Suppose X reduced and the point y normal in Y. For $n(y) = n$, it is necessary and sufficient that there exist a neighbourhood U of y in Y such that the restriction $f^{-1}(U) \to U$ of f is an étale and finite morphism.

Proof. A) Reduction to the case where f is quasi-finite. — One knows (13.1.4) that the set Z of isolated points $x \in X$ of $f^{-1}(f(x))$ is open in X and by hypothesis Z contains $f^{-1}(\eta)$. Since $f^{-1}(\eta)$ is finite, Z is an increasing filtered union of quasi-compact open sets V_λ containing $f^{-1}(\eta)$ (dense in X by hypothesis). If $n_\lambda(y)$ is the sum of the separable degrees over $k(y)$ of the residue fields of the points of $V_\lambda \cap f^{-1}(y)$, we have $n(y) = \sup n_\lambda(y)$; to prove (i), it will therefore suffice to show that $n_\lambda(y) \leqslant n_s$. On the other hand, if $n_\lambda(y) = n$, suppose that we have shown that there exists a neighbourhood U of y in Y such that the restriction $V_\lambda \cap f^{-1}(U) \to U$ of f is an étale and finite morphism; since f is separated, the canonical injection $V_\lambda \cap f^{-1}(U) \to f^{-1}(U)$ is then also a finite morphism (**II**, 6.1.5), (v)), hence $V_\lambda \cap f^{-1}(U)$ is closed in $f^{-1}(U)$; but being dense everywhere in $f^{-1}(U)$, it is necessarily equal to it.

B) Reduction to the case where X is reduced, f finite, and $Y = \mathrm{Spec}(\mathscr{O}_{Y,y})$. — We note that the hypothesis on f implies that it is also satisfied for f_{red} and that the conclusion of (i) does not concern f_{red}. One can therefore replace f everywhere by f_{red} and thus suppose X reduced.

On the other hand, by virtue of (18.12.13) [1], one can (after having replaced Y by an open affine **IV-4** | 164
neighbourhood V of y and f by its restriction $f^{-1}(V) \to V$) factorise f as $X \xrightarrow{j} X' \xrightarrow{g} Y$, where g is *finite* and j an open immersion. Since X is reduced, one can replace X' by the reduced closed sub-prescheme with underlying space $\overline{j(X)}$ (**I**, 5.2.2), hence suppose X' reduced and such that every irreducible component of X' dominates Y. Moreover, $j(X)$ is an open dense set in X' and has only a finite number of irreducible components by virtue of the hypothesis; hence (**0**, 1.2.7) the irreducible components of X' are the closures of those of $j(X)$, and by construction the fibre $g^{-1}(\eta)$ identifies with $f^{-1}(\eta)$. If one supposes the theorem proved for f finite, one can apply it to g, and as $f^{-1}(y)$ identifies with a sub-prescheme of $g^{-1}(y)$, the relation (i) for g implies the same relation for f. Suppose on the other hand that $n(y) = n$; then, by virtue of what precedes, one can apply the conclusion of (ii) to the morphism g, and replacing Y as needed by a neighbourhood of y, one can suppose g étale; it follows in addition, on the one hand that $f^{-1}(y) = g^{-1}(y)$. As the morphism g is closed and $j(X)$ is open in X', there exists a neighbourhood U of y in Y such that $j(X) \cap g^{-1}(U) = j(f^{-1}(U))$, which proves that (ii) is also true for f.

We are thus reduced to the case where f is moreover a *finite* morphism. It is immediate that, to prove (i), one can restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$. Let us show that the same is true for proving (ii). Indeed, suppose we have shown that the condition $n(y) = n$ implies that f is flat and formally unramified at the points of $f^{-1}(y)$, and that Y is *integral*, we conclude from (18.4.13) that f is *étale at the points* of $f^{-1}(y)$, hence also in a neighbourhood V of $f^{-1}(y)$ in X; but since f is a finite morphism, hence closed, V contains an open set of the form $f^{-1}(\mathrm{U})$, where U is a neighbourhood of y in Y.

C) Reduction to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$ *and* $\mathscr{O}_{\mathrm{Y},y}$ *is strictly local.* — Set $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$, and let $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$ be the strict henselisation of A (18.8.7), which is a geometrically unibranch local integral ring (18.8.15). Let $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, and let y' and η' be the closed point and the generic point of Y'; y' is above y and since the morphism $g : \mathrm{Y}' \to \mathrm{Y}$ is flat (18.8.8), η' is above η (2.3.4). Set $\mathrm{X}' = \mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$; f' is finite and since g is flat, every irreducible component of X' dominates Y' (2.3.7). If $n'(y')$, n' and n'_s are the numbers defined for f' in the same way as $n(y)$, n and n_s for f, we have $n'_s = n_s$ and $n'(y') = n(y)$ (**I**, 6.4.8), and it therefore amounts to proving the assertion (i) for f or for f'. Moreover, if X is reduced and y a normal point of Y (hence A integrally closed (18.8.12)). On the other hand, it follows from the definition (18.8.7), from the remark (8.1.2), *a*)) and from the theorem of the double inductive limit, that A' is the inductive limit of rings A_α such that the morphisms $\mathrm{Spec}(\mathrm{A}_\alpha) \to \mathrm{Spec}(\mathrm{A}) = \mathrm{Y}$ are étale; consequently X' is the projective limit of the $\mathrm{X}_\alpha = \mathrm{X} \otimes_{\mathrm{A}} \mathrm{A}_\alpha$, which are étale over X, hence reduced since X is (17.5.7); by passage to the limit,

we deduce that X' is reduced (8.7.1). This being so, if $n(y) = n$, on has $n_s = n$ and consequently **IV-4** | 165
the residue fields at the points of X_η are necessarily separable over $k(\eta)$; the residue fields at the points of $\mathrm{X}'_{\eta'}$ are then algebraic and separable over $k(\eta')$ (4.6.1), in other words $n'_s = n' = n$. The morphism f' has therefore the same properties as f, and if one proves that f' is étale, it will follow by (17.7.1).

D) End of the proof. — Suppose therefore A strictly local. Since A is henselian and the morphism f finite, it follows from (18.5.11) that if x_j ($1 \leqslant j \leqslant m$) are the distinct points of $f^{-1}(y)$, X is the *sum* of open sub-preschemes $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j}) = \mathrm{X}_j$, which are the connected components of X. By hypothesis, each of them dominates Y, hence every irreducible component of X meets $f^{-1}(\eta)$, and so the number of points of $f^{-1}(\eta)$ is $\geqslant m$; *a fortiori* we have $n_s \geqslant m$. Suppose now the hypotheses of (ii) satisfied; the relations $n(y) = n_s = n$ imply first of all, since every irreducible component of X dominates Y, that each of the X_j is *irreducible*; moreover, as the X_j are *reduced*, they are *integral*; finally the relations $m = n_s = n$ imply that for the generic point ξ_j of X_j, $k(\xi_j)$ is a separable extension of $k(\eta)$ of degree 1, hence isomorphic to $k(\eta)$; in other words, the restriction $f|\mathrm{X}_j : \mathrm{X}_j \to \mathrm{Y}$ of f is *finite and birational*; moreover, X_j is *integral* and by hypothesis Y is *normal*; hence (8.12.10.1) $f|\mathrm{X}_j$ is an *isomorphism*, which proves that f is étale, and finishes the proof of (18.10.16). □

Remark (18.10.17). — The method of "étale localisation" used in the proof of (18.10.16) also allows one to improve the results of (15.5.1) by eliminating the Noetherian hypothesis. One has first of all the following result which generalises (15.5.2):

(18.10.17.1). Let $f : \mathrm{X} \to \mathrm{Y}$ be a separated and quasi-finite morphism, y a point of Y such that f is universally open at the points of $f^{-1}(y)$. Then, for every generisation y' of y, we have $n(y) \leqslant n(y')$.

Proof. One can replace Y by the reduced closed sub-prescheme $\overline{\{y'\}}$ and X by $f^{-1}(\mathrm{Z})$, hence suppose Y integral with generic point y'. Using (18.12.13) (as in (18.10.16), B)), one can suppose that X is an open dense subset of a prescheme X' and that f is the restriction to X of a finite morphism $g : \mathrm{X}' \to \mathrm{Y}$. Let us show that $g^{-1}(y') = f^{-1}(y')$. Indeed, let $i : g^{-1}(y') \to \mathrm{X}'$ be the canonical injection, which is a flat morphism since y' is the generic point of Y (**I**, 3.6.5); one can write $f^{-1}(y') = i^{-1}(\mathrm{X})$. On the other hand, the canonical injection $j : \mathrm{X} \to \mathrm{X}'$ is a morphism of finite type since the composite $g \circ j = f$ is of finite type and g is separated (1.5.4); hence X is a retrocompact open subset of X', and consequently pro-constructible in X' (**I**, 1.9.5), (v)). One concludes therefore from (2.3.10) that $i^{-1}(\overline{\mathrm{X}}) = \overline{i^{-1}(\mathrm{X})}$, in other words $f^{-1}(y')$ is dense in $g^{-1}(y')$; but as $g^{-1}(y')$ is discrete, we necessarily have $f^{-1}(y') = g^{-1}(y')$.

[8](1) The reader will verify that (18.10.16) is not used in the proof of (18.12.13). If f is supposed locally of finite presentation (resp. quasi-projective), one can replace (18.12.13) by (8.12.8) (resp. (8.12.6)).

We are thus reduced to proving that the sum of the numbers $[k(x):k(y)]_s$ where x ranges over the points of $g^{-1}(y)$ where g is universally open, is at most equal to $n(y')$ (for the morphism g). **IV-4** | 166
Using then the fact that the property of being universally open at a point is preserved by base change, one shows, as in (18.10.16), C)), that one can reduce to the case where $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, with $\mathscr{O}_{\mathrm{Y},y}$ *strictly local*. Applying (18.5.11), *c*)), we see that if x_j $(1 \leqslant j \leqslant n)$ are the points of $f^{-1}(y)$, X is the *sum* of n sub-preschemes $\mathrm{X}_j = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j})$; the hypothesis that g is open in each of the points x_j implies that the sub-prescheme X_j corresponding dominates Y (1.10.3); one concludes as in (18.10.16), D)).

We then deduce from (18.10.17.1) that *the conclusions of* (15.5.1) *are still valid when one no longer supposes* Y *locally Noetherian, but only that f is separated, quasi-finite, and of finite presentation*. □

Indeed, to prove assertion (i) of (15.5.1), one remarks that, since f is of finite presentation, the set E of points $z \in \mathrm{Y}$ such that $n(z) \geqslant n(y)$ is locally constructible (9.7.9); to show that y is interior to E, it suffices, by virtue of (1.10.1), to see that every generisation y' of y belongs to E, which is nothing other than (18.10.17.1).

To prove assertion (ii) of (15.5.1), one remarks that, since f is of finite presentation, using the method of (8.10.5), (xii)) applied following (8.1.2), *a*)), and one can therefore suppose that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$, and the ring $\mathscr{O}_{\mathrm{Y},y}$ is *strictly local*. Applying (18.5.11), *c*)), we see then that if x_j $(1 \leqslant j \leqslant n)$ are the points of $f^{-1}(y)$, X is the *sum* of n open sub-preschemes $\mathrm{X}_j = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x_j})$ which are *finite* over Y, and of an open prescheme X''. If one proves that $\mathrm{X}'' = \varnothing$, we will have shown that f is a finite morphism, and the conclusion follows from the fact that, since X is open in each of the points x_j, the restriction of f to X_j, being a finite morphism (**II**, 6.1.10), is *surjective*; the hypothesis that $z \mapsto n(z)$ is constant implies (17.4.1), hence one terminates as in (18.10.16), D)).

Finally, the proof of (iii) reproduces without change that given in (15.5.1).

(18.10.17.2). By analogous methods using strict henselisation, one can, in most of the results of §§ 14 and 15, eliminate the Noetherian hypotheses.

Lemma (18.10.18). — *Let* X, Y *be two preschemes, $f : \mathrm{X} \to \mathrm{Y}$ a birational morphism* [1]*, x a point of* X*. For f to be an isomorphism local at the point x, it is necessary and sufficient that f be étale at the point x. For f to be an open immersion, it is necessary and sufficient that f be étale and separated.*

Proof. The conditions stated being trivially necessary, all that is needed is to see that they are sufficient. For the first assertion, the question is local on X and Y, hence one can suppose X and Y affine, hence f *separated*, and we are thus reduced to proving the second assertion. By virtue of (17.9.1), it is a question of proving that f is *radicial*. Let $y \in \mathrm{Y}$, and let us prove that $f^{-1}(y)$ is radicial on $k(y)$. The base change $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Y}$ not changing the fact that f is étale, separated, and birational, one can restrict to the case where $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, with $\mathrm{A} = \mathscr{O}_{\mathrm{Y},y}$. Let $\mathrm{A}' = {}^{\mathrm{hs}}\mathrm{A}$, which is a local ring **IV-4** | 167
and an $\mathscr{O}_{\mathrm{Y},y}$-module faithful flat. Set $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A}')$, $f' = f_{(\mathrm{Y}')} : \mathrm{X}' \to \mathrm{Y}'$, so that f' is étale and separated; moreover, as the morphism $\mathrm{Y}' \to \mathrm{Y}$ is flat, the reasoning of (6.15.4.1) shows that f' is also birational.

Suppose therefore $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, A strictly local, f separated, étale and birational, and let us show that $f^{-1}(y)$ can contain only a single point. Indeed, if $f^{-1}(y)$ contained two distinct points x_1, x_2, there would exist two Y-sections u_1, u_2 of f such that $u_1(y) = x_1$ and $u_2(y) = x_2$ (18.8.1); but since f is étale and separated, u_1 and u_2 would be isomorphisms of Y onto two connected components (open) of X (17.9.4) and there would then be two maximal points of X at least above every maximal point of Y, which contradicts the hypothesis that f is *birational*. □

Proposition (18.10.19). — *Let* Y *be a reduced prescheme whose set of irreducible components is locally finite, $f : \mathrm{X} \to \mathrm{Y}$ a separated morphism of finite type and formally unramified, g a* Y*-rational section of f (i.e. a* Y*-rational map from* Y *to* X*),* U *the domain of definition of g* (**I**, 7.2.1)*, and let* Z *be the closure of $g(\mathrm{U})$ in* X*. Then, for all points $y \in \mathrm{Y} - \mathrm{U}$ such that y is geometrically unibranch, $\mathrm{Z}_y = \mathrm{Z} \cap f^{-1}(y) = \varnothing$. In particular, if* Y *is geometrically unibranch, $g(\mathrm{U})$ is a subset of* X *that is both open and closed. If, for every closed irreducible* T *of* X*, $f(\mathrm{T})$ is closed in* Y*, g is defined at every geometrically unibranch point of* Y*.*

[8] [1] We understand by this the notion defined in (6.15.4), but where we do not suppose X and Y reduced.

Proof. Since Y is reduced and f separated, it follows from (**I**, 7.2.2) that there exists a U-section u of $f^{-1}(\mathrm{U})$ belonging to the class g; by (17.4.1.2), moreover, f is formally unramified, hence u is an isomorphism of U onto the prescheme induced on the open set $u(\mathrm{U})$ of X. If one denotes also by Z the reduced sub-prescheme of X having Z for underlying space, $g(\mathrm{U}) = u(\mathrm{U})$, being reduced, is induced also by Z, and by construction $f' = f|\mathrm{Z}$ is a *birational* morphism of Z to Y, moreover formally unramified since f is (17.1.3). The question being local over Y at the point y, one can restrict (replacing Y by a neighbourhood of y) to the case where Y is *integral*; then $u(\mathrm{U})$ is irreducible, hence also Z, and by construction Z is also integral. Then (**I**, 8.2.7) since f' is dominant, the homomorphism $\mathscr{O}_{\mathrm{Y},y} \to \mathscr{O}_{\mathrm{Z},x}$ is injective for all $x \in \mathrm{Z}_y$, and one concludes from (18.10.2), (iii)) that f' is an étale isomorphism local at the point x; but this would imply that the U-section u would be extendable to an open set distinct from U, contrary to the definition of the latter. We therefore necessarily have $\mathrm{Z}_y = \varnothing$, which establishes the first assertion; the second is an obvious consequence. Finally, under the last hypotheses of the statement, as Z is closed and irreducible, $f(\mathrm{Z})$ is closed in Y, hence equal to Y, i.e. $\mathrm{Z}_y \neq \varnothing$ for all $y \in \mathrm{Y}$, which finishes the proof. □

Remark (18.10.20). — Let Y be a locally Noetherian prescheme. One says that a morphism $f : \mathrm{X} \to \mathrm{Y}$ is *essentially proper* if it is locally of finite type and if, for every base change $\mathrm{Y}'' \to \mathrm{Y}$, where Y'' is *locally Noetherian*, the image by $f_{(\mathrm{Y}'')} : \mathrm{X}_{(\mathrm{Y}'')} \to \mathrm{Y}''$ of every closed irreducible subset of X'' is *closed*. To say that f is essentially proper for every Y-scheme $\mathrm{Y}' = \mathrm{Spec}(\mathrm{A})$ where A is **IV-4** | 168
a discrete valuation ring, the canonical map (**II**, 7.3.2.2) relative to the Y'-prescheme $\mathrm{X} \times_{\mathrm{Y}} \mathrm{Y}'$, is bijective. The reasoning of (**II**, 7.3.8) proves that this is the same as saying that f (supposed locally of finite type) is *separated* and that for every base change $\mathrm{Y}'' \to \mathrm{Y}$, where Y'' is *locally Noetherian*, the image by $f_{(\mathrm{Y}'')} : \mathrm{X}_{(\mathrm{Y}'')} \to \mathrm{Y}''$ of every closed irreducible subset of X'' is *closed*. To say that f is *proper* (for Y locally Noetherian) means therefore that f is *essentially proper and of finite type* (**II**, 7.3.8); but one encounters important examples of essentially proper morphisms that are not of finite type (for example certain "Picard preschemes" or certain "preschemes of Néron-Severi" (chap. VI)). It evidently follows from (18.10.19) that if Y is *locally Noetherian, reduced and geometrically unibranch*, and if $f : \mathrm{X} \to \mathrm{Y}$ is *essentially proper and not ramified*, then every Y-rational section of f is *defined everywhere*.

Proposition (18.10.21). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a locally of finite presentation morphism, x a point of* X*, $y = f(x)$, and suppose the ring $\mathscr{O}_{\mathrm{Y},y}$ integral and geometrically unibranch. Let r be the minimum number of generators of the $\mathscr{O}_{\mathrm{X},x}$-module $(\Omega^1_{\mathrm{X}/\mathrm{Y}})_x$ (equal to the rank of the $k(x)$-vector space $(\Omega^1_{\mathrm{X}/\mathrm{Y}})_x \otimes_{\mathscr{O}_{\mathrm{X},x}} k(x)$). Then, for f to be smooth at the point x, it is necessary and sufficient that, if y' is the generic point of the unique irreducible component of* Y *containing y, there exists a generisation x' of x such that $f(x') = y'$ and* $\dim_{x'} f^{-1}(y') \geqslant r$.

Proof. If f is smooth at the point x, it is also so in an open neighbourhood U of x, hence in every generisation x' of x, and $\dim_{x'}(f^{-1}(f(x'))) = r$ in each of these points, by virtue of (17.10.2). As f is moreover flat in U, there exists a generisation of x above every generisation of y (2.3.4), which proves that the condition is necessary. To show that it is sufficient, let us first note that by (17.5.1), it suffices to prove that the morphism deduced from f by the base change $\mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y}) \to \mathrm{Y}$ is smooth at the point x; one can therefore suppose that $\mathrm{Y} = \mathrm{Spec}(\mathscr{O}_{\mathrm{Y},y})$.

The question is local on X, hence (16.4.22) and (**0**, 5.2.2)) one can suppose that X is affine, and that there exist r sections s_i ($1 \leqslant i \leqslant r$) of $\mathscr{O}_{\mathrm{X}}$ above X such that the sections $d_{\mathrm{X}/\mathrm{Y}}(s_i)$ generate $\Omega^1_{\mathrm{X}/\mathrm{Y}}$. Let

$$g : \mathrm{X} \longrightarrow \mathrm{Z} = \mathrm{Y}[\mathrm{T}_1, \ldots, \mathrm{T}_r] = \mathbf{V}_{\mathrm{Y}}$$

be the Y-morphism corresponding to the homomorphism $\mathscr{O}^r_{\mathrm{Y}} \to f_*(\mathscr{O}_{\mathrm{X}})$ of $\mathscr{O}_{\mathrm{Y}}$-Modules defined by the s_i (considered as sections of $f_*(\mathscr{O}_{\mathrm{X}})$ above Y) (**II**, 1.2.7). We have the exact sequence

$$g^*(\Omega^1_{\mathrm{Z}/\mathrm{Y}}) \longrightarrow \Omega^1_{\mathrm{X}/\mathrm{Y}} \longrightarrow \Omega^1_{\mathrm{X}/\mathrm{Z}} \longrightarrow 0.$$

As the $d\mathrm{T}_i$ ($1 \leqslant i \leqslant r$) generate $\Omega^1_{\mathrm{Z}/\mathrm{Y}}$, it follows from the definition of g that the homomorphism $g^*(\Omega^1_{\mathrm{Z}/\mathrm{Y}}) \to \Omega^1_{\mathrm{X}/\mathrm{Y}}$ is surjective, hence $\Omega^1_{\mathrm{X}/\mathrm{Z}} = 0$, and g is consequently *formally unramified* (17.4.1). We shall now see that g is *étale* at the point x, and taking into account (17.3.8), this will prove that f is smooth at the point x. As $\mathscr{O}_{\mathrm{Y},y}$ is supposed integral and geometrically unibranch, so is $\mathscr{O}_{\mathrm{Z},z}$ **IV-4** | 169 by (17.5.7) where $z = g(x)$, since the structural morphism $Z \to Y$ is smooth (17.3.8). By virtue of (18.10.1), it suffices to show that the homomorphism $\mathscr{O}_{Z,z} \to \mathscr{O}_{X,x}$ is injective; it amounts to the same thing to see that the morphism $\operatorname{Spec}(\mathscr{O}_{X,x}) \to \operatorname{Spec}(\mathscr{O}_{Z,z})$ corresponding is *dominant*, since $\mathscr{O}_{Z,z}$ is integral (**I**, 1.2.7). Let y' be the generic point of $Y = \operatorname{Spec}(\mathscr{O}_{Y,y})$ and let z' be the generic point of $Z = \operatorname{Spec}(\mathscr{O}_{Y,y}[T_1, \dots, T_r])$ which is integral; if $h : Z \to Y$ is the structural morphism, on has $h(z') = y'$. It suffices to prove that there exists in $f^{-1}(y')$ a generisation x' of x such that the image of x' by the morphism $g_{y'} : f^{-1}(y') \to h^{-1}(y')$ equals z'. Moreover z' is also the generic point of $h^{-1}(y') = \operatorname{Spec}(k(y')[T_1, \dots, T_r])$; as the morphism $g_{y'}$ is not ramified, hence quasi-finite (17.4.2), it follows from (5.4.1) that the restriction of $g_{y'}$ to every irreducible component of *dimension* $\geqslant r$ of $f^{-1}(y')$ is a dominant morphism. Now, there exists by hypothesis a component containing a generisation x' of x; *a fortiori* its generic point is a generisation of x, whence the conclusion. □

18.11. Application to Noetherian local algebras complete over a field The following lemma generalises (**0**, 21.9.1) and (**0**, 21.9.2):

Lemma (18.11.1). — *Let k be a field, A a Noetherian complete local ring which is a k-algebra, K the residue field of A.*

(i) If the K-vector space $\Omega^1_{K/k}$ is of finite rank, the A-module $\widehat{\Omega}^1_{A/k}$ is of finite type.

(ii) If moreover A is a k-algebra formally smooth (for the adic topology), $\widehat{\Omega}^1_{A/k}$ is a free A-module of rank equal to

$$\dim(A) + \operatorname{rg}_K(\Omega^1_{K/k}) - \operatorname{rg}_K(\Upsilon_{K/k}).$$

Furthermore, for every subfield k_0 of k such that Ω^1_{K/k_0} is a k-vector space of finite rank, $\widehat{\Omega}^1_{A/k_0}$ is a free A-module of rank equal to

$$\dim(A) + \operatorname{rg}_K(\Omega^1_{K/k_0}) - \operatorname{rg}_K(\Upsilon_{K/k_0}) + \operatorname{rg}_K(\Omega^1_{k/k_0}).$$

Proof. The assertion (i) is noted as a reminder, having been proved in (**0**, 20.7.15). To prove (ii), note that this assertion has been in fact established by the demonstration of (**0**, 21.9.2), making only intervene the finiteness of the transcendence degree of K over k (by way of the Cartier equality (**0**, 21.7.1)). □

Lemma (18.11.2). — *Let k be a field of characteristic $p > 0$, C a k-algebra, local Noetherian, complete, whose residue field is a finite extension of k, and which is integral; let n be its dimension, L its field of fractions. Then $\Omega^1_{L/k}$ and $\Upsilon_{L/k}$ are L-vector spaces of finite rank, and one has*

(18.11.2.1) $$\operatorname{rg}_L(\Omega^1_{L/k}) - \operatorname{rg}_L(\Upsilon_{L/k}) \geqslant n.$$

Suppose furthermore that $[k : k^p] < +\infty$. Then one has the equality **IV-4** | 170

(18.11.2.2) $$\operatorname{rg}_L(\Omega^1_{L/k}) - \operatorname{rg}_L(\Upsilon_{L/k}) = n.$$

Furthermore, $\Omega^1_{C/k}$ is then isomorphic to $\widehat{\Omega}^1_{C/k}$, hence (18.11.1) is a C-module of finite type.

Proof. We know in fact (**0**, 19.8.9) that there exists a sub-k-algebra C_0 of C isomorphic to a formal power series algebra in n variables $k[[T_1, \dots, T_n]]$ and such that C is a C_0-algebra finite. Hence, L is a *finite* extension of the field of fractions $L_0 = k((T_1, \dots, T_n))$ of C_0. Now, one has the exact sequence of L-vector spaces obtained by applying (**0**, 20.6.17.1) to the three fields $k \subset L_0 \subset L$ (and taking into account (**0**, 20.4.13)):

$$0 \longrightarrow \Upsilon_{L/k} \otimes_{L_0} L \longrightarrow \Upsilon_{L/k} \longrightarrow \Upsilon_{L/L_0} \longrightarrow \Omega^1_{L/k} \otimes_{L_0} L \longrightarrow \Omega^1_{L/k} \longrightarrow \Omega^1_{L/L_0} \longrightarrow 0.$$

Since L is a finite extension of L_0, Ω^1_{L/L_0} and Υ_{L/L_0} are L-vector spaces of finite rank having the same rank, by the Cartier equality (**0**, 21.7.1); from the preceding exact sequence we therefore already deduce that

$$\operatorname{rg}_L(\Omega^1_{L/k}) - \operatorname{rg}_L(\Upsilon_{L/k}) = \operatorname{rg}_{L_0}(\Omega^1_{L_0/k}) - \operatorname{rg}_{L_0}(\Upsilon_{L_0/k}).$$

To prove (18.11.2.1) or (18.11.2.2), one can therefore restrict to proving these relations by replacing C and L by C_0 and L_0. As L_0 is separable over k (**0**, 21.7.1), we have $\Upsilon_{L_0/k} = 0$ (**0**, 20.6.3); one deduces hence already that in all cases $\operatorname{rg}_{L_0}(\Omega^1_{L_0/k}) \geqslant n$, with equality when $[k : k^p] < +\infty$; this already proves (18.11.2.1) and (18.11.2.2).

Finally, to see that $\Omega^1_{C/k}$ is isomorphic to $\widehat{\Omega}^1_{C/k}$ when $[k:k^p]<+\infty$, it suffices to prove that $\Omega^1_{C/k}$ is a C-module of finite type, since C is a Noetherian complete local ring (**0**, 7.3.5), taking into account (**0**, 20.4.5). Since $\Omega^1_{C/k}$ is isomorphic to $\widehat{\Omega}^1_{C/k}$ and that $k[C_0^p]\subset k[C^p]$, the result follows that $\Omega^1_{C/k}$ is a C-module of finite type and C_0 a $k[C_0^p]$-module of finite type in the hypothesis $[k:k^p]<+\infty$. □

Lemma (18.11.3). — *Let* A *be a Noetherian local ring,* $\mathfrak{p}$ *a prime ideal of* A *such that* $A_\mathfrak{p}$ *is a Cohen-Macaulay ring. For every system of parameters* $(t_i)_{1\leqslant i\leqslant s}$ *of* $A_\mathfrak{p}$*, there exists a sequence* $(x_i)_{1\leqslant i\leqslant s}$ *of elements of* $\mathfrak{p}$ *such that the* x_i *form part of a system of parameters of* A *and that, for all* i*,* $x_i/1$ *is congruent to* t_i *modulo the ideal* $t_1A_\mathfrak{p}+\ldots+t_{i-1}A_\mathfrak{p}$*. In particular, if* $A_\mathfrak{p}$ *is regular and if the* t_i *form a regular system of parameters of* $A_\mathfrak{p}$*, so it is also the case of the* $x_i/1$*.* **IV-4 | 172**

Proof. This lemma itself will be a consequence of the following: □

Lemma (18.11.3.4). — *Let* A *be a Noetherian local ring,* $\mathfrak{p}$ *a prime ideal of* A*,* x *an element of* A*; then there exists an element* $x'\in$ A *such that* $x'/1=x/1$ *in* $A_\mathfrak{p}$ *and that* x' *does not belong to any of the minimal prime ideals of* A *not contained in* $\mathfrak{p}$*.*

Proof. Let us first show how (18.11.3.4) implies (18.11.3). Multiplying the t_i if necessary by invertible elements of $A_\mathfrak{p}$, one can suppose that they are of the form $x_i/1$, with $x_i\in\mathfrak{p}$ for $1\leqslant i\leqslant s$. We reason by induction on s. As t_1 forms part of a system of parameters of $A_\mathfrak{p}$ (**0**, 16.5.5), there is no $x'_1\in$ A such that $x'_1/1=t_1$ that can belong to a minimal prime ideal of $A_\mathfrak{p}$, hence it cannot belong to any of the minimal prime ideals of A contained in $\mathfrak{p}$. By virtue of (18.11.3.4), one can moreover choose x'_1 such that $x'_1/1=t_1$ (hence $x'_1\in\mathfrak{p}$) and that x'_1 does not belong to *any* minimal prime ideal of A, hence x'_1 forms part of a system of parameters of A (**0**, 16.3.4) and (16.3.7). One reasons then by induction, considering the ring $A'=A/x'_1A$ and the prime ideal $\mathfrak{p}'=\mathfrak{p}/x'_1A$; as $A'_{\mathfrak{p}'}\cong A_\mathfrak{p}/t_1A_\mathfrak{p}$, $A'_{\mathfrak{p}'}$ is also a Cohen-Macaulay ring (**0**, 16.5.5), and the images t'_i $(2\leqslant i\leqslant s)$ of the t_i in $A'_{\mathfrak{p}'}$ form a system of parameters of this ring; it suffices to apply A' and to the t'_i $(2\leqslant i\leqslant s)$ the induction hypothesis. The last assertion of (18.11.3) follows from the fact that in $A_\mathfrak{p}$, a system of parameters that generates the maximal ideal is a regular system of parameters (**0**, 17.1.1). □

Proof of (18.11.3.4). Let $(p'_k)_{1\leqslant k\leqslant r}$ be the suite of minimal prime ideals of A not contained in $\mathfrak{p}$, and let $(\mathfrak{p}_j)_{1\leqslant j\leqslant n}$ be the suite of other minimal prime ideals of A; one can suppose that $r\geqslant 1$. As $\mathfrak{p}'_k$ does not contain $\bigcap_{1\leqslant i\leqslant n}\mathfrak{p}_j$, there exists an element $y\in\bigcap_{1\leqslant i\leqslant n}\mathfrak{p}_j$ that does not belong to any of the $\mathfrak{p}'_k$ (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2); moreover $y/1$ belongs to all the minimal prime ideals of $A_\mathfrak{p}$, hence is nilpotent; if $(y/1)^h=0$, the element $x'=x+y^h$ will satisfy the requirement, since $x'/1=x/1$ by definition of the $\mathfrak{p}'_k$, on has $x'\notin\mathfrak{p}'_k$ for $1\leqslant k\leqslant r$, and on the other hand, if $\mathfrak{p}_j$ is a prime ideal of A not contained in $\mathfrak{p}$ but such that $x\notin\mathfrak{p}_j$, we also have $x'\notin\mathfrak{p}_j$ since $y\in\mathfrak{p}_j$. □

Proof of (18.11.3)*, continued.* We return to the proof of (18.11.3) when $p=1$. By virtue of (18.11.3.4), there exists a regular system of parameters $(t_i)_{1\leqslant i\leqslant s}$ of $A_\mathfrak{p}$ where the x_i $(1\leqslant i\leqslant s)$ belong to $\mathfrak{p}$ and form part of a system of parameters $(x_j)_{1\leqslant j\leqslant n}$ of A. Set $A_0=k[[T_1,\ldots,T_n]]$; as A is a k-algebra complete and the x_j belong to the maximal ideal $\mathfrak{m}$ of A, there exists a local k-homomorphism $u:A_0\to A$ such that $u(T_j)=x_j$ for $1\leqslant j\leqslant n$ (Bourbaki, *Alg. comm.*, chap. III, § 4, n° 5, prop. 6); if n is the ideal of A generated by the x_j $(1\leqslant j\leqslant n)$, it is by hypothesis an ideal of definition of A; one deduces therefore from (**0**, 7.4.4) and (7.4.3) that u makes A an A_0-algebra *finite*.

Set $\mathfrak{p}_0=\sum_{j=1}^s A_0T_j$, $B_0=(A_0)_{\mathfrak{p}_0}$ and $B=A\otimes_{A_0}B_0$, so that $u_{\mathfrak{p}_0}:B_0\to B$ makes B a B_0-algebra finite; moreover, if $\mathfrak{p}'$ is the ideal of B generated by $\mathfrak{p}$, on has $B_{\mathfrak{p}'}=A_\mathfrak{p}$; as $\mathfrak{p}'$ contains $\mathfrak{p}_0B_0$ by construction, it is above the maximal ideal $\mathfrak{p}_0B_0$ of B_0. Let us show that the morphism $\mathrm{Spec}(B)\to\mathrm{Spec}(B_0)$ is *not ramified* at the point $\mathfrak{p}'$: indeed this follows from the fact that $k(\mathfrak{p}')$ is a finite extension of the field $k(\mathfrak{p}_0)$ of characteristic 0, hence is *necessarily separable*, and that $B_{\mathfrak{p}'}/\mathfrak{p}_0B_{\mathfrak{p}'}=k(\mathfrak{p}')$ by virtue of the choice of the x_i for $1\leqslant j\leqslant s$ (17.4.1). **IV-4 | 173**

Let us now note the following lemma:

Lemma (18.11.3.5). — *Let* k *be a field,* R*,* S *two Noetherian complete local* k*-algebras, such that, if* K *is the residue field of* S*,* $\Omega^1_{K/k}$ *is of finite rank over* K*; let* $u:R\to S$ *be a* k*-homomorphism making* S *an* R*-algebra finite; then* $\widehat{\Omega}^1_{S/R}=\Omega^1_{S/R}$*, and the sequence*

(18.11.3.6) $$\widehat{\Omega}^1_{R/k}\otimes_R S\xrightarrow{v}\widehat{\Omega}^1_{S/k}\xrightarrow{w}\Omega^1_{S/R}\longrightarrow 0$$

(cf. (**0**, 20.7.17.3)*) is exact.*

Proof. The first assertion follows from the fact that $\Omega^1_{S/R}$ is an S-module of finite type (**0**, 20.4.7) and that w is surjective; but it follows from (18.11.1) that $\widehat{\Omega}^1_{S/k}$ is an S-module of finite type; on concludes that every sub-S-module of $\widehat{\Omega}^1_{S/k}$ is closed (**0**, 7.3.5), whence the lemma. □

Applying this lemma to the case where $R = A_0$, $S = A$, and noting that $\widehat{\Omega}^1_{A_0/k}$ is an A_0-module free of rank n (**0**, 21.9.3); hence on a $\widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A = \widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A$ and this is an A-module free of rank n. Since Spec(B) is not ramified on Spec(B_0) at the point $\mathfrak{p}'$, on a $(\Omega^1_{A/A_0})_{\mathfrak{p}} = \Omega^1_{B_{\mathfrak{p}'}/B_0} = 0$ (16.4.15) and (17.4.1). If one localises the exact sequence (18.11.3.6) (applied to A_0 and A) at $\mathfrak{p}$, one obtains therefore a surjective homomorphism $(\widehat{\Omega}^1_{A_0/k} \otimes_{A_0} A)_{\mathfrak{p}} \to (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$, and the conclusion follows that $\widehat{\Omega}^1_{A/k}$ is a B-module of rank n, and the ring being complete and integral by the Cohen theorem (**0**, 19.8.8), (ii)) and the fact that the field of fractions of A is of characteristic 0. Hence the $A_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ admits a system of n generators. But the $A_{\mathfrak{p}}$-module $\widehat{\Omega}^1_{A/k}$ is of *rank n*, by virtue of (**0**, 21.9.5), which is applicable to the ring being complete and integral by the Cohen theorem (**0**, 19.8.8), (ii)) and the fact that the field of fractions of A is of characteristic 0. Hence the $A_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ admits also a system of n generators, and since it is of rank n, this quotient is necessarily *free*, whence the conclusion. □

Lemma (18.11.4). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k. Let k' be an arbitrary extension of k, and set $A' = A\widehat{\otimes}_k k'$. Then:*

- *(i) A' is a semi-local Noetherian ring, complete, composed as a direct product of complete local rings A'_i ($1 \leqslant i \leqslant r$) which are A-modules faithfully flat, and whose residue fields are finite extensions of k'; if $\mathfrak{m}$ is the maximal ideal of A, $\mathfrak{m}A'$ is an ideal of definition of A'; and $\dim(A'_i) = \dim(A)$ for all i.*
- *(ii) For all i, $\widehat{\Omega}^1_{A/k}$ is canonically isomorphic to $\widehat{\Omega}^1_{A/k} \otimes_A A'_i$.*

Proof. The first assertions follow immediately from (7.5.5) and (**0**, 6.6.2); the fact that $\mathfrak{m}A'$ is an ideal **IV-4** | 174
of definition of A' also follows from (7.5.5), since K is the residue field of A, $K \otimes_k k'$ is a finite k'-algebra. Finally $A'_i/\mathfrak{m}A'_i$ is one of the Artinian local rings whose direct product is $A' \otimes_k k'$; hence as A'_i is an A-module flat, the equality of dimensions of A and A'_i follows from (6.1.2).

(ii) As $\widehat{\Omega}^1_{A/k}$ is, by the hypothesis on K, an A-module of finite type (18.11.1), $\widehat{\Omega}^1_{A/k} \otimes_A A'$ is complete and identifies with the completed tensor product $\widehat{\Omega}^1_{A/k}\widehat{\otimes}_A A'$; by virtue of the associativity of the completed tensor product (**0**, 7.7.4), $\widehat{\Omega}^1_{A/k} \otimes_A A' = \widehat{\Omega}^1_{A/k} \otimes_A (A\widehat{\otimes}_k k')$ identifies with $\widehat{\Omega}^1_{A/k}\widehat{\otimes}_k k'$. But $\Omega^1_{A/k} \otimes_k k'$ identifies with $\Omega^1_{A''/k'}$, where $A'' = A \otimes_k k'$ (**0**, 20.5.5), and as A' is by definition the separated completion of A'', the separated completion $\widehat{\Omega}^1_{A''/k'}$ identifies by construction to $\widehat{\Omega}^1_{A'/k'}$ (**0**, 20.7.4); furthermore, if $\mathfrak{r}$ is the radical of A', the $\mathfrak{r}$-preadique topology on $\Omega^1_{A'/k'}$ identifies with the product of the $\mathfrak{m}'_i$-preadique topologies on the $\Omega^1_{A'_i/k'}$ (where $\mathfrak{m}'_i$ is the maximal ideal of A'_i); finally, it follows that the separated complete $\widehat{\Omega}^1_{A'/k'}$ for the $\mathfrak{r}$-preadique topology identifies with the product of the separated completions $\widehat{\Omega}^1_{A'_i/k'}$ for the $\mathfrak{m}'_i$-preadique topologies, and it suffices to use (**0**, 20.4.5).

The conclusion of (ii) follows now from the fact that $\Omega^1_{A'/k'}$ is a direct sum of the $\Omega^1_{A'_i/k'}$ (**0**, 20.4.13), and if $\mathfrak{r}$ is the radical of A', the $\mathfrak{r}$-preadique topology on $\Omega^1_{A'/k'}$ identifies with the product of the $\mathfrak{m}'_i$-preadique topologies on the $\Omega^1_{A'_i/k'}$ (where $\mathfrak{m}'_i$ is the maximal ideal of A'_i); the separated complete $\widehat{\Omega}^1_{A'/k'}$ for the $\mathfrak{r}$-preadique topology identifies therefore with the product of the separated completions $\widehat{\Omega}^1_{A'_i/k'}$ for the $\mathfrak{m}'_i$-preadique topologies, hence one has a canonical isomorphism

$$\widehat{\Omega}^1_{A'/k'} \xrightarrow{\sim} \widehat{\Omega}^1_{A/k} \otimes_A A'.$$

□

Proposition (18.11.5). — *Let k be a field, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal $\mathfrak{m}$, such that there exists a* *unique minimal prime ideal* $\mathfrak{q} \subset \mathfrak{p}$ *for which* $\dim(A/\mathfrak{q}) = \dim(A)$ *(which will in particular hold when* A *is equidimensional), m an integer* $\geqslant 0$*. The following conditions are equivalent:*

- *a) The* $A_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ *admits a system of m generators.*
- *b) There exists a local k-homomorphism* $u : B \to A$*, where* $B = k[[T_1, \dots, T_m]]$*, making* A *a* B*-algebra finite, and such that the corresponding morphism* $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ *is étale at the point* $\mathfrak{p}$*.*

Proof. To prove that *b*) implies *a*), let us note that by virtue of the lemma (18.11.3.5), on has the exact sequence

$$\widehat{\Omega}^1_{B/k} \otimes_B A \longrightarrow \widehat{\Omega}^1_{A/k} \longrightarrow \Omega^1_{A/B} \longrightarrow 0$$

since A is a B-algebra finite; localising at $\mathfrak{p}$ and noting that by hypothesis $(\Omega^1_{A/B})_{\mathfrak{p}} = 0$ (17.4.1), one obtains a surjective homomorphism $(\widehat{\Omega}^1_{B/k} \otimes_B A)_{\mathfrak{p}} \to (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$, and the conclusion follows that **IV-4** | 175 $\widehat{\Omega}^1_{B/k}$ is a B-module of rank m (**0**, 21.9.3).

To prove that *a*) implies *b*), let us first prove the following lemmas. □

Lemma (18.11.5.1). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k,* $\mathfrak{p}$ *a prime ideal of* A*,* $\mathfrak{q}$ *a minimal prime ideal of* A *contained in* $\mathfrak{p}$*. Then the minimum number of generators of the* $A_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ *is at least equal to* $\dim(A/\mathfrak{q})$*.*

Proof. Set $n = \dim(A/\mathfrak{q})$, and let m be the minimum number of generators of the $A_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$, which is equal to $\mathrm{rg}_{k(\mathfrak{p})}((\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p}))$ (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, cor. 2 of prop. 4). Note that $(\widehat{\Omega}^1_{A/k})_{\mathfrak{q}} = (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{q}}$, hence the minimum number of generators of the $A_{\mathfrak{q}}$-module $(\widehat{\Omega}^1_{A/k})_{\mathfrak{q}}$ is *at most* equal to m. It suffices therefore to consider the case where $\mathfrak{p} = \mathfrak{q}$ is minimal. In second place, let us show one can restrict to the case where k is *algebraically closed*. Indeed, let k' be an algebraic closure of k, and set $A' = A\widehat{\otimes}_k k'$, which is composed as a direct product of k'-local algebras A'_i $(1 \leqslant i \leqslant r)$, whose residue fields are isomorphic to k', and which are A-modules faithfully flat (18.11.4). Applying (2.3.4) and (6.1.1), one sees that in a given A'_i, there exists a minimal prime ideal $\mathfrak{q}'_i$ above $\mathfrak{q}$ such that $n' = \dim(A'_i/\mathfrak{q}'_i) \geqslant \dim(A/\mathfrak{q}) = n$. On the other hand by (18.11.4), the minimum m' of generators of the $(A'_i)_{\mathfrak{q}'_i}$-module $(\widehat{\Omega}^1_{A'_i/k'})_{\mathfrak{q}'_i}$ is *at most* equal to m; whence our assertion.

Suppose therefore k algebraically closed and $\mathfrak{q} = 0$. In the case where $\mathfrak{q} = 0$, one can moreover restrict to the case where A is integral and K separable over the algebraically closed field k, hence A is a geometrically regular algebra (6.7.6); one can therefore apply (18.11.3) for $\mathfrak{p} = 0$ since $k = k^p$ and in this case $m = n$. □

Lemma (18.11.5.2). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k,* $\mathfrak{m}$ *its maximal ideal,* $\mathfrak{p} \subset \mathfrak{m}$ *a prime ideal of* A*. Let m be the minimum number of generators of the* $A_{\mathfrak{p}}$*-module* $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$*, and let* $(x_i)_{1 \leqslant i \leqslant m}$ *be a family of elements of* $\mathfrak{m}$ *not belonging to* $\mathfrak{p}$*. Then there exist m invertible elements* u_i $(1 \leqslant i \leqslant m)$ *of* A *such that if* $y_i = u_i x_i$*, the canonical images of the* dy_i *in* $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ *generate this* $A_{\mathfrak{p}}$*-module.*

Proof. For all $x \in A$, denote by $\delta(x)$ the canonical image of dx $(= d_{A/k}x)$ in $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p}) = \widehat{\Omega}^1_{A/k} \otimes_A k(\mathfrak{p})$; as this is a $k(\mathfrak{p})$-vector space of dimension m by hypothesis, it suffices (by virtue of the Nakayama lemma) to prove that one can determine the u_i such that the $\delta(u_i x_i)$ form a free system. We reason by induction and suppose for an integer $r < m$, we have determined the u_i $(1 \leqslant i \leqslant r)$ such that the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$ are linearly independent over $k(\mathfrak{p})$; if $\delta(x_{r+1})$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$, one takes $u_{r+1} = 1$ to continue the induction. In the contrary case, note that for $u \in A$, $\delta(u x_{r+1})$ is the canonical image of $(du)x_{r+1} + u(dx_{r+1})$; as $u(dx_{r+1})$ has a canonical image which is a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$, there **IV-4** | 176 exists therefore $z \in A$ such that $\delta(z)$ is not a linear combination of the $\delta(u_i x_i)$ for $1 \leqslant i \leqslant r$; if $z \notin \mathfrak{m}$, one takes $u = z$; otherwise, $1 + z$ is invertible and $d(1+z) = d(z)$ since $(du)x_{r+1} = \delta(z)$; one takes then $u = 1 + z$, which finishes the proof of (18.11.5.2). □

Proof of (18.11.5), *continued.* We now return to the proof of the implication *a*) ⇒ *b*) in (18.11.5). Let us first note that since $\mathfrak{p} \neq \mathfrak{m}$, there exists a system of parameters $(x_i)_{1 \leqslant i \leqslant n}$ of A (with $n = \dim(A)$) such that $x_i \notin \mathfrak{p}$ for all i, without which $A/\mathfrak{p}$ would be of finite length, and $\mathfrak{p}$ would be maximal, contrary to the hypothesis. But if for example $x_1 \notin \mathfrak{p}$, it suffices, for each index i such that $x_i \in \mathfrak{p}$, to replace x_i by $x_1 + x_i$, to have a system of parameters none of whose elements belong to $\mathfrak{p}$. The relation $\dim(\mathrm{A}/\mathfrak{q}) = n$ implies, by virtue of (18.11.5.1), that $m \geqslant n$; one can therefore consider a family $(x_i)_{1\leqslant i\leqslant m}$ of elements $x_i \notin \mathfrak{p}$, whose n first form a system of parameters of A. Multiplying furthermore the x_i by invertible elements u_i of A, one can, by (18.11.5.2), suppose that the images of dx_i $(1 \leqslant i \leqslant m)$ in $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ generate this $\mathrm{A}_{\mathfrak{p}}$-module, and the multiplication by the u_i has not altered the fact that the x_i for $1 \leqslant i \leqslant n$ form a system of parameters. We shall consider the local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$ such that $u(\mathrm{T}_i) = x_i$ for $1 \leqslant i \leqslant m$ (Bourbaki, *Alg. comm.*, chap. III, § 4, n$^{\mathrm{o}}$ 5, prop. 6); as the x_i generate an ideal of definition of A, it follows from (**0**, 7.4.4) and (7.4.3) that u makes A a B-algebra *finite*.

But the $d_{\mathrm{A}}x_i$ are the canonical images by v of the elements $d_{\mathrm{B}}\mathrm{T}_i \otimes 1$ (**0**, 20.5.2.6). If one localises the exact sequence preceding at $\mathfrak{p}$, we see therefore that $v_{\mathfrak{p}} : (\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}} \to (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is *surjective*, and consequently $0 = (\Omega^1_{\mathrm{A}/\mathrm{B}})_{\mathfrak{p}} = \Omega^1_{\mathrm{A}_{\mathfrak{p}}/\mathrm{B}_{\mathfrak{r}}}$ (where $\mathfrak{r}$ is the inverse image of $\mathfrak{p}$ in A, cf. (16.4.15)), thus the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at $\mathfrak{p}$ by (17.4.1), and the property b) of (18.11.5) follows. □

Remark (18.11.6). — In the statement of (18.11.5), one can drop the hypothesis $\mathfrak{p} \neq \mathfrak{m}$. Indeed, for every local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$, where $\mathrm{B} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_r]]$ (r any integer) making A a B-algebra finite, $\mathfrak{m}$ is the only point of Spec(A) above the maximal ideal $\mathfrak{n}$ of B; the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ cannot be unramified at $\mathfrak{m}$ unless $\mathrm{A}/\mathfrak{n}\mathrm{A}$ is a field, an extension separable of k (17.4.1), which implies that the residue field K of A is a separable extension of k. If this condition is not satisfied, the conclusion of (18.11.5) can never be satisfied for $\mathfrak{p} = \mathfrak{m}$, whatever the integer m.

Corollary (18.11.7). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, equidimensional, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of* A*. Then the minimum number of generators of the $\mathrm{A}_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is at least equal to* $\dim(\mathrm{A})$.

Proof. It suffices to set $\mathfrak{p} = 0$ in (18.11.7). □

Corollary (18.11.8). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, integral, whose residue field is a finite extension of k. If* K *is the field of fractions of* A*, one has* $\mathrm{rg}_{\mathrm{K}}((\widehat{\Omega}^1_{\mathrm{A}/k}) \otimes_{\mathrm{A}} \mathrm{K}) \geqslant \dim(\mathrm{A})$.

Proof. It suffices to set $\mathfrak{p} = 0$ in (18.11.7). □

IV-4 | 177

Corollary (18.11.9). — *Let k be a field,* A *a k-algebra, local Noetherian and complete, of dimension n, whose residue field is a finite extension of k. Let $\mathfrak{p}$ be a prime ideal of* A *distinct from the maximal ideal, and containing a minimal prime ideal $\mathfrak{q}$ such that* $\dim(\mathrm{A}/\mathfrak{q}) = n$*. Suppose that the $\mathrm{A}_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ admits n generators. Then:*

- *(i) The $\mathrm{A}_{\mathfrak{p}}$-module $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is free of rank n.*
- *(ii) There exists a local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$, where $\mathrm{B} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$, making* A *a* B*-algebra finite, and such that the corresponding morphism* $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ *is étale at the point $\mathfrak{p}$.*
- *(iii) The k-algebra $\mathrm{A}_{\mathfrak{p}}$ is geometrically regular over k.*

Proof. Let us prove first of all (ii); by virtue of (18.11.5), there exists a local k-homomorphism $u : \mathrm{B} = k[[\mathrm{T}_1, \ldots, \mathrm{T}_n]] \to \mathrm{A}$ making A a B-algebra finite, and such that the corresponding morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at the point $\mathfrak{p}$; set $\mathfrak{r} = u^{-1}(\mathfrak{p})$. The hypothesis on $\mathfrak{q} \subset \mathfrak{p}$ and the fact that A is a quotient ring of a regular ring (**0**, 19.8.8) imply that $\dim(\mathrm{A}_{\mathfrak{p}}) = n - \dim(\mathrm{A}/\mathfrak{p})$ (**0**, 16.5.12). Similarly $\dim(\mathrm{B}_{\mathfrak{r}}) = n - \dim(\mathrm{B}/\mathfrak{r})$. Finally, since the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at the point $\mathfrak{p}$, the fibre of this morphism at the point $\mathfrak{r}$ is of dimension 0, hence (**0**, 16.3.9) $\dim(\mathrm{A}/\mathfrak{p}) \leqslant \dim(\mathrm{B}/\mathfrak{r})$ and consequently $\dim(\mathrm{B}_{\mathfrak{r}}) \leqslant \dim(\mathrm{A}_{\mathfrak{p}})$. We conclude therefore from (18.10.1) that the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is étale at $\mathfrak{p}$.

The assertion (iii) follows from the fact that $\mathrm{A}_{\mathfrak{p}}$ is formally smooth and that $\mathrm{B}_{\mathfrak{r}}$ is formally smooth on k for their preadique topologies (**0**, 19.3.4) and (19.3.5)), hence $\mathrm{A}_{\mathfrak{p}}$ is formally smooth on k for its preadique topology, and consequently geometrically regular over k (**0**, 22.5.8).

Let us finally prove (i). Let k' be an algebraically closed extension of k, and let us consider the k'-algebra semi-local $A' = A\widehat{\otimes}_k k'$. As a B-module of finite type via u, A is considered as a B-algebra finite; since the morphism $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ is finite and étale at every point $\mathfrak{p}'$ above $\mathfrak{p}$ (17.3.3); furthermore, A' is composed as a direct product of local rings A'_i of dimension n (18.11.4) and $\mathfrak{p}'$ identifies with a prime ideal $\mathfrak{p}'_i$ of one of the A'_i. The same reasoning as above proves that $(A'_i)_{\mathfrak{p}'_i}$ is geometrically regular over k', and as k' is perfect, the fact d) implies b), hence by faithful flatness (18.11.4) and (2.5.2), $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of finite type. But $(\widehat{\Omega}^1_{A'_i/k'})_{\mathfrak{p}'_i} = (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} (A'_i)_{\mathfrak{p}'_i}$ (18.11.4) and $(A'_i)_{\mathfrak{p}'_i}$ is a faithfully flat $A_{\mathfrak{p}}$-module; one sees that $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module *free* of finite type (2.5.2); this finishes the proof. □

Theorem (18.11.10). — *Let k be a field of characteristic exponent p, A a k-algebra, local Noetherian and complete, whose residue field is a finite extension of k, $\mathfrak{p}$ a prime ideal of A distinct from the maximal ideal. The following conditions are equivalent:* **IV-4 | 178**

- *a) For every extension k' of k, and every prime ideal $\mathfrak{p}'$ of $A' = A\widehat{\otimes}_k k'$ above $\mathfrak{p}$, $A_{\mathfrak{p}'}$ is a regular ring.*
- *a′) There exists a perfect extension k' of k and a prime ideal $\mathfrak{p}'$ of $A' = A\widehat{\otimes}_k k'$ above $\mathfrak{p}$, such that $A_{\mathfrak{p}'}$ is regular.*
- *b) Let n be the largest of the dimensions of the irreducible components of* $\mathrm{Spec}(A)$ *to which $\mathfrak{p}$ belongs; then $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of rank n.*

Suppose furthermore that $n = \dim(A)$ (which will be the case if A is equidimensional); then the preceding conditions are also equivalent to:

- *c) There exists a local k-homomorphism $u : B \to A$, where $B = k[[T_1, \dots, T_n]]$, making A a B-algebra finite, and such that the corresponding morphism* $\mathrm{Spec}(A) \to \mathrm{Spec}(B)$ *is étale at the point $\mathfrak{p}$.*
- *d) The ring $A_{\mathfrak{p}}$ is geometrically regular over k.*

Furthermore, if $[k : k^p] < +\infty$, the condition d) is equivalent to a′), b), and c).

Proof. The fact that d) implies b) when $[k : k^p] < +\infty$ is nothing other than (18.11.3). It is clear that a) implies a'). Let us show c) implies a) when $\dim(A) = n$. One says that A is an A-module flat (18.11.4) and $\dim(A') = \dim(A) = n$; it follows from (2.3.4) and (6.1.1) that there exists a minimal prime ideal $\mathfrak{q}'$ of A' contained in $\mathfrak{p}'$, above $\mathfrak{q}$ and such that $\dim(A'/\mathfrak{q}') = \dim(A/\mathfrak{q}) = n$. Furthermore, by (18.11.4), $(\widehat{\Omega}^1_{A'/k'})_{\mathfrak{p}'} = (\widehat{\Omega}^1_{A/k})_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A'_{\mathfrak{p}'}$; as k' is perfect, the hypothesis implies that it is geometrically regular over k', one can apply to A' and $\mathfrak{p}'$ the fact d) implies b), hence $(\widehat{\Omega}^1_{A'/k'})_{\mathfrak{p}'}$ is an $A_{\mathfrak{p}'}$-module free of rank n; by faithful flatness (18.11.4) and (2.5.2), $(\widehat{\Omega}^1_{A/k})_{\mathfrak{p}}$ is an $A_{\mathfrak{p}}$-module free of rank n. This shows the equivalence of a), a'), b), and c) when $\dim(A) = n$. It remains to prove that a), a'), and b) are still equivalent in the general case. Let $\mathfrak{J}$ be the kernel of A, the noyau of the canonical homomorphism $A \to A_{\mathfrak{p}}$, and set $A_1 = A/\mathfrak{J}$; we necessarily have $\mathfrak{J} \subset \mathfrak{p}$, and if we set $\mathfrak{p}_1 = \mathfrak{p}/\mathfrak{J}$, the canonical homomorphism $A_{\mathfrak{p}} \to (A_1)_{\mathfrak{p}_1}$ is *bijective*; it follows from (**I**, 6.5.4) that the canonical injection $\mathrm{Spec}(A_1) \to \mathrm{Spec}(A)$ is an *isomorphism local* at the point $\mathfrak{p}_1$, and one has $\mathfrak{J}_{\mathfrak{p}} = 0$. One sees as in (18.11.3) that the sequence is exact, and that one concludes that the condition a) (resp. a'))) for the ring A and the prime ideal $\mathfrak{p}$ is *equivalent* to the condition a) (resp. a'))) for the ring A_1 and the prime ideal $\mathfrak{p}_1$. Now, all the minimal prime ideals of A contained in $\mathfrak{p}$ contain $\mathfrak{J}$ since $\mathrm{Spec}(A_1) \to \mathrm{Spec}(A)$ is an isomorphism local at $\mathfrak{p}_1$; on the other hand the $\mathrm{Ass}_A(A/\mathfrak{J})$ are the prime ideals of $\mathrm{Ass}(A)$ which are contained in $\mathfrak{p}$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 2, prop. 6); hence all the minimal prime ideals of A_1 are contained in $\mathfrak{p}_1$, and consequently $\dim(A_1) = n$. It suffices then to apply to A_1 and $\mathfrak{p}_1$ which has been proved above. □

Remark (18.11.11). — **IV-4 | 179**

(i) The equivalence of conditions d) and b) in (18.11.10) is no longer valid when one no longer supposes that $[k : k^p] < +\infty$. Indeed, by the example of (**0**, 22.7.7), (ii)), the ring $B = A/\mathfrak{q}$ is integral and of dimension 1; on the other hand, the sequence

$$(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L \xrightarrow{j} \widehat{\Omega}^1_{A/k} \otimes_A L \longrightarrow \widehat{\Omega}^1_{B/k} \otimes_B L \longrightarrow 0$$

is exact (same demonstration as in (18.11.3)), and we know that *j is not injective*, hence $j = 0$ since $(\mathfrak{q}/\mathfrak{q}^2) \otimes_A L$ is of rank 1. We conclude that $\mathrm{rg}_L(\widehat{\Omega}^1_{B/k} \otimes_B L) = 2$; as $B_{\mathfrak{p}}$ is a k-algebra geometrically regular, one sees that the condition d) does not imply here the condition b).

(ii) With the notations of (18.11.10) and supposing that $n = \dim(\mathrm{A})$ and $(x_i)_{1\leqslant i\leqslant n}$ a system of parameters of A not belonging to $\mathfrak{p}$; one deduces a local k-homomorphism $u : \mathrm{B} \to \mathrm{A}$ such that $u(\mathrm{T}_i) = x_i$ for $1 \leqslant i \leqslant n$, making A a B-algebra finite. For the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ to be *étale* at the point $\mathfrak{p}$, it is necessary and sufficient that the images of $d_{\mathrm{A}}x_i$ in $(\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ form a *system of generators* of this $\mathrm{A}_{\mathfrak{p}}$-module. Indeed, as one has seen in the proof of (18.11.5) that if the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at $\mathfrak{p}$, the canonical homomorphism $(\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}} \to (\widehat{\Omega}^1_{\mathrm{A}/k})_{\mathfrak{p}}$ is surjective and the images of the elements $d_{\mathrm{B}}\mathrm{T}_i \otimes 1$, which generate $(\widehat{\Omega}^1_{\mathrm{B}/k} \otimes_{\mathrm{B}} \mathrm{A})_{\mathfrak{p}}$, are the $d_{\mathrm{A}}x_i$; whence the *necessity* of the condition. Conversely, the same reasoning shows that if this condition is satisfied, the morphism $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is not ramified at $\mathfrak{p}$, and the reasoning of (18.11.9) proves in fact that this morphism is étale at $\mathfrak{p}$.

Corollary (18.11.12). — *Let k be a field of characteristic exponent p such that $[k : k^p] < +\infty$, A a k-algebra, local Noetherian and complete and integral, which is not a field, and whose residue field is a finite extension of k. Then there exists a finite radicial extension k' of k such that, setting $\mathrm{A}' = \mathrm{A} \otimes_k k'$, the k'-algebra $\mathrm{A}'_{\mathrm{red}}$ and the prime ideal $\mathfrak{o}$ of this algebra satisfy the equivalent conditions a), a′), b), c), and d) of (18.11.10). In particular, if $n = \dim(\mathrm{A}) = \dim(\mathrm{A}')$, there exists a k'-homomorphism local $\mathrm{B}' = k'[[\mathrm{T}_1, \dots, \mathrm{T}_n]] \to \mathrm{A}'_{\mathrm{red}}$* **IV-4** | 180 *making $\mathrm{A}'_{\mathrm{red}}$ a B'-algebra finite, and such that the field of fractions K' of $\mathrm{A}'_{\mathrm{red}}$ is a finite separable extension of the field of fractions $k'((\mathrm{T}_1, \dots, \mathrm{T}_n))$ of B'.*

Proof. The morphism $\mathrm{Spec}(\mathrm{A}') \to \mathrm{Spec}(\mathrm{A})$ being complete, radicial, and finite, A' is a local ring, complete and its nilradical is the only prime ideal above $\mathfrak{o}$ of A; by flatness, A' identifies with a sub-ring of $\mathrm{K} \otimes_k k'$, which is also the total ring of fractions of A'; on has therefore $\mathrm{K}' = (\mathrm{K} \otimes_k k')_{\mathrm{red}}$. It is a question of proving, by the equivalence $d) \Leftrightarrow c)$, that there exists a finite radicial extension k' of k such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a separable extension of k'. Or one knows (**0**, 19.8.9) that, under all the hypotheses made, there exists a sub-k-algebra $\mathrm{C} = k[[\mathrm{T}_1, \dots, \mathrm{T}_n]]$ of A such that A is a finite C-algebra; hence, K is a *finite* extension of the field of fractions $\mathrm{K}_1 = k((\mathrm{T}_1, \dots, \mathrm{T}_n))$ of C, which is *separable* on k (**0**, 21.9.6.4). If p is the characteristic exponent of k, one can therefore write $\mathrm{K} \otimes_k k^{p^{-\infty}} = \mathrm{K} \otimes_{k_1} (\mathrm{K}_1 \otimes_k k^{p^{-\infty}})$ and $\mathrm{K}_1 \otimes_k k^{p^{-\infty}}$ is a *field*, a radicial extension of K_1; on concludes that $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is a finite algebra over the field $\mathrm{K}_1 \otimes_k k^{p^{-\infty}}$, hence is *Artinian*. The conclusion follows therefore from the following more general lemma: □

Lemma (18.11.12.1). — *Let k be a field of characteristic $p > 0$, K an extension of k. The following conditions are equivalent:*

a) The ring $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is Artinian.
b) There exists an extension k' of k of finite type such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a k'-algebra separable.
c) There exists a finite radicial extension k' of k such that $(\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is a separable extension of k'.

Proof. Let us first show that $c)$ implies $a)$. The ring $\mathrm{A} = \mathrm{K} \otimes_k k'$ is then a local Artinian ring, and if $\mathfrak{N}$ is nilradical, the residue field $\mathrm{L} = \mathrm{A}/\mathfrak{N}$ is separable over k'; hence $\mathrm{L} \otimes_{k'} k^{p^{-\infty}}$ is separable over k', and as it is equal to $(\mathrm{A} \otimes_{k'} k^{p^{-\infty}})/(\mathfrak{N} \otimes_{k'} k^{p^{-\infty}})$, one sees that $\mathfrak{N} \otimes_{k'} k^{p^{-\infty}}$ is the nilradical of $\mathrm{K} \otimes_k k^{p^{-\infty}}$; as $\mathfrak{N}$ is a nilpotent ideal of finite type, the nilradical of $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is of finite type, hence the ring $\mathrm{K} \otimes_k k^{p^{-\infty}}$ is Artinian.

Conversely, let us prove that $a)$ implies $c)$. Let $\mathfrak{N}$ be the nilradical of the local Artinian ring $\mathrm{B} = \mathrm{K} \otimes_k k^{p^{-\infty}}$, which is by hypothesis generated by a finite number d of elements of the form $y_i = \sum_j x_{ij} \otimes \xi_{ij}$, $x_{ij} \in \mathrm{K}$, $\xi_{ij} \in k^{p^{-\infty}}$. Let k' be the finite radicial extension of k generated by the ξ_{ij}, $\mathfrak{N}_0$ the ideal of $\mathrm{B}_0 = \mathrm{K} \otimes_k k'$ generated by the y_i; it is clear that $\mathfrak{N}_0 \otimes_{k'} k^{p^{-\infty}} = \mathfrak{N}$, and hence $\mathfrak{N} \cap \mathrm{B}_0 = \mathfrak{N}_0$, and so it is equal to $\mathrm{B}_0/\mathfrak{N}_0 = (\mathrm{K} \otimes_k k')_{\mathrm{red}}$ is separable over k (4.6.1).

It is clear that $c)$ implies $b)$. Conversely, suppose $b)$ is satisfied, and let us note that there exists a separable extension k_1 of k such that k' is a finite radicial extension of k_1; set $\mathrm{K}_1 = \mathrm{K} \otimes_k k_1$, which is a field. Applying the equivalence of $a)$ and $c)$ to the extension K_1, on sees that $\mathrm{K}_1 \otimes_{k_1} k_1^{p^{-\infty}}$ is **IV-4** | 181 also Artinian (Bourbaki, *Alg. comm.*, chap. I, § 3, nº 5, cor. of prop. 8); we have thus shown that $b)$ implies $a)$, which finishes the proof. □

18.12. Applications of étale localisation to quasi-finite morphisms The results of this section were communicated by P. Deligne.

Theorem (18.12.1). — *Let $f : X \to Y$ be a locally of finite type morphism, x a point of X, $y = f(x)$. Suppose that x is an isolated point of the space $f^{-1}(y)$. Then there exist an étale morphism $Y' \to Y$, a point x' of $X' = X \times_Y Y'$ above x and a neighbourhood V' of x' in X' such that, if $f' = f_{(Y')} : X' \to Y'$, $f'|V'$ is a finite morphism. Furthermore V' is both open and closed, and X' is therefore the sum of two sub-preschemes induced on open subsets of X', of which one is finite on Y' and contains x'.*

Proof. The last assertion follows from the fact that, if f is separated, it is the same for f'; hence, if $j' : V' \to X'$ is the injection canonical, and the fact that f' is finite implies that it is the same for j' (**II**, 6.1.5), and $V' = j'(V')$ is closed in X' (**II**, 6.1.10).

The question being local on X, one can suppose $Y = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$ affine, B being an A-algebra of finite type, and one can then suppose $f^{-1}(y) = \{x\}$. Hence the question is local on Y, hence of the form $C/\mathfrak{J}$, where $C = A[T_1, \dots, T_r]$ and $\mathfrak{J}$ is an ideal of type finite of E; we then have $A' = E'/\mathfrak{J}E'$, where $E' = C'[T_1, \dots, T_r]$. Let $(\mathfrak{J}_\lambda)$ be the family of ideals of type fine of C contained in $\mathfrak{J}$, and denote the corresponding $X_\lambda = \mathrm{Spec}(C/\mathfrak{J}_\lambda)$ considered as closed sub-preschemes of $Z = \mathrm{Spec}(C)$, so that $\mathfrak{J}$ is the filtered union of the $\mathfrak{J}_\lambda$. If X and the $X_\lambda = \mathrm{Spec}(C/\mathfrak{J}_\lambda)$ are considered as closed sub-preschemes, $X = \bigcap X_\lambda$; if $f_\lambda : X_\lambda \to Y$ is the structural morphism, and as the sets $f_\lambda^{-1}(y)$ are closed in the Noetherian space $Z_y = \mathrm{Spec}(k(y)[T_1, \dots, T_r])$, there exists an index λ such that $f_\lambda^{-1}(y) = f^{-1}(y) = \{x\}$. One can then suppose that for a given X_λ, the condition holds that X is at point x; if one proves the proposition for such an X_λ, it will follow for X, since x' is a point above x in $Z' = Z \times_Y Y'$, $X'_\lambda = p^{-1}(X_\lambda)$, where $p : Z' \to Z$ is the projection (from the restriction of f_λ to X_λ), the same will hold for the open set V'_λ of x' in X'_λ which will be finite over Y'. Furthermore V' and X' are sums of two open sub-preschemes, V' and X'_λ; one answers the question by taking V' as a sum of U' and V'. In particular, note that the set of isolated points of X in their fibre et open in X (13.1.4), hence f is *quasi-finite*. Set $A'' = \mathscr{O}_{Y,y}$, and set $Y'' = \mathrm{Spec}(A'')$. Set $X'' = X \times_Y Y''$, $f'' = f_{(Y'')} : X'' \to Y''$; $X'' \to Y''$ is *separated*, quasi-finite and of presentation finite; A'' is henselian, for every $x'' \in X''$ above x an open and closed neighbourhood V'' of x'' in X'' which is finite on Y'' (18.5.11), *c*)); as the immersion $V'' \to X''$ is open and closed, it is also quasi-compact, hence $f''|V''$ is of presentation finite since $V'' \to X''$ is open, and $X'' \to Y''$ is of presentation finite. f'' is finite and quasi-compact over Y'' (1.6.2), hence $f''|V''$ is of presentation finite.

Since, by definition, A'' is the inductive limit of a filtered inductive family (A_λ) of $\mathscr{O}_{Y,y}$-algebras strictly étales ((18.6.5)) and (8.1.2), *a*))); also B_{M_λ} can be assumed to be a $\mathscr{O}_{Y,y}$-algebra of presentation finite, and it follows from (18.6.1) and (8.1.2), *a*)) and the result of (17.7.8) that B_{M_λ} is a $\mathscr{O}_{Y,y}$-algebra étale for all λ; Namely, Y'' is the inductive limit projective of a family (Y_α) of affine Y-schemes étale. If x''_α is the projection of x'' on $X'_\alpha = X \times_Y Y_\alpha$, on can suppose that $X'_\alpha \cap f'^{-1}(U)$ where V'_α is a neighbourhood of x''_α in X'_α, is finite on Y_α (8.2.11). Finally, V'' being of presentation finite over Y'', on concludes from (8.10.5), (x)) that for a suitable α, V'_α is finite over Y'_α. □

Remark (18.12.2). — In the preceding proof, one sees (taking into account (18.6.2)) that one has constructed Y' answering the question, and in addition such that if $y' = f'(X')$, the homomorphism $k(y) \to k(y')$ is bijective.

Corollary (18.12.3). — *Let $f : X \to Y$ be a locally of finite type morphism and separated, y a point of Y such that the underlying space $f^{-1}(y)$ is finite and discrete; in particular, a proper and quasi-finite morphism. Then there exist an étale morphism $Y' \to Y$, a point $y' \in Y'$ above y such that $k(y') = k(y)$, and a decomposition of $X' = X \times_Y Y'$ as a sum of two sub-preschemes X'_1, X'_2 induced on open subsets of X', such that the restriction $f' = f_{(Y')} : X'_1 \to Y'$ is a finite morphism and that $X'_2 \cap f'^{-1}(y') = 0$.*

Proof. Indeed, one can apply the corollary (18.12.3) at any point whatsoever y of Y. Since f' is a closed morphism, hence, since X'_2 is closed in X', $f'(X'_2)$ is closed in Y' and does not contain y'; hence there exists by construction an open neighbourhood U' of y' in Y' such that $f'^{-1}(U') = X \times_Y U'$ is finite on Y', and the morphism $U' \to Y$ is étale; the morphism $f'^{-1}(U') \to U'$ is faithfully flat and quasi-compact, hence the conclusion. □ **IV-4** | 182

Corollary (18.12.4). — *Let $f : X \to Y$ be a finite morphism, it is necessary and sufficient that it be separated, universally closed and, for all $y \in Y$, $f^{-1}(y)$ is a finite and discrete space; in particular, a proper and quasi-finite morphism.*

Proof. Indeed, one can apply the corollary (18.12.3) at any point whatsoever y of Y. Since f' is a closed morphism, hence, since X'_2 is closed in X', $f'(X'_2)$ is closed in Y' and does not contain y'; hence there exists by construction an open neighbourhood U' of y' in Y' such that $f'^{-1}(U') = X \times_Y U'$ is finite over Y', and the morphism $U' \to Y$ is étale; the morphism $U' \to Y$ is faithfully flat and quasi-compact, hence by (2.7.1), (xv)), the restriction of f to X_λ, being a finite morphism (**II**, 6.1.10), hence the conclusion results from (11.3.1), since f restricted is finite and quasi-compact. □

Remark (18.12.5). — One notes that in the proof of (18.12.4) one has not used the hypothesis (18.12.4) in its general form, but only that $f_{(Y')}$ is a universally closed morphism for every étale morphism $Y' \to Y$.

Corollary (18.12.6). — *Let $f : X \to Y$ be a morphism of preschemes. The following conditions are equivalent:*

a) f is a closed immersion.
b) f is a proper monomorphism.
c) f is proper and, for all $y \in Y$, the $k(y)$-prescheme $X_y = f^{-1}(y)$ is radicial and geometrically reduced on $k(y)$ (i.e. empty or $k(y)$-isomorphic to $\mathrm{Spec}(k(y))$*).*

Proof. This is deduced from (18.12.4) as (18.11.5) is deduced from (8.11.1). □

Proposition (18.12.7). — *Let $f : X \to Y$ be a locally of finite type morphism, y a point of Y. For there to exist a neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is a closed immersion, it is necessary and sufficient that X_y be a $k(y)$-prescheme radicial and geometrically reduced on $k(y)$ (i.e. empty or $k(y)$-isomorphic to* $\mathrm{Spec}(k(y))$*) and that there exist a neighbourhood V of y such that the restriction $f^{-1}(V) \to V$ of f is a universally closed morphism.*

Proof. The conditions being evidently necessary, let us prove that they are sufficient. One can, by the second condition, already suppose f universally closed. Let us show, moreover, hence separated; this follows from the following lemma: □

Lemma (18.12.7.1). — *Let $f : X \to Y$ be a closed morphism, $y \in Y$ a point such that every neighbourhood of X_y contains an affine neighbourhood of X_y (condition always satisfied when X_y is empty or reduced to a single point). Then there exists a neighbourhood U of y such that the restriction $f^{-1}(U) \to U$ of f is an affine morphism.*

Proof. Indeed, let U_0 be an affine open neighbourhood of y in Y and let V be a neighbourhood of X_y contained in $f^{-1}(U_0)$ such that $f^{-1}(U) \subset V$, which is affine. As f is closed and V open, there exists a neighbourhood $U \subset U_0$ of y such that $f^{-1}(U) \subset V$. As the restriction $f^{-1}(U) \to U$ is also an affine morphism (**II**, 1.2.5). □

Suppose therefore f affine and universally closed, and let us show that for all $y \in Y$ such that the restriction $f^{-1}(U) \to U$ of f is a closed immersion, it suffices to prove that f is quasi-finite (18.12.4), and since X_y is empty or reduced to a single point, f is quasi-finite (18.12.4), taking into account (18.12.6).

Proposition (18.12.8). — *Let $f : X \to Y$ be a morphism of preschemes. For f to be entire, it is necessary and sufficient that f be affine and universally closed.*

Proof. We can suppose $X = \mathrm{Spec}(B)$, and it suffices to prove that every element $b \in B$ is entire on A (**II**, 6.1.1). Let B' be the sub-A-algebra of B generated by b, which is an A-algebra of finite type; set $X' = \mathrm{Spec}(B')$, so that $f : X \to Y$ factorises as $X \xrightarrow{g} X' \xrightarrow{h} Y$, where h is of finite type, and g is *dominant* since the homomorphism $B' \to B$ is injective (**I**, 1.2.7). As h is separated and $h \circ g$ universally closed, g is universally closed (**II**, 5.4.3) and (5.4.9)); hence h is also universally closed, **IV-4** | 183
therefore proper since it is of finite type and separated. But then, for all $y \in Y$, the morphism $h^{-1}(y) \to \mathrm{Spec}(k(y))$ deduced from h is proper and affine, hence *finite* (**III**, 4.4.2), in other words

h is *quasi-finite*; one deduces therefore from (18.12.4) that h is finite, and consequently b is entire on A. ☐

Remark (18.12.9). — One can say that a morphism $f : \mathrm{X} \to \mathrm{Y}$ is entire when one supposes only separated, universally closed, and such that for all $y \in \mathrm{Y}$, the morphism $f^{-1}(y) \to \mathrm{Spec}(k(y))$ deduced from f is entire. It would suffice to prove that these conditions imply that f is affine, or again that every fibre X_y is contained in an affine open neighbourhood.

Corollary (18.12.10). — *A morphism $f : \mathrm{X} \to \mathrm{Y}$ which is injective and universally closed is entire.*

Proof. It suffices, by (18.12.8), to prove that f is affine, which follows from the lemma (18.12.7.1) and the hypothesis. ☐

Corollary (18.12.11). — *Let $f : \mathrm{X} \to \mathrm{Y}$ be a morphism of preschemes (resp. a locally of finite type morphism). For f to be a universal homeomorphism* (2.4.2), *it is necessary and sufficient that f be entire (resp. finite), radicial, and surjective.*

Proof. One knows that the conditions are sufficient (2.4.5); they are necessary by virtue of (18.12.10) and (18.12.4). The last proposition improves (8.11.2). ☐

Proposition (18.12.12). — *Every morphism $f : \mathrm{X} \to \mathrm{Y}$ which is quasi-finite and separated is quasi-affine.*

Proof. Set $\mathscr{A} = f_*(\mathscr{O}_\mathrm{X})$, which is an $\mathscr{O}_\mathrm{Y}$-Algebra quasi-coherent and separated (1.7.4); and $\mathrm{Z} = \mathrm{Spec}(\mathscr{A})$ (**II**, 1.3.1), so that f factorises canonically as $\mathrm{X} \xrightarrow{g} \mathrm{Z} \xrightarrow{h} \mathrm{Y}$, h being affine and g corresponding to the automorphism identical to $\mathscr{A}$ (**II**, 1.2.7); it suffices to prove that g is an *open immersion*, or, what comes to the same (17.9.1), that the morphism g is étale and *radicial*. It suffices evidently to prove that, for all $y \in \mathrm{Y}$, the morphism g is étale and radicial at each point of $f^{-1}(y)$. One can apply the result to X, f the restriction and Y by X, Y, and the sections of f of f; hence f is quasi-compact and separated, and the fact that f is étale and radicial at each point of $f^{-1}(y)$ is part of (18.10.2), (iii)), quasi-separable, from X, f, and g as in (18.12.3), whose X' is the extension $\mathrm{Z}' = \mathrm{Z} \times_\mathrm{Y} \mathrm{Y}'$ a $\mathscr{O}_{\mathrm{Y}'}$-Module $\mathscr{O}_{\mathrm{X}'}/\mathfrak{J}\mathrm{E}'$, so that as X and Y are both Y-schemes étale (17.3.2); hence by (**II**, 1.2.7) f factorises as $\mathrm{X} \xrightarrow{g} \mathrm{Z} \xrightarrow{h} \mathrm{Y}$, where g is the canonical injection from the identity on $\mathscr{A}$, and g is an *open immersion*, and g is *radicial*, whence the result. ☐

Corollary (18.12.13). — *("Main Theorem" of Zariski). Let Y be a prescheme quasi-compact and quasi-separated. Then there exists a factorisation of f*

$$\mathrm{X} \xrightarrow{g} \mathrm{Z} \xrightarrow{u} \mathrm{Y}$$

where g is an open immersion (necessarily quasi-compact) and u a finite morphism.

Proof. Indeed, it follows from (18.12.12) that f is quasi-affine, hence quasi-projective. One can therefore apply (8.12.8), where the hypothesis on $\mathscr{O}_\mathrm{Y}$-Module ample can in fact be replaced by the sole hypothesis that Y is quasi-compact and quasi-separated: indeed, it follows from (8.12.3) that the existence of the factorisation announced is a *local* property on Y, and it suffices to prove it when Y is affine. ☐

Remark (18.12.14). — One can give a proof of (18.12.13) analogous to that of (18.12.12) (but not using (8.12.3), hence (18.5.11)), which itself uses the "Main Theorem" under its local form (8.12.9)). Keeping indeed the notations of the proof of (18.12.12), and let $\mathscr{B}$ be the $\mathscr{O}_\mathrm{Y}$-Algebra quasi-coherente, integral closure of $\mathscr{O}_\mathrm{Y}$ in $\mathscr{A}$ (**II**, 6.3.2). If one sets $\mathrm{T} = \mathrm{Spec}(\mathscr{B})$, it suffices, by virtue of (8.12.3), to prove that $g : \mathrm{X} \to \mathrm{T}$ corresponding to the canonical injection $\mathscr{B} \to \mathscr{A}$ is an immersion, and radicial. With the notations of (18.12.12), and one can suppose that $\mathrm{Y} = \mathrm{Spec}(\mathrm{C})$ and $\mathrm{Y}' = \mathrm{Spec}(\mathrm{C}')$ are affine, C' being a C-algebra étale (17.3.2); hence $\mathscr{A} = \overline{\mathrm{A}}$, where A is a C-algebra, B the integral closure of C in A. The algebra A' is a C'-algebra, $\mathrm{A}' = \mathrm{A} \otimes_\mathrm{C} \mathrm{C}'$, and $\mathscr{B} = \overline{\mathrm{B}}$, where B is the integral closure of C in A. As $\mathrm{Z}' = \mathrm{Z} \times_\mathrm{Y} \mathrm{Y}'$ and the X'_λ and X_λ are étale over X, the conclusion follows in this case.

When C is Noetherian, the demonstration of (6.14.4) no longer requires the use of (6.14.1). Indeed, the reductions successives of the proof of (6.14.4) lead back, without using (6.14.1), to the case where C is *integral*, A the field of fractions of C, and the morphism $\mathrm{Spec}(\mathrm{B}') \to \mathrm{Spec}(\mathrm{B})$ is étale, it follows from (17.5.7) that B' is a normal ring, and the reasoning terminates by making appeal only to the easy lemma (6.14.1), and not to the difficult part of the proof of (6.14.1).

IV-4 | 184

Proposition (18.12.15). — *Let* G *be a ring,* C' *a* C*-algebra étale,* A *a* C*-algebra,* B *the integral closure of* C *in* A*; on sets* $\mathrm{A}' = \mathrm{A} \otimes_{\mathrm{C}} \mathrm{C}'$, $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{C}} \mathrm{C}'$, B' *identifying to a sub-algebra of* A'*; then* B' *is the integral closure of* C' *in* A'*.*

Proof. Considering the filtered increasing family of sub-C-algebras of type finite of A and using the fact that A is a C-algebra of type fine, hence of the form $\mathrm{E}/\mathfrak{J}$, where $\mathrm{E} = \mathrm{C}[\mathrm{T}_1, \dots, \mathrm{T}_r]$ and $\mathfrak{J}$ is an ideal of type fine of E; we then have $\mathrm{A}' = \mathrm{E}'/\mathfrak{J}\mathrm{E}'$, where $\mathrm{E}' = \mathrm{C}'[\mathrm{T}_1, \dots, \mathrm{T}_r]$. Let $(\mathfrak{J}_\lambda)$ be the increasing filtered family of the ideals of type fine of C contained in $\mathfrak{J}$ and denote B_λ the integral closure of C in $\mathrm{E}/\mathfrak{J}_\lambda$; if B_λ is the inductive limit of the integral closures B_λ, B is the inductive limit of the B_λ, as one sees by the reasoning of (5.13.4); same reasoning for C and the $\mathrm{B}_\lambda \otimes_{\mathrm{C}} \mathrm{C}'$ for $\beta \geqslant \alpha$. The same, A' is the inductive limit of the integral closures of C' in $\mathrm{E}'/\mathfrak{J}_\lambda \mathrm{E}'$, hence the same reasoning as the preceding shows that $\mathrm{B}' = \mathrm{B} \otimes_{\mathrm{C}} \mathrm{C}'$ is the integral closure of C' in A' once we have reduced to the case where C is Noetherian, the proposition becomes a particular case of (6.14.4). It is necessary however to observe that, when C' is supposed *étale* on C Noetherian, the demonstration of (6.14.4) no longer requires that one use (6.14.1). Indeed, the successive reductions in the proof of (6.14.4) lead back, *without using* (6.14.1), to the case where C is *integral*, A the field of fractions of C, hence B is a normal ring and the morphism $\mathrm{Spec}(\mathrm{B}') \to \mathrm{Spec}(\mathrm{B})$ is étale, it follows from (17.5.7) that B' is a normal ring, and the reasoning terminates by making appeal only to the elementary lemma (6.14.1), and not the difficult part of the proof of (6.14.1). ☐

Proposition (18.12.16). — *Let* $g : \mathrm{Y} \to \mathrm{S}$ *be a quasi-compact morphism,* $f : \mathrm{X} \to \mathrm{Y}$ *a separated and quasi-finite morphism. For every* $\mathscr{O}_{\mathrm{Y}}$*-Module invertible* $\mathscr{L}$*, ample relative to* S (**II**, 4.6.1)*, the* $\mathscr{O}_{\mathrm{X}}$*-Module* $f^*(\mathscr{L})$ *is ample relative to* S*.*

Proof. Indeed, by virtue of (18.12.12), the morphism f is quasi-affine, hence (**II**, 5.1.6) the $\mathscr{O}_{\mathrm{X}}$-Module $\mathscr{O}_{\mathrm{X}}$ is ample for f. We deduce from (**II**, 4.6.13), (iii)) that $\mathscr{O}_{\mathrm{X}} \otimes_{\mathscr{O}_{\mathrm{X}}} f^*(\mathscr{L}^{\otimes n}) = f^*(\mathscr{L}^{\otimes n})$ is ample relative to S for n large enough. But as $f^*(\mathscr{L}^{\otimes n}) = (f^*(\mathscr{L}))^{\otimes n}$, this implies that $f^*(\mathscr{L})$ is ample relative to S (**II**, 4.6.9), (i)). ☐

Corollary (18.12.17). — *Let* Z *be an affine scheme,* $h : \mathrm{X} \to \mathrm{Z}$ *a morphism of finite type,* $\mathscr{L}$ *an* $\mathscr{O}_{\mathrm{X}}$*-Module invertible. With the notations of* (**II**, 4.5.2)*, the conditions of* (**II**, 4.5.2) *(which are equivalent to saying that* $\mathscr{L}$ *is ample) are also equivalent to each of the following:*

- b″) *h is separated, one has* $\mathrm{G}(\varepsilon) = \mathrm{X}$ *and the canonical morphism* $u : \mathrm{G}(\varepsilon) \to \mathrm{Proj}(\mathrm{S})$ *has finite and discrete fibres.*
- b‴) *One has* $\mathrm{G}(\varepsilon) = \mathrm{X}$ *and the canonical morphism u is radicial.*

If $\mathrm{Z} = \mathrm{Spec}(\mathrm{A})$*, the ring* S *is canonically equipped with a structure of graded* A*-algebra, hence* $h = g \circ u$*. Since every radicial morphism is separated* (**I**, 1.8.7.1)*, b‴) implies b″). As the condition b) of* (**II**, 4.5.2) *implies evidently b″), it remains to see that b″) implies* $\mathscr{L}$ *ample. Since* $h = g \circ u$ *is of finite type by hypothesis, u is also quasi-finite and separated* (**I**, 6.3.4)*, hence by the hypothesis b″) implies that u is quasi-finite and separated* (**I**, 5.5.1)*. To prove that* $\mathscr{L}$ *is ample, we shall now apply* (18.12.16)*. Set* $\mathrm{Y} = \mathrm{Proj}(\mathrm{S})$*. Since* X *is quasi-compact, there exists an integer* $n > 0$ *and a finite number of elements* $f_j \in \mathrm{S}_n$ *such that the inverse images* $u^{-1}(\mathrm{D}_+(f_j))$ *cover* X*; one can therefore consider u as a quasi-finite and separated morphism of* X *in the open* $\mathrm{Y}' = \bigcup \mathrm{D}_+(f_j)$ *of* Y*. Let us consider then the* $\mathscr{O}_{\mathrm{Y}'}$*-Module invertible* $\mathscr{L}' = \mathscr{O}_{\mathrm{Y}}(n) \,|\, \mathrm{Y}'$ (**II**, 2.5.8)*; it is very ample relative to* Z (**II**, 4.4.3)*, and consequently it is the same for* $u^*(\mathscr{L}')$ (**II**, 4.4.3).

$\mathscr{L}' = \mathscr{O}_{\mathrm{Y}}(n) \,|\, \mathrm{Y}'$ (**II**, 2.5.8)*; it is very ample relative to* Z (**II**, 4.4.3)*, and as* $u^*(\mathscr{L}') = \mathscr{L}^{\otimes n}$ *by* (**II**, 4.4.3)*,* $\mathscr{L}^{\otimes n}$ *is thus ample relative to* S *(this amounts to saying that it is ample* (**II**, 4.6.6)*), and consequently the same is true of* $\mathscr{L}$ (**II**, 4.5.6)*, (i)).*

We leave it to the reader to state analogous conditions equivalent to those of (**II**, 4.6.3) *for the* $\mathscr{O}_{\mathrm{X}}$*-Modules relatively ample.*